

NURSING FOUNDATIONS

Objectives

- *On completion of this chapter, the audience should be able to:*
  1. *Find the nursing origins.*
  2. *Identify Nightingale's Contributions.*
  3. *Identify nursing definitions mainly Virginia Henderson.*
  4. *Determine the nursing theory concepts.*
  5. *Identify the educational ladder.*
  6. *Determining the Nursing roles.*
  7. *Ability to define the delegation needs.*
  8. *Define the unique nursing skills*

ORIGINS OF NURSING

- Nursing is one of the youngest professions but one of the oldest arts.
- It evolved from the familial roles of nurturing and caretaking.
- Early responsibilities included:
  - *assisting women during childbirth,*
  - *suckling healthy newborns, and ministering to the ill, aged, and*
  - *helpless within households and surrounding communities.*
  - *Its hallmark was caring more than curing.*

Nightingale's Contributions

- *Nightingale changed the negative image of nursing to a positive one.*
- *She is credited with the following:*
  1. *Training women for future work*
  2. *Selecting only those with upstanding characters as potential nurses*
  3. *Improving sanitary conditions for the sick and injured*
  4. *Significantly reducing the death rate of British soldiers by (20%)*
  5. *Providing classroom education and clinical teaching*
  6. *Advocating that nursing education should be lifelong*

Nursing Definitions

- *Nursing is; an **A**rt, **S**cience & **S**pirit*
  - ***A**rt ; At first, the training of nurses consisted of learning the art (ability to perform an act skillfully) of nursing.*
  - ***S**cience ; A science (body of knowledge unique to a particular subject)*
  - ***S**pirit ; the fine sense in side nurse as human (patient is as me, my mom, my dad, my brother …)*
  - **The combination of art, science and spirit make Super Nurse**

Nursing Definitions

- *Virginia Henderson (1897 to 1996).*
- *Her definition, adopted by the International Council of Nurses, broadened the description of nursing to include health promotion, not just illness care. As stated in 1966,*

Virginia Henderson Definition

- *As stated in 1966,*

The unique function of the nurse is to assist the individual, sick or well, in the performance of those

activities contributing to health or its recovery (or to a peaceful death) that he could perform unaided if

he had the necessary strength, will or knowledge. And to do this in such a way as to help him gain

independence as rapidly as possible.

Virginia Henderson Definition

- *As stated in 1966,*

إن وظيفة التمريض الأساسية هي مساعدة الفرد إن كان مريضا أو سليما، ومساعدته على أساس القيام بالنشاطات الصحية التي تؤدي إلى تعزيز الصحة أو الوصول إلى الشفاء (أو حتى الوصول إلى الموت بأمان). وتلك النشاطات التي كان يقوم بها الفرد دون مساعدة من أحد إن كان يمتلك العلم والقدرة والرغبة، ونساعد الانسان بالطريقة المثلى للوصول بالإنسان للقيام بتلك النشاطات باستقلالية بأسرع وقت ممكن.

النشاطات هي:

الأكل والشرب، الحمام، الحركة ... ونسميها النشاطات الصحية اليومية

Activities of daily livings (ADLs)

Integrating Nursing Theory

- **The word theory (opinion, belief, or view) comes from a Greek word that means vision.**
- Nursing has undergone a similar scientific review. Florence Nightingale and others have examined the relationships among
  - *humans,*
  - *health,*
  - *the environment, and*
  - *nursing.*
- The outcome of such analysis becomes the basis for nursing theory (proposed ideas about what is involved in the process called nursing).

Integrating Nursing Theory cont.

TABLE 1-2 Nursing Theories and Applications

THEORIST	THEORY	EXPLANATION
Florence Nightingale 1820–1910	**Environmental Theory**	
	Man	An individual whose natural defenses are influenced by a healthy or unhealthy environment
	Health	A state in which the environment is optimal for the natural body processes to achieve reparative outcomes
	Environment	All the external conditions capable of preventing, suppressing, or contributing to disease or death
	Nursing	Putting the client in the best condition for nature to act
	Synopsis of theory	External conditions such as ventilation, light, odor, and cleanliness can prevent, suppress, or contribute to disease or death.
	Application to nursing practice	Nurses modify unhealthy aspects of the environment to put the client in the best condition for nature to act.
Virginia Henderson 1897–1996	**Basic Needs Theory**	
	Man	An individual with human needs that have unique meaning and value
	Health	The ability to independently satisfy human needs composed of 14 basic physical, psychological, and social elements
	Environment	The setting in which a person learns unique patterns for living
	Nursing	Temporarily assisting a person who lacks the necessary strength, will, and knowledge to satisfy one or more of 14 basic needs
	Synopsis of theory	People have basic needs that are components of health. The significance and value of these needs are unique to each person.
	Application to nursing practice	Nurses assist in performing those activities that the client would perform if he or she had the strength, will, and knowledge.

THE EDUCATIONAL LADDER

#	Degree	Years
0	Aid Nurse	6 – 8 months training
1	Practical Nurse (LPN)	2 Academic years
2	Registered Staff Nurse (RN)	3 Academic years
3	Bachelor Degree in Nursing (BSN)	4 Academic years
4	Master Degree in Nursing (MSN)	4 Academic years + 2 Academic years
5	Philosophy of Doctorate (Ph.D.)	4 Academic years + 2 + 3 – 5 Academic years
6	Post Doctorate Level Scientist	Ph.D. +
7	Scientist	Not bounded and my begin in any stage from BSN
	Unlicensed assistive personnel (UAP) like family members	

Type of nurse	What they will do	Length of training	Training site
Nursing aide; paramedic	Work under direction of registered nurses (RN) in hospital setting.	6 months certificate	New training institutes
Junior RN	Work under direction of doctors and RNs in hospital setting; may work under direction of doctors with stable patients in community setting.	2-year diploma	Government nurse training institutes
Intermediate RN-midwife	Work under direction of doctors and senior RNs in hospital setting; may work independently with stable patients in hospital setting.	3-year associate degree plus additional midwifery training	Government nurse training institutes
Senior RN-midwife	As Head Nurse, lead groups of nurses under direction of doctors in hospital setting; may work independently with stable and unstable patients in hospital and community settings.	4 year BSC or BSN degree	Private and public sector universities

Delegating Nursing Work

- Regardless of whether it is an RN delegating to an LPN or a UAP or an LPN delegating to an UAP, delegation requires adhering to the following five guidelines:
- Right task: matching the client's needs with the caregiver's skills
- Right circumstance: ensuring that the situation is appropriate
- Right person: knowing the unique competencies of the caregiver
- Right directions and communication: providing sufficient information
- Right supervision and evaluation: being available for assistance; validating

Roles of a Nurse

- Coordinator
- Communicator
- Teacher
- Counselor
- Manager
- Leader
- Team player
- Motivator
- Delegator
- Critical thinker
- Innovator
- Researcher
- Advocate

UNIQUE NURSING SKILLS

- *Assessment Skills*
- *Caring Skills*
- *Counseling Skills*
- *Comforting Skills*

Assessment Skills

- *Before the nurse can determine what care a person requires, he or she must determine the client's needs and problems.*
- *This requires the use of assessment skills (acts that involve collecting data), which include interviewing, observing, and examining the client, and in some cases, the client's family*

Miss. /Mr. Assessment (her You are)



Caring Skills

- *Caring skills (nursing interventions that restore or maintain a person's health) may involve actions as simple as assisting with activities of daily living (ADLs).*
- *(ADLs), the acts that people normally do every day, for example, bathing, grooming, dressing, toileting, and eating.*

Counseling Skills

- ***Counseling Skills***
- ***Nurses implement counseling skills (interventions that include:***
  - **communicating with clients,**
  - **actively listening during exchanges of information,**
  - **offering pertinent health teaching,**
  - **and providing emotional support)**

Because clients do not always communicate their feelings to strangers, nurses use empathy (intuitive awareness of what the client is experiencing) to perceive the client's emotional state and need for support. This skill differs from sympathy (feeling as emotionally distraught as the client).

Empathy helps the nurse become effective at providing for the client's needs while remaining compassionately detached.

Comforting Skills

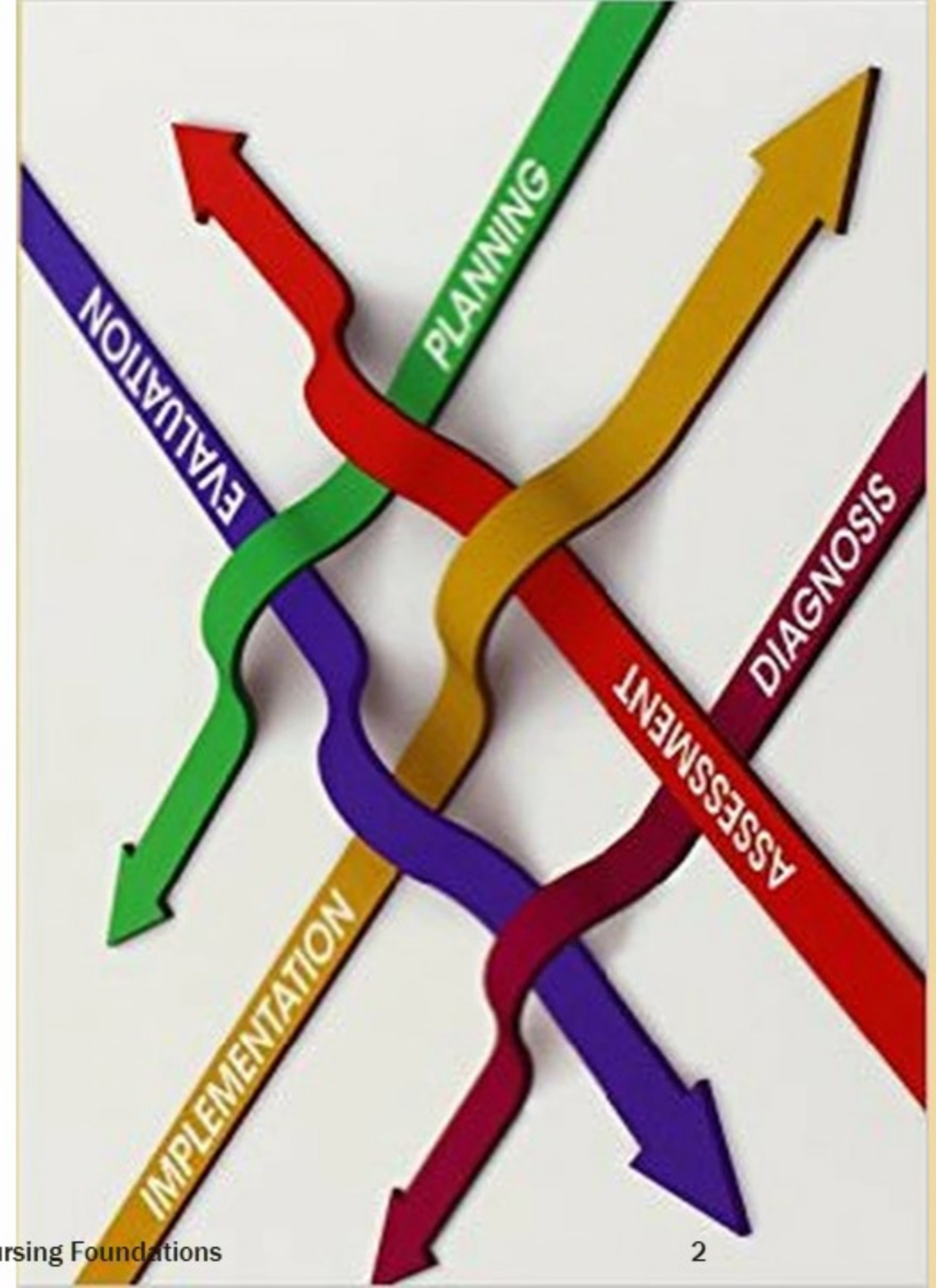
- *Nightingale's presence and the light from her lamp communicated comfort to the frightened British soldiers in the 19th century.*
- *The nurse uses comforting skills (interventions that provide stability and security during a health-related crisis)*



اللّٰه يحفظك

God Bless You


ASSESSMENT
What data is collected?
DIAGNOSIS
What is the problem?
THE NURSING PROCESS
Nurseslabs
PLANNING
How to manage the problem?
IMPLEMENTATION
Putting the plan into action.
EVALUATION
Did the plan work?

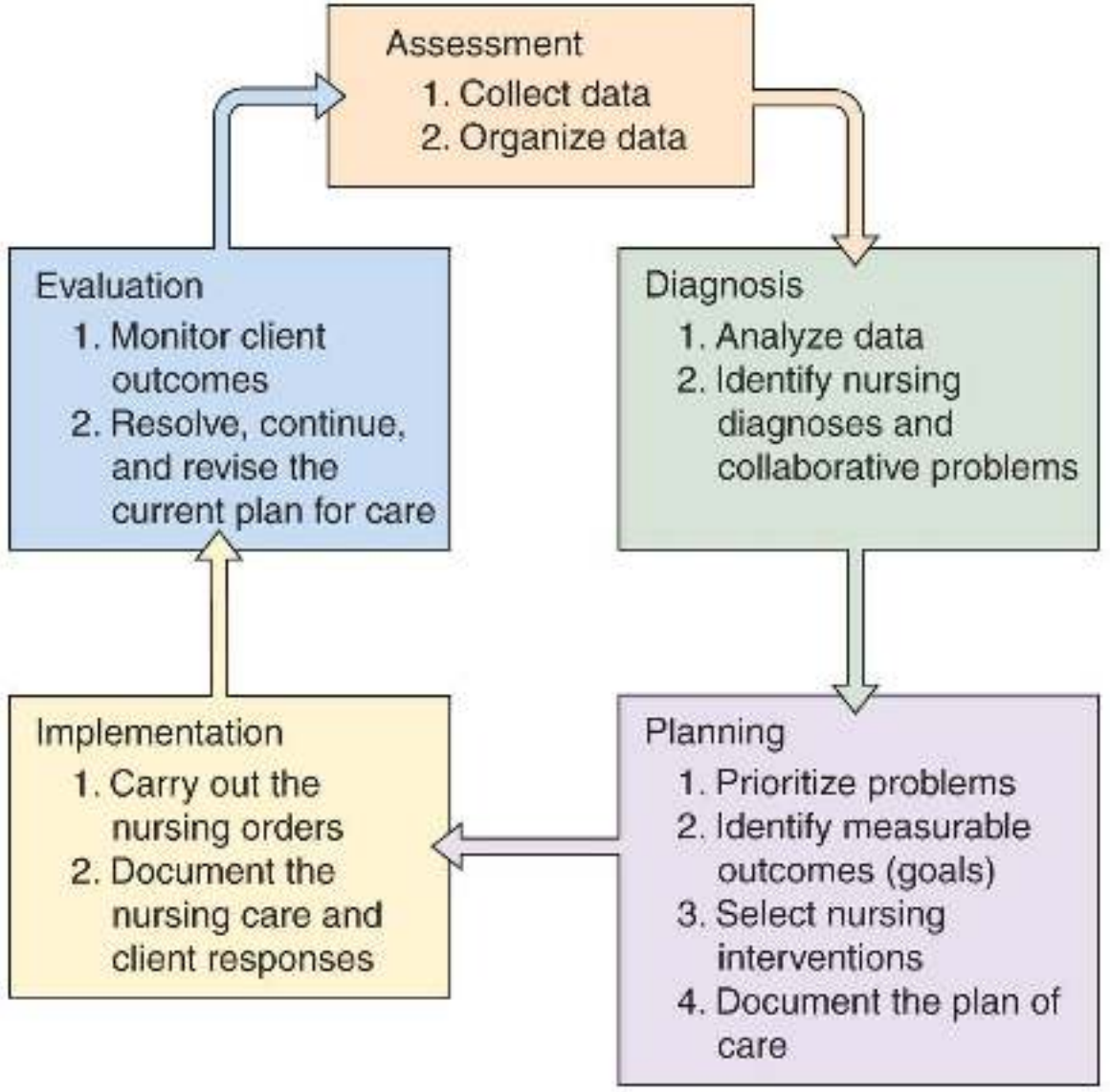

PLANNING
DIAGNOSIS
ASSESSMENT
IMPLEMENTATION
EVALUATION

Objectives

- *On completion of this chapter, the audience should be able to:*
  1. *Define the term nursing process.*
  2. *Describe seven characteristics of the nursing process.*
  3. *List five steps in the nursing process.*
  4. *Identify sources of assessment data.*
  5. *Differentiate between database, focus, and functional assessments.*
  6. *List three parts of a nursing diagnostic statement.*
  7. *Describe & setting priorities & short- and long-term goals.*
  8. *Identify ways to document a plan of care.*
  9. *Discuss outcomes that result from an evaluation.*

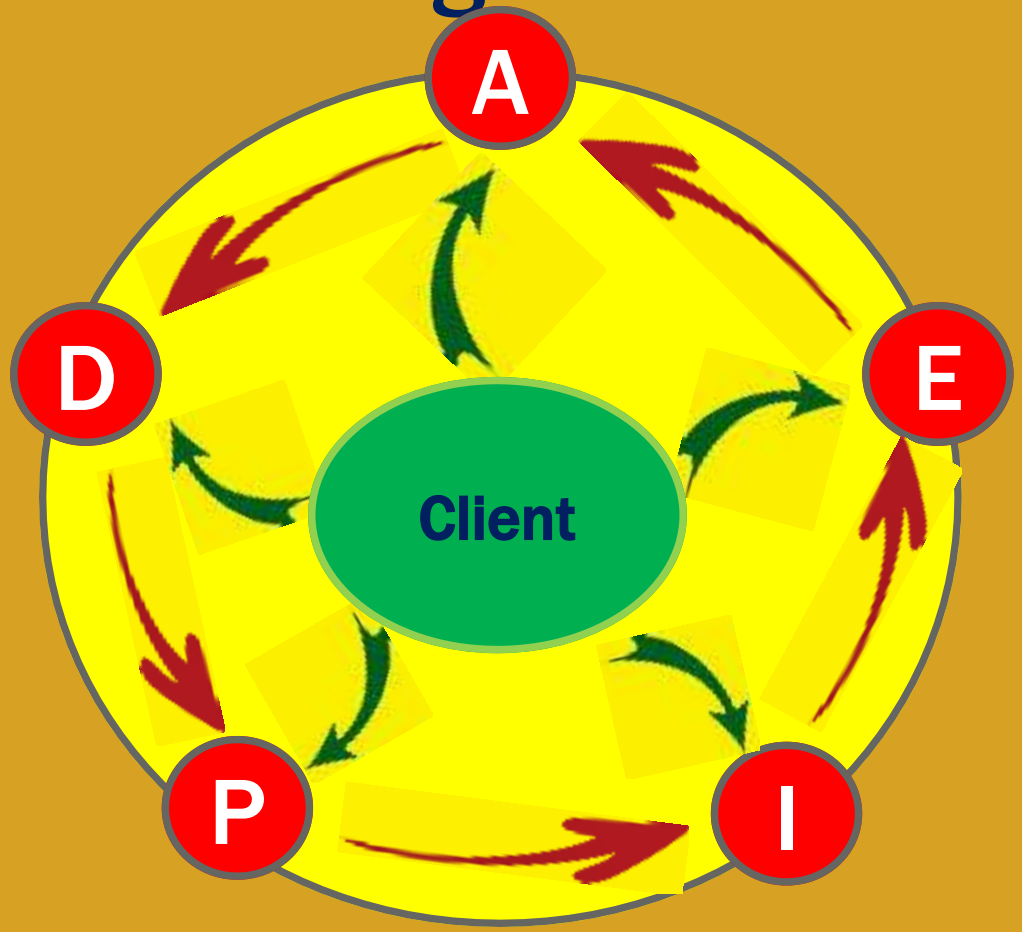
Definition of Nursing process

- **A process;** is a set of actions leading to a particular goal.
- **The nursing process;** is an organized sequence of problem-solving steps used to identify and manage the health problems of clients.
  - *Steps of Nursing process* (ADPIE أدبي)*:*
    - Assessment
    - Diagnosis
    - Planning
    - Implementation
    - Evaluation



CHARACTERISTICS OF THE NURSING PROCESS

1. Within the legal scope of nursing.
2. Based on knowledge.
3. Planned.
4. Client-centered.
5. Goal-directed.
6. Prioritized.
7. Dynamic.



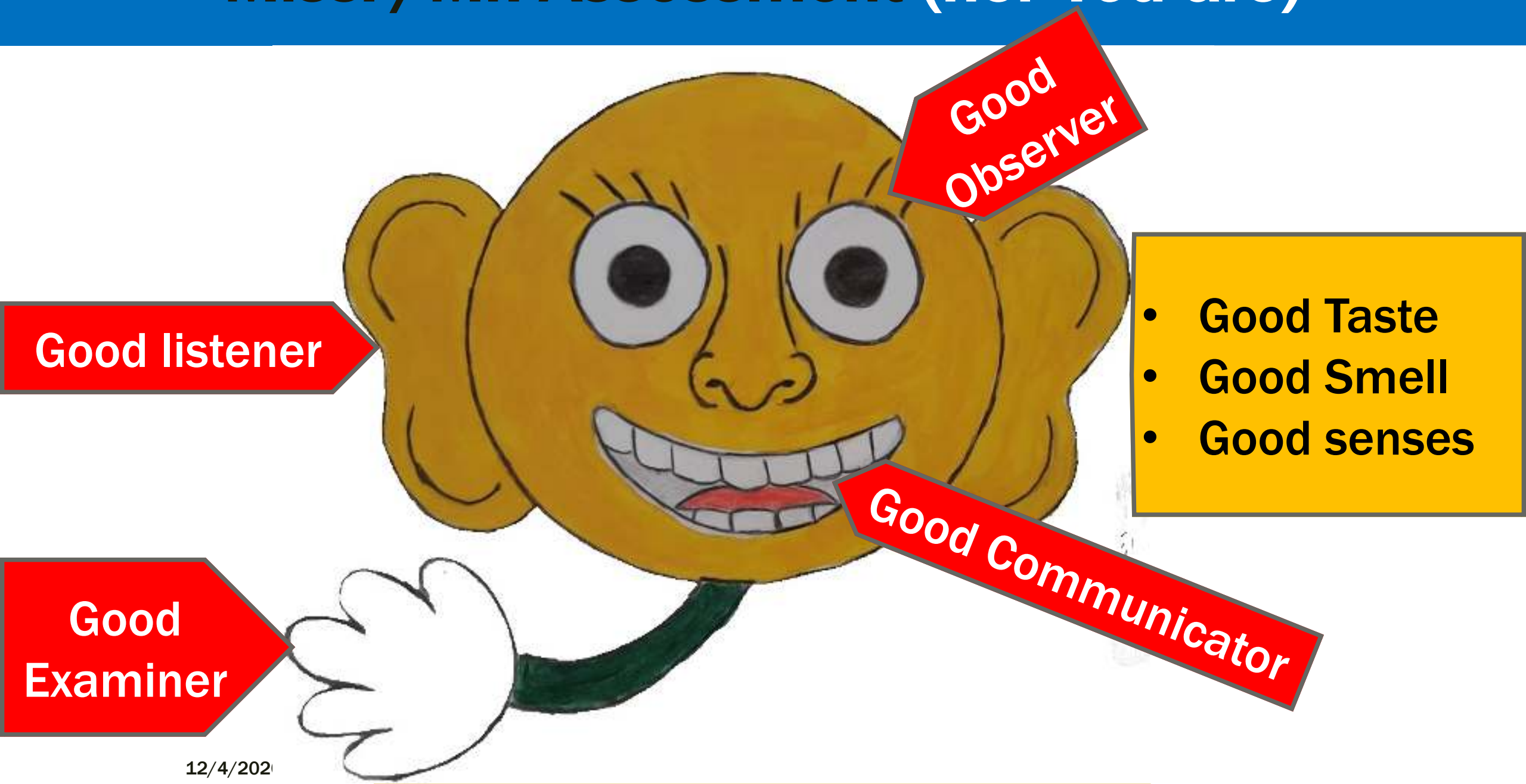
I- Assessment

- **Assessment**, the first step in the nursing process, is the systematic collection of facts or data.
- Assessment begins with the nurse's first contact with a client and continues as long as a need for health care exists.

- **Types of Data**
- **Objective data** are observable and measurable facts and are referred to as signs of a disorder.
- **Subjective data** consist of information that only the client feels and can describe, and are called symptoms.

NB. Signs and Symptoms referred as (S&S)

Miss. /Mr. Assessment (her You are)



OBJECTIVE DATA	SUBJECTIVE DATA
Weight	Pain
Temperature	Nausea
Skin color	Depression
Blood cell count	Fatigue
Vomiting	Anxiety
Bleeding	Loneliness

Sources for Data

- The primary source of information is the client **(the best source).**
- Secondary sources include the: **(any source other than client)**
  - Client's family
  - Reports
  - Test results,
  - Medical records, and Discussions with other health care providers.

Types of Assessments

- **Database Assessment : is the** initial information about the client's physical, emotional, social, and spiritual health.
  - *Printed or available on a computer for use as a guide Information*
  - *Comparisons of ongoing or future assessments with baseline data*
- **Focus Assessment : is the** information that provides more details about specific problems and expands the original database.

Types of Assessments cont.

- **Focus Assessment :** To do focus assessment for problem do the following:
  1. *Location of the problem (exactly were is the problem)*
  2. *Duration of the problem (time that problem is exist)*
  3. *Frequency (how many attacks occur in a specific period)*
  4. *Severity (mild, moderate, sever on untolerable)*
  5. *Consistency (if it soft, hard, mobile, bloody ...)*
  6. *Other problems related (any other problem occurs congruent with the main complaint)*

Types of Assessments cont.

- **Functional Assessment: is the** comprehensive evaluation of a client's physical strengths and weaknesses in areas such as:
  - (1) the performance of activities of daily living
  - (2) cognitive abilities, and
  - (3) social functioning.

Organization of Data

Interpreting data is easier if information is organized. Organization involves grouping related information. Related Clusters llike:

- *Lassitude, fever*
- *Weak cough, thick sputum*
- ***Distended abdomen; dry, hard stool passed with difficulty***

II- Diagnosis

- *Diagnosis, the second step in the nursing process, is the identification of health related problems.*
- *Nurses analyze data to identify one or more nursing diagnoses.*
- *A nursing diagnosis is a health issue that can be prevented, reduced, resolved, or enhanced through independent nursing measures.*

Diagnosis cont.

Actual Problem

Potential Problem

Actual / Potential

Potential Issue

TYPE	EXPLANATION AND EXAMPLE
Problem-focused diagnosis	A problem that currently exists *Impaired Physical Mobility related to pain as evidenced by limited range of motion, reluctance to move*
Risk diagnosis	A problem the client is uniquely at risk for developing *Risk for Deficient Fluid Volume related to persistent vomiting*
Syndrome diagnosis	Cluster of problems related to an event or situation that can be managed together (Carpenito-Moyet, 2012) *Rape Trauma Syndrome, Disuse Syndrome*
Health promotion diagnosis	A concern with which a healthy person desires nursing assistance to maintain or achieve a higher level of wellness *Readiness for Enhanced Immunization Status*

Wellness diagnosis

Diagnostic Statements

(1) Problem / Issue	(ثابت) Connection	(2) Etiology (cause)	(ثابت) Connection	(3) Manifestations S&S
Fever (Actual Problem)	Related to	Tonsillitis	As evidenced by	• Headache • Flushing • Hot skin • Sweating
(1) ✓		(2) ✓		(3) ✓

(1) Problem / Issue	(ثابت) Connection	(2) Etiology (cause)	(ثابت) Connection	(3) Manifestations S&S
Risk for drowsiness and comma (Risk Problem)	Related to	Sever drop in blood glucose level (Hypoglycemia)		
(1) ✓		(2) ✓		(3) ✗

Diagnostic Statements

(1) Issue	(ثابت) Connection	(2) Etiology (cause)	(ثابت) Connection	(3) Manifestations S&S
Enhancing dental health among school children (Health promotion Potential)				
(1) 		(2) 		(3) 

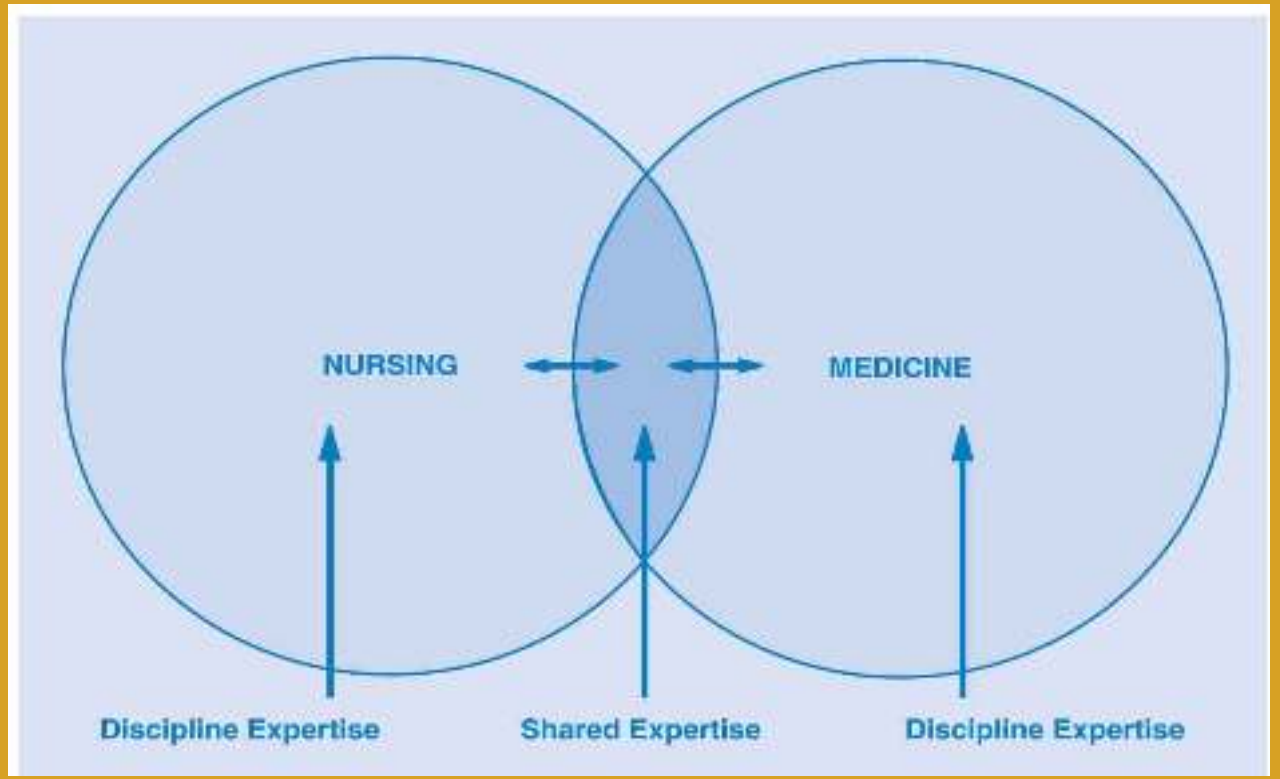
- *Diagnostic Statements (PES)*

*Nursing diagnostic statement contains **3 parts**, sometimes referred to as PES.*

1. ***P:** Name of health-related issue or problem as identified in the NANDA-I list*
2. ***E:** Etiology (its cause)*
3. ***S:** Signs and symptoms also called defining characteristics*

Collaborative problems

- **Collaborative problems** are those potential complications from a disorder, test, or treatment that the nurse cannot treat independently, for example, hemorrhage.



III- Planning

- The third step in the nursing process is **planning** through:
  1. ***Setting Priorities;*** *the process of prioritizing nursing diagnoses and collaborative problems,*
  2. ***Identifying measurable expected outcomes,***
  3. ***Selecting appropriate interventions, and***
  4. ***Documenting the plan of care.***
- Whenever possible, the nurse consults the client while developing and revising the plan.

Setting Priorities

- ***Since many clients' problems take time to resolve, it is important to determine which problems require the most immediate attention.***
- ***Prioritization involves ranking, from those that are most serious or immediate to those of lesser importance.***
- ***There is more than one way to determine priorities. One method nurses frequently use is Maslow's Hierarchy of Human Needs***

Setting Priorities cont.



Maslow's Hierarchy of Needs

Serious Dangerous

Setting priorities way

Less Dangerous

TABLE 2-5 Prioritizing Nursing Diagnoses

HUMAN NEED	EXAMPLES OF NURSING DIAGNOSES
Physiologic	Imbalanced nutrition: less than body requirements
A - Airway	Ineffective breathing pattern
B - Breathing	Pain
C - Circulation	Impaired swallowing
D - Delirium state	Urinary retention
Safety and security	Risk for injury
	Impaired verbal communication
	Disturbed thought processes
	Anxiety
	Fear
Love and belonging	Social isolation
	Impaired social interactions
	Interrupted family processes
	Parental role conflict
Esteem and self-esteem	Disturbed body image
	Powerlessness
	Caregiver role strain
	Ineffective breast-feeding
Self-actualization	Delayed growth and development
	Spiritual distress

Establishing Outcome Criteria

- *Outcome criteria, sometimes called goals*

> A goal is a specific and measurable objective designed to reflect the patient highest level of wellness and independence in function.

- *Statement contains objective evidence for verifying that the client has improved.*

- **Short-Term Goals;** Nurses use **short-term goals (outcomes achievable in seconds, minutes, hours, or days to 1 week) EG.**
  - *The client will have a bowel movement in 2 days (specify date)*
- **Long-Term Goals; Nurses generally identify long-term goals (outcomes that take weeks or months to accomplish) EG.**
  - *For the client with a stroke is the return of full or partial function to a paralyzed limb.*

Goals are patient-centered and SMART

- Specific
- Measurable
- Attainable
- Relevant
- Time Bound

Problem statement of Nsg Diagnoses	Goal/Expected outcome
Pain	Client reports absence/diminished pain within 8 hrs
Imbalanced nutrition more than body requirement	Client reaches target weight of 60 kg within a week.
Impaired physical mobility	Client walks along the hallway independently before discharge

e.g.

Pt walks 50 ft.

Pt eats 75% of meal

Pt maintains HR<100

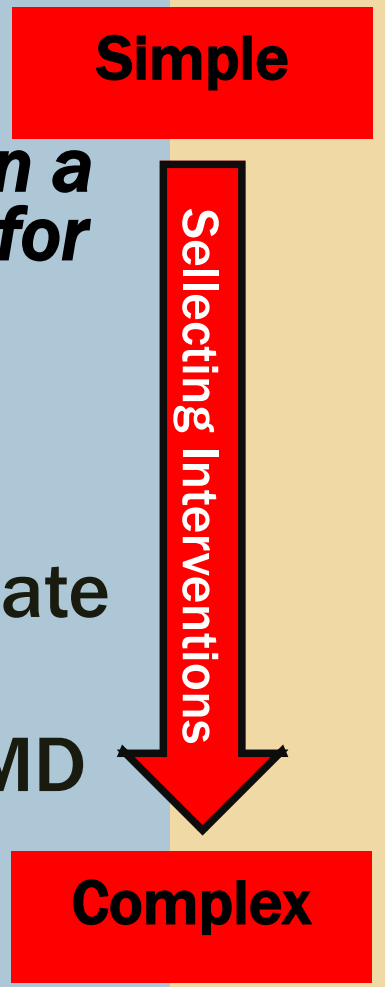
Pt states pain level is acceptable 6 (0-10)

Selecting Nursing Interventions

- ***Planning the measures that the client and nurse will use to accomplish outcome criteria involves critical thinking.***

  (Intervention selected from simple to complex)

- ***Writing Nursing Orders; (directions for a client's care) within a nursing care plan identify the what, when, where, and how for performing nursing interventions***



Interventions are 3 types:

1. **Independent (Nurse initiated)-** any action the nurse can initiate without direct supervision
2. **Dependent (Physician initiated)-**nursing actions requiring MD orders
3. **Collaborative-** nursing actions performed jointly with other health care team members

IV- Implementation

- The **fourth** step in the Nursing Process
- This is the “Doing” step
- Carrying out nursing interventions (orders) selected during the planning step
- This includes monitoring, teaching, further assessing, reviewing **Nursing care plan (NCP)**, incorporating physicians orders and monitoring cost effectiveness of interventions

V- Evaluation

- **Evaluation**, the fifth and final step in the nursing process, is the way by which nurses determine whether a client has reached a goal.

When determining whether a goal has been achieved, the nurse can draw one of the three possible conclusions:

- **The goal was met, that is the client response is the same as the desired outcomes.** → ☑ Well done
- **The goal was partially met, that is either a short term goal was achieved but the long term was not, or the desired outcome was only partially attained.** → Monitoring
- **The goal was not met.** → Revision NCP

Wat to do if so

Nursing care Plan (NCP)

After assessment, organizing and data analysis

Plan (NCP)

Diagnosis	Goal / expected out comes	Implementation /Intervention	Evaluation
1. Fever (39.5c) related pharyngeal infection as manifested by; Tachypnea Headache Flushing Hot skin 2. Pain … 3. Diarrheal ……	Temp will be subsided to (37.5) within 2 hours	1. Room ventilation / switch fan on 2. Lightening clothes 3. Give cold juice 4. Measuring temp each 20 minutes 5. Monitor patient vial signs 6. Make shower 7. Apply cold compresses 8. Give anti pyretic (Acamol) as ordered 9. Call the Dr. if there is no response	After one and a half hour if • Temp is 37.7 our work is well • Temp still high revise the plan and may need re-planning

Dangerous

Less Dangerous

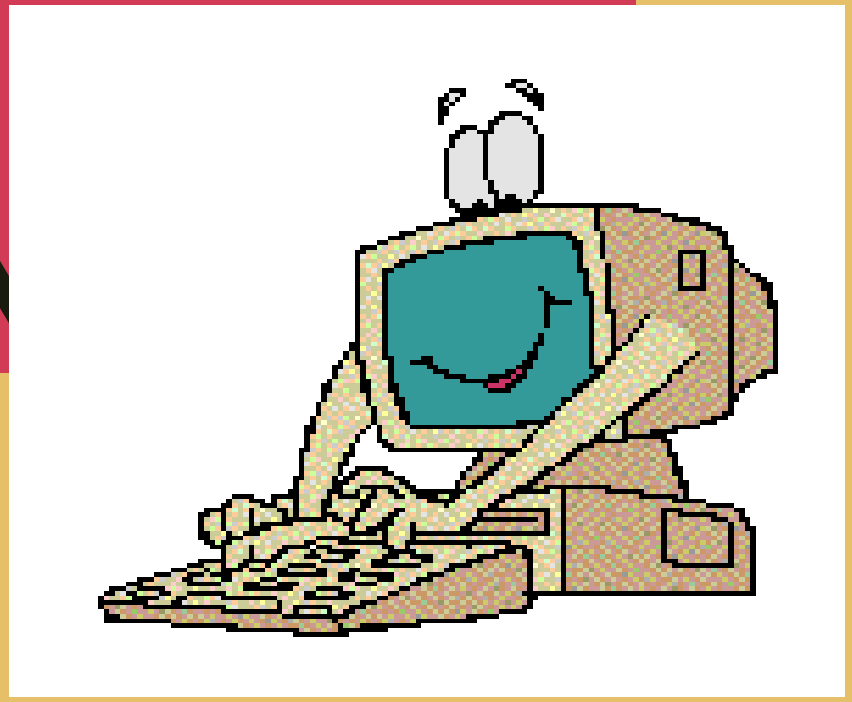
Simple

Complex


Thank
You
Merci
Obrigado
Obrigado
Asante
Spasibo
Maake
Grazie
Kiitos
Multumesc
Vinaka
Arigato
Dank Je
Matondo
Juspaxar
Mamana
Nirringrazzjak
Ua Tsaug Rau Koj
Mochchakkeram
Terma Kasih
Welalin
Kia Ora
Chokrane
Danken
Salamat
Matondo
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Cam on ban
Asante
Grazie
Kiitos
Merci
Welalin
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Chokrane
Danken
Kiitos
Multumesc
Welalin
Merci



RECORDING
REPORTIN



Objectives

- By the end of this lecture the student will be able to:
  1. *Describe the components of a medical record*
  2. *List the uses of a medical record*
  3. *Explain & differentiate between the 2 types of client records-source-oriented & problem-oriented*
  4. *List and define the methods of charting*
  5. *List aspects of documentation required in the medical records.*
  6. *Discuss why it is important to use approved abbreviations when charting*
  7. *Explain how to convert traditional time & military time*
  8. *Discuss the written forms used to communicate information re: a patient*
  9. *Discuss ways for exchange client information other than the medical record.*
  10. *Discuss the Do's & Don'ts of written/typed documentation*

Medical Records/Health Records/Client Records

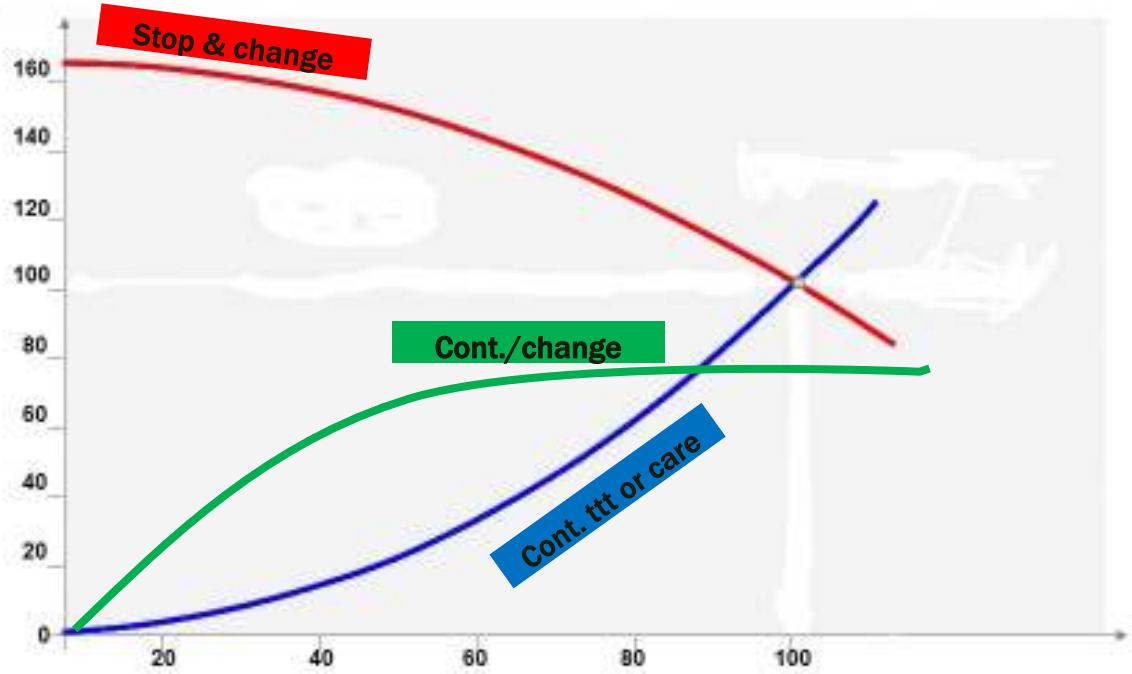
- **Written/typed collections:** information about a client's health problems, care provided by health practitioners and client's progress, by charting, recording, documenting.
- **Forms either in a chart or on computer:** containing Nursing notes & assessments, MD orders, progress notes, admission data, care plan, graphic sheet, Medication sheets, x-ray & lab reports, teaching notes, discharge plan
- **Hard copy paper forms are placed in a chart (a binder or folder that promotes the orderly collection, storage, and safekeeping of a person's medical record).**
- **The paper forms in the chart are color-coded or separated by tabbed sheets.**





Uses of Medical Records

1. **Permanent account,** old and new information are filed and maintained for future reference. (جواز السفر الطبي)
2. **Sharing information:**
   - *All health care providers to meet and exchange information*
   - *Prevents duplication of care and helps reduce the chance of error or omission.*
3. **Quality Assurance:**
   - *To maintain a high level of care, quality assurance (QA), continuous quality improvement (CQI), or total quality improvement (TQI).*



Uses of Medical Records cont.

4. **Accreditation** (التذكرة وثيقة معتمدة ويستند لها فى الدوائر والمؤسسات)
5. **Reimbursement**
   - *The costs calculated up on chart to be payed by client or insurance companies.*
6. **Education and research** (الكتب والأبحاث مجازا نصفها بأنها عصير تذاكر مركز)
   - *Published references are primary resources for health education.*
7. **Legal evidence** (تعتبر التذكرة مستند قانونى يؤخذ به فى المحاكم والقضايا)

Client Access to Records

Historically, clients were not allowed to see their medical records.

Clients now have the right to see their own medical records, request changes to anything they feel is inaccurate, and be informed about who has seen their medical records.

Criteria for Legally Defensible Charting

- Ensure that the client's name on each page.
- Never chart for someone else.
- Use the specified color for each data
- Date and time each entry as it is made.
- Chart promptly after providing care.
- Make entries in chronologic order.
- Write or print legibly.
- Use correct grammar and spelling.
- Reflect the plan of care.
- Describe the outcomes of care.
- Record relevant details.
- Use only approved abbreviations
- Never scribble over entries or use correction fluid
- Draw a single line through erroneous information.
- Record facts, not subjective interpretations.
- Quote the client's verbal comments.
- Write "duplicate" or "recopied" on documentation that is not original; include the date, time, initials, and reason for the duplication.
- Never imply criticism of another's care.
- Document the circumstances for notifying a physician, the specific data reported, and the physician's recommendations.
- Identify specific information provided when teaching a client and the evidence that indicates the client has understood the instructions.
- Leave no empty spaces between entries and signature.
- Sign each entry by name and title.

Types of Client Records

- **Source-Oriented Records:**
  - *The traditional type (records organized according to the source of documented information).*
  - *This type of record contains separate forms on which (physicians, nurses, dietitians, physical therapists, and other) make entries about their own specific activities in relation to the client's care.*
- **Problem-Oriented Records**
  - *A second type of client record (records organized according to the client's health problems).*
  - *All health care providers make entries in the same chart*

TABLE 9-2 Common Components of a Problem-Oriented Record

COMPONENT	DESCRIPTION
Database	Contains initial health information
Problem list	Consists of a numeric list of the client's health problems
Plan of care	Identifies methods for solving each identified health problem
Progress notes	Describes the client's responses to what has been done and revisions to the initial plan

THE WELLINGTON ADULT MEDICAL HIGH DEPENDENCY CHART

Patient Label Here

DATE:

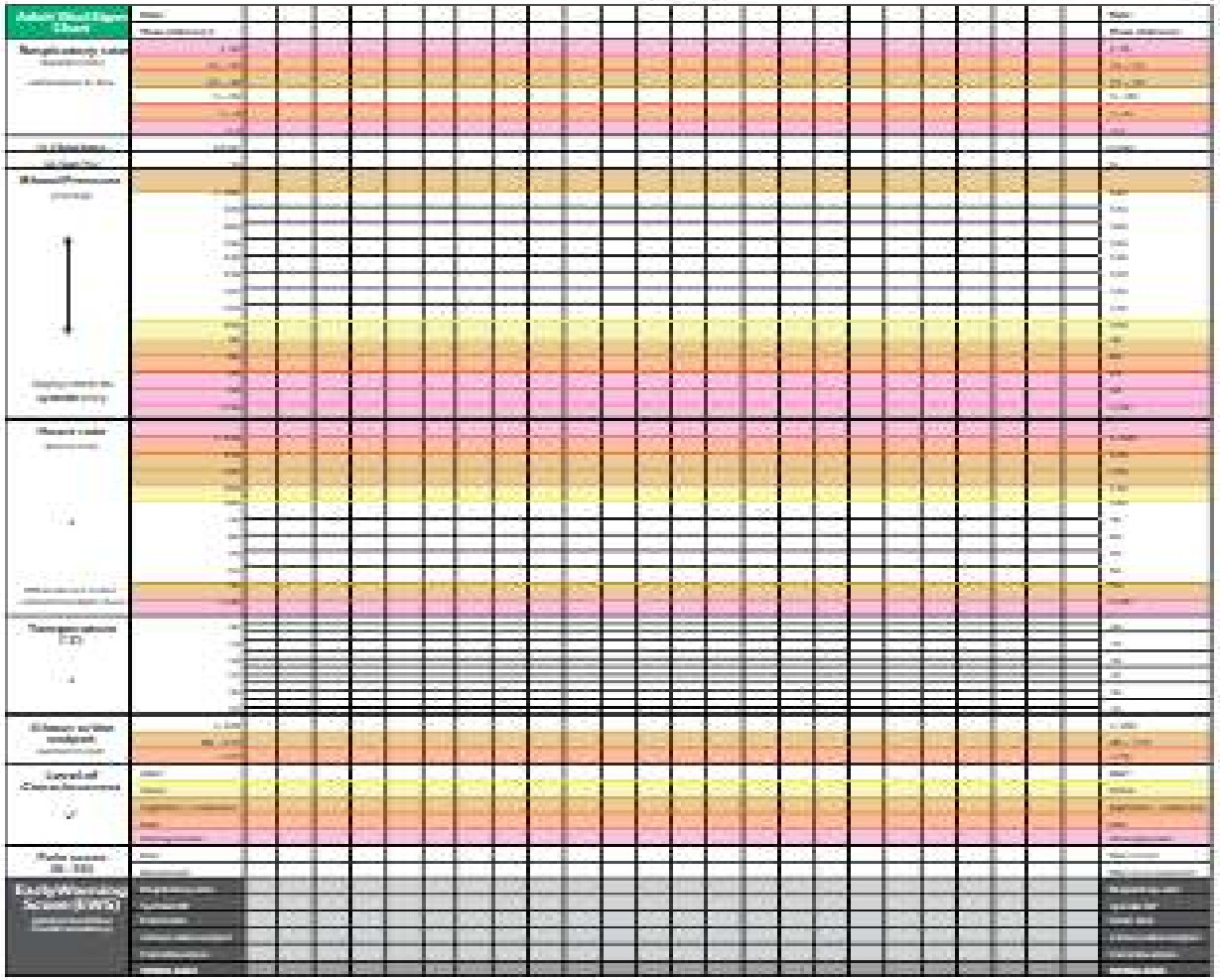
DAY NO:

Height (cm): Weight (kg):

Diagnosis:

Past Medical History:

Allergies:

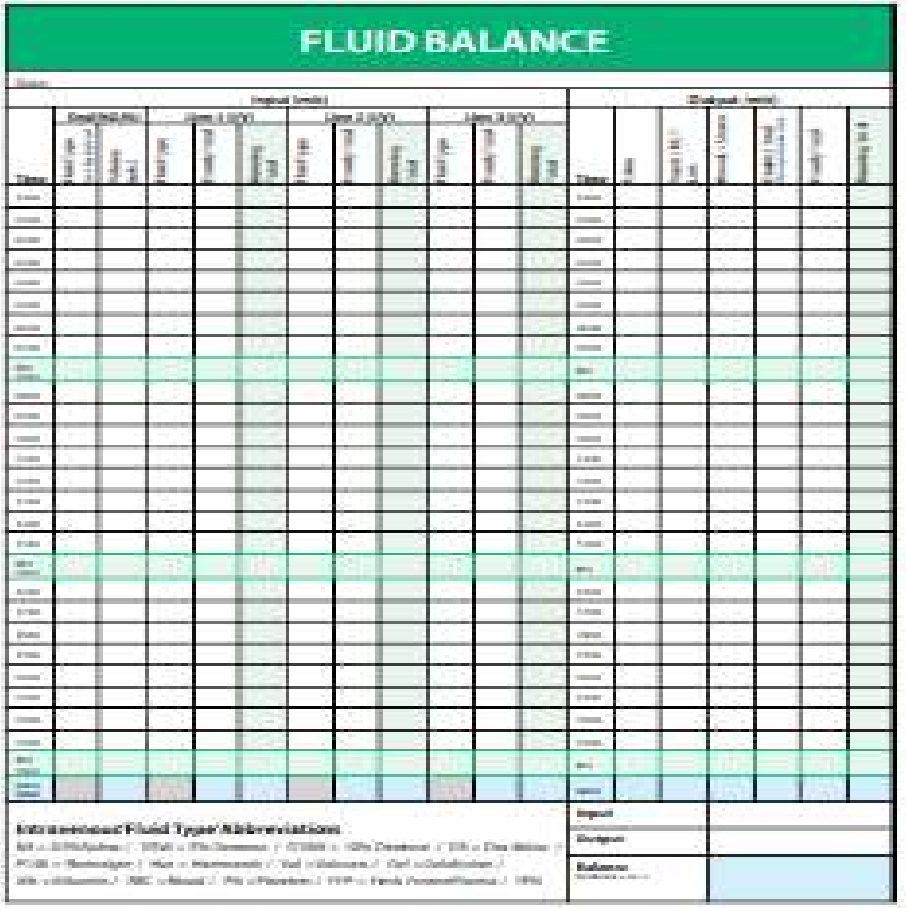


Capital & Coast District Health Board

Date	Total Input	Total Output	Balance

PLAN

FLUID BALANCE



Intravenous Fluid Type Abbreviations

Ab = Albumin / Ald = Aldomet / Ca = Calcium / Cef = Cefotaxime / Clx = Clexane / Dex = Dextrose / FFP = Fresh Frozen Plasma / Hep = Heparin / Hyd = Hydrocortisone / K = Potassium / Mag = Magnesium / Man = Mannitol / Nac = Acetylcysteine / Nif = Nifedipine / Nit = Nitrolingual / Pab = Pabrinex / PCA = Patient Controlled Analgesia / Pip = Piperacillin / Pot = Potassium / RBC = Blood / Ris = Risperidone / Sal = Salbutamol / Tzp = Tazocin / Vit = Vitamins

Input

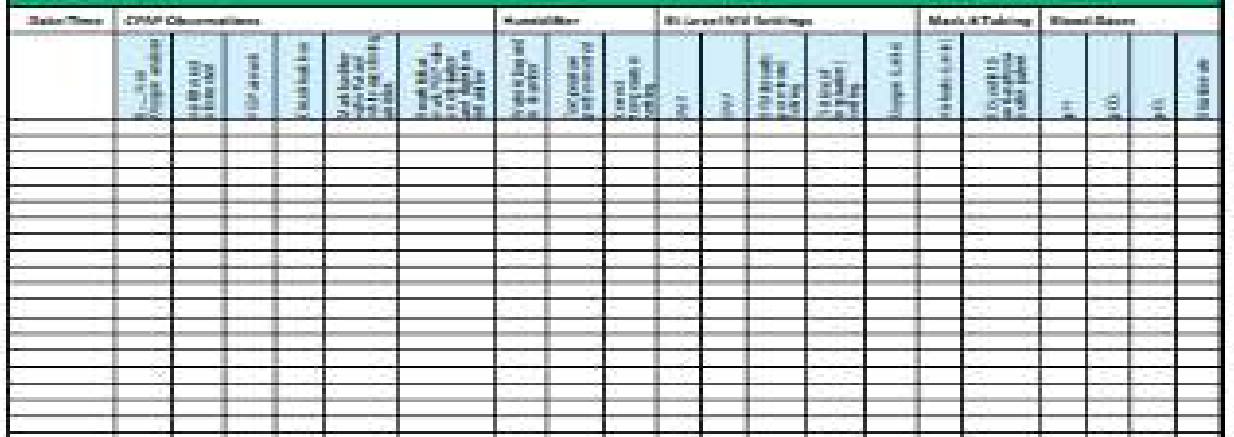
Output

Balance

CPAP AND HUMIDIFIER DELIVERY CHECKLIST

- [ ]
- [ ]
- [ ]
- [ ]

COMBINED BI-LEVEL & CPAP OBSERVATIONS



NURSING ACTION REQUIRED FOR PATIENTS TRIGGERING EARLY WARNING SCORE

Early Warning Scores (EWS) should be calculated when any vital sign falls into a shaded area (see below for interpretation). Vital signs should be recorded in the beginning of each shift unless the patient is deteriorating – for patients with a critical condition.

Any vital sign in the pink zone or total score of 4 or more	Dial 777 and state 'Medical Emergency Team' and communicate clearly
Any vital sign in the orange zone or total score of 3	Nursing care requires urgent intervention. Inform RN/Nurse Leader (CCU, Primary Clinical Nurse or Charge)
Any vital sign in the gold zone or total score of 2	Nursing care requires intervention. Inform RN/Nurse Leader (CCU, Primary Clinical Nurse or Charge)
Any vital sign in the yellow zone or total score of 1	Nursing care requires observation. Consider increasing frequency of vital sign observations and observation unit to Nurse in charge/without limitations

CALL 777 MET FOR ANY PATIENT YOU ARE SERIOUSLY CONCERNED ABOUT REGARDLESS OF TOTAL EARLY WARNING SCORE

EARLY WARNING SCORE KEY

0	1	2	3	777 MET

MEDICAL STAFF MODIFICATION TO EWS

If the patient is seen by Medical Emergency Team call (777) Not For Resuscitation (please document in clinical notes and inform by completing the box on the right and below.

Any Early Warning Score (EWS) modification must be made by a Doctor and should be triggered by assessing the clinical parameters.

	Yes	No
Hemoglobin g/dl		
Systolic BP		
Heart rate		

NOT FOR MET

MET FOR CPR

METHODS OF CHARTING cont.

- **Narrative charting: used in source-oriented records (التسجيل القصصى الانشائى)**
  - *Written information-chronologic order, content resembles a log or journal*
  - *Time consuming, may include information not needed*
- **SOAP or SOAPIER charting: Used in problem-oriented record**
  - *S = **Subjective data** -what client says; nurse quotes client's words*
  - *O = **Objective data-** information that is measured/observed by use of the senses (ex.=vital signs, lab & x-ray results)*
  - *A = **Assessment- analysis** interpretation/conclusions drawn from subjective/objective information — problem list created*
  - *P = **Plan-** plan of care*
  - *I = **Intervention-**specific interventions that have actually been performed by caregiver*
  - *E= **Evaluation-** includes client responses to nursing interventions & medical treatments*
  - *R = **Revision-** reflects care plan modifications to plan of care*

METHODS OF CHARTING cont.

- **Narrative charting: used in source-oriented records (التسجيل القصصى الانشائى)**
- **Focus charting:** Modified form of SOAP charting
  - *Uses the word focus instead of problem*
  - *Intended to make the client and client concerns & strengths the focus of care*
- **Progress notes organized into DAR;** *reflects phases of nursing process*
  - ***(D) data****-assessment phase of nursing process consists of observations – objective & subjective- EX: data from flow sheets (vital signs, pupil reaction)*
  - ***(A) Action-*** *reflects planning & implementation*
  - ***(R) response*** *- reflects evaluation.*

METHODS OF CHARTING cont.

- **PIE charting:** nursing care is documented using **PIE; P** = problems; **I** = interventions; **E** = evaluation
- **Charting by Exception:** Documentation -in which only abnormal assessment findings or care that is different – deviates from the standard are recorded
- **Computerized Documentation:** documenting information electronically
  - *Information quickly accessible – labs, x-ray reports, medications, procedure instructions, admission & discharge information, VS etc.*
  - *Advantages: information legible, records date & time of documentation, fewer omissions, saves time-eliminates charting, information easy to retrieve, abbreviations & terms consistent*
  - *In Long-term care setting –information put into Nursing Minimum Data Set –(MDS) source of patient care and electronically sent for payment*

Three Rivers Area Hospital

214 SPRING STREET
THREE RIVERS, MICHIGAN 49093

NAME

DOCTOR

NURSING NOTES

Date Time	NURSES REMARKS	Signature	Date Time	
1330	States "I'm having chest pain. It's like an elephant is sitting on me." ——— B. Zack, RN		1440	
1340	BP 150/90, P-122 and irregular. Skin is pale and moist. O2 started at 5L/min. ———			

TABLE 9-3 SOAPIER Charting Format

LETTER	EXPLANATION	EXAMPLE OF RECORDING
S = Subjective information	Information reported by the client	S "I don't feel well."
O = Objective information	Observations made by the nurse	O—Temperature 102.4°F
A = Analysis	Problem identification	A—Fever
P = Plan	Proposed treatment	P—Offer extra fluids and monitor body temperature
I = Intervention	Care provided	I—750 mL of fluid intake in 8 hours; temperature assessed every 4 hours
E = Evaluation	Outcome of treatment	E—Temperature reduced to 101°F
R = Revision	Changes in treatment	R—Increase fluid intake to 1,000 mL per shift until temperature is ≤100°F

NURSING NOTES

Date Time	NURSES REMARKS	Signature
6/19	P#1 Crackles heard on inspiration in	
0750	the bases of R and L lungs. ———	
	I#1 Incision splinted with pillow.	
	Instructed to breathe deeply,	
	open mouth, and cough at the	
	end of expiration ———	
	E#1 Lungs clear with coughing.	
	——— A. Walker, LPN	

6/30/2007	D(ata) –	Bladder distended 2 fingers above pubis.
1015		Has not urinated in 8 hrs. since
		catheter was removed. ———
	A(ction) –	Assisted to toilet. Water turned on at
		faucet. Instructed to press over bladder
		with hands. ———
	R(esponse) –	Voided 525 mL of clear urine. L. Cass, SN



METHODS OF CHARTING

1. Narrative
2. SOAPIER
3. Focus
4. PIE
5. DAR
5. Carting by Exceptions
6. Computerized

Protecting Health Information-Privacy Standards

- All health care agencies-required to protect health information
- Client given a written notice-identifying uses & disclosures of their health information
- Privacy Standards; Workplace Applications:
- The names of clients on charts can no longer be visible to the public.
- Clipboards must obscure identifiable names of clients and private information about them.
- Whiteboards must be free of information linking a client with a diagnosis, procedure, or treatment.
- Computer screens must be oriented away from public view.
- Conversations regarding clients must take place in private places.
- Facsimile (fax) machines, must be located in areas off-limits to the public.

Documenting Information

- Assessment data, Client care needs, routine care such as hygiene.
- Safety precautions that have been used
- Nursing interventions described in the care plan
- Medical treatments prescribed by the physician, medication administration
- Outcomes of treatment and nursing interventions
- Client activity, percentage of food consumed at each meal
- Visits or consults by physicians or other health professionals, Referrals to other health care agencies.
- Reasons for contacting the physician and the outcome of the communication
- Transportation to other departments (Radiography dept., diagnostic tests, and time of return)
- Client teaching and discharge instructions

Abbreviations used in documentations

TABLE 9-4 Commonly Used Abbreviations

ABBREVIATION	MEANING
abd.	abdomen
a.c.	before meals
ad lib	as desired
AMA	against medical advice
amt.	amount
approx.	approximately
b.i.d.	twice a day
BM	bowel movement
BP	blood pressure
bpm	beats per minute
BRP	bathroom privileges
c̄	with
C	Centigrade
CCU	coronary care unit
c/o	complains of
dc	discontinue
ED	emergency department
et	and
H_2O	water
I & O	intake and output
IM	intramuscular
IV	intravenous
kg	kilogram
L and Lt	left
L	liter
lb	Pound
NKA	no known allergies

TABLE 9-4 Commonly Used Abbreviations

ABBREVIATION	MEANING
NPO	nothing by mouth
NSS	normal saline solution
O_2	oxygen
OB	obstetrics
OOB	out of bed
OR	operating room
per	by or through
P	pulse
p.c.	after meals
p.o.	by mouth
postop.	postoperative
preop.	preoperative
pt.	patient
PT	physical therapy
q	every
q.i.d.	four times a day
q.s.	quantity sufficient
R and Rt	right
R	respirations
s̄	without
SS	soap suds
stat	immediately
t.i.d.	three times a day
TPR	temperature, pulse, respirations
UA	urinalysis
via	by way of
WC	wheelchair
WNL	within normal limits
Wt.	weight

Indicating Documentation Time

- Traditional time (time based on two 12-hour revolutions on a clock)
- Military time (time based on a 24-hour clock)

TABLE 9-5 Examples of Military Time Conversions

TRADITIONAL TIME	MILITARY TIME
Midnight	0000 or 2400
12:01 AM	0001
1:30 AM	0130
Noon	1200
1:00 PM	1300
3:15 PM	1515
7:59 PM	1959
10:47 PM	2247

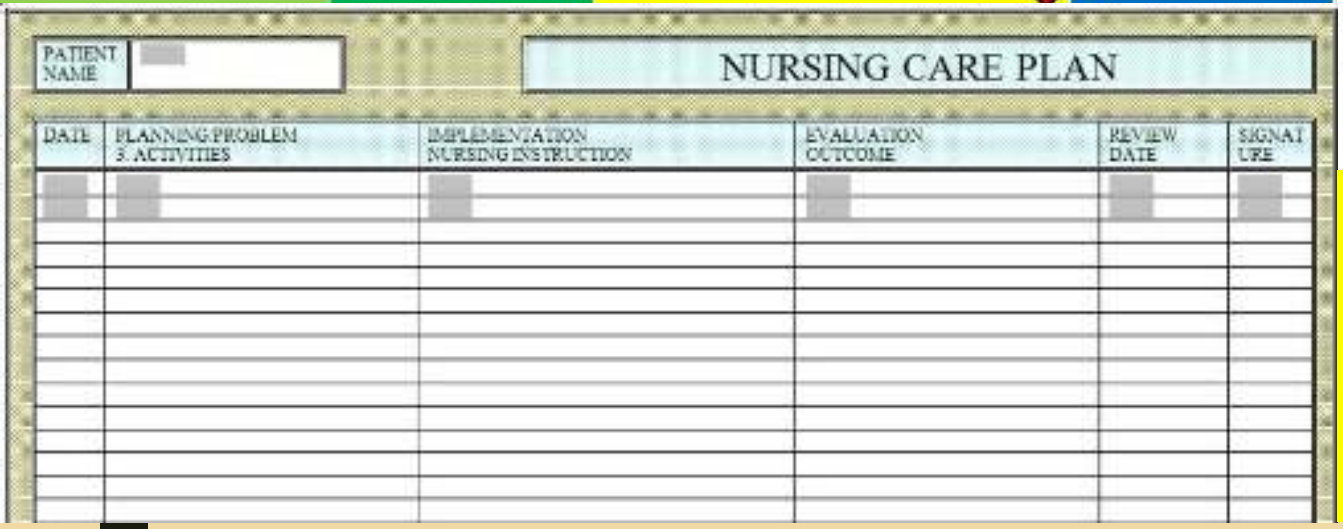
Military = Civilian	Military = Civilian
0001 = 12:01 am	1300 = 1:00 pm
0100 = 1:00 am	1400 = 2:00 pm
0200 = 2:00 am	1500 = 3:00 pm
0300 = 3:00 am	1600 = 4:00 pm
0400 = 4:00 am	1700 = 5:00 pm
0500 = 5:00 am	1800 = 6:00 pm
0600 = 6:00 am	1900 = 7:00 pm
0700 = 7:00 am	2000 = 8:00 pm
0800 = 8:00 am	2100 = 9:00 pm
0900 = 9:00 am	2200 = 10:00 pm
1000 = 10:00 am	2300 = 11:00 pm
1100 = 11:00 am	2400 = 12 Midnight
1200 = Noon	

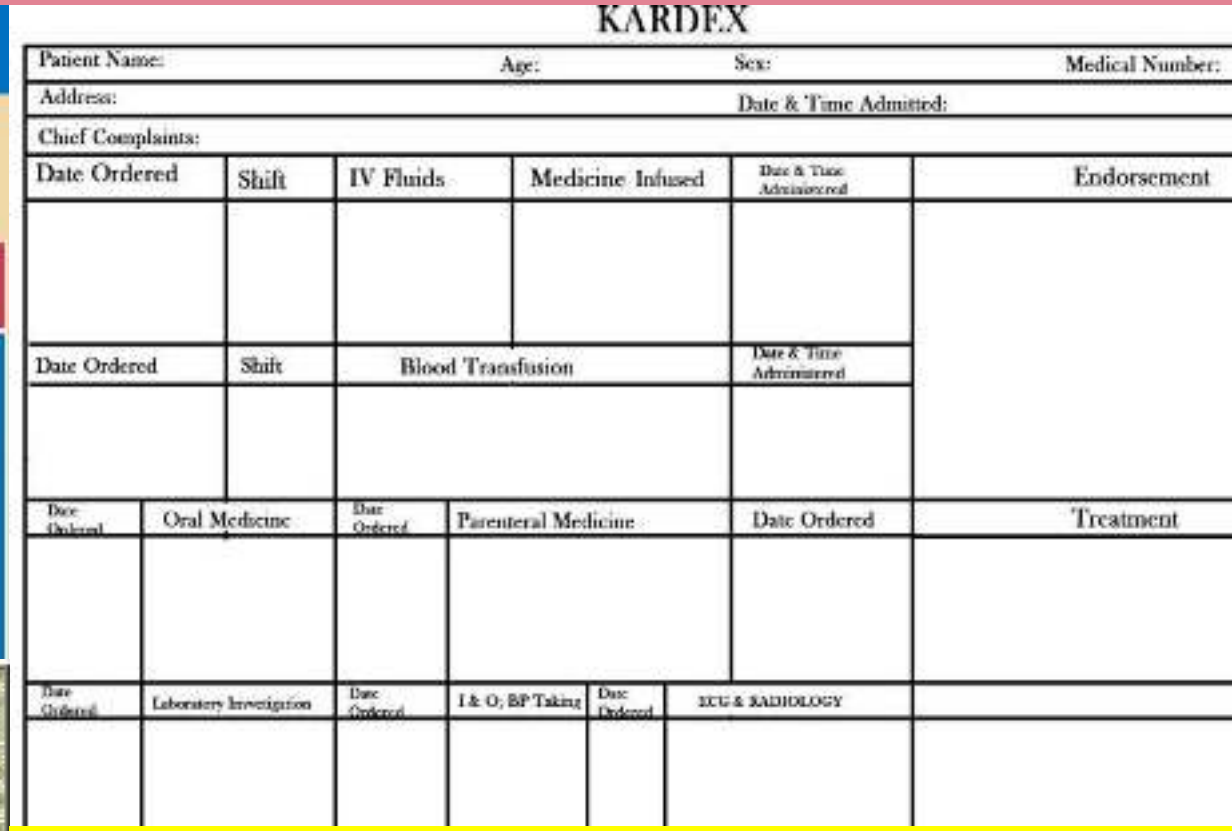
Written Forms of Communication

- ***Nursing Care Plan*** – ***Permanent record***
  - List of Client's problems, goals, nursing interventions, evaluations
  - Promotes-prevention, reduction or resolution of problems
- ***Nursing Kardex –*** ***Not a permanent record***
  - Quick reference- Current information about client and care
  - Information changes frequently
- ***Checklists-*** ***Permanent record***
  - Form of documentation-initials or a check mark used when care is similar each day
- ***Flow Sheets -*** ***Permanent record***
  - Recording for frequently repeated assessment data
  - Graphic record (vital signs) & Fluid Balance Sheet (I & O)
- ***Other Sheets:*** ***Permanent record***
  - Medication Administration Record
  - Skin & Safety/Fall Assessments
  - Admissions, Transfer, Referral, Discharge
  - Other facility and or state and federal required documentation

Written Forms of Communication







Written Forms of Communication

Written Forms of Communication

Orientation
- ☐ Person _____
- ☐ Place _____
- ☐ Time _____
- ☐ Situation _____

Vitals
- ☐ Temp _____ ☐ R _____
- ☐ BP _____ Pulse Ox _____

Head
- ☐ Hair _____
- ☐ PERLA _____ mm
- ☐ Nose _____
- ☐ Ears _____
- ☐ Mouth _____
  - ○ Midline tongue _____
  - ○ Moist _____
  - ○ Lesions _____
  - ○ Dentition _____

Neck
- ☐ Carotid pulse ____ ☐ JVD + ☐ Trachea midline

Chest
- ☐ Apical Pulse _____ ☐ Muffled ☐ Arrhythmia
- ☐ Breath Sounds - Anterior _____ Posterior _____ Lateral _____
- ☐ Chest Symmetry _____
- ☐ Skin Turgor (clavicle) _____

Abdomen
- ☐ Inspection _____
- ☐ Ausculation
  - ○ LUQ (active / hyper / absent)
  - ○ RUQ (active / hyper / absent)
  - ○ LLQ (active / hyper / absent)
  - ○ RLQ (active / hyper / absent)
- ☐ Palpation _____

Upper Extremities
- ☐ Radial pulses equal, +2
  - ○ Other: _____
- ☐ Temp vs. trunk (warm / cool)
- ☐ Grip equal and strong _____
- ☐ Capillary refill <3 sec
- ☐ Vein filling rapid

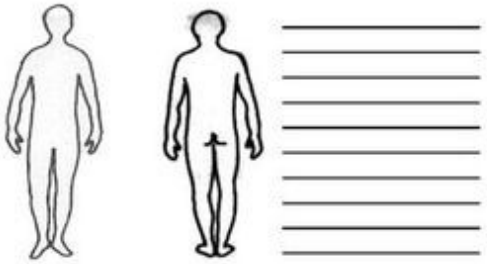
- ☐ Edema _____
- ☐ Foot strength
- ☐ Homain's (+ / -) Claudication (+ / -)
- ☐ Temp vs. Trunk (warm / cool)
- ☐ Nails ☐ Yellowed ☐ Thickened ☐ Ingrown
- ☐ Pedal pulse R(palp / doppler) L(palp / doppler)

ROM / Strength
- ☐ Upper R _____ ☐ Upper R _____
- ☐ Upper L _____ ☐ Upper L _____
- ☐ Lower R _____ ☐ Lower R _____
- ☐ Lower L _____ ☐ Lower L _____
- ☐ Sensation _____

General Assessment
- ☐ Weight/Height _____
- ☐ BM _____

Pain Assessment
- ☐ Acute/Chronic ☐ Intensity (0-10) _____
- ☐ Location _____
- ☐ Duration _____
- ☐ Characteristics _____
- ☐ Precipitation _____
- ☐ Frequency _____
- ☐ Non-verbals _____
- ☐ Relief factors _____
- ☐ Sleep _____

Skin Assessment
- ☐ Description: _____



Courtesy of http://nursing.arizela.com

Fluid Restriction Timetable for _____ - _____ Cups or _____ ounces (_____ cc)							
Times	Amount	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
8a B'fast								
9								
10								
11								
Noon								
1p								
2								
3								

DATE	FL in	FL out	OBS	FL in	FL out	OBS	FL in	FL out	OBS	FL in	FL out	OBS	FL in	FL out	OBS	FL in	FL out	OBS	FL in	FL out	OBS
2400-0200																					
0200-0400																					
0400-0600																					
0600-0800																					
0800-1000																					
1000-1200																					
1200-1400																					
1400-1600																					
1600-1800																					
1800-2000																					
2000-2200																					
2200-2400																					
TOTAL																					

PATIENTS NAME

Use Millilitres(Mils) 1 Fluid oz = 30Mils(approx)

Flow charts

CNA ADL DOCUMENTATION

Page One (#230/10-06)

*Code/Description:

0 - INDEPENDENT
I didn't have to help the resident at all

1 - SET-UP/SUPER-VISION The only help I had to give was watching or verbal encouragement for safety or to get him/her to move along with the activity

2 - LIMITED The resident did need for me to touch him/her to get the activity done, but it was to physically guide him/her rather than to actually take on their weight or complete part of the activity for them

3 - EXTENSIVE The resident needed me to take on some of his/her weight while the activity was being accomplished, or needed me to complete part of the activity for them

4 - TOTAL The resident wasn't able to do anything at all to help with this activity. Staff had to do the whole thing for them

8 - DIDN'T OCCUR The activity did not occur, not even once

Year: 2006 2007 2008

	Day	1			2			3			4			5			6			7		
	Date																					
	Shift	N	D	P	N	D	P	N	D	P	N	D	P	N	D	P	N	D	P	N	D	P
Bed Mobility: How resident moves to and from lying position, turns side to side, and positions body while in bed	*code																					
	# staff																					
Transfer: How resident moves between surfaces—to/from bed, chair, wheelchair, standing position (exclude to/from bath/toilet)	*code																					
	# staff																					
Walk In Room: How resident walks between locations in his/her room	*code																					
	# staff																					
Walk In Corridor: How resident walks in hallway on unit	*code																					
	# staff																					
Locomotion On Unit: How resident moves between locations in his/her room and adjacent corridor on same floor. If in wheelchair, self-sufficiency once in chair	*code																					
	# staff																					
Locomotion Off Unit: How resident moves to and returns from off unit locations. If in wheelchair, self-sufficiency once in chair	*code																					
	# staff																					
	INITIALS →																					

Interpersonal Communication

- **Change of shift report** - review client information to on-coming shift nurse
- **Client care assignments** - made out at the beginning of shift
- **Team conferences** - used to exchange information-includes personnel caring for client
  - *For doing assignments*
  - *In education*
  - *Solving problems*
- **Client rounds** - visit with client one-on or a group



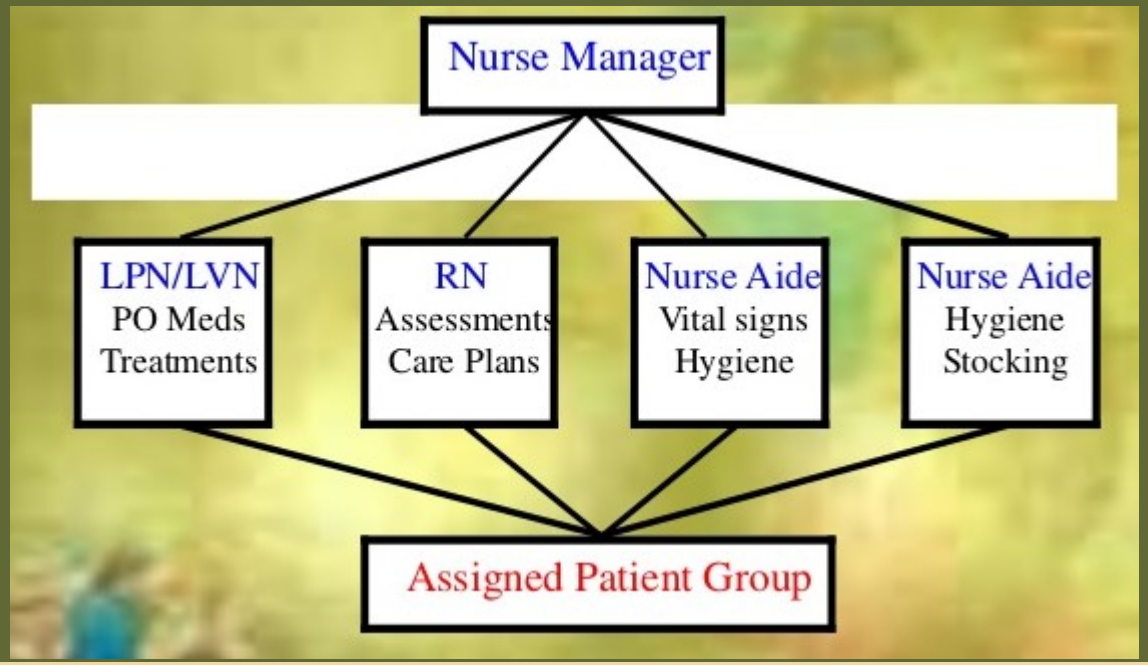
- **Telephone** - communicate information-when not able come together
  - *Repeat what was told to you back before documenting- to ensure correct information given & received*

Client care assignments

- **A change-of-shift report is a discussion between a nursing spokesperson from the shift that is ending and the arriving personnel.**
- *To maximize the efficiency of change-of-shift reports do following:*
  - *Be prompt so that the report can start and end on time.*
  - *Come prepared with a pen and paper or clipboard.*
  - *Avoid socializing during reporting sessions.*
  - *Take notes.*
  - *Clarify unclear information.*
  - *Ask questions about unclear pertinent information.*

Change of shift report

- Made out at the beginning of shift
- Assignments are posted, discussed with team members, or written on a worksheet
- Each assignment identifies the clients for whom the staff person is responsible and describes their care.
-



Client Care Assignments

- Used to exchange information-includes personnel caring
- **Why Team conferences?**
  - *For doing assignments*
  - *In education*
  - *Solving problems*

Client Round

- Rounds (visits to the bedside of clients on an individual basis or as a group).
- Client round used as a means of learning firsthand about clients





Conference

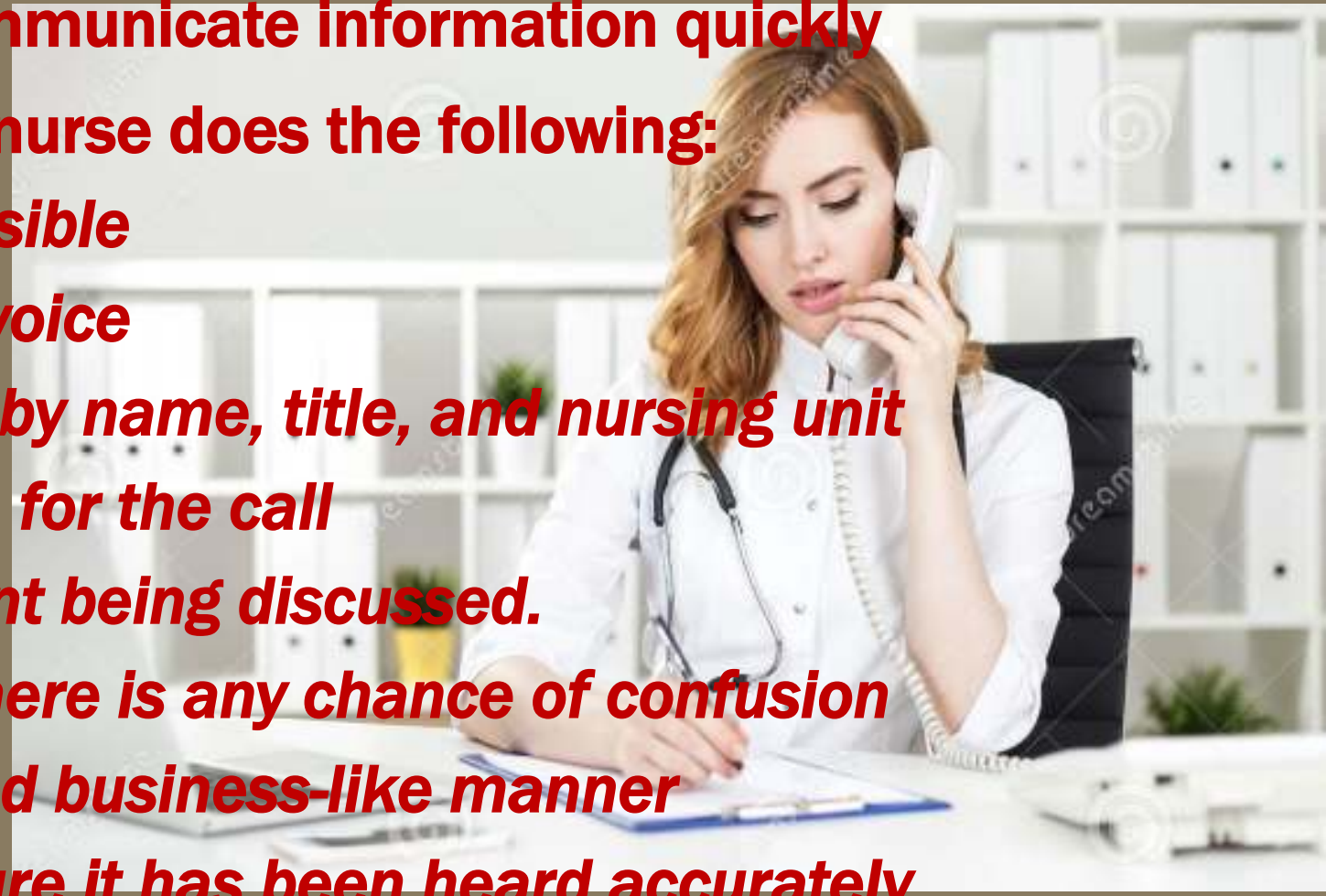
Shift report





Change of shift report

- Telephone used to exchange information when it is difficult for people to get together or when they must communicate information quickly
- When using the telephone, the nurse does the following:
  - *Answers as promptly as possible*
  - *Speaks in a normal tone of voice*
  - *Identifies himself or herself by name, title, and nursing unit*
  - *Obtains or states the reason for the call*
  - *Discretely identifies the client being discussed.*
  - *Spells the client's name if there is any chance of confusion*
  - *Converses in a courteous and business-like manner*
  - *Repeats information to ensure it has been heard accurately*



Interpersonal Communication

- **Change of shift report -** review client information to on-coming shift nurse
- **-** made out beginning of shift
- **Team conferences -** used to exchange information-includes personnel caring for client
  - *For doing assignments*
  - *In education*
  - *Solving problems*
- **Client rounds -** visit with client one-on or a group
- **Telephone -** communicate information-when not able come together
  - *Repeat what was told to you back before documenting- to ensure correct information given & received*

Do's for Documentation

- All entries must be legible & easy to read
  - Permanent- written in dark ink
  - Document date and time
  - Correct spelling
  - Your Signature with title
  - Proper sequence
  - Appropriateness-Completeness
  - Accuracy

The No-No's of Documentation

- Don't chart a symptom "c/o pain" without also charting what you did for it.
- **Don't change a client's record...criminal offense**
- Don't use shorthand or abbreviations that are not accepted.
- Don't write vague descriptions (appears or seems)
- Don't chart your opinions
- Don't use negative language to describe your client
- Don't chart ahead of time

The No-No's of Documentation cont.

- Never use ditto marks (= = علامات التكرار)
- Don't record staffing problems or record staff conflicts
- No white out or an eraser if using paper
- No empty lines or spaces
- No writing in margins
- Don't mention an incident report in the chart-document only the facts of the incident/accident
- Don't name a second client in chart
- Do not chart for someone else


Thank
You
Merci
Obrigado
Asante
Spasibo
Maake
Kiitos
Grazie
Matondo
Mochchakkeram
Dank Je
Vinaka
Ariigato
Juspaxar
Multumesc
Nirringrazzjak
Ua Tsaug Rau Koj
Terma Kasih
Cam on ban
Welalin
Kia Ora
Salamat
Danken
Chokrane
Mamana
Matondo
Obrigado
Grazie
Kiitos
Asante
Merci
Spasibo
Maake

ADMISSION, DISCHARGE, TRANSFER, AND REFERRALS

Objectives

- *On completion of this chapter, the audience should be able to:*
  1. *List four major steps involved in the admission process.*
  2. *Identify common psychosocial responses when clients are admitted to a health agency.*
  3. *List the steps involved in the discharge process.*
  4. *Give three examples of the use of transfers in client care.*
  5. *Explain the difference between transferring clients and referring clients.*

THE ADMISSION PROCESS

- Admission means entering a health care agency for nursing care and medical or surgical treatment. It involves the following:
  - *Authorization from a physician.*
  - *Collection of billing information by the admitting dept.*
  - *Completion of the admission database by nursing personnel.*
  - *Documentation of the client's medical history and findings from physical examination*
  - *Development of an initial nursing care plan*
  - *Initial medical orders for treatment*

Types of Admission

TYPE	EXPLANATION	EXAMPLE
Inpatient	Length of stay generally more than 24 hours	Acute pneumonia
Planned (nonurgent)	Scheduled in advance	Elective or required major surgery
Emergency admission	Unplanned; stabilized in emergency department and transferred to nursing care unit	Unrelieved chest pain, major trauma
Direct admission	Unplanned; emergency department bypassed	Acute condition such as prolonged vomiting or diarrhea
Outpatient	Length of stay less than 24 hours; possible return on a regular basis for continued care or treatment	Minor surgery, cancer therapy, physical therapy
Observational	Monitoring required; need for inpatient admission determined within 23 hours	Head injury, unstable vital signs, premature or early labor

Main hospital Departments

Department	القسم	Acronym
Emergency room / department	الطوارئ	ER or ED
Surgical Men Department	قسم جراحة رجال	SMD
Surgical Women Department	قسم جراحة حريم	SWD
Medical Men Department	قسم باطنة رجال	MMD
Medical Women Department	قسم باطنة حريم	MWD
Chest men department	قسم الصدرية رجال	CMD
Chest women department	قسم الصدرية حريم	CWD
Operating Room / Theater	غرفة العمليات	OR
The PACU, or post-anesthesia unit	وحدة الإفاقة	PACU
Intensive care unit / Critical care unit	العناية المركزة	ICU or CCU
Surgical intensive care unit	وحدة العناية المكثفة للجراحة	SICU
Medical intensive care unit	وحدة العناية المكثفة للباطنة	MICU
Cardiac/coronary care unit	وحدة القلب	CCU

Department	القسم	Acronym
Emergency room /Emergency department	الطوارئ	ER or ED
Intensive care unit / Critical care unit	العناية المركزة	ICU or CCU
The PACU, or post-anesthesia unit	وحدة الإفاقة	PACU
Operating Room / Theater	غرفة العمليات	OR
Surgical Men Department	قسم جراحة رجال	SMD
Surgical Women Department	قسم جراحة حريم	SWD
Surgical intensive care unit	وحدة العناية المكثفة للجراحة	SICU
Medical Men Department	قسم باطنة رجال	MMD
Medical Women Department	قسم باطنة حريم	MWD
Medical intensive care unit	وحدة العناية المكثفة للباطنة	MICU
Cardiac/coronary care unit	وحدة القلب	CCU
Cardiac/ Intensive coronary care unit	العناية المكثفة للقلب	CICU
Nursery Intensive care unit	حضانة الأطفال والعناية المكثفة	NICU
Nursery care Unit	حضانة الأطفال	NCU
Pediatric intensive care unit	العناية المكثفة للأطفال	PICU
Trauma intensive care unit	العناية المكثفة للإصابات	TICU

Department	القسم	Acronym
Chest men department	قسم الصدرية رجال	CMD
Chest women department	قسم الصدرية حريم	CWD
Delivery room	كشك الولادة	DR
Delivery room	غرفة العمليات	OR
Obstetric and Gynecology department	قسم الولادة وأمراض النساء	Ob.D
Oncology department	قسم الأورام	
Pediatric department	قسم الأطفال	PD
Pediatric surgical department	قسم جراحة أطفال	PSD
Ear Nose Throat department	قسم الأنف والأذن والحنجرة	ENT
Orthopedic department	قسم جراحة العظام	
Neurosurgery Department	قسم جراحة الأعصاب	NSD
Nephrology Department	قسم أمراض الكلى	
Hemodialysis unit	وحدة غسيل الكلي	
Radiology department /Reontgenography	قسم الأشعة	X-RAY
Computerized Tomography	التصوير المقطعي	CT
Magnetic Resonance Imaging	الرنين المغناطيسي	MRI

Nursing Admission Activities

- *Preparing the Client's Room*
- *Welcoming the Client and family*
- *Orienting the client and family*
- *Safeguarding Valuables and Clothing*
- *Helping the Client Undress & Dress*
- *Compiling the Nursing Database*
- *Reconciling Medications &*
- *Administration of the initial drug dose*

Nursing Admission Activities cont.

- **Preparing the Client's Room**
  - Preparing the specific room for the client after information from the admission department and before the client arrival to department.
  - Room prepared generally by all basic supplies and
  - Any special supplies needed for the special care to the client before arrival.

Nursing Admission Activities cont.

- **Preparing client room, each bedside stand is generally stocked with the following:**
  - A wash basin
  - A soap dish
  - An emesis basin
  - A water carafe
  - A bedpan and a urinal









Nursing Admission Activities cont.

Welcoming the Client and family

- One of the most important steps in admission is to make the client feel welcome.
- On arrival, the admitting nurse greets the client warmly with a smile and a Handshake.
- Nurse wears a name tag, introduces himself or herself, and also introduces other personnel in the area. Being treated courteously helps the client relax.
- A client who feels unwanted is likely to have a poor, and lasting, negative first impression of the unit.





Inappropriate Welcoming









Nursing Admission Activities cont.

- **Orienting the client & family,** helping a person become familiar with a new environment, facilitates comfort and adaptation. The nurse describes the following:
  - *The location of the nursing station, toilet, shower or bathing area, and lounge available to the client.*
  - *Where to store clothing and personal items*
  - *How to call for nursing assistance from the bed and bathroom*
  - *How to adjust the hospital bed. How to regulate the room lights*
  - *How to use the telephone. How to operate the television*
  - *The daily routine such as meal times., when the doctor usually visits*
  - *When surgery is scheduled, When laboratory or diagnostic tests are performed*
- NB. Some hospitals provide booklets with information about the agency

Nursing Admission Activities cont.

Safeguarding Valuables and Clothing

- *Nurses give certain items, such as prescription and nonprescription medications, and*
- *Valuable jewelry, and large sums of money, to family members to take home.*

Helping the Client Undress& Dress

- *Provides privacy*
- *Has the client sit on the edge of the bed, or helps the client lie down if weak or tired*
- Gathers each stocking, sliding it down the leg and over the foot
- *Releases fasteners such as zippers and buttons and removes clothing in comfortable way*
- *Lifts the client's head to guide garments over it*
- Rolls the client from side to side to remove clothes
- Rolls the client from side to side to remove clothes
- *Puts a hospital gown on the client, explaining that hospital gowns fasten in the back*

Nursing Admission Activities cont.

- **Compiling the Nursing Database**
  - *On admission, the nurse begins assessing the client and collecting information for the database.*
  - *Physical assessment skills, which include taking vital signs*
- **Reconciling Medications**
  - *Medication reconciliation refers to obtaining and verifying the medications a client is currently taking, dosage, frequency, and route are necessary , and reporting for physician.*
- **Administering initial treatment (ttt) dose**
  - *The nurse must give the drug ordered immediately by the end of admission process.*
  - *Then programing the frequency of administration of drug up on hospital policy.*

Care plan for client

Initial Nursing Plan for Care

- *Once all admission data are collected, the nurse develops an initial plan for the*
- *client's care as soon as possible but no later than 24 hours following admission.*

Medical Admission Responsibilities

- *After completing admission if client needs medical care notify physician for:*
- *Physician must take medical assessment.*
- *Doing tests if needed.*
- *Prescription of medications.*
- *Prescription for formulas*
- *Activities and rest*
- *Physician must sign the medical sheets and stamped it.*

Common Responses to Admission

- *Anxiety*
- *Fear*
- *Decreased Privacy*
- *Decisional conflict*
- *Situational low self-esteem*
- *Powerlessness*
- *Social isolation & Loneliness*
- *Ineffective self-health management*
- *Loss of Identity*

Client Caring for Responses to Admission

- *Let client express all his feelings and concerns*
- *Discuss wit him his situation, listen carefully.*
- *Build trust with client and family.*
- *Supporting the client and family.*
- *Tell client and family success stories fore previous case like his situation.*
- *let client feel that he in place like home.*
- *Treat client as human not as a file or number.*
- *Tell the client that visits not restricted except in situations affect his health status.*
- *Tell the client that he is a partenner and encourage him/her in sharing the plan of care.*

THE DISCHARGE PROCESS

- **Discharge; is the termination of care from a health care agency: generally consists of discharge planning,**
  - ***obtaining a written medical order,***
  - ***completing discharge instructions,***
  - ***notifying the business office,***
  - ***helping the client leave the agency,***
  - ***writing a summary of the client's condition at discharge, and***
  - ***requesting that the room be cleaned.***

Discharge Planning

- **Discharge planning is a process that improves client outcomes by:**
  - ***(1) predetermining his or her post-discharge needs in a timely manner and***
  - ***(2) coordinating the use of appropriate community resources to provide a continuum of care.***
- **Obtaining Authorization for Medical Discharge**
- **Providing Discharge Instructions**
- **Notifying the Business Office**
- **Discharging a Client**
- **Writing a Discharge Summary**
- **Terminal Cleaning**

Discharge Process cont.

- **Obtaining Authorization for Medical Discharge**
  - The physician determines when the client is well enough for discharge.
  - Generally, he or she waits to write the medical order until examining the client.
  - The physician writes the discharge order provides:
  - written prescriptions for the client, and
  - indicates when and where a follow-up appointment should occur.
  - Leaving against medical advice (AMA) is a term that applies to situations in which the client leaves before the physician authorizes the discharge.
- **Providing Discharge Instructions**
  - When the nurse anticipates that a client will be discharged home,
  - he or she establishes the anticipated knowledge, skills, and community resources that the client will need to maintain a safe level of self-care.

Discharge Process cont.

Providing Discharge Instructions

TABLE 11-2 The METHOD Discharge Planning Guide

TOPIC	NURSING ACTIVITY	EXAMPLE
M—Medications	Instruct the client about drugs that will be self-administered.	Insulin
E—Environment	Explore how the home environment can be modified to ensure the client's safety.	Remove scatter rugs
T—Treatments	Demonstrate how to perform skills involved in self-care and provide opportunities for returning the demonstration.	Dressing changes
H—Health teaching	Identify information that is necessary for maintaining or improving health.	Signs and symptoms of complications
O—Outpatient referral	Explain what community services are available that may ease the client's transition to independent living.	Physical therapy
D—Diet	Arrange for the dietitian to provide verbal and written instructions on modifying or restricting certain foods or suggestions for altering their methods of preparation.	Low-fat diet

Medication and Treatment use 5Ws for instructions

- Where to access medication
- When take ttt & frequency
- What to do in administration
- Why have this ttt
- With what take the ttt

In out patient referral tell pt. the:

- Day of referral
- Time of referral
- Place and Phone

Discharge Process cont.

- **Notifying the Business Office**
  - *Before the client leaves the agency, the nurse notifies the business office for verification of insurance or for future payments.*
- **Discharging a Client**
  - *Gathering Belongings*
  - *Arranging Transportation*
  - *Escorting the Client*

Discharge Process cont.

- **Writing a Discharge Summary**
  - ***After the client has left the health care agency, the nurse documents the discharge activities and client's condition as:***
    - Vital signs before leaving
    - Discharge reasons, date and time
    - General patient condition (consciousness, activity, feeding …)
    - Each instructions you give to the patient about his condition.
- **Terminal Cleaning**
  - ***Except in unusual circumstances, housekeeping personnel prepare the client's room for the next admission.***

THE TRANSFER PROCESS

- **A transfer**
  - *discharging a client from one unit or agency and admitting him or her to another without going home in the interim.*
  - *may occur when a client's condition improves or worsens.*
  - *Advantage for the client, It may facilitate more specialized care in a life-threatening situation, or it may reduce health care costs.*

Transfer Activities

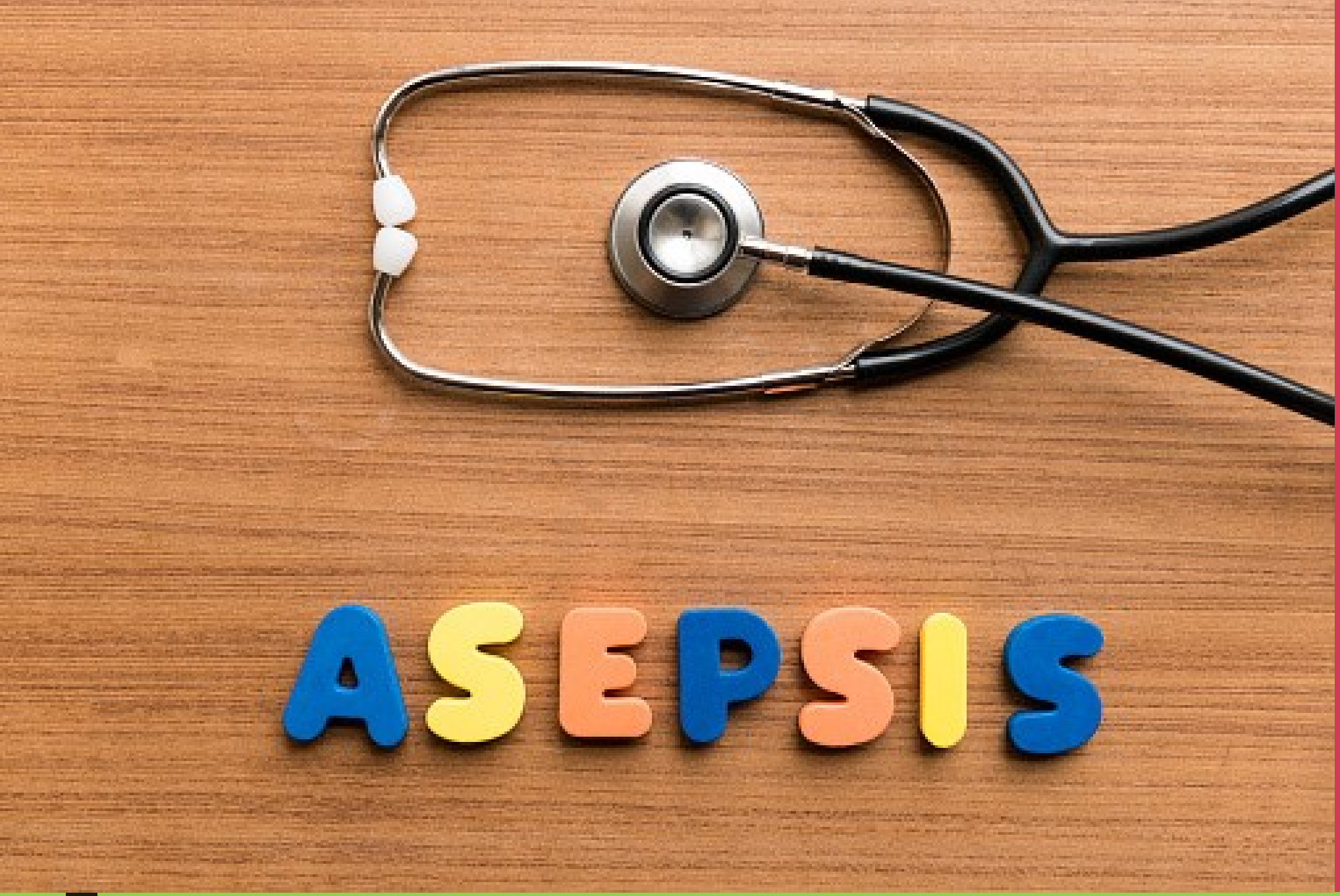
- **Transferring a client to a different nursing unit is less complex than to another agency. In a transfer within the same agency, the nurse does the following:**
  - ***Informs the client and family about the transfer***
  - ***Completes a transfer summary.***
  - ***Speaks with a nurse on the transfer unit to coordinate the transfer.***
  - ***Transports the client and his or her belongings, medications, nursing supplies, and chart to the other unit.***

THE REFERRAL PROCESS

- **A referral**
  - *is the process of sending someone to another person or agency for special services.*
  - *Referrals generally are made to private practitioners or community agencies.*
- **In referral the client must be oriented fore:**
  - *Day of referral*
  - *Time of referral*
  - *Place and*
  - *Phone to ask and clarify*


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God Bless You



ASEPSIS

PART I

Objectives

- By the end of this lecture the student will be able to:
  1. *Name and describe eight specific types of microorganisms.*
  2. *Differentiate between nonpathogens and pathogens. and aerobic and anaerobic microorganisms.*
  3. *Name the six components in the chain of infection.*
  4. *Discuss the concept of asepsis.*
  5. *Differentiate between medical and surgical asepsis.*
  6. *Identify at least three principles of medical asepsis, and examples.*
  7. *Name at least three techniques for sterilizing equipment.*
  8. *Identify principles of surgical asepsis.*
  9. *List at least three nursing activities that require application of the principles of surgical asepsis.*

Microorganisms

- **Microorganisms**, living animals or plants visible only through a microscope, are commonly called “microbes” or “germs.”
- **Microorganisms are present everywhere:**
  - ***in the air***
  - ***soil, and water,***
  - ***on and within virtually everything***
  - ***and everyone.***

Types of Microorganisms

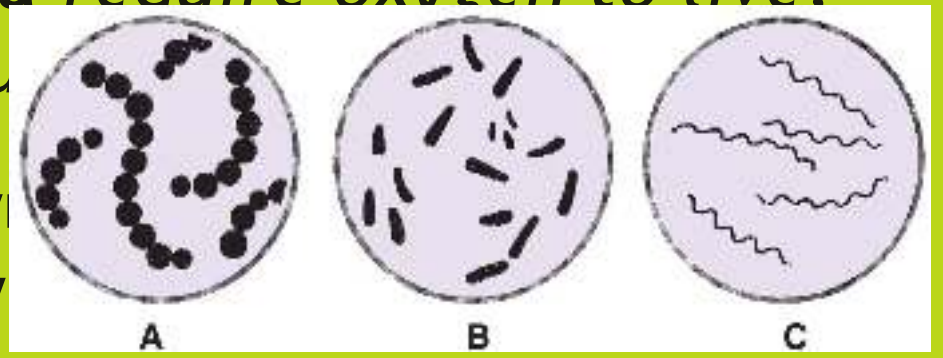
Microorganisms are divided into two main groups:

- Nonpathogens, or normal flora (harmless, beneficial microorganisms), and
- Found in areas of the body exposed to the external environment, such as the skin, nose, mouth, throat, lower urethra, and intestines.
- Pathogens (microorganisms that cause illness).
- Pathogens have a high potential for causing infectious diseases that can be transmitted to other people or also called:
  - ***communicable diseases***
  - ***contagious diseases*** *and*
  - ***community-acquired infectio***

Some examples of communicable diseases are measles, streptococcal sore throat, sexually transmitted infections, and tuberculosis (TB).

Types of Microorganisms cont.

1. Bacteria; are single-celled microorganisms.
   - *They appear in various shapes: round (cocci), rod-shaped (bacilli), and spiral (spirochetes). **Aerobic bacteria** require oxygen to live, whereas **anaerobic bacteria** exist withou*



Classification of bacteria according to shape: cocci (A), bacilli (B), and spirochetes (C).

2. Viruses; the smallest microorganisms know
   to cause infectious diseases, are visible only
   through an electron microscope.
3. Fungi; include yeasts and molds. Only a few
   types of fungi produce infectious diseases in humans.
4. Rickettsiae; resemble bacteria; like viruses

Types of Microorganisms cont.

5. Protozoans; are single-celled animals classified according to their ability to move. Some use amoeboid motion (Amoeba).
6. Mycoplasmas; lack a cell wall. They are referred to as pleomorphic because they assume various shapes.
7. Helminths; are infectious worms, some of which are microscopic.
8. Prions; A prion is a protein containing no nucleic acid.

CHAIN OF INFECTION

- The six essential components in the **chain of infection** (the sequence that enables the spread of disease-producing microorganisms), must be in place if pathogens are to be transmitted from one location or person to another:
  1. An infectious agent
  2. A reservoir for growth and reproduction
  3. An exit route from the reservoir
  4. A means of transmission
  5. A portal of entry
  6. A susceptible host

Chain of Infection

Next Sick Person

(Susceptible Host)

- Babies
- Children
- Elderly
- People with a weakened immune system
- Unimmunized people
- Anyone



Germs

(Agent)

- Bacteria
- Viruses
- Parasites
  - Fungi
  - Rickettsiae
  - Protozoans
  - Mycoplasmas
  - Helminths
  - Prions



How Germs Get In

(Portal of Entry)

- Mouth
- Cuts in the skin
- Eyes
- Mucous membrane
- GI tract
- GU tract
- Respiratory tract
- Broken skin



Where Germs Live

(Reservoir)

- People
- Animals/Pets (dogs, cats, reptiles)
- Wild animals
- Food
- Soil
- Water



Germs Get Around

(Mode of Transmission)

- Contact (hands, toys, sand)
- Droplets (when you speak, sneeze or cough)



How Germs Get Out

(Portal of Exit)

- Mouth (vomit, saliva)
- Cuts in the skin (blood)
- During diapering and toileting stool)



Routes of Transmission

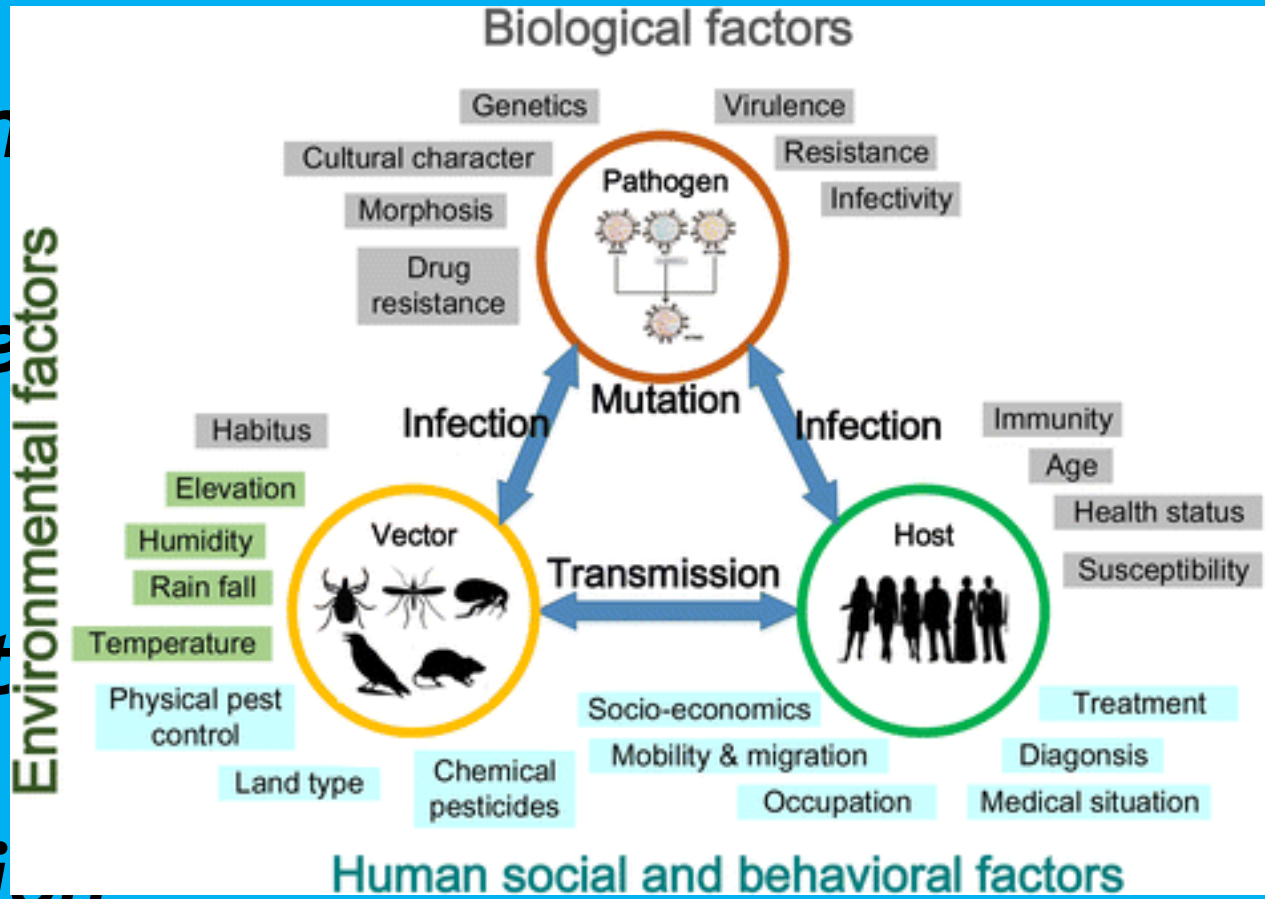
- Transmission of pathogens to humans occurs through many different routes. However, many of the clinically relevant pathogens have mainly six modes of

Methods of Transmission

ROUTE	DESCRIPTION	EXAMPLE
Contact transmission		
Direct contact	Actual physical transfer from one infected person to another (body surface to body surface contact)	Sexual intercourse with an infected person
Indirect contact	Contact between a susceptible person and a contaminated object	Use of a contaminated surgical instrument
Droplet transmission	Transfer of moist particles from an infected person who is within a radius of 3 feet	Inhalation of droplets released during sneezing, coughing, or talking
Airborne transmission	Movement of microorganisms attached to evaporated water droplets or dust particles that have been suspended and carried over distances greater than 3 feet	Inhalation of spores
Vehicle transmission	Transfer of microorganisms present on or in contaminated items such as food, water, medications, devices, and equipment	Consumption of water contaminated with microorganisms
Vector transmission	Transfer of microorganisms from an infected animal carrier	Diseases spread by mosquitoes, fleas, ticks, or rats
Infection via blood and tissue	Transfer of microorganism Infection via blood and tissue, body secretions, e.g. saliva, sweat, pus, or semen.	Hepatitis B, HIV, for example, is transmitted through sexual contact.

Factors Affecting Susceptibility to Infections

- *Inadequate nutrition*
- *Poor hygiene practices*
- *Suppressed immune syste*
- *Chronic illness*
- *Insufficient white blood ce*
- *Prematurity*
- *Advanced age*
- *Compromised skin integrit*
- *Weakened cough reflex*
- *Diminished blood circulatio*



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- Health care institutions are teeming reservoirs of microorganisms because of the sheer numbers of confined sick people.
- Nurses must understand and practice methods to prevent health care-associated infections
- Nosocomial Infection / Cross Infections; (infections acquired while a person is receiving care in a health care agency).
- Asepsis; refers to those practices that decrease or eliminate infectious agents, their reservoirs, and vehicles for transmission.
- It is the major method for preventing and controlling infection. Health care providers use:
  - *Medical and*
  - *Surgical asepsis to accomplish this goal.*

Medical Asepsis

- **Medical asepsis** means those practices that **confine or reduce the numbers of microorganisms. Also called *clean technique***
- **The following principles underlie medical asepsis:**

1. Microorganisms exist everywhere except on sterilized equipment.
2. Frequent hand washing is the best method for reducing the transmission of microorganisms.
3. Blood, body fluids, cells, and tissues are considered major reservoirs of microorganisms.
4. Personal protective equipment (PPE) such as gloves, gowns, masks, goggles, hair and shoe covers serve as barriers to microbial transmission.
5. A clean environment reduces microorganisms.
6. Certain areas —floor, toilets, & the insides of sinks—are more contaminated than others.
7. Cleaning should be done from cleaner to dirtier areas.

Using Antimicrobial Agents

- **Antimicrobial agents;** are chemicals that destroy or suppress the growth of infectious microorganisms.
- Some antimicrobial agents are used to clean equipment, surfaces, and inanimate objects.
- Others are applied directly to the skin or administered internally.
- Examples are antiseptics:
  - *disinfectants,*
  - *anti-infective drugs.*

TABLE 10-2 Antimicrobial Agents

TYPE	MECHANISM	EXAMPLE	USE
Soap	Lowers the surface tension of oil on the skin, which holds microorganisms; facilitates removal during rinsing	Dial, Safeguard	Hygiene
Detergent	Acts as soap, except detergents do not form a precipitate when mixed with water	Dreft, Tide	Sanitizing eating utensils, laundry
Alcohol	Injures the protein and lipid structures in the cellular membrane of some microorganisms (70% concentration)	Isopropyl ethanol	Cleansing skin, instruments
Iodine	Damages the cell membrane of microorganisms and disrupts their enzyme functions, not effective against *Pseudomonas*, a common wound pathogen	Betadine	Cleansing skin
Chlorine	Interferes with microbial enzyme systems	Bleach, Clorox	Disinfecting water, utensils, blood spills
Chlorhexidine	Damages the cell membrane of microorganisms, but is ineffective against spores and most viruses	Hibiclens	Cleansing skin and equipment
Glutaraldehyde	Inactivates cellular proteins of bacteria, viruses, and microbes that form spores	Cidex	Sterilizing equipment

Antimicrobial Agents

- **Antiseptics;** also known as bacteriostatic agents, inhibit the growth of, but do not kill, microorganisms. An example is alcohol.
  - *Antiseptics generally are applied to the skin or mucous membranes.*
  - *Some are also used as cleansing agents.*
- **Disinfectants;** also called germicides and bactericides, destroy active icroorganisms but not spores. Phenol, household bleach, and formaldehyde are examples.
  - *rarely applied to the skin because they are very strong;*
  - *they are used to kill and remove microbes from equipment, walls, and floors.*

Antimicrobial Agents cont.

- **Anti-infective Drugs** The two groups of drugs used most often to combat infections are antibacterials and antivirals.
  - *The chemical actions of antibacterials, which consist of antibiotics and*
  - *sulfonamides, alter the metabolic processes of bacteria but not viruses.*
  - *Antiviral agents were developed more recently, most likely in response to the rising incidence of influenza and blood-borne viral diseases such as hepatitis and AIDS.*

Hand Hygiene

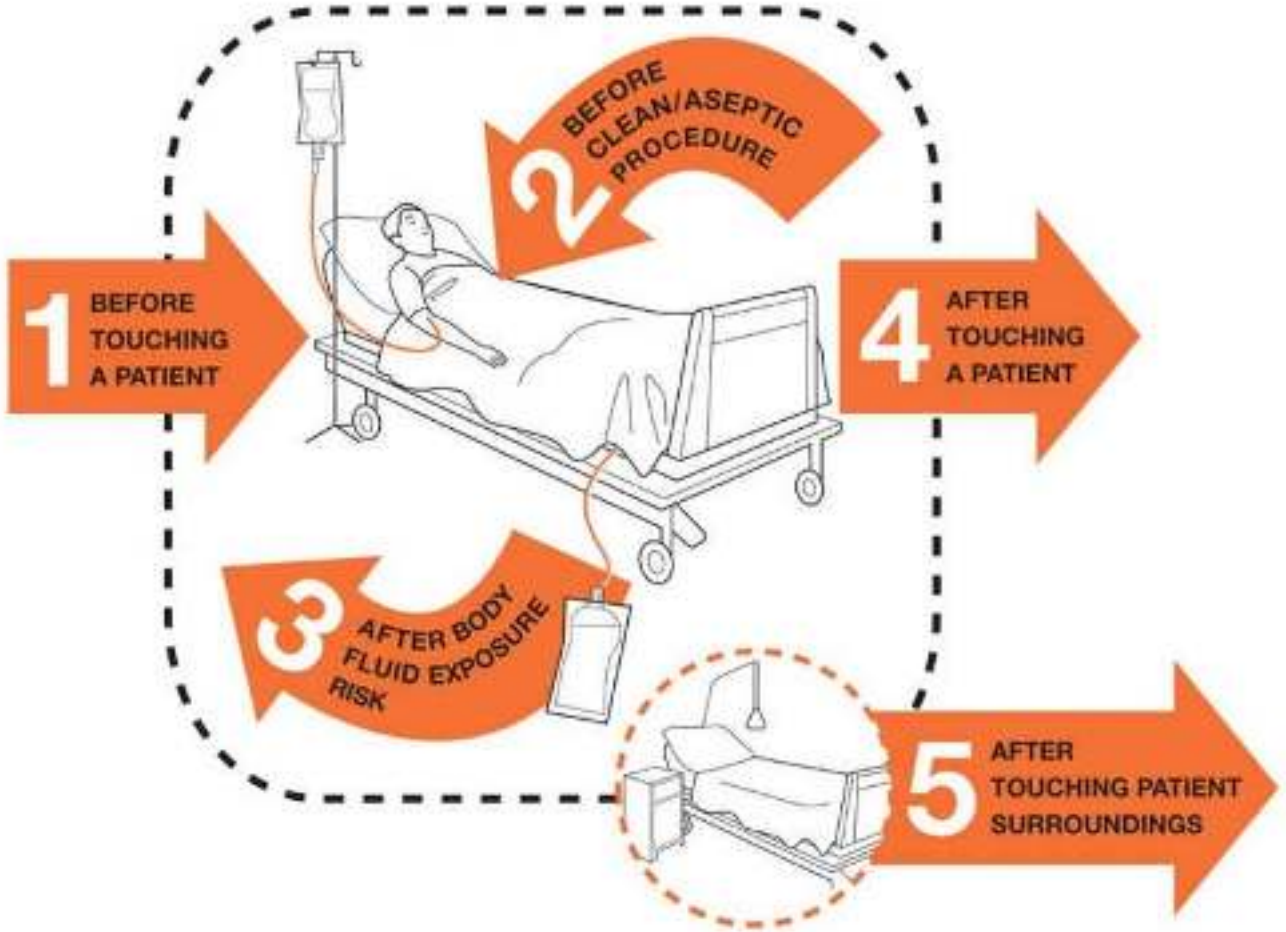
- **Hand hygiene** refers to removing surface contaminants on the skin by either hand washing or hand antisepsis.
  - *Is a medical aseptic practice that involves cleaning hands with soap, water, and friction to remove dirt and organic substances.*
  - *Removes resident microorganisms (nonpathogens constantly present on the skin) and transient microorganisms (pathogens picked up during brief contact with contaminated reservoirs).*
- **Hand antisepsis** means
  - *Removal and destruction of transient microorganisms without soap and water.*
  - *It involves products such as alcohol-based liquids, thick gels, and foams.*

Hand Washing & Hand Antisepsis Guidelines

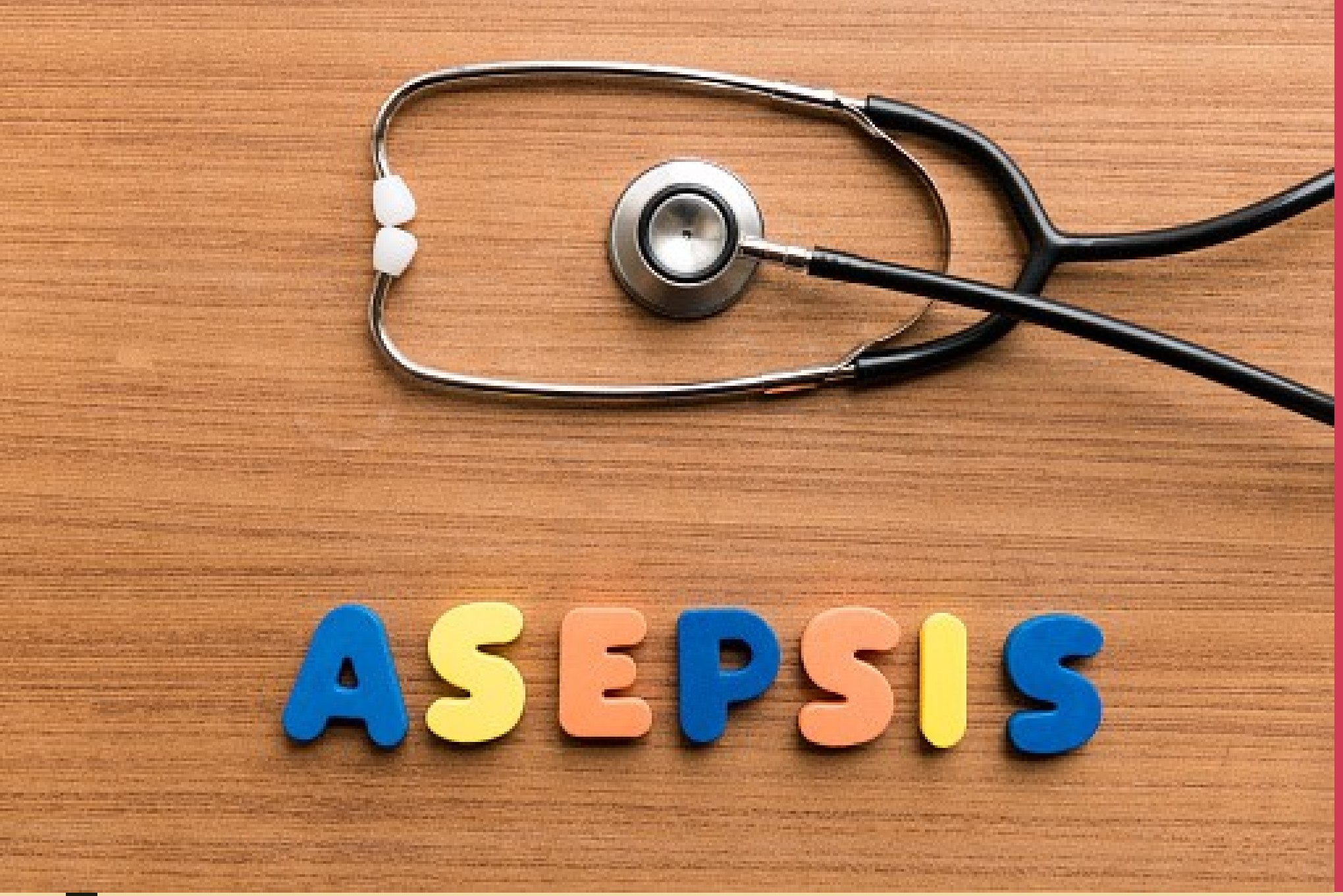
- When hands are visibly dirty
- When hands are contaminated with proteinaceous material
- When hands are visibly soiled with blood or other body fluids
- Before eating and after using the restroom
- If exposure to spore-forming pathogens is suspected or proven
- Hand antisepsis with an alcohol-based hand rub can be for hand washing:
  - *Before having direct contact with clients*
  - *After contact with a client's intact skin.*
  - *Before donning sterile gloves to insert invasive devices such as urinary catheters.*
  - *After contact with body fluids or excretions.*
  - *If moving from a contaminated body site to a clean body site during client care*
  - *After taking off gloves because gloves are not an impervious barrier*



Your 5 Moments for Hand Hygiene




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PART II

Performing Surgical Hand Antisepsis

- Surgical hand antisepsis, previously referred to as a surgical scrub, is a medically aseptic hand-hygiene procedure that is performed before donning
- sterile gloves and garments when the nurse is actively involved in an operative
- or obstetric procedure.
- To maximally reduce the number of microorganisms, the fingernails must be short, not extend beyond the tip of the fingers.
- Artificial nails are prohibited.
- Nail polish is discouraged, because it is conducive to harboring an increased number of microorganisms.
- All rings, watches, and jewelry are removed before surgical hand antisepsis.

Differences Between Hand Washing and Surgical Hand Antisepsis

HAND WASHING	SURGICAL HAND ANTISEPSIS
Plain wedding band may be worn.	All hand jewelry, including watches, are removed.
Faucets with hand controls are used; elbow, knee, or foot controls are preferred.	Faucets are regulated with elbow, knee, or foot controls.
Liquid, bar, leaflet, or powdered soap or detergent is used.	Liquid antibacterial soap is used; devices such as sponges may be incorporated with antibacterial soap.
Washing lasts a minimum of 20 seconds.	Antisepsis lasts 2–6 minutes, depending on the antibacterial agent and time interval between subsequent repetitions.
Hands are held lower than the elbows during washing, rinsing, and drying.	Hands are held higher than the elbows during washing, rinsing, and drying.
Areas beneath fingernails are washed.	Areas beneath fingernails are cleaned with an orange stick or similar nail cleaner.
Friction is produced by rubbing the hands together.	Friction is produced by scrubbing with a brush and/or sponge.
Hands are dried with paper towels; the paper is used to turn off hand-regulated faucet controls.	Hands are dried with sterile towels.
Clean gloves are donned if the nurse has open skin or if there is a potential for contact with blood or body fluids.	Sterile gloves are donned immediately after the hands are dried.

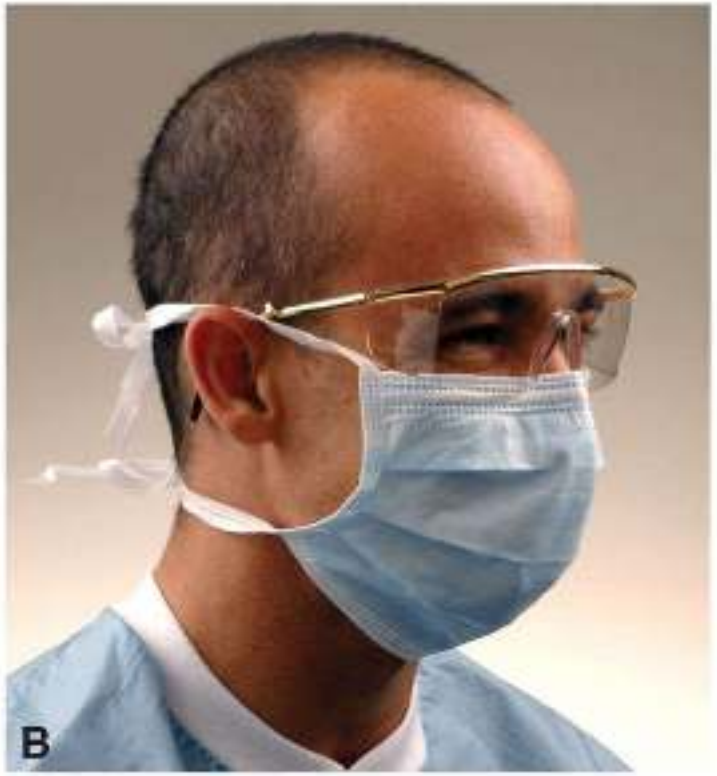
Wearing Personal Protective Equipment (PPE)

- **Uniforms**
  - ***Health care providers should wear their uniforms only while working with clients.***
- **Scrub Suits and Gowns**
  - *Scrub suits and gowns are hospital garments instead of a uniform.*
  - *Their use is mandatory in some areas of a hospital, such as the nursery, the operating room, and the delivery room.*
  - *These garments prevent health care providers from bringing microorganisms on their clothes into the working environment.*

Wearing Personal Protective Equipment (PPE) cont.

- ***Masks***
  - ***Masks are disposable, loose-fitting covers for the nose and mouth .***
  - ***They help prevent droplet and airborne transmission of microbes, but not necessarily viruses***
- **Respirators**
  - *Air filter respirator that fits snugly to the face.*
  - *The minimum specification for a particulate air filter respirator is N95.*
  - *N95 air respirator can filter very small particles that may contain viruses.*
  - *The minimum efficiency of 95%.*

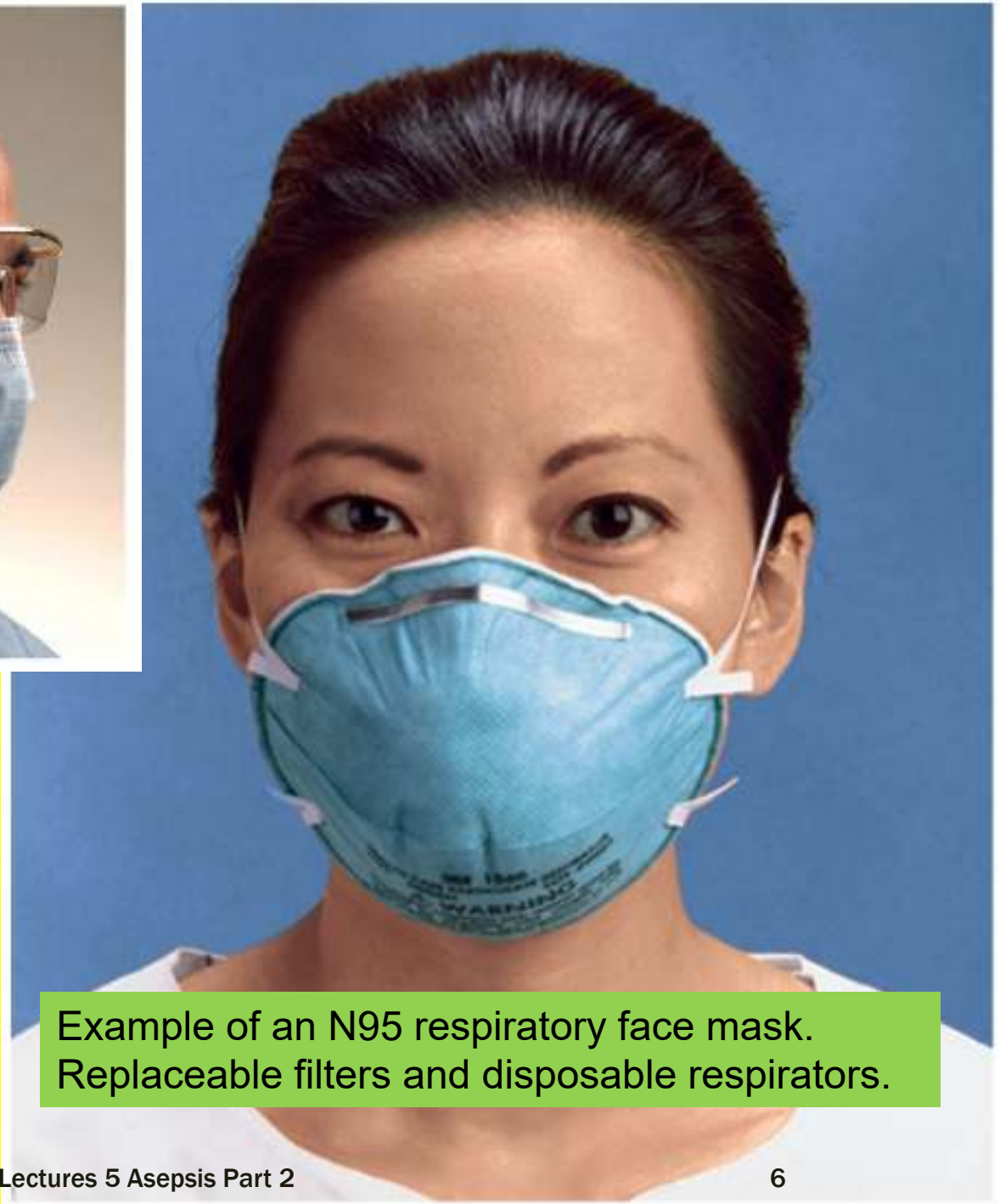






Various types of face masks.
(A) Earloop style.
(B) Tie-on style.

(C) Molded cup style.



Example of an N95 respiratory face mask.
Replaceable filters and disposable respirators.

Wearing Personal Protective Equipment (PPE) cont.

- **Gloves**

  Nurses wear clean gloves, sometimes called "examination gloves," in the following circumstances:

  - *As a barrier to prevent direct hand contact with blood, body fluids, MM, and nonintact skin.*
  - *As a barrier to protect clients from microorganisms transmitted from nursing personnel when performing procedures*
  - *When there is a potential transfer of microorganisms from one client or object to another during subsequent nursing care.*

Wearing Personal Protective Equipment (PPE) cont.

Hair and Shoe Covers:

- *Hair and shoe covers reduce the transmission of pathogens present on the hair or shoes. Health care providers generally wear these garments during surgical or obstetric*

Protective Eyewear

- *Protective eyewear is essential when there is a possibility that body fluids will splash into the eyes.*
- *Goggles are worn along with a mask, or a multipurpose face shield is used.*



Eye Goggles

Confining Soiled Articles

Utility Rooms

- *Health care agencies have at least two utility rooms:*
- *Clean utility room and the other considered dirty utility room.*
- *Personnel must not place soiled articles in the clean room.*
- *The dirty or soiled utility room contains covered waste receptacles, large laundry hamper, and a flushable hopper.*
- *This room also houses equipment for testing stool or urine.*
- *A sink is located in the soiled room for hand washing.*



Waste Receptacles

- *Wastebaskets generally are lined with plastic.*
- *Suction and drainage containers are kept covered and emptied at least once during each shift.*



Keeping the Environment Clean

- ***Housekeeping personnel*** *are responsible for collecting and disposing of accumulated trash and for performing concurrent and terminal disinfection.*
- *Housekeepers who follow the* ***principles of medical asepsis,*** *disinfection or measures that keep the client environment clean on a daily basis:*

1. *They clean less soiled areas before grossly dirty ones.*
2. *They wet-mop floors and damp-dust furniture to avoid dispersing microorganisms on dust particles and air currents.*
3. *They discard solutions used a flushable hopper.*
4. *They never place clean & uncleaned items on the floor.*


A sharps container.

Keeping the Environment Clean cont.

- ***Terminal disinfection is more thorough than concurrent disinfection.***
- ***Measures used to clean a client's environment after discharge.***
- ***It includes scrubbing the mattress surface and the insides of drawers and bedside stands.***
- ***Nurses who work in home health care can teach the client and the family simple aseptic practices for cleaning contaminated articles.***

Surgical Asepsis

- **Surgical asepsis** refers to those measures that render supplies and equipment totally free of microorganisms.
- **Sterile techniques** include practices that avoid contaminating microbe-free items.

_ Both begin with the process of sterilization. _

- **Sterilization**

Consists of physical and chemical techniques that destroy all microorganisms, including spores.

Sterilization

- **Physical Sterilization**
  - *Microorganisms and spores are destroyed physically through radiation or heat, boiling water, free-flowing steam, dry heat, and steam under pressure.*
  - *The autoclave is a type of pressure steam sterilizer that most health care agencies use.*
- **Chemical Sterilization**
  - *Both gas and liquid chemicals are used to sterilize invasive equipment.*
  - *Gas sterilization, using ethylene oxide gas, is a traditional method for destroying microorganisms if heat or moisture is likely to damage items or if no better method is available.*

Autoclave for Under steam pressure sterilization







Pressurized saturated steam at 121 °C for around 15–20 minutes

Principles of Surgical Asepsis

following principles known as sterile technique:

- They preserve sterility by touching one sterile item with another that is sterile.
- Once a sterile item touches something that is not sterile, it is considered contaminated.
- Any partially unwrapped sterile package is considered contaminated.
- If there is a question about the sterility of an item, it is considered unsterile.
- The longer the time since sterilization, the more likely it is that the item is no longer sterile.
- A commercially packaged sterile item is not considered sterile after expiration date.

Principles of Surgical Asepsis cont.

- ***following principles known as sterile technique:***

- Once a sterile item is opened or uncovered, it is only a matter of time before it becomes contaminated.
- The outer 1-inch margin of a sterile area is considered a zone of contamination.
- A sterile wrapper, if it becomes wet, considered contaminated.
- Any opened sterile item or sterile area is considered contaminated if it is left unattended.
- Coughing, sneezing, or excessive talking over a sterile field causes contamination.
- Reaching across an area that contains sterile equipment has a high potential for causing contamination.
- Sterile items that are located or lowered below waist level are considered

Sterile Techniques

Creating a Sterile Field

- *A **sterile field** means a work area free of microorganisms.*
- *It is formed using the inner surface of a cloth that holds sterile items, such like a table cloth.*





- ***Adding Items to a Sterile Field***
- ***Sterile Solutions***
- ***Sterile Items,*** *Agency-sterilized items or those that have been commercially prepared may be added to the sterile field.*



NURSING IMPLICATIONS

- ***Everyone is susceptible to infections, especially if sources of microorganisms among health care providers, clients, equipment, and the agency are not controlled.***

Nurses generally identify pertinent nursing diagnoses like those that follow when caring for particularly susceptible clients:

- Risk for infection
- Ineffective protection
- Delayed surgical recovery
- Deficient knowledge



الله يحفظك

God Bless You


vital signs

Objectives

- *On completion of this chapter, the audience should be able to:*
  1. *Recommendations for Measuring Vital Signs*
  2. *List four physiologic components measured in an assessment of vital signs.*
  3. *List temp. assessment sites, indication & contraindications.*
  4. *Discuss fever and hyperthermia, and the care plan for them.*
  5. *Identify pulse assessment sites. technique and normal and abnormal measures.*
  6. *Explain respiratory process, and abnormal breathing characteristics.*
  7. *Identify blood pressure, sites, techniques and normal and abnormal ranges.*
  8. *Name pieces of equipment for assessing blood pressure.*
  9. *Identify alternative techniques for assessing blood pressure.*

Recommendations for Measuring Vital Signs

- Vital signs are taken:
  - *On admission, when obtaining database assessments*
  - *According to written medical orders*
  - *Once per day when a client is stable*
  - *is unstable or at risk for rapid physiologic changes such as At least every 4 hours when one or more vital signs are abnormal*
  - *Every 5 to 15 minutes when a client after surgery*
  - *Whenever a client's condition appears to have changed*
  - *A second time, or more, when there is a significant difference from the previous measurement*
  - *When a client is feeling unusual*
  - *Before, during, and after a blood transfusion*
  - *Before administering medications that affect any of the vital signs and after to monitor the drug's effect*

BODY TEMPERATURE

- Body temperature refers to the warmth of the human body.
- Body heat is produced from exercise, & metabolism of food.
- Heat is lost through the skin, the lungs, and the body's waste products through the processes of radiation, conduction, convection, and evaporation.
- Shell temperature (warmth at the skin surface) is usually lower than its core temperature (warmth in deeper sites within the body like the brain and heart).
- Core temperature is much more significant than shell temperature.

BODY TEMPERATURE Mechanisms of heat transfer

	RADIATION	CONVECTION	EVAPORATION	CONDUCTION
Definition	The diffusion or dissemination of heat by electromagnetic waves	The dissemination of heat by motion between areas of unequal density	The conversion of a liquid to a vapor	The transfer of heat to another object during direct contact
Example	The body gives off waves of heat from uncovered surfaces.	An oscillating fan blows currents of cool air across the surface of a warm body.	Body fluid in the form of perspiration and insensible loss is vaporized from the skin.	The body transfers heat to an ice pack, causing the ice to melt.
Illustration				

Temperature Measurement

- The Centigrade scale is also known as Celsius. is used more often in scientific research and in countries that use the metric system.
- The Fahrenheit scale generally is used in the United States to measure and report body temperature.

Nurses are required to use both scales

Centigrade & Fahrenheit

occasionally and to convert between the two measurements

To convert Fahrenheit to centigrade, use the formula:

$$°C = \frac{(°F - 32)}{1.8}$$

Example: Step 1: $98.6°F - 32 = 66.6$

Step 2: $66.6 \div 1.8 = 37°C$

To convert centigrade to Fahrenheit:

$$°F = (°C \times 1.8) + 32$$

Example: Step 1: $15°C \times 1.8 = 27$

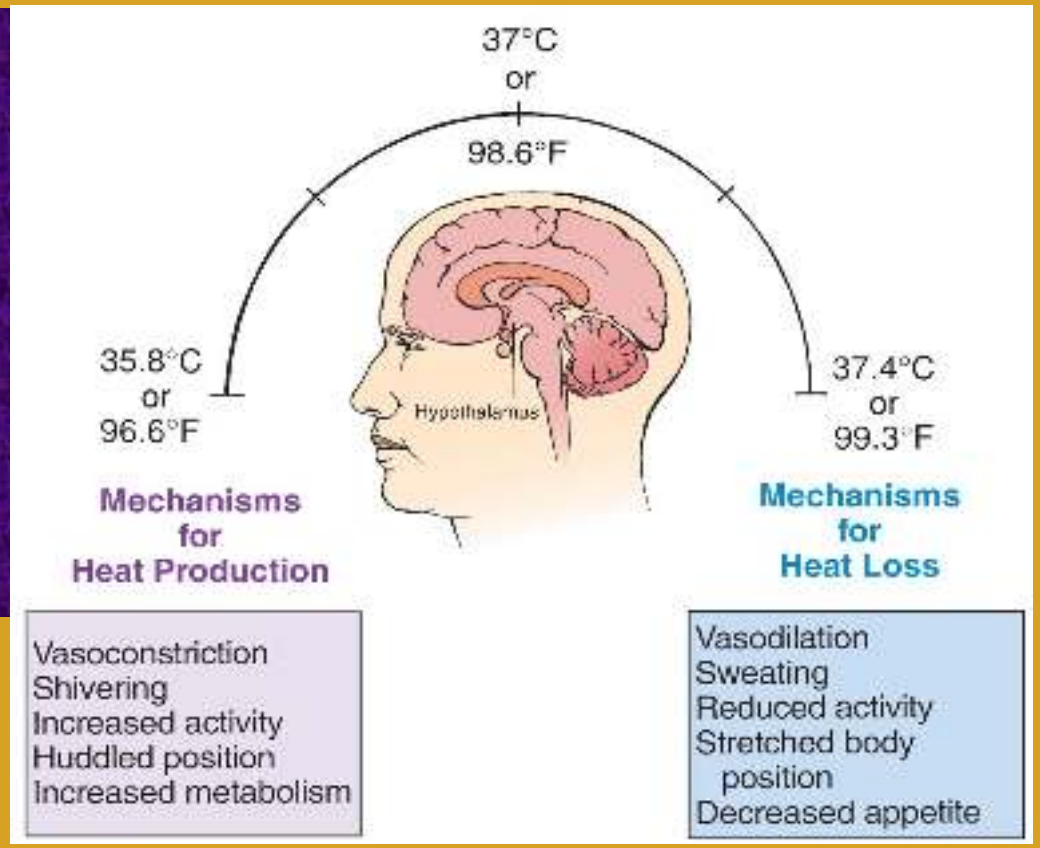
Step 2: $27 + 32 = 59°F$

Temperature regulation

- Humans, are homeothermic; that is, various structural and physiologic adaptations keep their body temperature within a narrow stable range regardless of the environmental temperature.

Regulatory system include:

1. Hypothalamus (Center of process)
2. Circulating blood
3. Skin
4. Fatty tissue



Factors Affecting Body Temperature

1. **Food Intake,** affects thermogenesis (heat production).
2. **Age,** Infants & elders have difficulty maintaining normal temp.
3. **Climate,** affects mechanisms for temperature regulation.
4. **Gender,** Body tem. slightly ↑ in women of childbearing age during ovulation.
5. **Exercise and Activity,** both involve muscle contraction.
6. **Circadian rhythms,** are physiologic changes, such as fluctuations in body temp. and other V.S. over 24-hour cycles.
7. **Emotions,** affect metabolic rate by triggering hormonal changes
8. **Illness or Injury,** that affect the function of the hypothalamus
9. **Medications,** Various medications that affect body temperature

Assessment Sites

1. **Oral,** affects thermogenesis (heat production).
2. **Axilla,** Infants & elders have difficulty maintaining normal temp.
3. **Rectal,** affects mechanisms for temperature regulation.
4. **The Ear (with the use of infra red thermometers)**
   A. **Tympanic membrane,** just (3.8 cm) from the hypothalamus.
   B. **Temporal artery,** receives blood from the aorta, lies less than 2 mm below the skin surface at the forehead.

Assessment Site/description	Normal set point	Time	Difference of core temp.	Contra Indications
Oral **Description** **The most convenient**	37° c	3-5 ms.	0.5° c	1. Vomiting 2. Oral infection 3. Mouth surgery 4. Infant children 5. Convulsions / chilling 6. Obligatory mouth berating 7. Unconsciousness 8. Uncooperative client
Axilla **Description** **The most safe & common**	36.5° c	5-10 ms.	1° c	1. Amputated upper limbs 2. Skin Infections 3. Surgeries 4. Burns 5. 20 ms. after baths
Rectal **Description** **The most accurate**	37.5° c	2-3 ms.	0° c	1. Diarrhea 2. Surgeries 3. If other site available 4. Infections and infestation

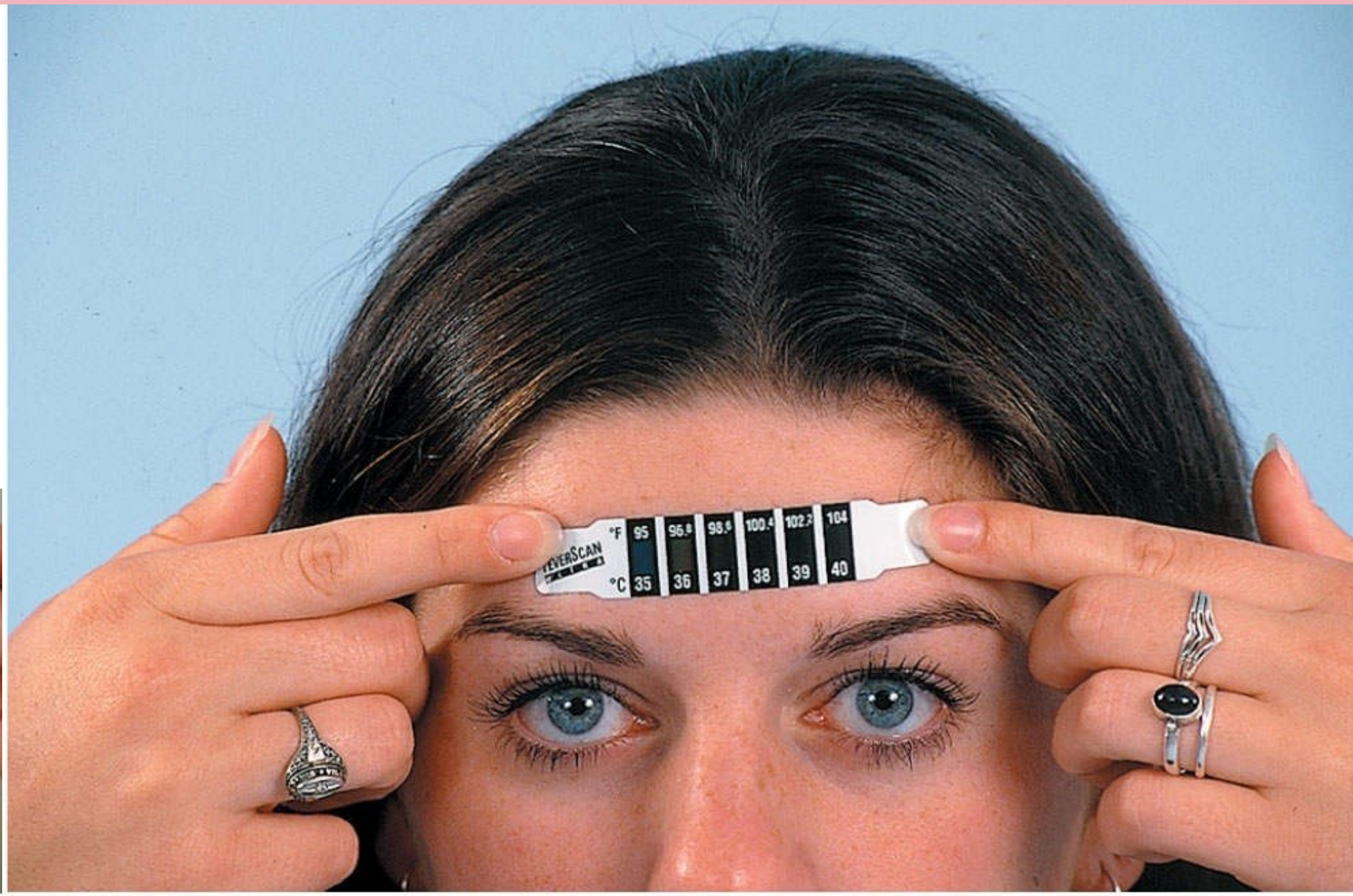
Types of clinical thermometers

TYPE	ADVANTAGES	DISADVANTAGES
Glass	Inexpensive Small Portable	Breakable Difficult to read Cleaning necessary before use by another client Cannot sterilize using heat Time-consuming Accuracy affected by eating, drinking, smoking, talking, mouth breathing, stool in rectum, vasoconstriction of skin, and mucous membranes Porous; possible inaccuracy from mercury evaporation High risk for injury if broken during use Environmental pollution from mercury is possible if not properly disposed.
Digital	Inexpensive Safe; no glass to break or potential mercury spill Memory displays last temperature. Fast; records in 1–3 minutes Audible signal during or after assessment Automatic shutoff to prolong battery Battery life of 200 hours Water resistant, which facilitates cleaning Large, lighted numerical display for ease of reading	Requires a battery (1.55 V) Accuracy of ±0.2°F compared with glass thermometer at 95–102.2°F Accuracy is ±0.4°F compared with glass thermometer at <95 or >102.2°F.
Chemical	Inexpensive Safe; nonbreakable Sanitary Temperature registers in approximately 45 seconds to 3 minutes. Resets in 30 seconds Cleans easily in hot soapy water Easily used by untrained people	Varying measurements at different body sites depending on blood flow and room temperature

Glass, Digital, and Chemical thermometers cont.







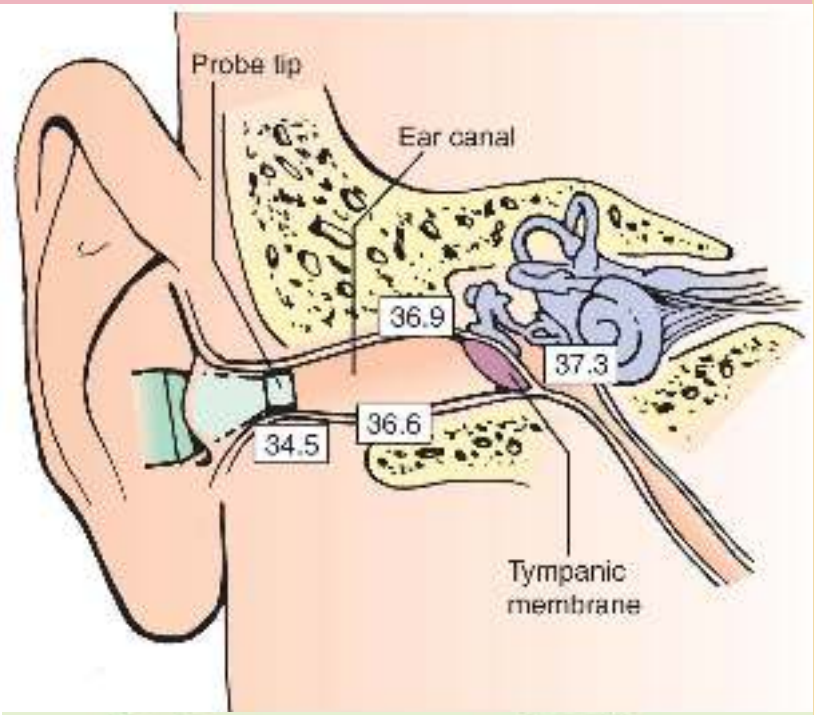
Types of clinical thermometers cont.

TYPE	ADVANTAGES	DISADVANTAGES
Electronic	Faster than glass Accurate No sterilization or disinfection needed Easy to use	Expensive Recharging is necessary. Probe needs to be held by the client or nurse. Interference with simultaneously taking the client's pulse while holding the probe with one hand and unit in the other
Infrared (tympanic)	Fast Convenient Close approximation of core temperature Less invasive Accuracy unaffected by eating, drinking, or breathing Sanitary	Expensive in comparison with others Battery recharging is necessary. Accuracy is affected by improper placement and probe size. Actual ear and core temperature ranges are slightly different from oral, rectal, and axillary sites. Tip requires cleaning with a paper tissue or alcohol swab. Extreme hot or cold environmental temperatures may affect electronics. No sterilization or disinfection is required.
Infrared (temporal artery)	Closest approximate of core temperature Most sanitary Most convenient for clients Records within 2 seconds Initial cost is similar to other types of electronic and tympanic membrane thermometers. Probe covers are not needed; decreases volume of disposal waste Can be used over the femoral artery or lateral thoracic artery if the temporal artery is inaccessible due to bandaging or trauma	User error if the thermometer is moved too quickly across the skin Hair, clothing, or bandages between the probe and the skin can result in falsely high readings. Infrared probe requires cleaning between uses with an alcohol prep pad and dry swab.

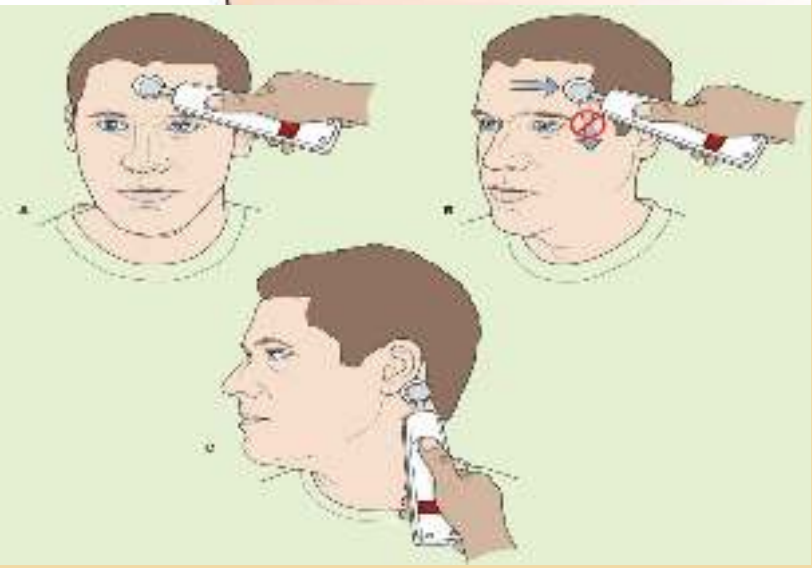
Electronic and Infra red thermometers cont.





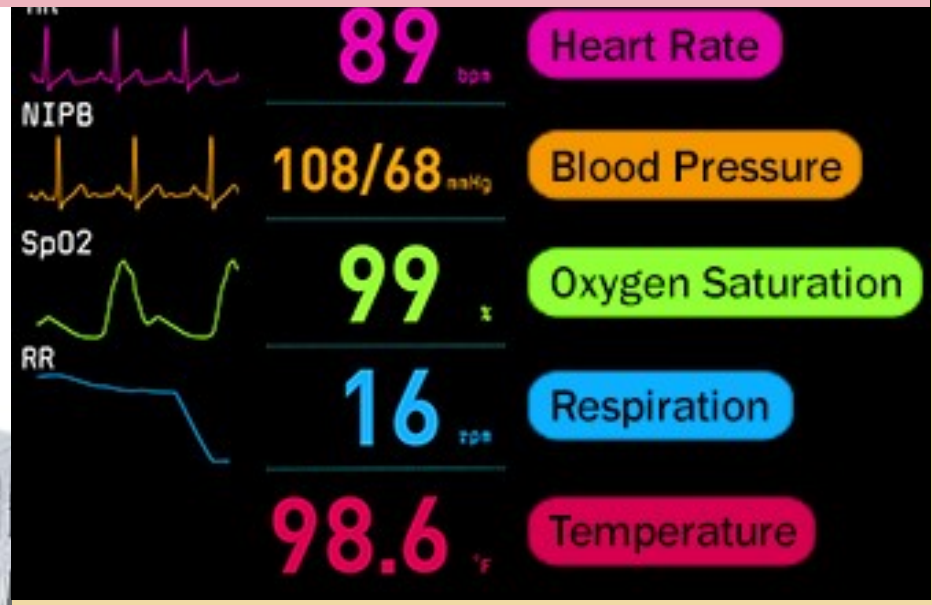


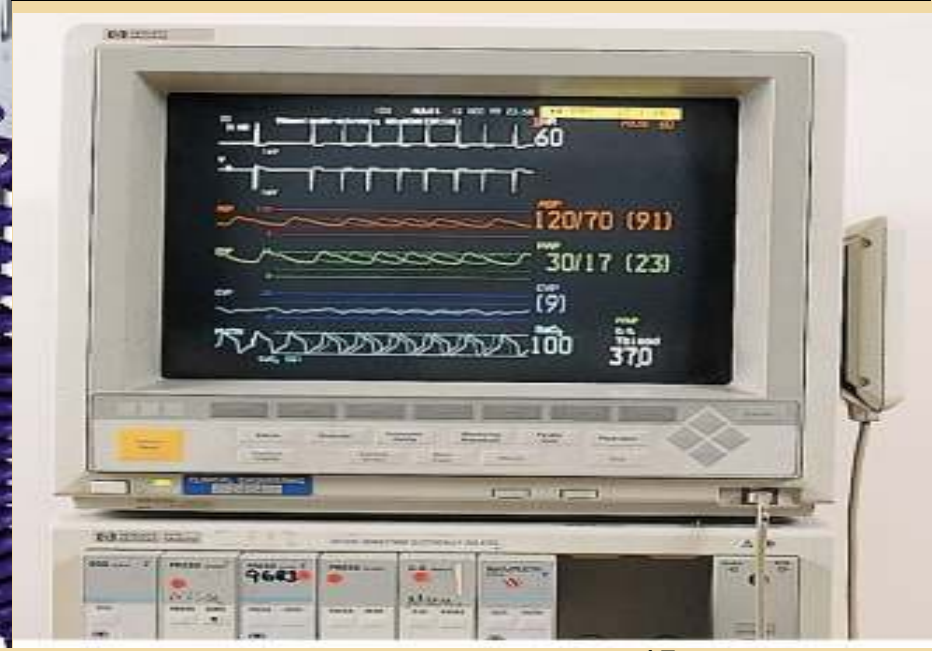




Continuous Vital signs monitoring devices cont.







Elevated Body Temperature

- *A fever (a body temperature that exceeds 37.4 °C is a common indication of illness.*
- *Pyrexia is a term used to describe a warmer than normal set point.*
- *A person with a fever is said to be **febrile** (a condition in which the temperature is elevated)*
- *as opposed to **afebrile** (no fever).*

Elevated Body Temperature; S&S of fever

- Pinkish, red (flushed) skin that is warm to the touch
- Restlessness or, in others, excessive sleepiness
- Irritability
- Poor appetite
- Glassy eyes and a sensitivity to light
- Increased perspiration
- Headache
- Above-normal pulse and respiratory rates (tachycardia and tachypnea)
- Disorientation and confusion (when the temperature is very high)
- Convulsions in infants and children (when the temperature is very high)
- Fever blisters around the nose or lips

After age of 5 there is other reason for convulsions

Elevated Body Temperature ↑ 40.6°C

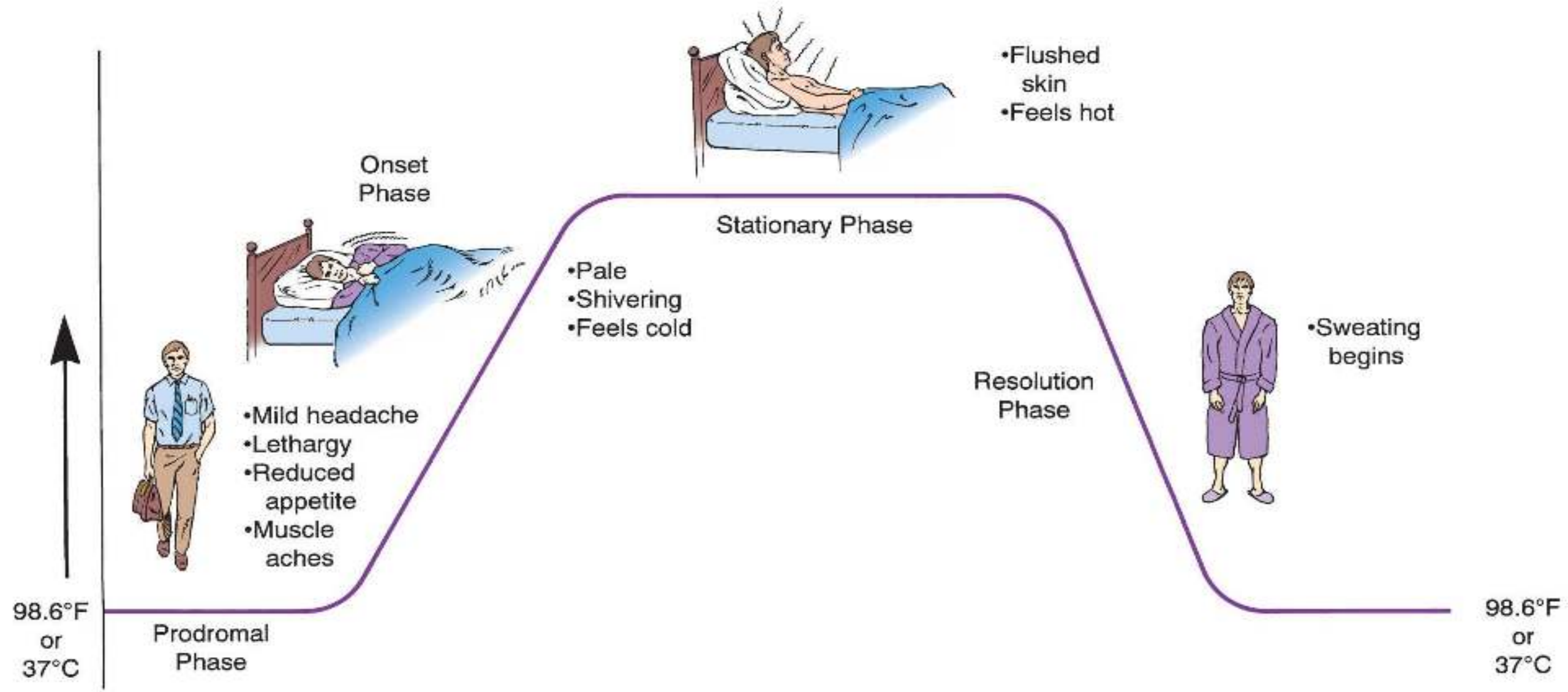
- **Hyperthermia** (excessively high core temperature)
- Temperature exceeds 40.6°C.
- At this level, the person is at extremely high risk for:
  - *brain damage or*
  - *death from complications associated with increased metabolic demands.*

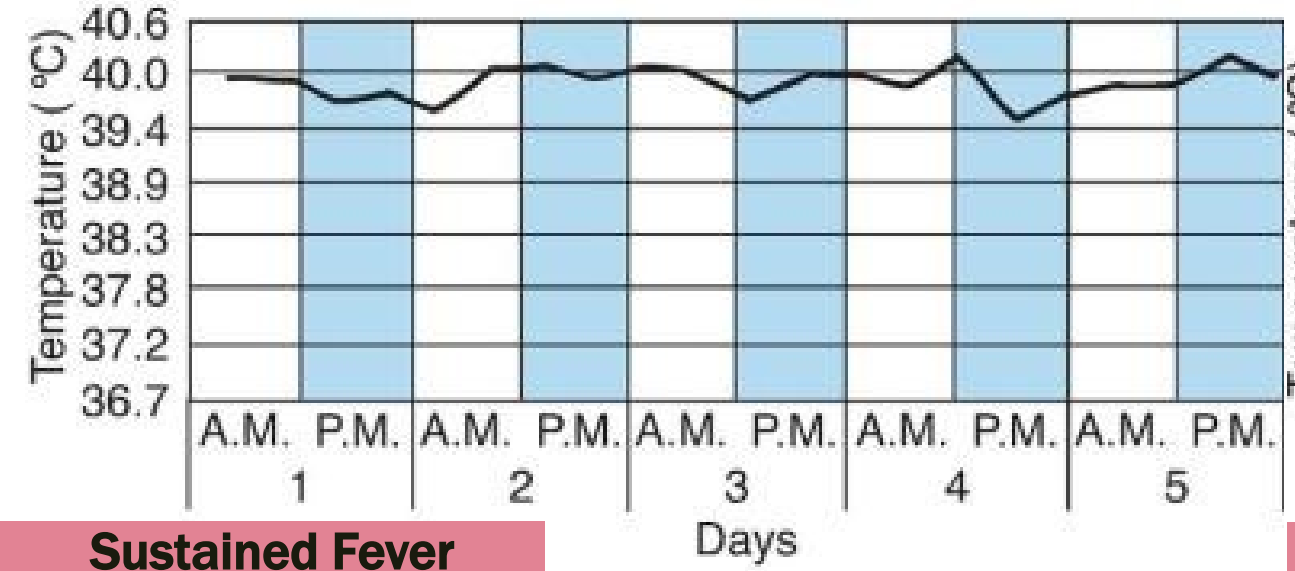


Phases of a Fever

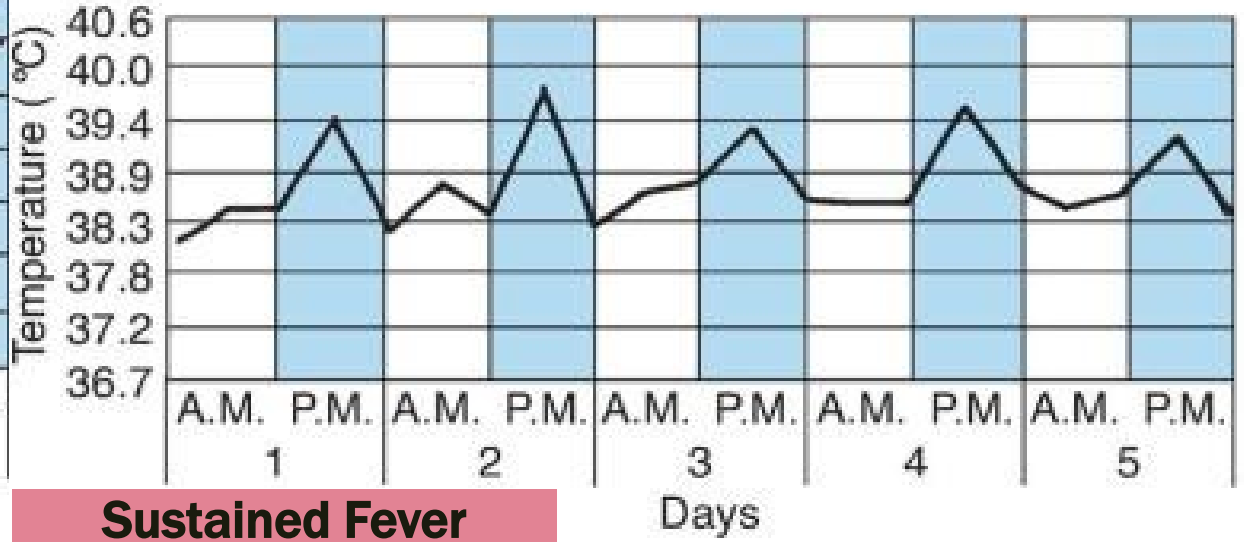
1. Prodromal phase: The client has nonspecific symptoms just before the temperature rises.
2. Onset or invasion phase: Obvious mechanisms for increasing body temperature, such as shivering, develop.
3. Stationary phase: The fever is sustained.
4. Resolution or defervescence phase: The temperature returns to normal

Phases of a Fever cont.

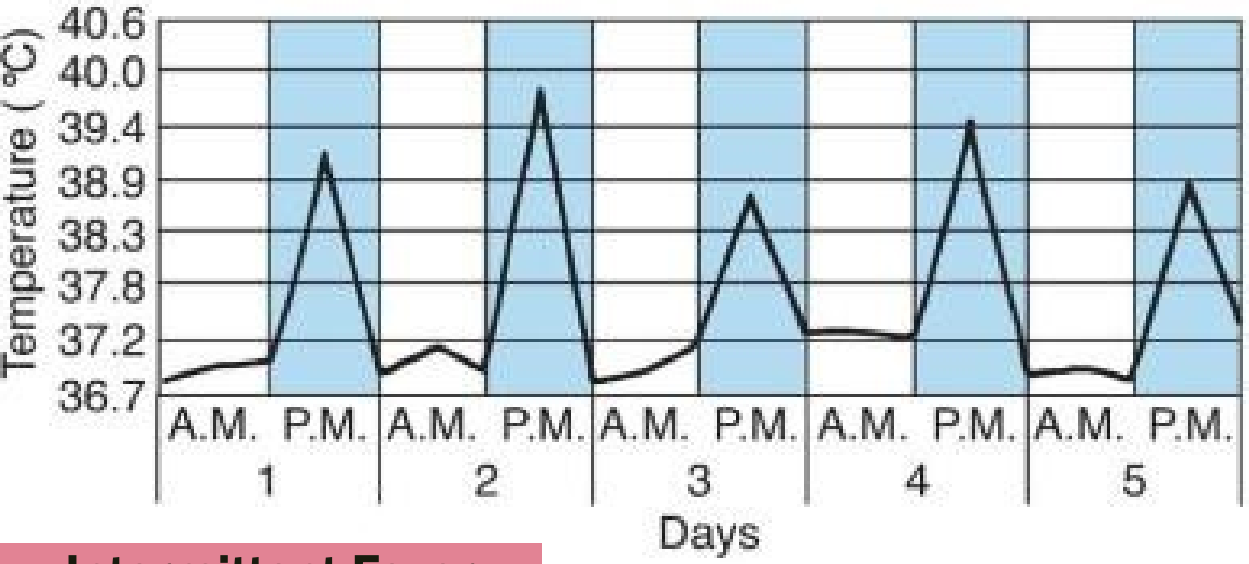




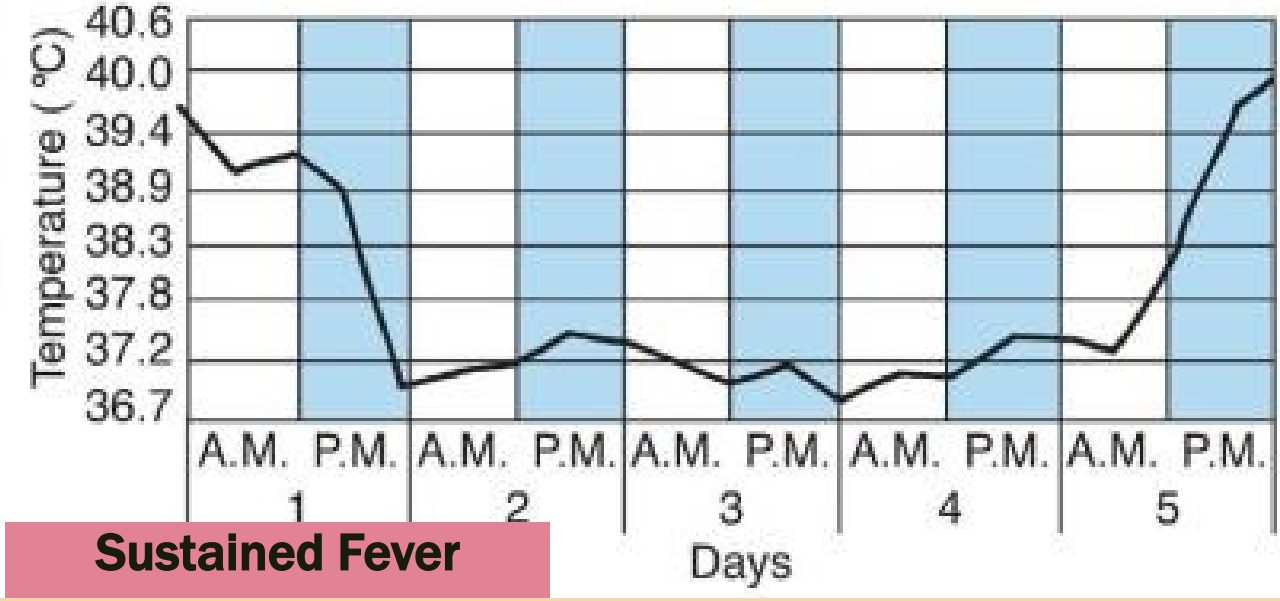
Sustained Fever



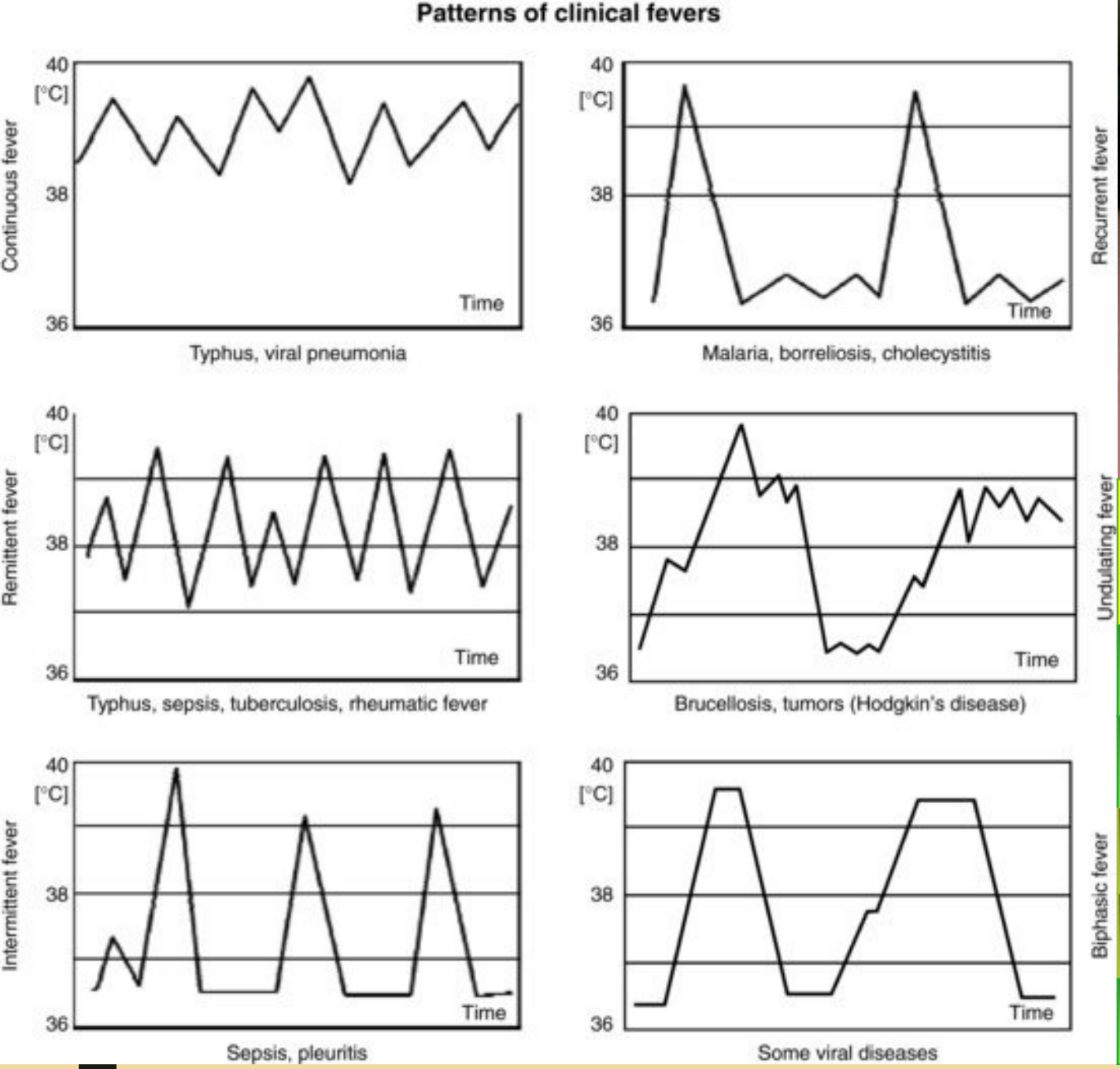
Sustained Fever



Intermittent Fever



Sustained Fever





BODY TEMPERATURE	°C
Normal	36.5 – 37.5
Mild/low grade fever	37.6 – 38.5
Moderate grade fever	38.6 – 39.5
High grade fever	39.6 – 40.5
Hyperthermia	40.6 +

Nursing management care plan for Fever

Diagnosis	Goal	Implementation	Evaluatio
Fever 39.5° c related to throat infection and manifested as: • Flush face • Headache • Irritability • Shivering	To subside temp. to 37° c within 1-2 hours	simple 1. Room ventilation 2. Switch fan on 3. Lightening clothes 4. Check temp. every 15 minutes 5. Give cold juice 6. Apply cold compresses 7. Apply shower with tap water 8. Give antipyretics as ordered 9. Call physician if there is no response complex	Temp. after 1.5 hour was 36.5° c  If not revision for the plan of care

Subnormal Body Temperature

- **Hypothermia, a core body less than (35°C).**
  1. *Mild hypothermia, temperature less than (35° to 34°C).*
  2. *Moderate hypothermia, temp. at (33.8° to 30°C).*
  3. *Severe hypothermia temp. below (30°C).*
- **Cold body temperatures are best measured with a tympanic thermometer for two reasons:**
  1. *First, other clinical thermometers do not have the capacity to measure temperatures in hypothermic ranges.*
  2. *Second, the blood flow in the mouth, rectum, or axillae generally is so reduced that measurements taken from these sites are inaccurate.*

Signs and symptoms associated with hypothermia

1. *Shivering until body temperature is extremely low*
2. *Pale, cool, and puffy skin*
3. *Impaired muscle coordination*
4. *Listlessness*
5. *Slow pulse and respiratory rates*
6. *Irregular heart rhythm*
7. *Decreased ability to think coherently and use good judgment*
8. *Diminished ability to feel pain or other sensations*

Nursing management care plan for Hypothermia

Diagnosis	Goal	Implementation	Evaluatio
Fever 35° c related to cold wither manifested as: • Pale face • Shivering • Irritability • Cold skin	To elevate temp. to 37° c within 1-2 hours	simple 1. Switch wormer on 2. Cover pt. with blanket 3. Check temp. every 15 minutes 4. Give worm juice 5. Apply shower with worm water 6. Give treatment as ordered 7. Call physician if there is no response complex	Temp. after 1.5 hour was 36. ° c  If not revision for the plan of care

PULSE

- A pulse, is a wave like sensation that can be palpated in a peripheral artery.
  - *Pulse Is produced by the movement of blood during the heart's contraction.*
  - *In most adults, the heart contracts 60 to 100 times per minute at rest.*
  - *Pulse Rate (the number of peripheral pulsations palpated in 1 minute).*
  - *Pulse is counted by compressing a superficial artery against an underlying bone with the tips of the fingers.*





PULSE

- **Rapid Pulse Rate Tachycardia:**
- The pulse rate of adults is considered rapid if it exceeds 100 beats per minute (bpm)
- The term **palpitation** (awareness of one's own heart contraction) can accompany tachycardia. Clients with rapid pulse rates are monitored closely, and the results are reported and recorded according to agency policy.





- **Slow Pulse Rate Bradycardia**
- The pulse rate of adults is considered slower than normal if it falls below 60 (bpm).

Factors Affecting Pulse and Heart Rates

- Age. Some common rates are listed in Table 12-5.
- Circadian rhythm. PR tend to be ↓ in the morning and ↑ later in the day.
- Gender. Men average 60 to 65 bpm at rest; for women is about 7 or 8 bpm faster.
- Body build, tall, slender people usually have slower PR. than short, stout people.
- Exercise and activity. Rates increase with exercise and activity and decrease
- Stress and emotions, emotions such as anger, fear, and excitement ↑ PR.
- Body temperature. For a 1°c increase causes a 15-bpm increase.
- Blood volume, excessive blood loss causes the heart and pulse rates to increase.
- Drugs. Certain drugs can slow or speed the rate of heart contraction.

Normal Pulse Rates per Minute at Various Ages

AGE	APPROXIMATE RANGE	APPROXIMATE AVERAGE
Newborn	120–160	140
1–12 months	80–140	120
1–2 years	80–130	110
3–6 years	75–120	100
7–12 years	75–110	95
Adolescence	60–100	80
Adulthood	60–100	80

Pulse rhythm

- **The pulse rhythm (the pattern of the pulsations and the pauses between them) is normally regular.**
- **That is, the beats and the pauses occur similarly throughout the time the pulse is palpated.**
- An arrhythmia or dysrhythmia (an irregular pattern of heartbeats) with a consequently irregular pulse rhythm is reported promptly.
- Pulse Volume
- Pulse volume (the quality of palpated pulsations) is usually related to the
- amount of blood pumped with each heartbeat, or the force of the heart's
- contraction.

Pulse Volume

- **Pulse volume** (the quality of palpated pulsations). is usually related to the amount of blood pumped with each heart beat, or the force of the heart's contraction.

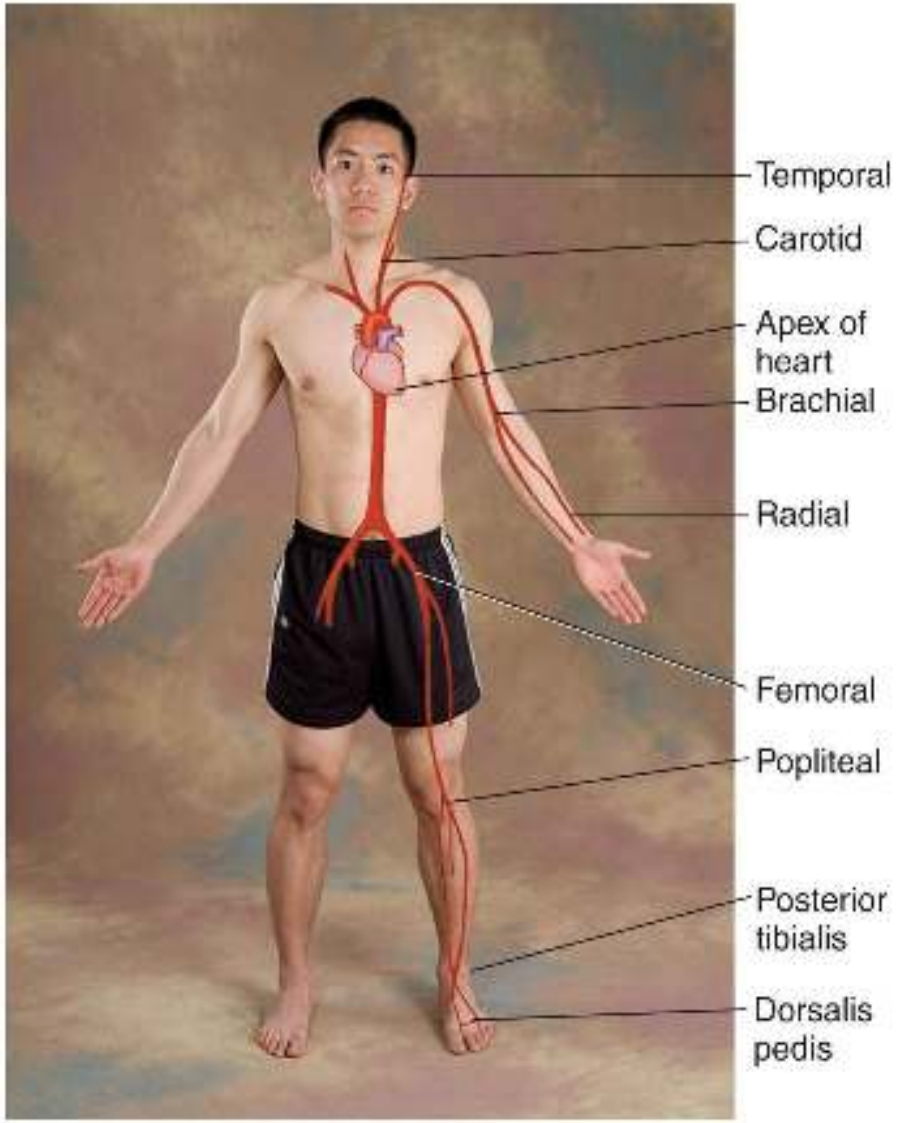
PULSE CHARACTER to be documented

1. Rate; Number of bpm (60-100)
2. Rhythm; regular / irregular
3. Pulse Volume
4. Equality; Are pulses the same on both sides of the body

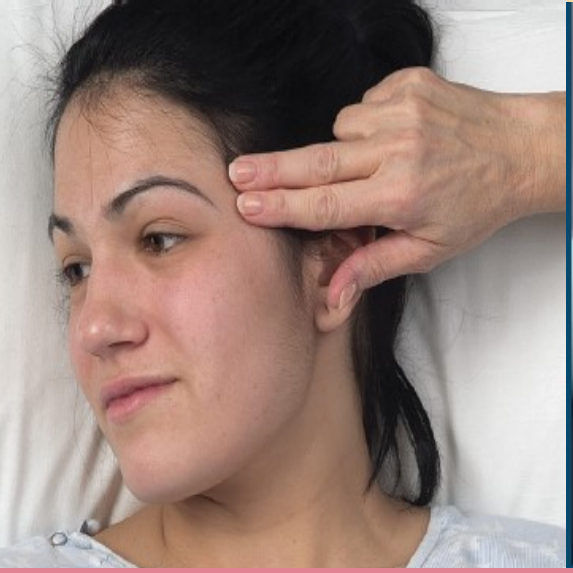
Identifying Pulse Volume

NUMBER	DEFINITION	DESCRIPTION
0	Absent pulse	No pulsation is felt despite extreme pressure.
1+	Thready pulse	Pulsation is not easily felt; slight pressure causes it to disappear.
2+	Weak pulse	Pulse is stronger than thready; light pressure causes it to disappear.
3+	Normal pulse	Pulsation is felt easily; moderate pressure causes it to disappear.
4+	Bounding pulse	Pulsation is strong and does not disappear with moderate pressure.

Pulse Assessment sites



1. The apical heart rate
2. Temporal,
3. Carotid, assessed mainly in critical situations
4. Brachial, for assessment for blood pressure BP.
5. Radial, is the most common used site,
6. Femoral
7. Popliteal, the alternative site to brachial for assessing BP.
8. Posterior tibialis
9. Dorsalis pedis, for assessment of blood flow to feet.

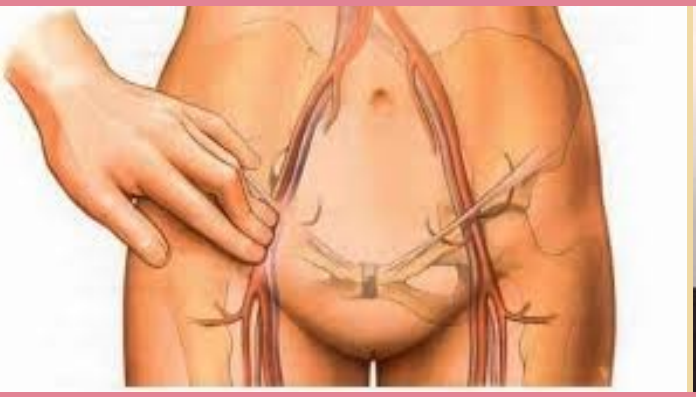








A, Temporal. B, Carotid. C, Brachial. D, Radial.

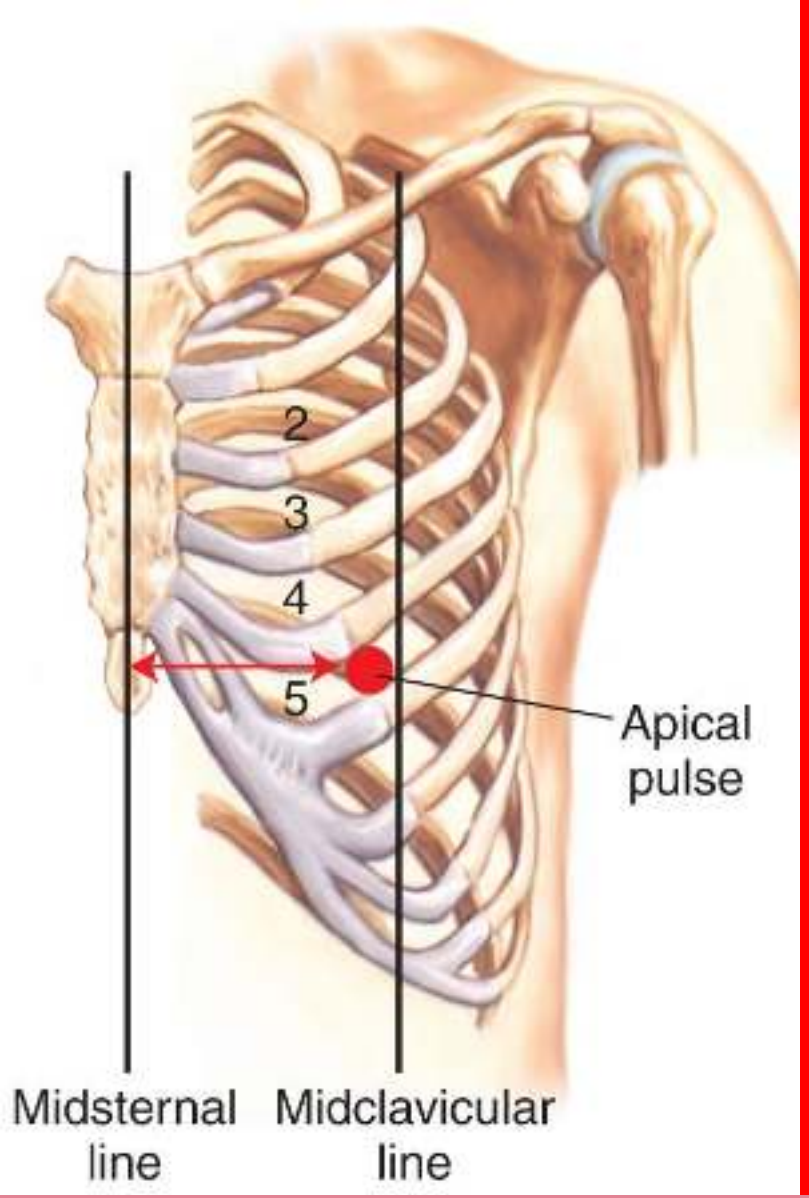








E, Femoral. F, Popliteal. G, Posterior Tibialis. H, Dorsalis Pedis.



The apex in a healthy adult is slightly below the left nipple in line with the middle of the clavicle



Palpation of arterial Apical pulse

Pulse Assessment sites cont.

- Apical radial (the number of sounds heard at the heart's apex and the rate of the radial pulse during the same period for 1 full minute).
- is counted by separate nurses at the same time using one watch or clock between them.



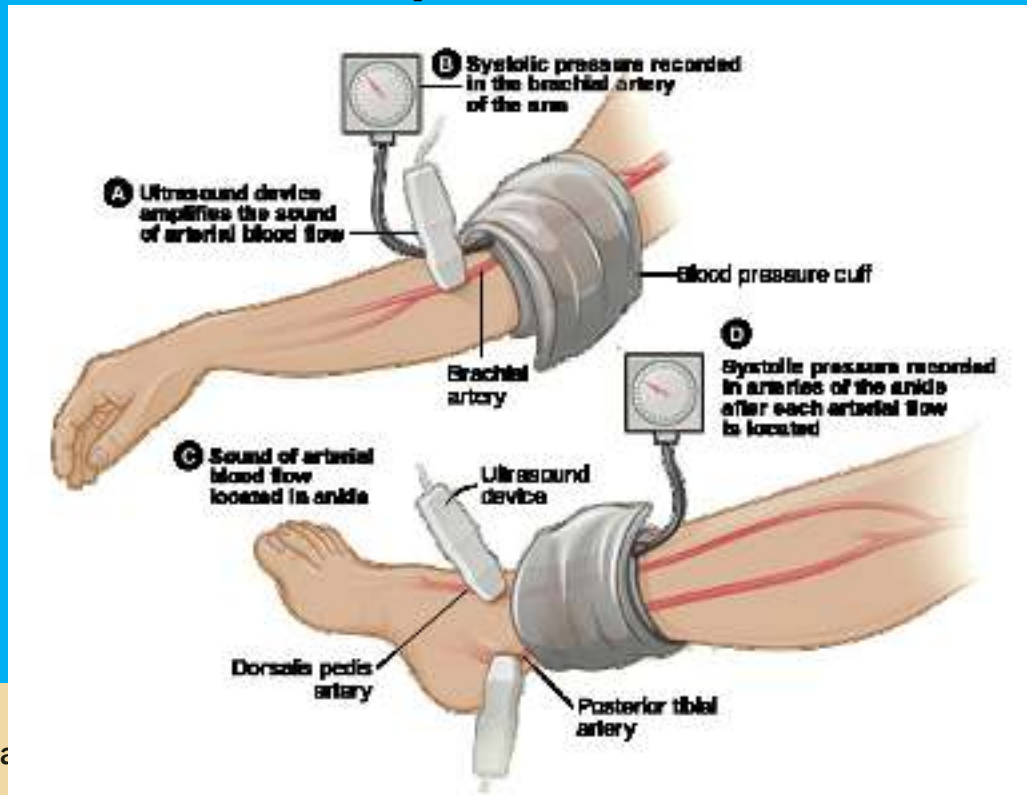
- The apical and radial rates should be the same, but in some clients, they are not.
- The pulse deficit (the difference between the apical and the radial pulse rates) is noted.
- If a pulse deficit is significant and the rates have been counted accurately the nurse

reports the findings promptly and documents them in the client's medical record.

Pulse Assessment sites cont.

- **DOPPLER ULTRASOUND DEVICE**
- A Doppler ultrasound device is an electronic instrument that detects the movement of blood through peripheral blood vessels and converts the movement to a sound.
- This instrument is most helpful when slight pressure occludes pulsations or when arterial blood flow is severely compromised.





RESPIRATION

- Respiration is the exchange of oxygen and carbon dioxide.
  - *When it occurs between the alveolar and the capillary membranes, it is called "external respiration."*
  - *The exchange of oxygen and carbon dioxide between the blood and the body cells is called "internal" or "tissue respiration."*
- **Ventilation (the movement of air in and out of the chest) involves**
  - ***Inhalation or inspiration (breathing in) and***
  - ***exhalation or expiration (breathing out).***

Respiratory Rate

- The respiratory rate (the number of ventilations per minute) varies considerably in healthy people, but normal ranges have been established

- Ventilat… e chest) involves
  - *Inhala…*
  - *exhala…*

Normal Respiratory Rates at Various Ages

AGE	AVERAGE RANGE
Newborn	30–80
Early childhood	20–40
Late childhood	15–25
Adulthood	
Men	14–18
Women	16–20

RESPIRATION

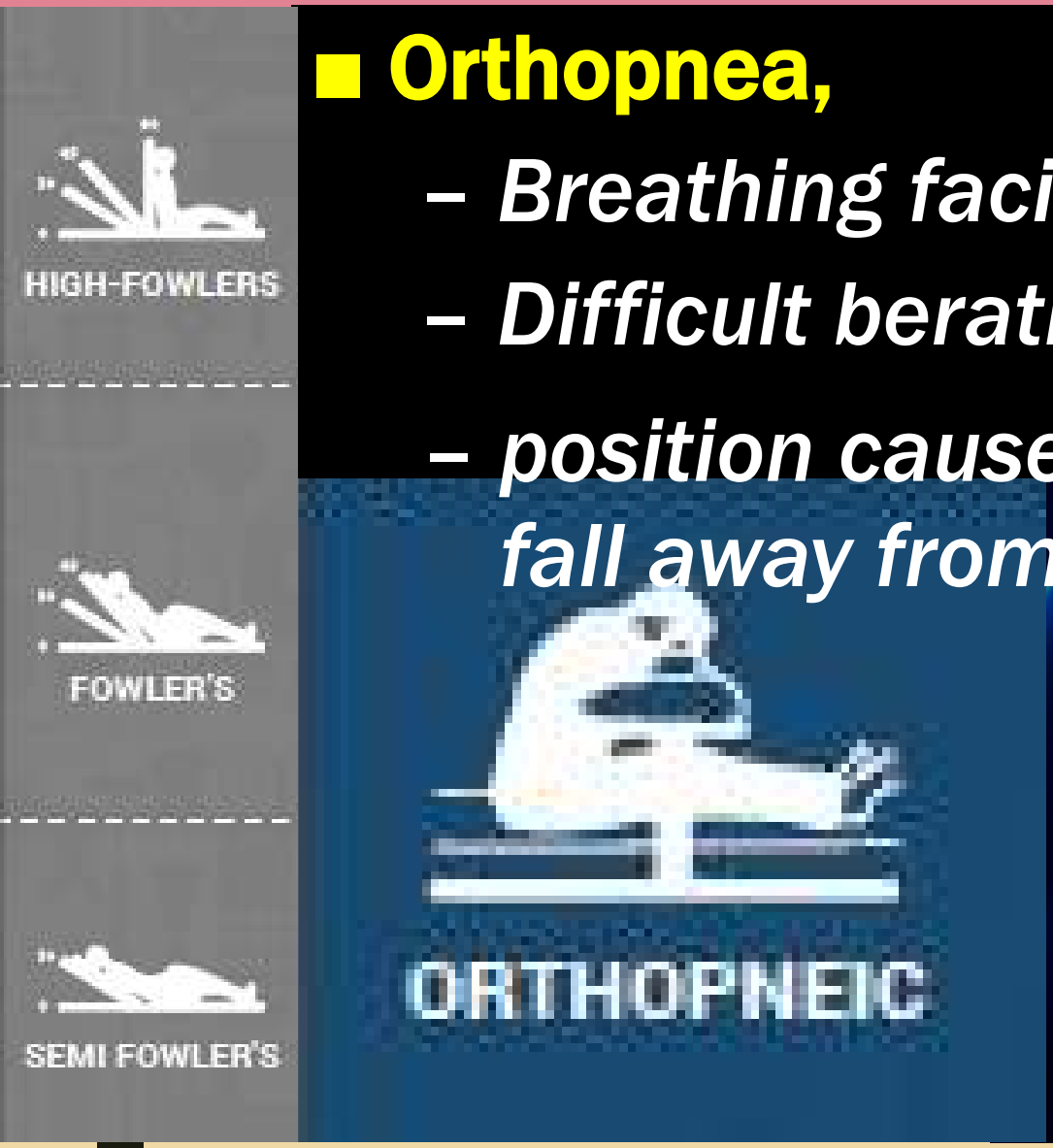
- Rapid Respiratory Rates
- Tachypnea (a rapid respiratory rate) often accompanies an elevated temperature or diseases that affect the cardiac and respiratory systems.
- **Slow Respiratory Rates**
- **Bradypnea (a slower-than-normal respiratory rate at rest) can result from**
- **Ventilation (the movement of air in and out of the chest) involves Inhalation or inspiration (breathing in) and exhalation or expiration (breathing out).**

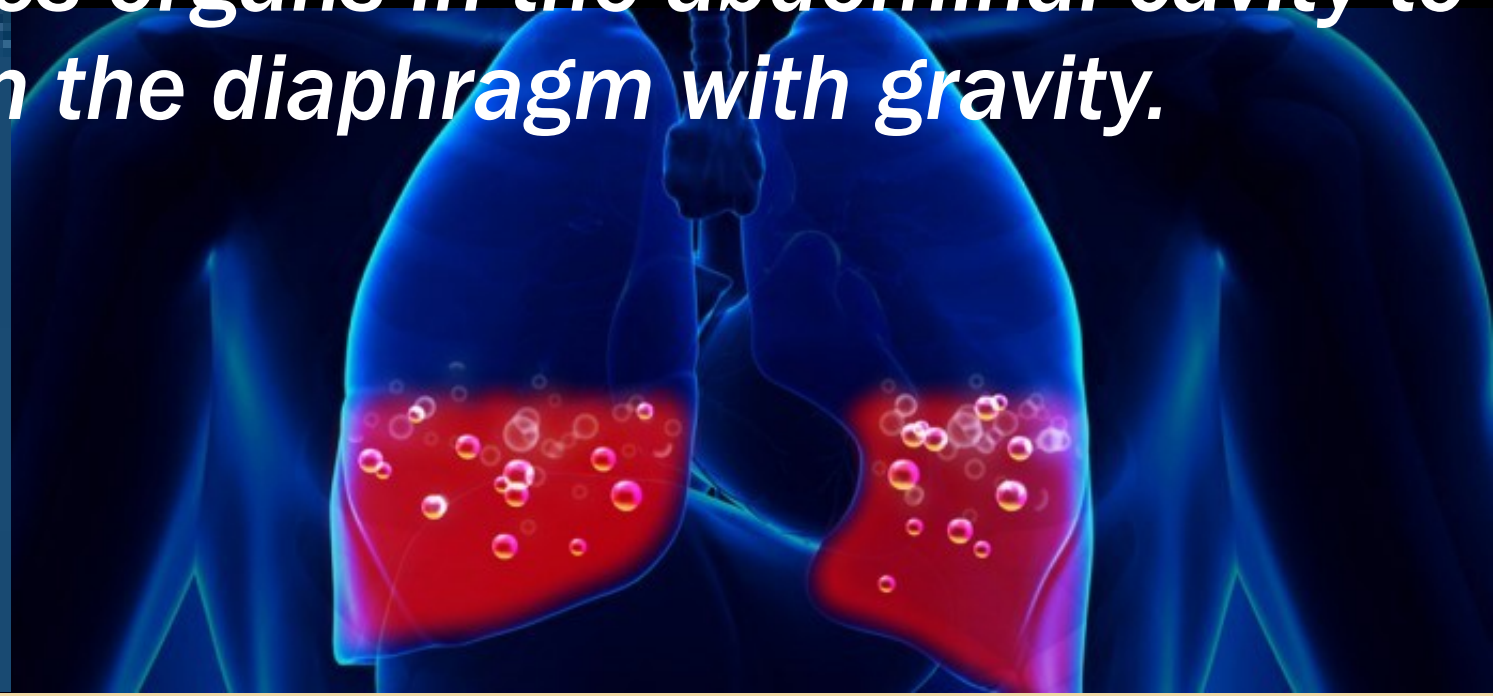
Breathing Patterns and Abnormal Characteristics

- **Cheyne–Stokes respiration,** refers to a breathing pattern in which the depth of respirations gradually increases, followed by a gradual decrease, and then stops briefly before resuming the pattern again.
- It's a serious sign that may occur as death approaches.
- **Hyperventilation,** (rapid or deep breathing or both)
- **Hypoventilation,** (diminished breathing) affect the volume of air entering and leaving the lungs.
- Dyspnea (difficult or labored breathing) is almost always accompanied by a rapid respiratory rate as clients work to improve the efficiency of their breathing.

Breathing Patterns and Abnormal Characteristics

- **Orthopnea,**
  - *Breathing facilitated by sitting up or standing.*
  - *Difficult berating in lying position (supine).*
  - *position causes organs in the abdominal cavity to fall away from the diaphragm with gravity.*





Breathing Patterns and Abnormal Characteristics

- **Apnea,** (the absence of breathing)
  - *is life-threatening if it lasts more than 4 to 6 minutes.*
  - *Prolonged apnea leads to brain damage or death.*
- **Stertorous breathing,** (noisy ventilation) and obstruction) are used to describe sounds that accompany breathing.
- Stridor, (a harsh, high-pitched sound heard on inspiration when there is laryngeal

BLOOD PRESSURE cont.

- **Blood pressure** is the force that the blood exerts within the arteries. Several physiologic variables create blood pressure:

1. **Circulating blood volume;** averages 4.5 to 5.5 L in adult women and 5.0 to 6.0 L in adult men.
   - *Lower-than-normal volumes decrease blood pressure; excess volumes increase it.*
2. **Contractility of the heart** is influenced by the stretch of cardiac muscle fibers.
   - ***Preload*** *(the volume of blood that fills the heart and stretches the heart muscle fibers during its resting phase).*

BLOOD PRESSURE cont.

- **Blood pressure** is the force that the blood exerts within the arteries. Several physiologic variables create blood pressure:

3. **Cardiac output,** (the volume of blood ejected from the left ventricle per minute) is approximately 5 to 6 L (slightly more than a gallon) in adults at rest.
   - ***stroke volume,*** *(amount of blood that leaves the heart with each contraction). The average stroke volume in adults is 70 mL.*
   - *With exercise, cardiac output can increase as much as five*
   - *times the resting volume.*
   - *Bradycardia can severely reduce cardiac output and thus, blood pressure.*

BLOOD PRESSURE cont.

- **Blood pressure** is the force that the blood exerts within the arteries. Several physiologic variables create blood pressure:

4. **Blood viscosity,** (thickness) creates a resisting force when the heart contracts.

5. **Peripheral resistance, (Elasticity of the arterial walls)** referred to as;
   - ***Afterload,*** *(the force against which the heart pumps when ejecting blood),*
   - *increases when the valves of the heart and arterioles (small subdivisions of arteries) are narrowed or calcified.*

Blood pressure & relations with physiological variables

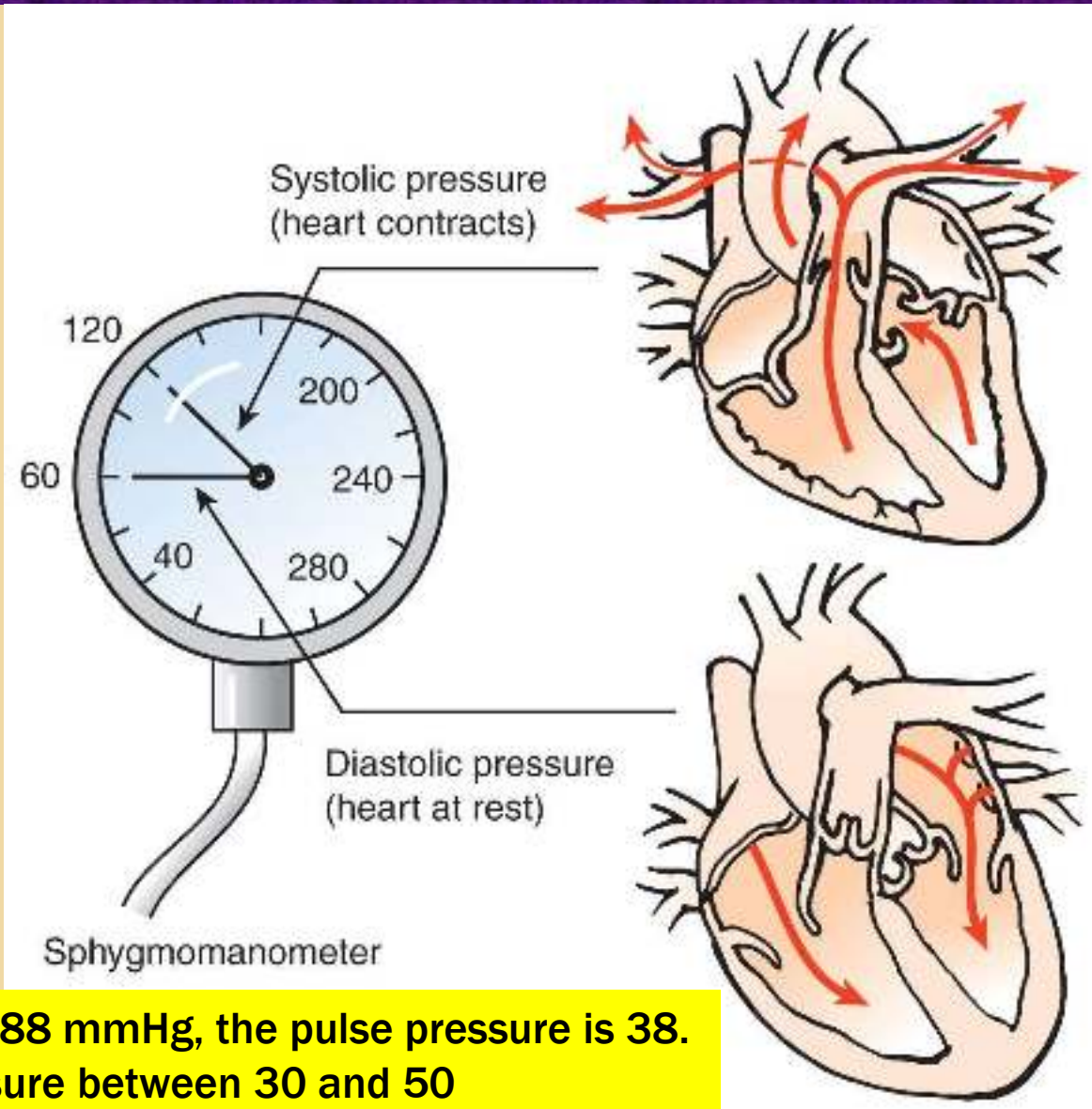
BODY TEMPERATURE	RELATION WITH BP	
1- Circulating blood volume	Positive	علاقة طردية
2- Contractility of the heart	Positive	علاقة طردية
3- Cardiac output	Positive	علاقة طردية
4- Blood viscosity	Positive	علاقة طردية
5- Peripheral resistance = ⇩ • Elasticity of blood vessels	Positive Inverse	علاقة طردية علاقة عكسية

Factors affecting blood pressure

- ***Age.*** **Blood pressure tends to become elevated with age**
- ***Circadian rhythm.*** **Blood pressure tends to be lowest after midnight, begins rising at approximately 4 or 5 AM, and peaks during late morning or early afternoon.**
- ***Gender. Women tend to have lower blood pressure than men of the same age.***
- ***Exercise and activity. Blood pressure rises during exercise and activity***
- ***Emotions and pain. Strong emotional experiences and pain tend to increase blood pressure***
- ***Miscellaneous factors.***
  - ***lower blood pressure when lying down***
  - ***Drugs that stimulate the heart such as nicotine, caffeine, cocaine***

Blood Pressure Measurements

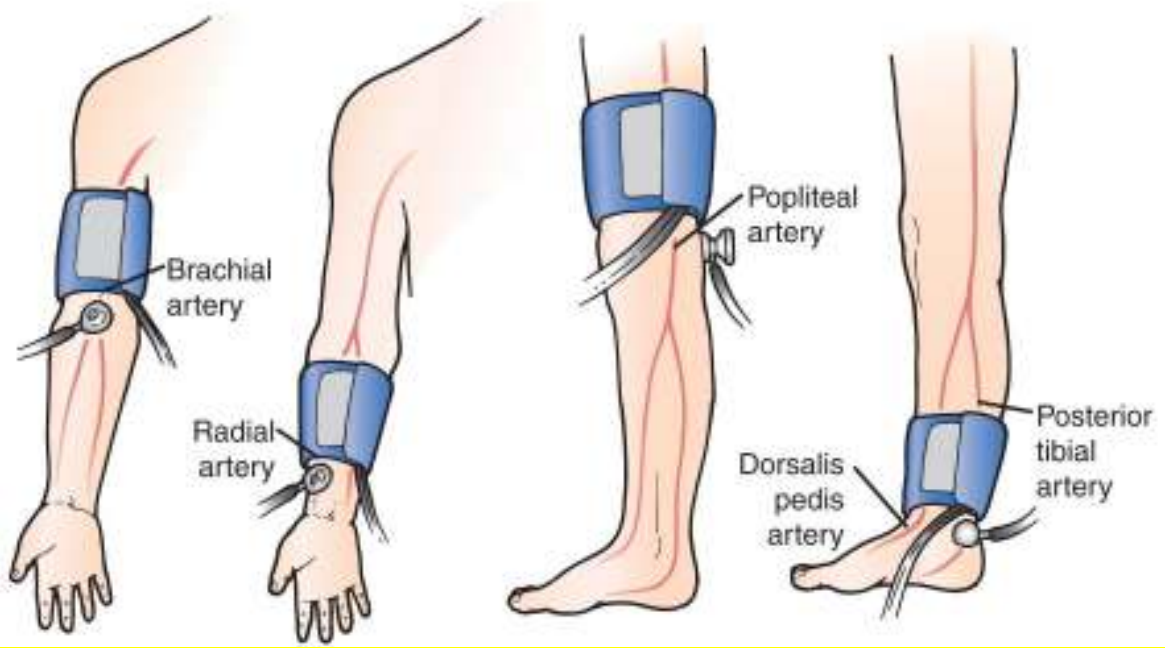
- **Blood pressure** When assessing blood pressure, nurses obtain both systolic and diastolic measurements.
- **Systolic pressure** (pressure within the arterial system when the heart contracts) is higher than,
- **Diastolic pressure** (pressure within the arterial system when the heart relaxes and fills with blood).
- The pulse pressure (the difference between systolic and diastolic blood pressure measurements)



e.g. BP. 126/88 mmHg, the pulse pressure is 38.
A pulse pressure between 30 and 50

Blood Pressure Measurements

- Because individual differences can be considerable, the usual ranges and patterns of BP. measurements for each person is important.
- A rise or fall of 20 to 30 mmHg in usual significant, even if it is well within the generally accepted range for normal.



- **Assessment Sites,**
  - *Blood pressure usually is assessed over the brachial artery at the inner aspect of the elbow.*
  - *It is also possible to use the lower arm and radial artery*

- **Alternative BP. Sites When ???**

1. When the client's arms are missing
2. When both of a client's breasts have been removed
3. When a client has had vascular surgery.
4. When dressings or plaster or fiberglass casts obscure the brachial and radial sites

Equipment for Measuring Blood Pressure

- Blood pressure most often is measured with:
  - *a sphygmomanometer (a device for measuring blood pressure),*
  - *an inflatable cuff, and*
  - *a stethoscope.*





- Sphygmomanometer Aneroid Manometer
  - *named from the French word aneroid, which means "no liquid," measures pressure using a spring mechanism.*



Equipment for Measuring Blood Pressure cont.

- **Electronic Oscillometric Manometer:**
  - *is battery operated or uses power from an electrical outlet*
  - *does not require a stethoscope for auscultating*





- **Inflatable Cuff**
- **The cuff contains an inflatable bladder to which two tubes are attached.**
- **One is connected to the manometer, which registers the pressure.**



Equipment for Measuring Blood Pressure cont.

- **Inflatable Cuff parameters:**
  - *Cuffs come in various sizes ranging from small – large adult arm*
  - *Including pediatric sizes also.*

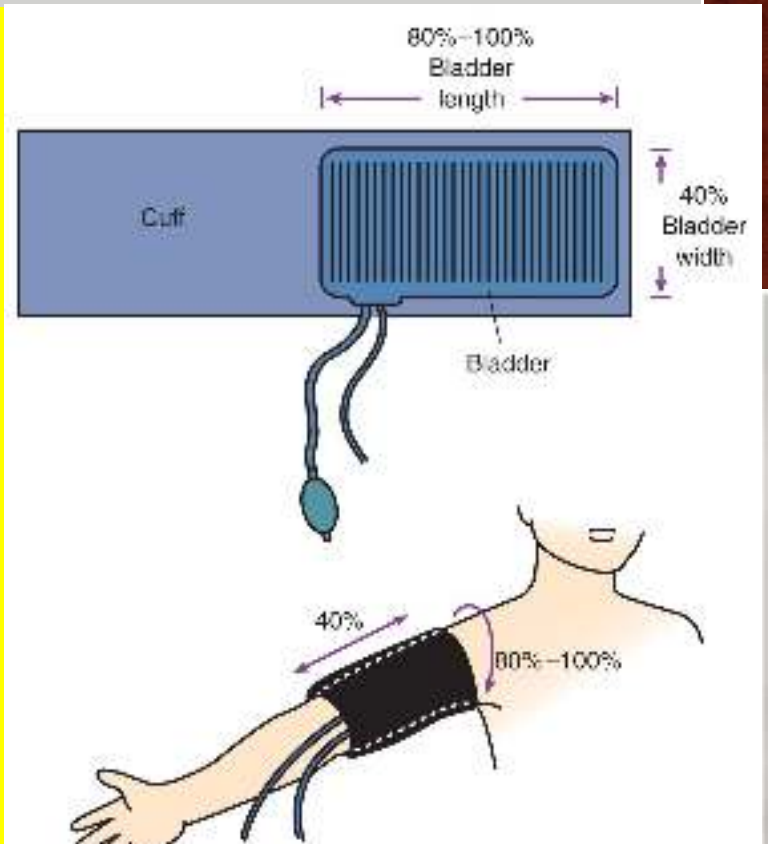




A common guide is to use a cuff that has a bladder width of at least 40% and a length that is

80% - 100% of

Mid limb circumference



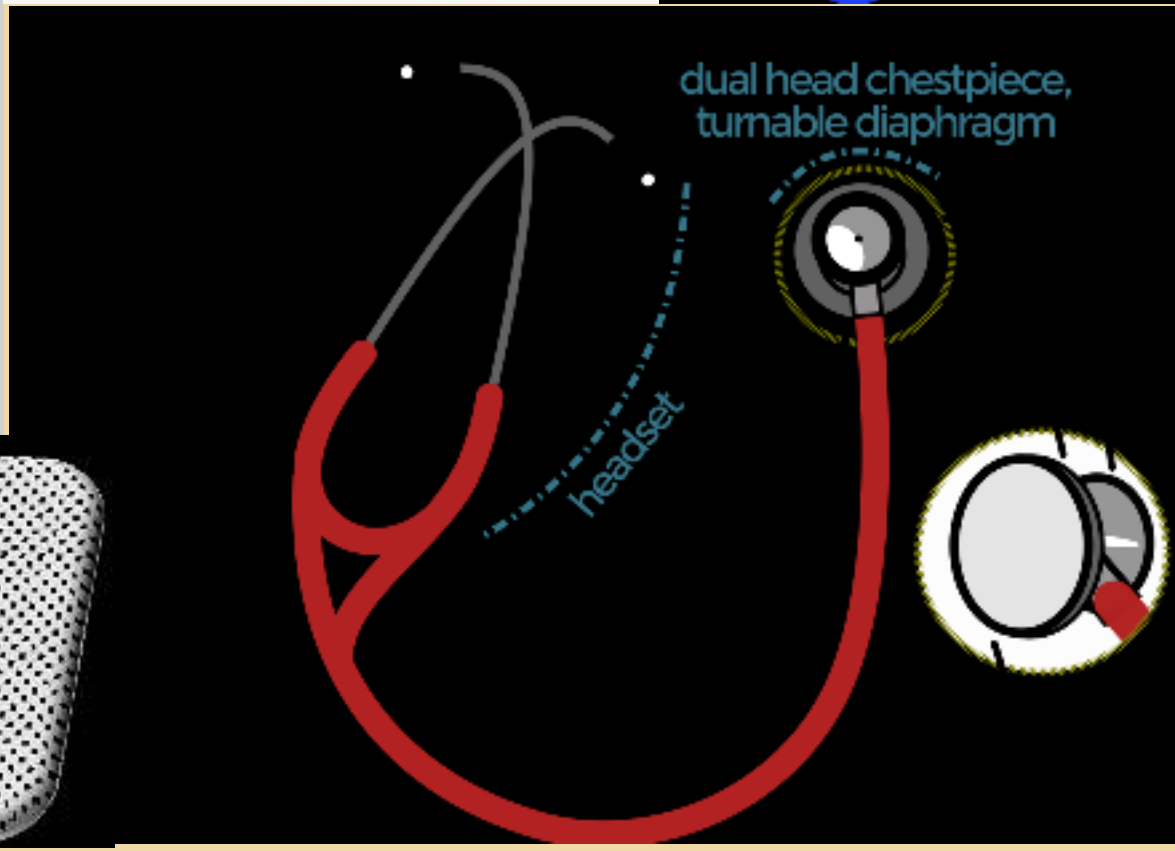


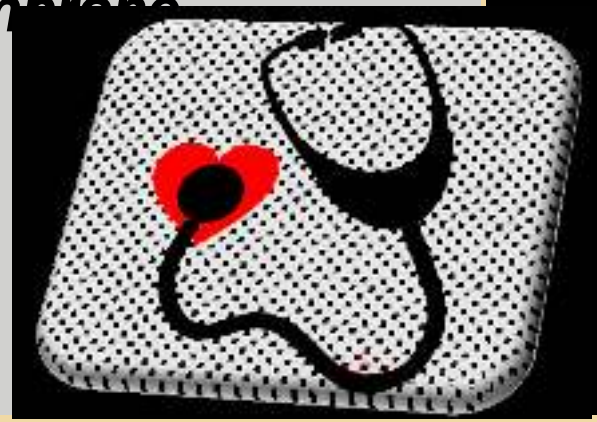
Equipment for Measuring Blood Pressure cont.

- **Stethoscope, (an instrument that carries sound to the ears),** composed of:
  - ***Eartips, (must be forwarded with auditory canal direction.***
  - ***a brace and binaurals, and***
  - ***tubing leading to***
  - ***a chest piece that may be a bell, diaphragm, or both.***
  - ***Diaphragm sensitive membrane***









Measuring Blood Pressure

- Client should be at rest and to have rest foe 15 - 20 ms. after arrival
- The first time blood pressure is measured, it is assessed in each arm. The two blood pressure measurements should not vary more than 5 to 10 mmHg unless pathology (disease) is present.
- Measure cuff to left forearm (preferable) client and fit it
- Raise the limb before inflation.
- Lower limb with the level of the heart and so the manometer.
- Inflate till pulse not hearing in brachial site
- Deflate pressure slowly and hear the Korotkoff sounds









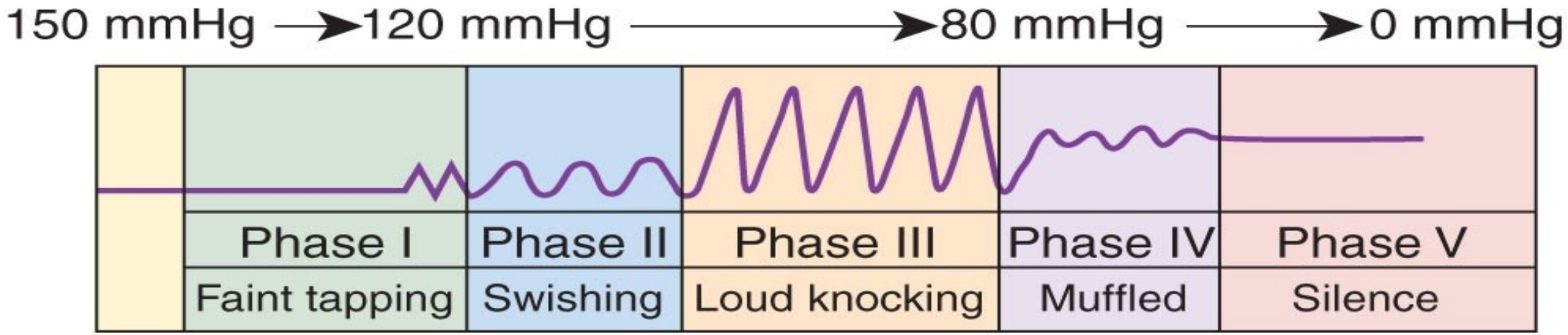


Remember:

- ✓ Welcome, introduce self
- ✓ Provide rest & privacy
- ✓ Wash hands
- ✓ Get all equipment's
- ✓ Describe procedure
- ✓ Assess response
- ✓ Documentation

Measuring Blood Pressure, Korotkoff sounds (the vibrations of blood within the arterial wall in blood flow)

- ***Phase I*** begins with the first faint but clear tapping, follows a period of silence as pressure released from the cuff.
- ***Phase II*** is characterized by a change from tapping sounds to swishing sounds.
- ***Phase III*** is characterized by a change to loud and distinct sounds described as crisp knocking sounds.
- ***Phase IV*** sounds are muffled and have a blowing quality.
- **Phase V** is the point at which the last sound is heard, or the second diastolic pressure measurement.



Alternative Assessment Techniques

Palpating the Blood Pressure

- When palpating the BP, the nurse applies a blood pressure cuff.
- Instead of using a stethoscope, however, he or she positions the fingers over the artery while releasing the cuff pressure.

Doppler Stethoscope

- A Doppler stethoscope hel
  detect sounds created by
  velocity of blood moving
  through a blood vessel.



Alternative Assessment Techniques cont.

Automatic Blood Pressure Monitoring

- An automatic electronic blood pressure monitoring device consists of a blood pressure cuff attached to a microprocessing unit.

Measuring Thigh Blood Pressure

- The thigh is a structure that corresponds anatomically to the upper arm.
- Nurses use this site for blood pressure assessment when they cannot obtain readings in either of the client's arms.



Abnormal Blood Pressure Measurements

- **Blood pressures above or below normal ranges may indicate significant health problems.**
  - *High Blood Pressure (Hypertension*
  - *Hypertension exists when the systolic pressure, diastolic pressure, or both are sustained above normal levels for the person's age.*
  - *Evaluation, and Treatment of High Blood Pressure considers a systolic pressure of 140 mmHg or greater and a diastolic pressure of 90 mmHg or greater to be abnormally high*

Abnormal Blood Pressure Measurements cont.

- **Blood pressures above or below normal ranges may indicate significant health problems.**
  - *occasional elevation in blood pressure does not necessarily mean a person has hypertension.*
  - *It does mean that the BP should be monitored at various intervals.*
  - *Monitoring is important to determine whether the elevated blood pressure is sustained or is the result of white-coat hypertension.*
  - *which the blood pressure is elevated when taken by a health care provider but normal at other times).*

High Blood Pressure cnt.

CLASSIFICATION OF ADULT PLOD PRESSURE MEASUREMENT

Category	Systolic mmHG	Diastolic mmHG
1- Normal	90 – 120	60 – 80
3- Prehypertension	121 – 139	81 – 89
4- Hypertension • Stage I • Stage II	 170 – 159 160 +	 90 – 99 100 +
• Hypertensive crisis	› 180	› 110

High Blood Pressure cnt.

CLASSIFICATION OF ADULT PLOD PRESSURE MEASUREMENT

Systolic	Diastolic	Recommended follow up
< 120	< 80	Recheck in 2 years
121 – 139	81 – 89	Recheck in 1 years
140 – 159	90 – 99	Confirm with 2 months
160 - 179	100 – 109	Evaluate or refer to source of care within one month
≥ 180	≥ 110	Evaluate or refer to source of care at once o within 1 week up on situation

High Blood Pressure cnt.

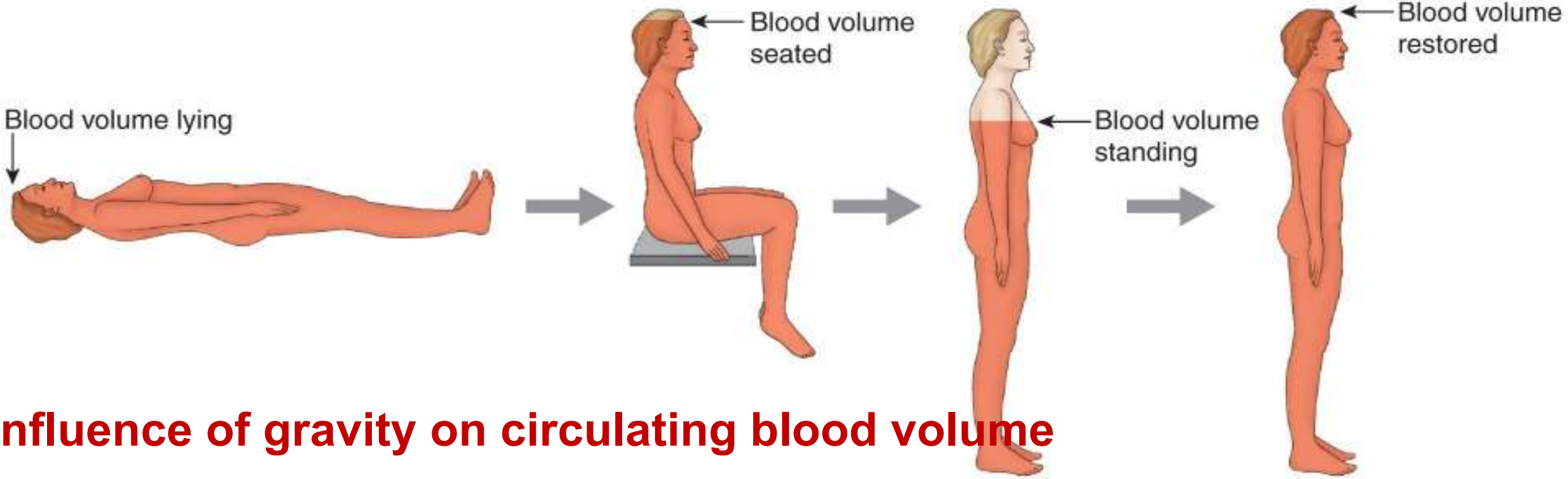
- Hypertensive blood pressure measurements often are associated with the following:
  - *Anxiety*
  - *Obesity*
  - *Vascular diseases*
  - *Stroke*
  - *Heart failure*
  - *Kidney diseases*

Low Blood Pressure

- **Hypotension (low blood pressure)**
  - *is when blood pressure measurements are below the normal systolic values for the person's age.*
  - *Having a consistently low pressure, 96/60 mmHg, for example, seems to cause no harm.*
  - *People with low BP, should continue to be monitored to evaluate its significance.*
  - *Low blood pressure measurements may indicate shock, hemorrhage, or side effects from drugs.*

Postural Hypotension

- ***Postural or orthostatic hypotension is a temporary drop in blood pressure when rising from a reclining position after <u>3 to 5 minutes</u> of rest.***



Influence of gravity on circulating blood volume

Postural Hypotension cnt.

- It is most common in those with circulatory problems especially common in older adults, those who are dehydrated, and those who take diuretics or other drugs that lower blood pressure.

- Once postural hypotension is confirmed, the nurse can reduce the client's risk for falling in the following ways:
  1) *ensure that the client remains seated after rising until dizziness passes*
  2) *restore adequate hydration if the client's fluid volume is low*
  3) *increase consumption of salty foods if the client is not hypertensive*
  4) *apply compression stockings to the lower extremities to reduce pooling of blood upon standing*
  5) *administering prescribed medications as ordered*

Postural Hypotension

- ***Postural or orthostatic hypotension is a temporary drop in blood pressure when rising from a reclining position after 3 to 5 minutes of rest.***
- ***It is most common in those with circulatory problems especially common in older adults, those who are dehydrated, and those who take diuretics or other drugs that lower blood pressure.***

DOCUMENTING VITAL SIGNS

- Once nurses have obtained vital sign measurements, they are documented in the
- medical record for analysis of patterns and trends.
- They also may be entered as data, along with any other subjective or objective information, elsewhere in the client's record such as in the narrative nursing notes.

Remember:

- **Documented Data must be:**
  - **Clear**
  - **Accurate &**
  - **concise**

Vital Signs Sheet

Date			Day1			Day2			Day3			Day4			Day5			Day6			Day7		
Shift			D	E	N	D	E	N	D	E	N	D	E	N	D	E	N	D	E	N	D	E	N
Time			6	14	20	6	14	20	6	14	20	6	14	20	6	14	20	6	14	20	6	14	20
R	P	T																					
40	160	41																					
35	140	40																					
30	120	39																					
25	100	38																					
20	80	37																					
15	60	36																					
10	40	35																					
BP																							
Notes																							


Thank
You
Merci
Obrigado
Obrigado
Asante
Spasibo
Maake
Grazie
Kiitos
Multumesc
Vinaka
Arigato
Dank Je
Matondo
Juspaxar
Mamana
Nirringrazzjak
Ua Tsaug Rau Koj
Mochchakkeram
Kia Ora
Chokrane
Welalin
Cam on ban
Terma Kasih
Danken
Salamat
Matondo
Obrigado
Grazie
Asante
Kiitos
Merci
Spasibo
Maake
Kia Ora
Chokrane
Welalin
Juspaxar
Mamana

PHYSICAL ASSESSMENT

Objectives

- *On completion of this chapter, the audience should be able to:*
  1. *List four purposes of a physical assessment.*
  2. *Name four assessment techniques.*
  3. *List at least five items needed when performing a basic physical assessment.*
  4. *Discuss at least three criteria for an appropriate assessment environment.*
  5. *Identify at least five assessments that can be obtained during the initial survey of clients.*
  6. *Differentiate a head-to-toe and a body systems approach to physical assessment.*
  7. *List six ways in which the body may be divided for organizing data collection.*
  8. *Identify two self-examinations that nurses should teach their adult clients.*

Components of Assessment



- **Identifying Data**
  - *Age, gender, marital status, living …*
  - *General appearance*
- ***Chief Complaint***
  - *Reason for seeking care (from client's perspective)*
- ***Present Illness***
  - *Chronologic description of onset, frequency, and duration of current signs and symptoms*
  - *Outcomes of earlier attempts at self-treatment and medical treatment*

Components of Assessment cont.



- **Personal History**
  - *Occupation, level of education, Religion, Residence*
  - *Primary language, travel …*
- ***Past Health History***
  - *Childhood disease, injuries, major illnesses and surgeries*
  - *Previous hospitalizations, drug history, tobacco use, allergy history*
- ***Family History***
  - *H. problems in close cause of death (especially parents)*
- ***Review of Body Systems,*** *Results of physical examination*
- ***Laboratory investigations,*** *all lab, done and client ranges with the comparison with normal ranges*

PHYSICAL ASSESSMENT

- *The first step in the nursing process is assessment, or gathering of information.*
- *The physical assessment (a systematic examination of body structures) is one method for gathering health data.*

OVERVIEW OF THE PHYSICAL ASSESSMENT

- Health care providers use various techniques and equipment to perform a physical assessment.
- Accurate data collection, and to be conducive to the client's privacy and comfort.
- Purposes
- The overall goal of a physical assessment is to gather objective data about a client.
- The purposes of assessment are as follows:
  1. *To evaluate the client's current physical condition*
  2. *To detect early signs of developing health problems*
  3. *To establish a baseline for future comparisons*
  4. *To evaluate the client's responses to medical and nursing interventions*

PHYSICAL ASSESSMENT TECHNIQUES cont.

- The four basic physical assessment techniques are inspection, percussion, palpation, and auscultation.

1- Inspection (purposeful observation) is the most frequently used assessment technique.

It involves examining particular body parts and looking for specific normal and abnormal characteristics



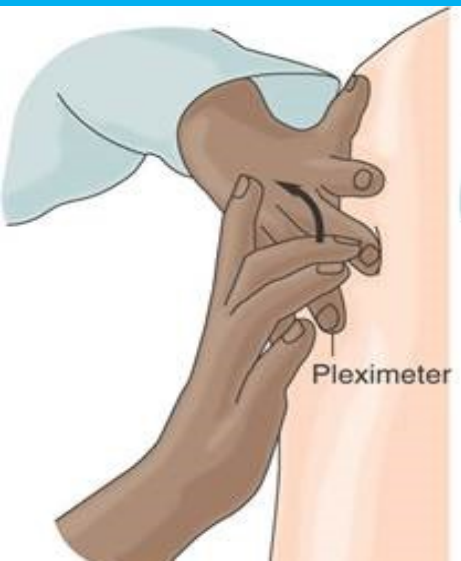



PHYSICAL ASSESSMENT TECHNIQUES cont.

- The four basic physical assessment techniques are inspection, percussion, palpation, and auscultation.

2- Percussion

Percussion, the least used assessment technique by nurses, is striking or tapping a part of the client's body with the fingertips to produce vibratory sounds.











PHYSICAL ASSESSMENT TECHNIQUES cont.

- The four basic physical assessment techniques are inspection, percussion, palpation, and auscultation.

3- Palpation involves lightly touching or applying pressure to the body.

- ***Light palpation*** *involves using the fingertips, the back of the hand, or the palm of the hand*
  - *It is best used when feeling the surface of the skin, structures that lie just beneath the skin, pulsations from peripheral arteries, and vibrations in the chest.*
- ***Deep palpation*** *is performed by depressing tissue approximately 1 inch (2.5 cm) with the forefingers of one or both hands*


A






B





PHYSICAL ASSESSMENT TECHNIQUES cont.

- The four basic physical assessment techniques are inspection, percussion, palpation, and auscultation.

4- Auscultation (listening to body sounds) is used frequently, most often to assess the heart, lungs, and abdomen.

A stethoscope is required to hear most internal sounds







PHYSICAL ASSESSMENT TECHNIQUES cont.

I. Inspection

II. Palpation

III. Auscultation

IV. Percussion

1. Size
2. Shape
3. Consistency
4. Mobility
5. Masses
6. Symmetry or asymmetry
7. Skin Warmth
8. Moisture
9. Tenderness
10. Sounds

1. Vibrations
2. Color
3. Smoothness
4. Range of motion (ROM)
5. Exudates
6. Wounds
7. Excretions
8. Gate
9. Consciousness
10. Grooming

PHYSICAL ASSESSMENT TECHNIQUES cont.

I. Inspection

1	2	3	4	5	6	12	14	15	16
17	18	19	20						

II. Palpation

1	2	3	4	5	6	7	8	9	13
14	15	16	17						

III. Auscultation

10	11								

IV. Percussion

10	11								

1. Size
2. Shape
3. Consistency
4. Mobility
5. Masses
6. Symmetry or asymmetry
7. Skin Warmth
8. Moisture
9. Tenderness
10. Sounds
11. Vibrations
12. Color
13. Smoothness
14. Range of motion (ROM)
15. Exudates
16. Wounds
17. Excretions
18. Gate
19. Consciousness
20. Grooming

Physical Assessment Equipment & Environment

For a basic physical assessment, the nurse needs the following:

- Gloves
- Examination gown
- Cloth or paper drapes
- Scale
- Stethoscope
- Sphygmomanometer
- Thermometer
- Pen light or flashlight
- Tongue blade
- Assessment form and pen

Nurses assess the special examination room or bedside for:

- Easy access to a restroom
- Door or curtain for privacy
- Adequate warmth
- padded, adjustable table or bed
- Sufficient area for moving to either side of the client
- Adequate lighting
- Facilities for hand hygiene
- Clean counter or surfacefor placing examination equipment
- lined receptacle for soiled articles.

PERFORMING A PHYSICAL ASSESSMENT

- **Gathering General Data during the first contact with the client.**
- **At this time, the nurse appraises the client's overall condition.**
- **By observing and interacting with the client before the actual physical examination, the nurse notes the following:**

- ***Physical appearance with regard to clothing and hygiene***
- ***Level of consciousness***
- ***Body size***
- ***Posture***
- ***Gait and coordinated movement (or lack of it)***
- ***Use of ambulatory aids***
- ***Mood and emotional tone***

PERFORMING A PHYSICAL ASSESSMENT cont.

- Obtaining Weight and Height
- Provide privacy
- Draping and Positioning, preferred by two drapes (a sheet of soft cloth or paper) up and down.





Selecting an Approach for Data Collection

- **Head-to-Toe Approach**
  - A head-to-toe approach means assessing the client from the top of the body to the feet. This has three advantages:

1. It prevents overlooking some aspect of data collection.
- 2. It reduces the number of position changes.
- 3. It generally takes less time than system review appeoach.

- **Body Systems Approach**
- A body systems approach means assessing the client according to the functional systems of the body.
- It involves examining the structures in each system separately.

DATA COLLECTION

- **Head and Neck**
  - Head

The nurse begins at the client's head by assessing his or her mental status and the symmetry and function of craniofacial structures (eyes, ears, nose, mouth).

- **Mental Status Assessment**
- **A technique for determining the level of a client's cognitive functioning) provides information about a client's attention, concentration, memory, and ability to think abstractly.**

DATA COLLECTION Mental Assessment

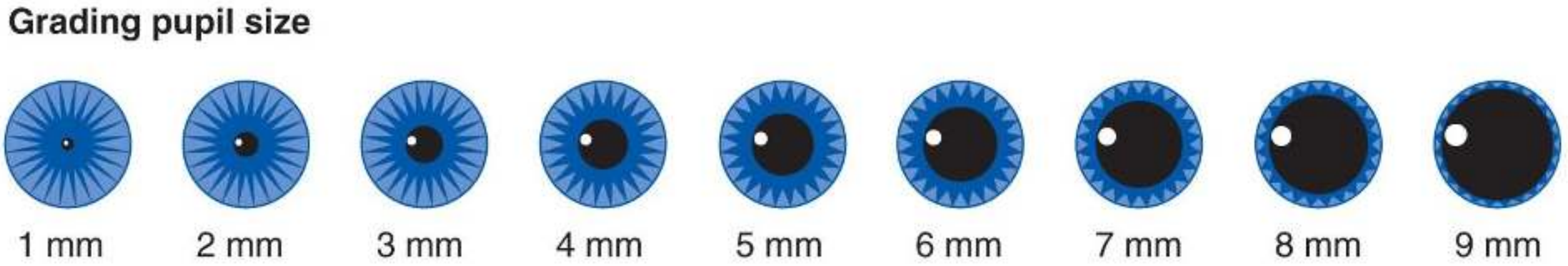
- More objective assessment data are important, however, when caring for the following clients:

- Clients who were previously unconscious
- Clients who were recently resuscitated
- Clients with periods of confusion
- Clients with a head injury
- Clients who took an overdose of drugs
- Clients with a history of chronic alcoholism
- Clients with a psychiatric diagnosis

DATA COLLECTION eyes

- Are both eyes similar
- Iris color
- Sclera color
- Use ophthalmoscope

- visual acuity,
- pupil size and response
- Use PERRLA: pupils equally round and react to light and accommodation







A gross examination of the visual field



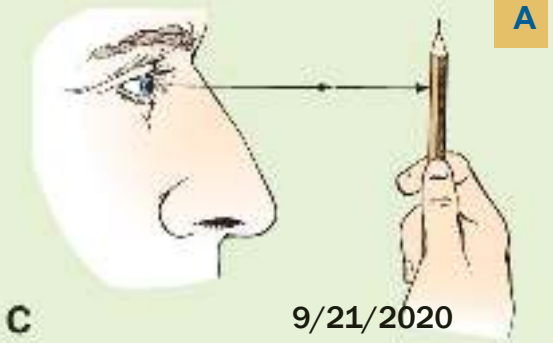
Ophthalmoscope

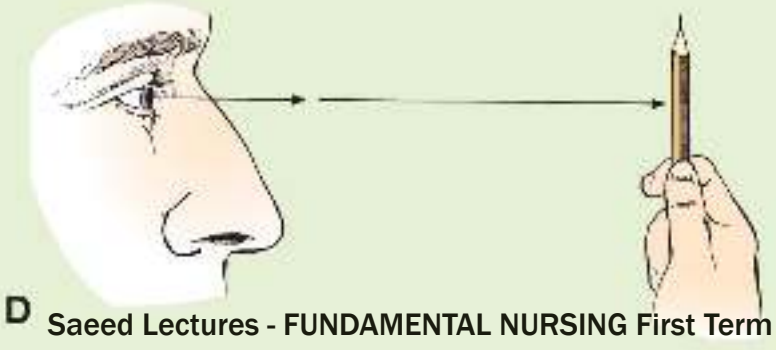


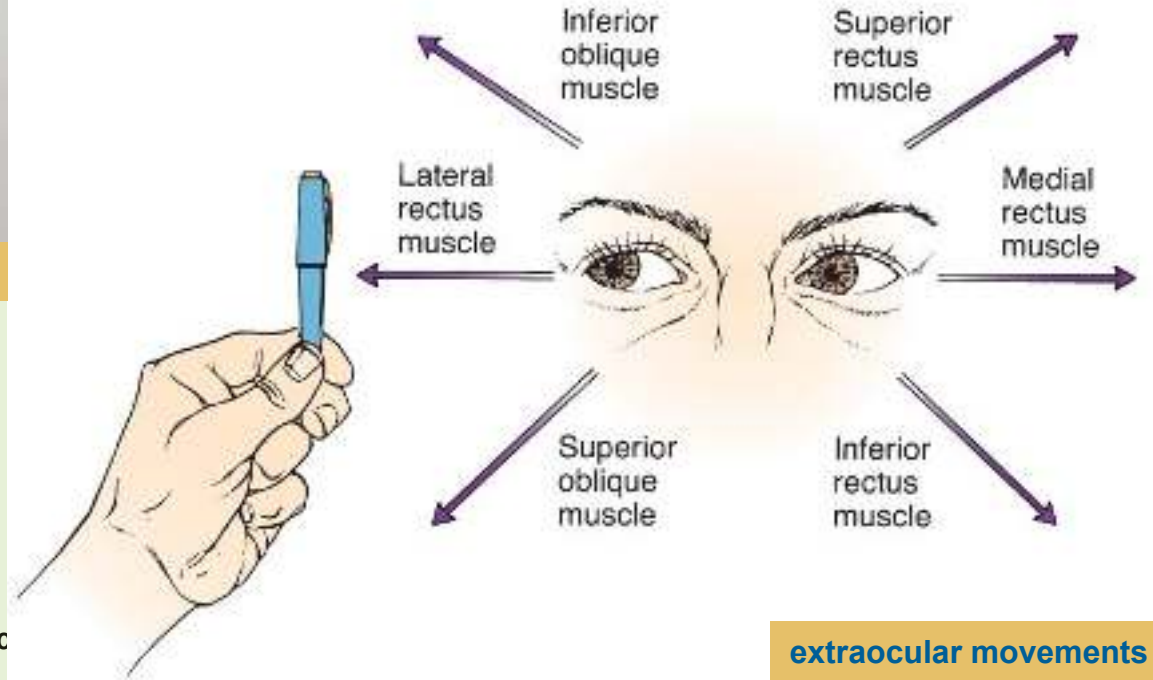
P
E
R
R
L
A



Snellen eye chart





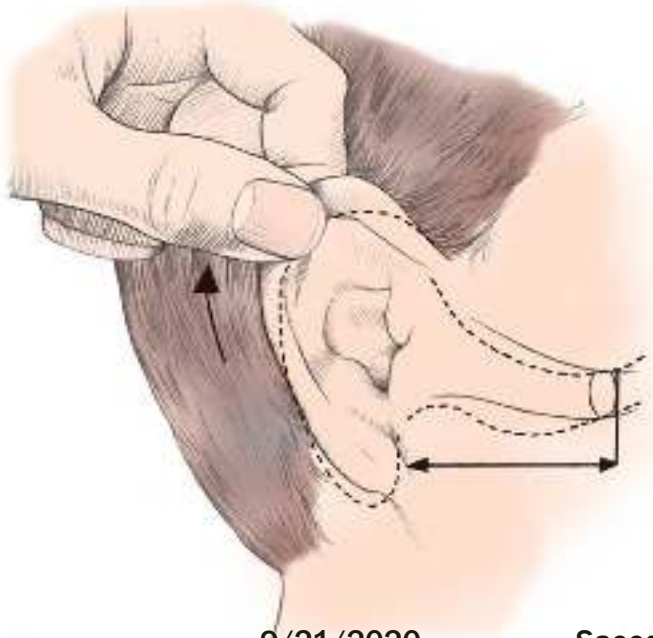


extraocular movements

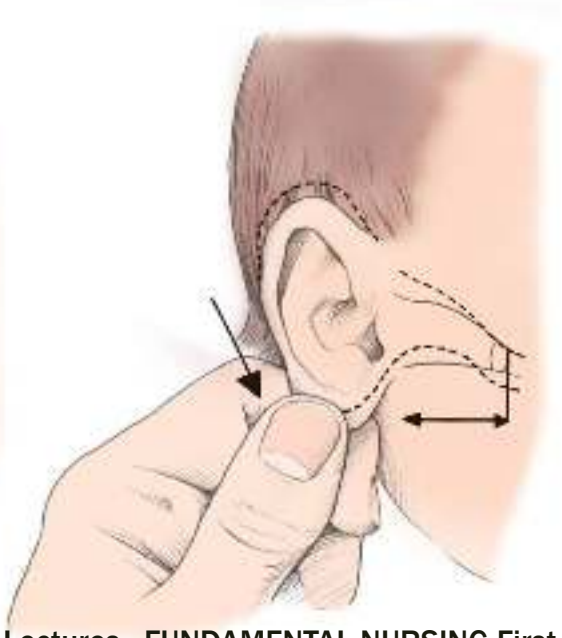
DATA COLLECTION Ears

- The nurse inspects and palpates the external ears.
- Use an otoscope, to examine the tympanic membrane, or eardrum.

- Hearing acuity
- Chick cerumen & color
- Refer client if there is any hearing problem to th audiologist

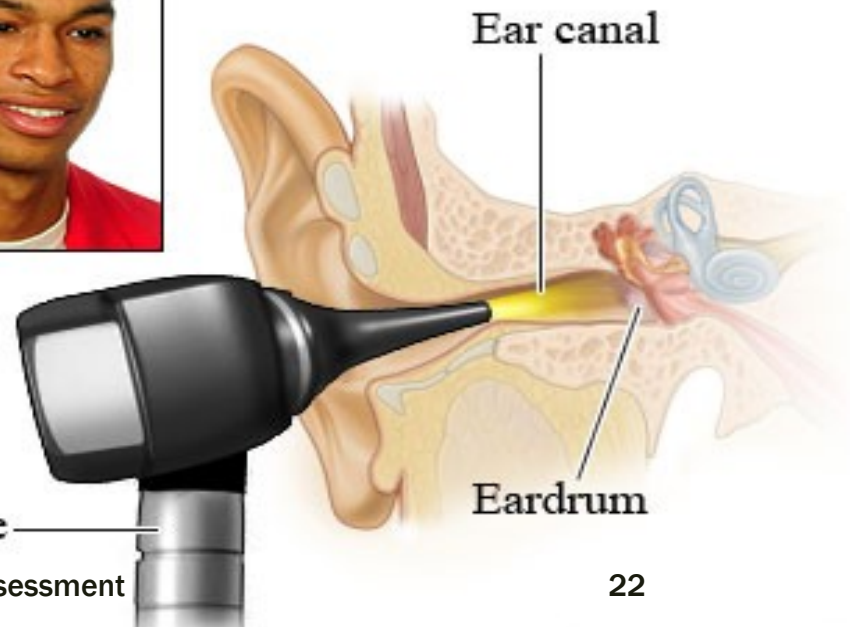


A



B

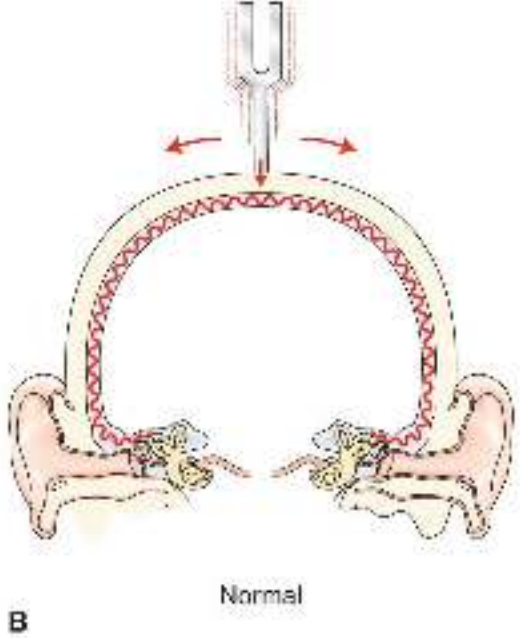






Voice/Whisper Test for Hearing Acuity



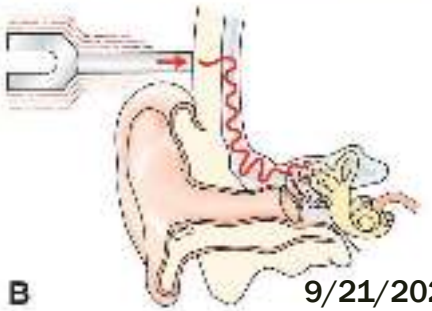


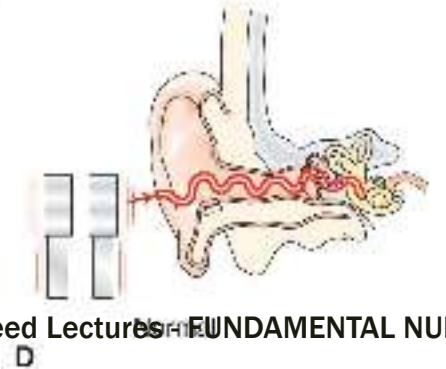
The Weber test assesses sound conducted through bone.

The Weber test assesses sound conducted through bone.









For the Rinne test, the tuning fork base is placed first on the mastoid process.

- (**A, B**) to test bone conduction of sound, after which the prongs are moved to the front of the external auditory canal.
- (**C, D**) to test air conduction of sound.

DATA COLLECTION Nose

- The nurse inspects the nose and nasal passages by having the client assume a “sniffing” position.
- The nurse documents a deviated septum, lesions, growths, flaring of the nostrils, or unusual drainage.
- Smelling acuity if impairment is suspected.
- To test smelling acuity:
- 1. Have the client occlude one nostril and close his or her eyes.
- 2. Place substances with strong odors, such as lemon, vanilla extract, coffee, peppermint, or alcohol, one at a time beneath the patient’s (open) nostril.



DATA COLLECTION Mouth and Oral Mucous Membranes

- Bu using torch and tongue depressor
- The nurse inspects, the lips, mouth, tongue and teeth.
- having the client open the mouth widely. protruding tongue is normally in midline.

- The nurse documents any dentures,
- missing or malpositioned teeth.
- Some unusual breath odors.
- Normal oral mucous membranes are pink, intact, and moist





DATA COLLECTION Facial Skin

- The nurse notes characteristics of the facial skin while assessing the head.
- While examining the skin, the nurse may detect one or more alterations in its
- integrity:

- A wound
- An ulcer
- An abrasion
- A laceration
- A fissure, a crack in skin
- A scar is a mark left by the healing of a wound or lesion.

Open wounds

Closed wounds

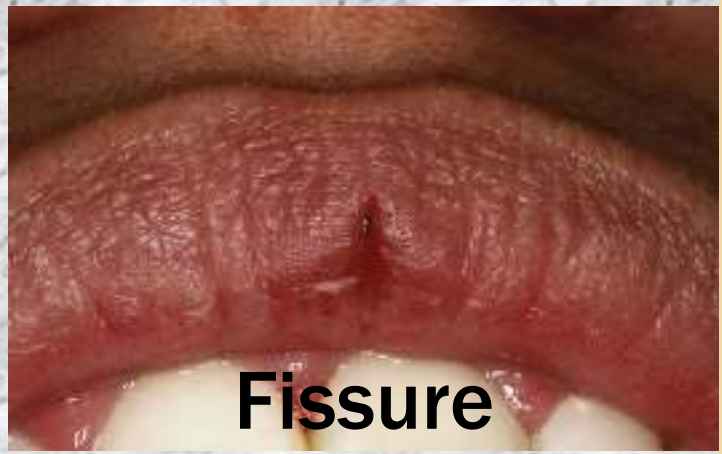








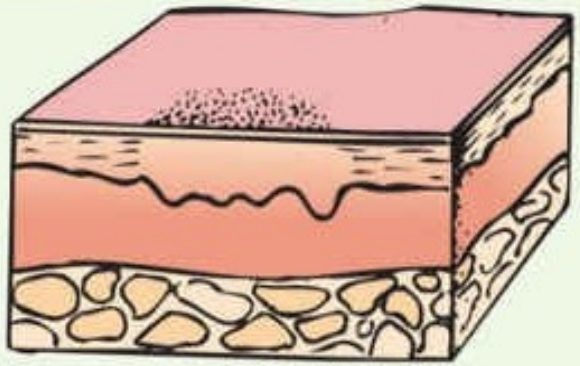
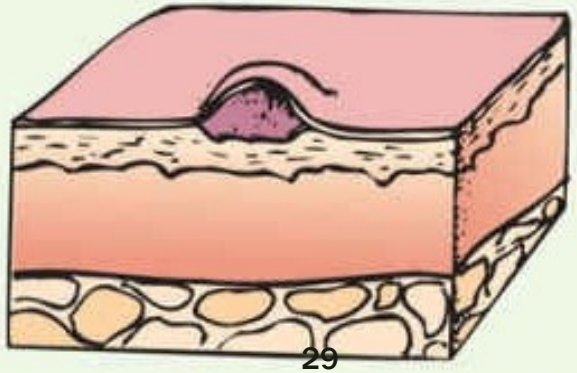




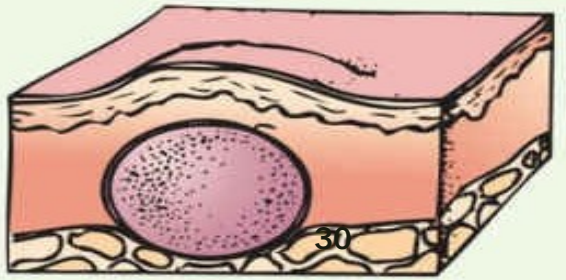
Common skin color variation

COLOR	TERM	Description
Pale	Pallor	Anemia, blood loss
Red	Erythema	Superficial burns, local inflammation
Pink	Flushed	Fever
Purple	Ecchymosis	Trauma to soft tissue
Blue	Cyanosis	Hypo oxygenation
Yellow	Jaundice	Liver disease, destruction of RBCs
Brown	Tan	Pregnancy, kidney disease

Common skin lesions

TYPE OF LESION	DESCRIPTION	EXAMPLE	ILLUSTRATION
Macule	Flat, round, colored, nonpalpable area	Freckles	
Papule	Elevated, palpable, solid	Wart	
Vesicle	Elevated, round, filled with serum	Blister	

Common skin lesions cont.

TYPE OF LESION	DESCRIPTION	EXAMPLE	ILLUSTRATION
Wheal	Elevated, irregular border, no free fluid	Hives	
Pustule	Elevated, raised border, filled with pus	Boil	
Nodule	Elevated, solid mass, deeper and firmer than papule	Enlarged lymph node	
Cyst	Encapsulated, round, fluid-filled or solid mass beneath the skin	Tissue growth	

DATA COLLECTION Hair Scalp

- Hair
- Assessment of the hair includes scalp hair, eyebrows, and eyelashes.
- The nurse notes the color, texture, and distribution (presence or absence in unusual locations for gender or age).

- Scalp
- The nurse assesses the scalp by randomly separating the hair and inspecting the skin surface. He or she looks for signs that the scalp is smooth, intact, and free of lesions.
- The nurse also palpates the skull for any unusual contour.

DATA COLLECTION Neck

- Assess the client ability to bend the head forward, backward, and to either side, as well as to rotate it 180 degrees.
- The trachea (windpipe) should be in the center of the neck.
- Pulsations in the carotid arteries are visible and easy to palpate.

- There should be no unusual bulges or fullness in the neck.
- Some nurses lightly palpate the lymph nodes in the neck area or assess anteriorly for an enlarged thyroid gland.




Pinned





DATA COLLECTION Chest Shape and Movement

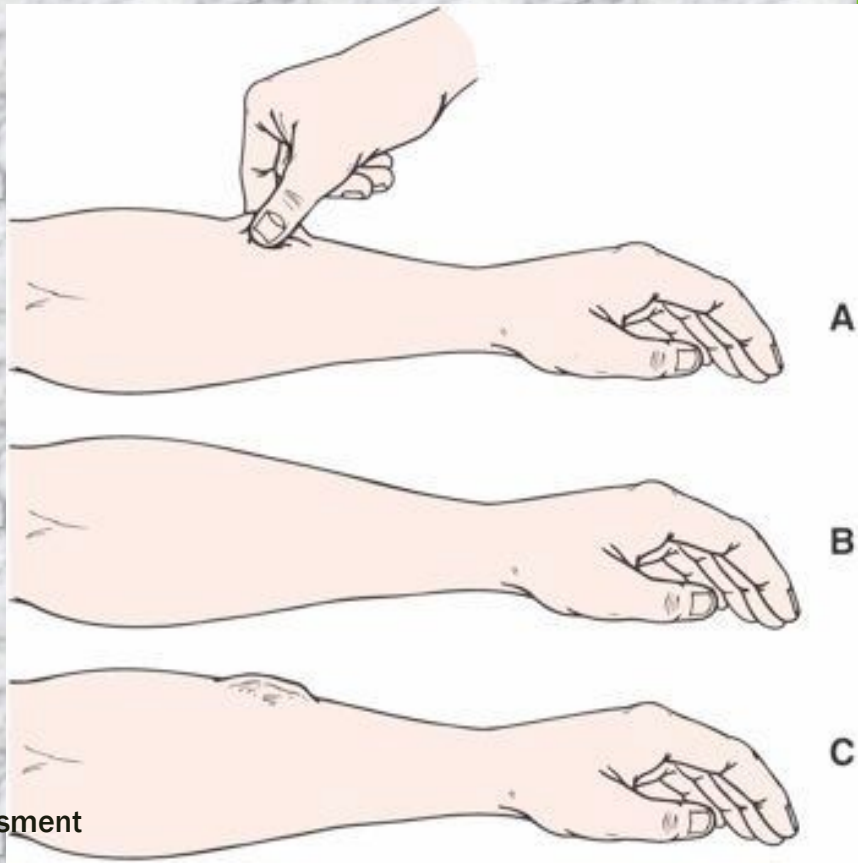
- Assess chest's shape and movement with breathing
- Notes the curved appearance of the spine
- Assesses skin turgor
- Breasts, heart sounds, and lung sounds.

- Skin turgor, the elastic quality of the skin and the pressure exerted on it by the fluid within.
- When skin is pinched will return to normal status

A pinched

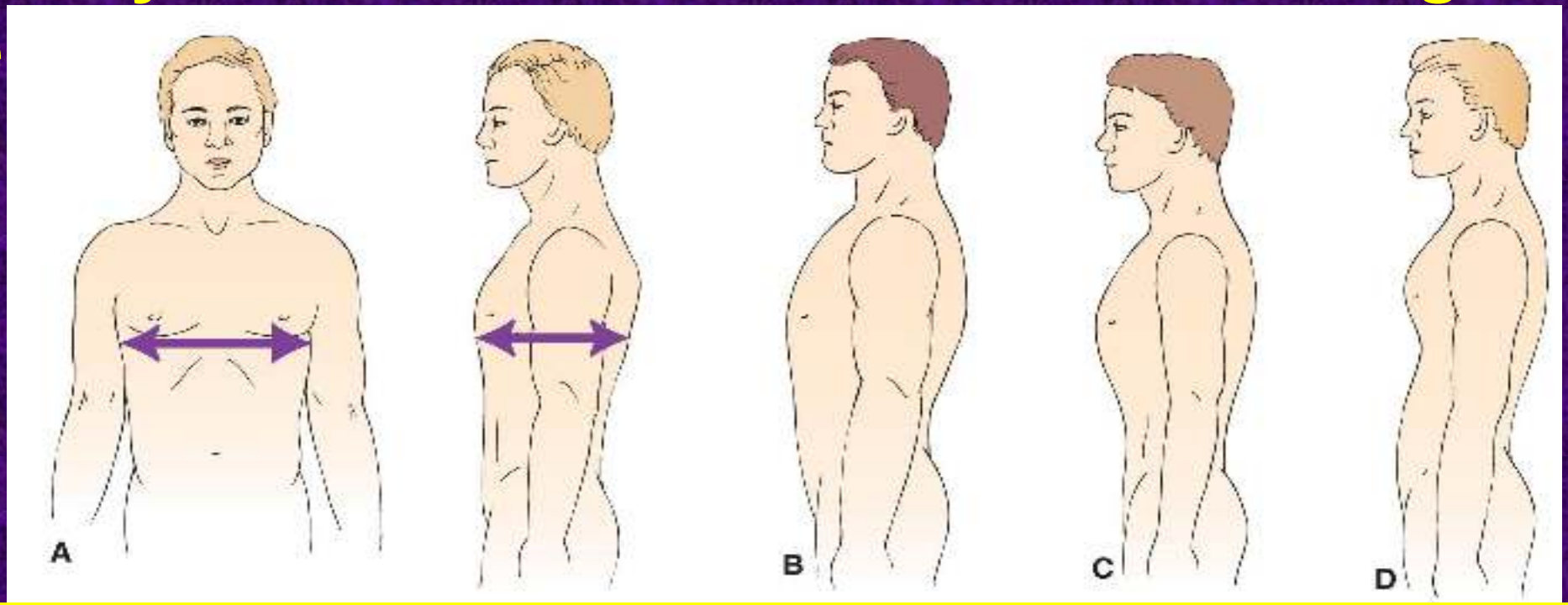
B Normal

C Pooe turgor



DATA COLLECTION Chest Shape and Movement

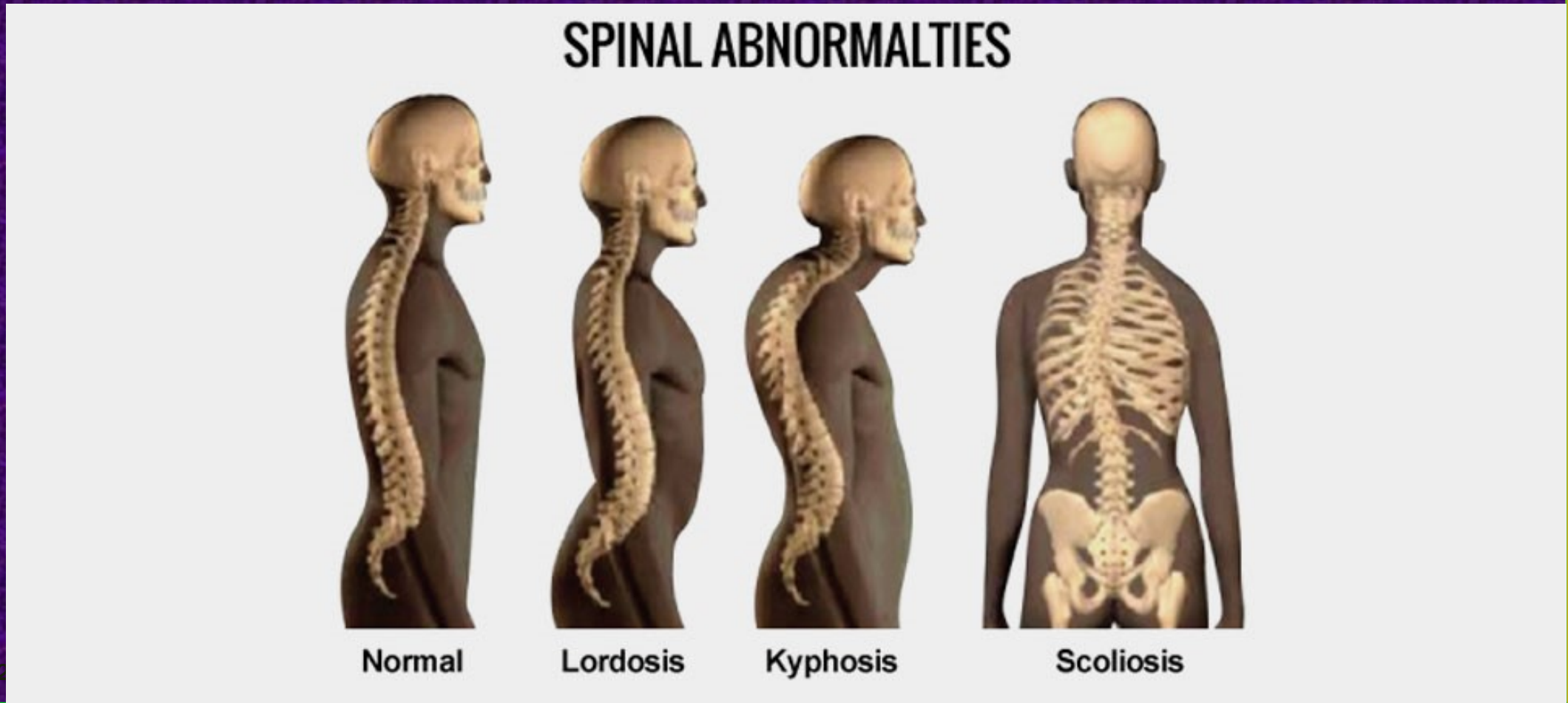
- In normal adults, the lateral dimension of the chest is approximately twice the anterior - posterior dimension.
- Various musculoskeletal abnormalities, cardiac or respiratory diseases, or trauma can cause changes in shape



A. Normal chest. **B.** Barrel chest. **C.** Pigeon chest. **D.** Funnel chest.

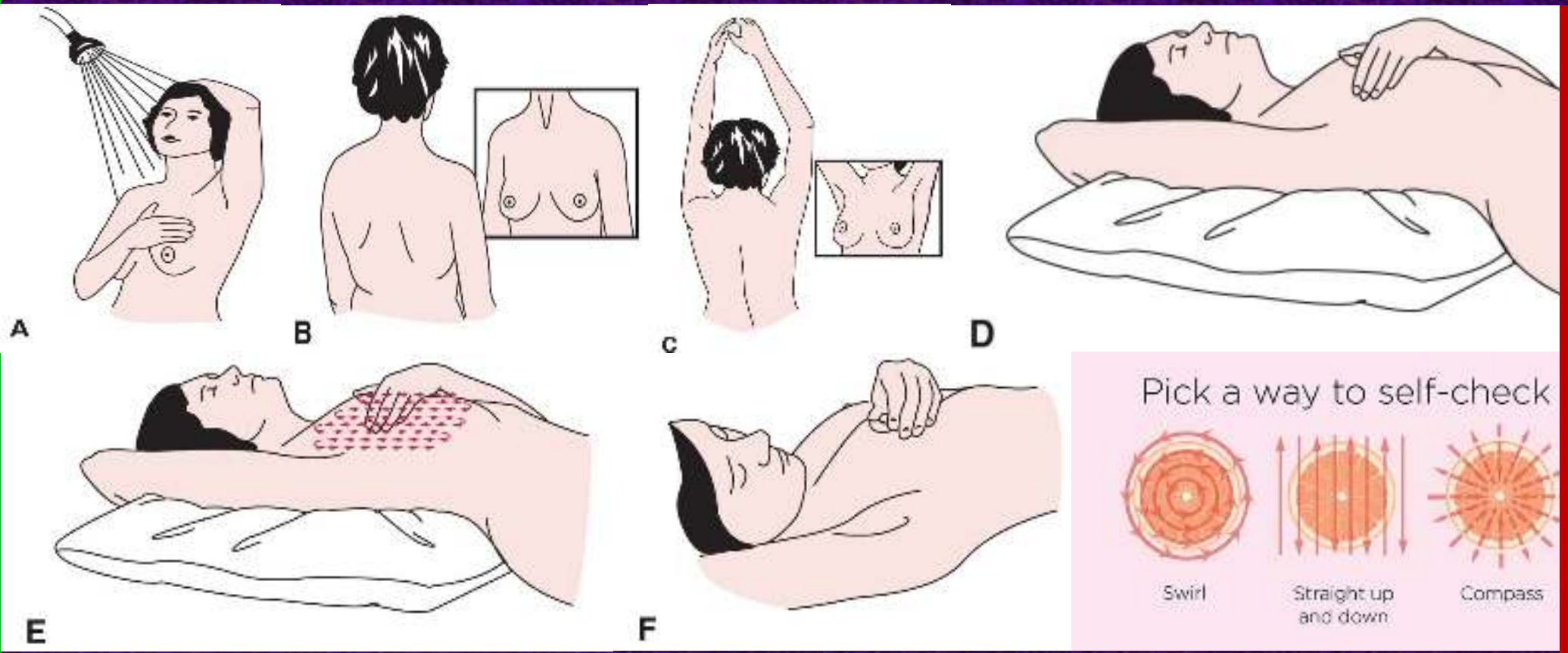
DATA COLLECTION Spine

- The spine, or vertebral column, appears in midline with gentle concave and convex curves when viewed from the side.



DATA COLLECTION Breasts

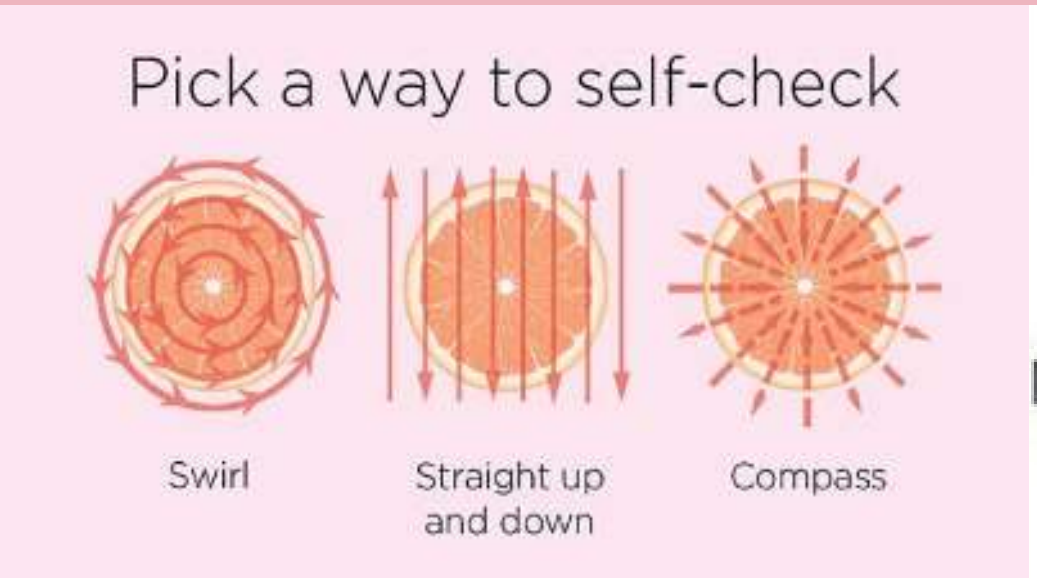
- Although breast abnormalities such as tumors can occur in men, they are more common in women.
- Teaching breast self-examination (BSE)



REPORT IF FIND

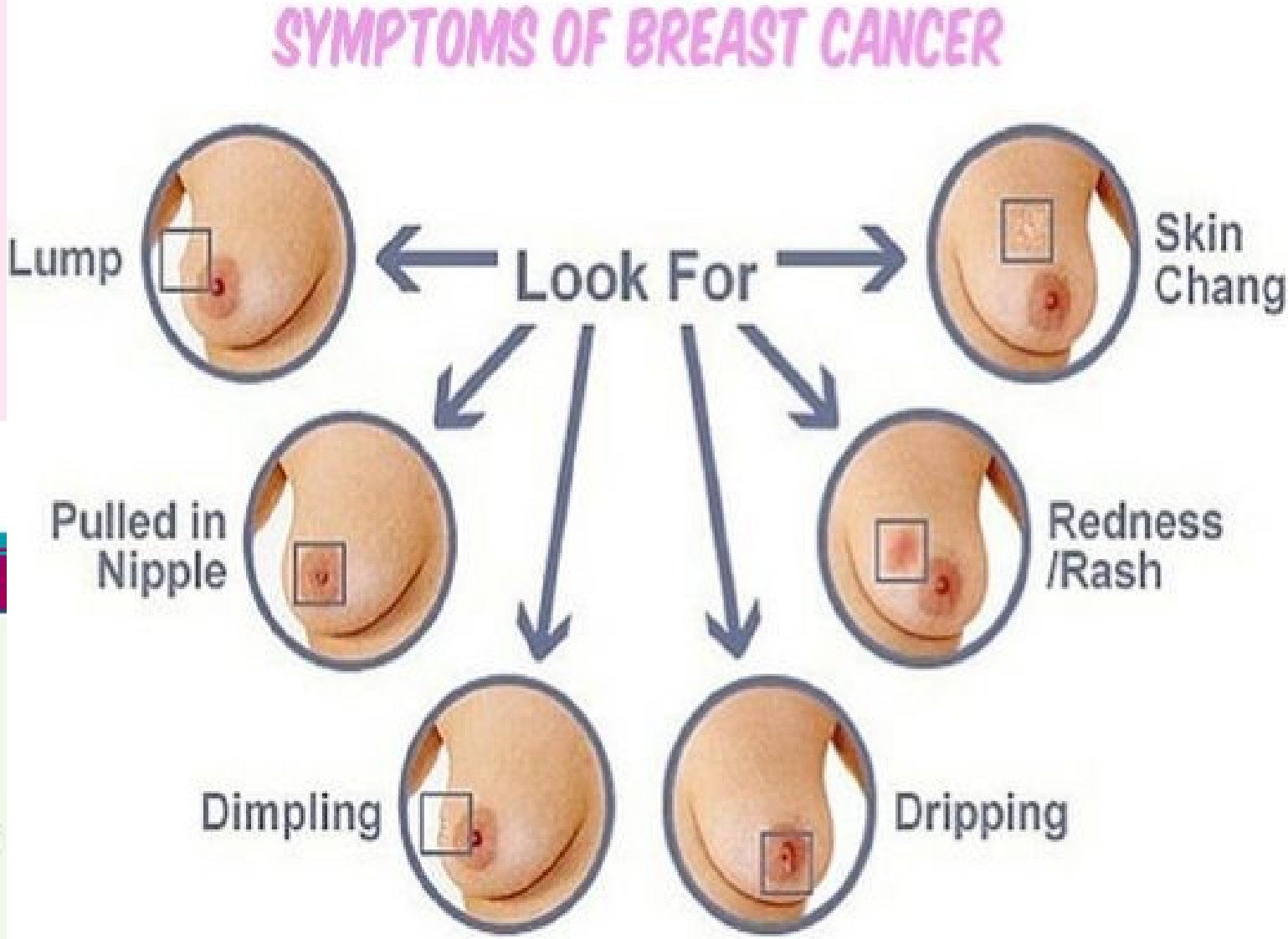
1. Lump
2. Nipple retract
3. Dripping
4. Dimpling
5. Redness
   Rash
6. Skin change

DATA COLLECTION Breasts cont.

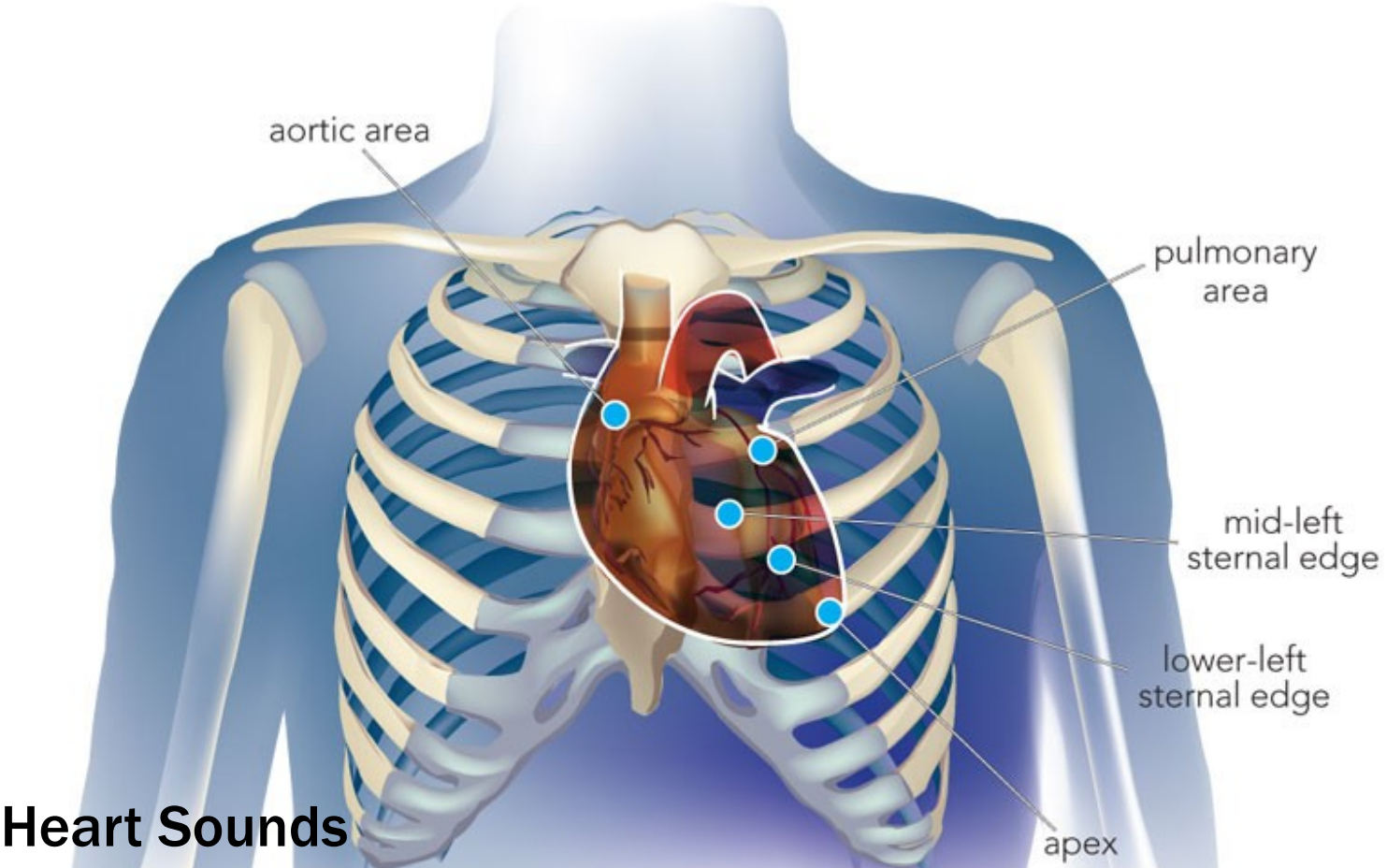


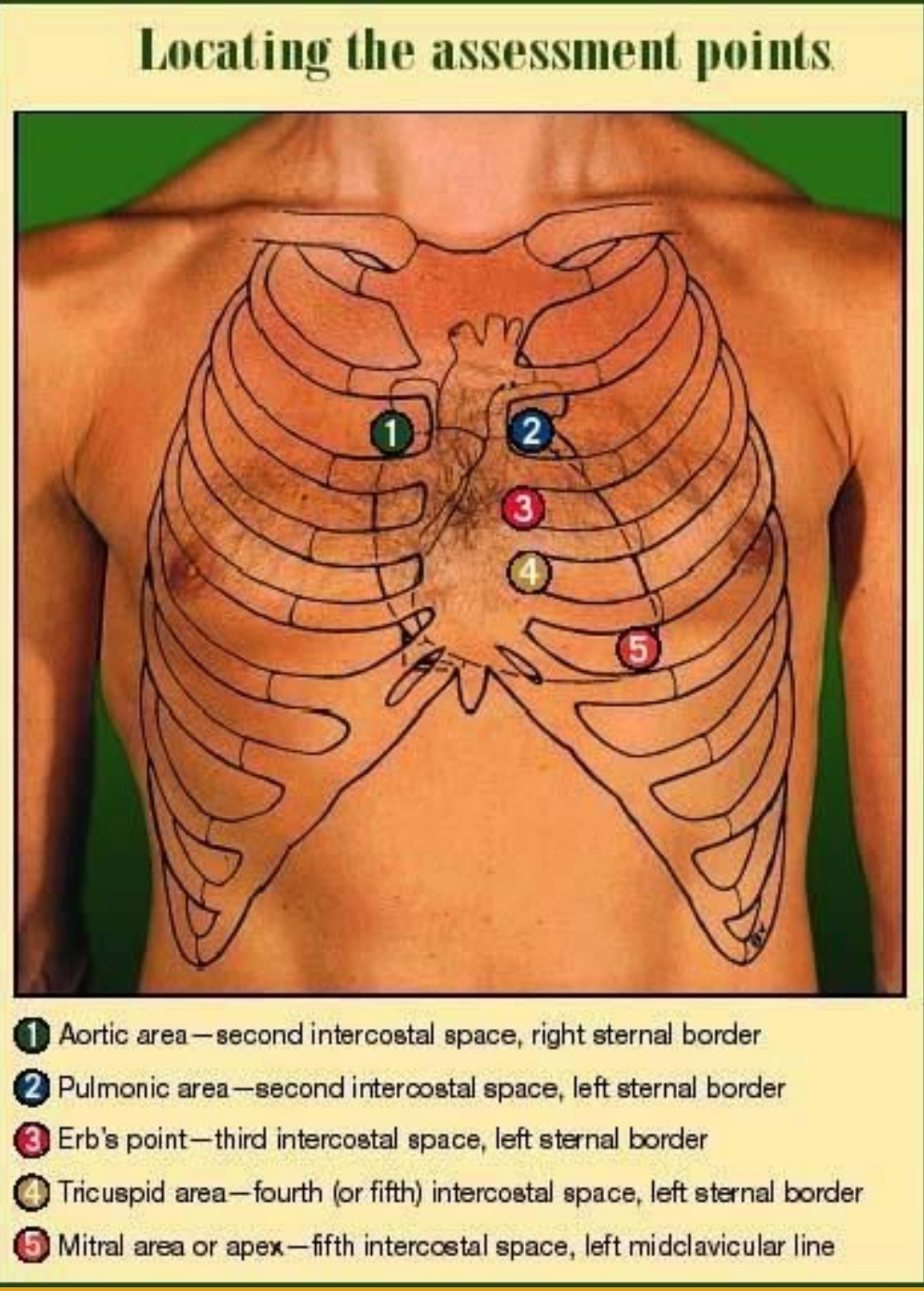
Breast Examination Guidelines

TECHNIQUE	AGE	FREQUENCY
Self-examination (optional)	≥20 years	Once per month
Clinical examination by a nurse or physician	20–39 years	Every 3 years
	≥40 years	Every year
Mammography	40 years	First examination and yearly thereafter
Clinical examination, mammography, magnetic resonance imaging (MRI)	Any age, high risk	Yearly



DATA COLLECTION Heart sounds





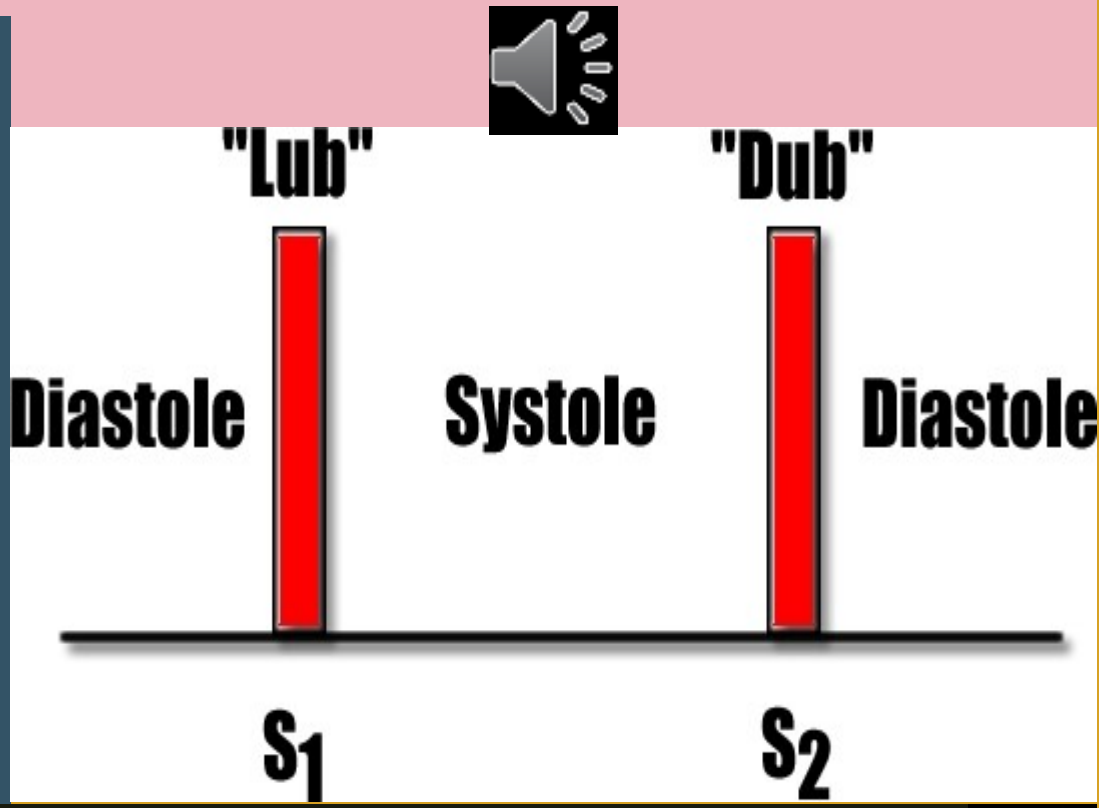
Heart Sounds

When assessing the anterior chest, the nurse listens to the heart sounds, which presumably are caused by the closing of the atrial and ventricular valves.

DATA COLLECTION Heart sounds

Normal Heart Sounds

- The two normal heart sounds are S1 and S2.
- The first heart sound, S1correlates with the “lub” sound and is louder at the apex or mitral area when using the diaphragm of a stethoscope.
- S2 or the “dub” sound, can be heard in the mitral area, it is louder over the aortic area.



Abnormal Heart Sounds

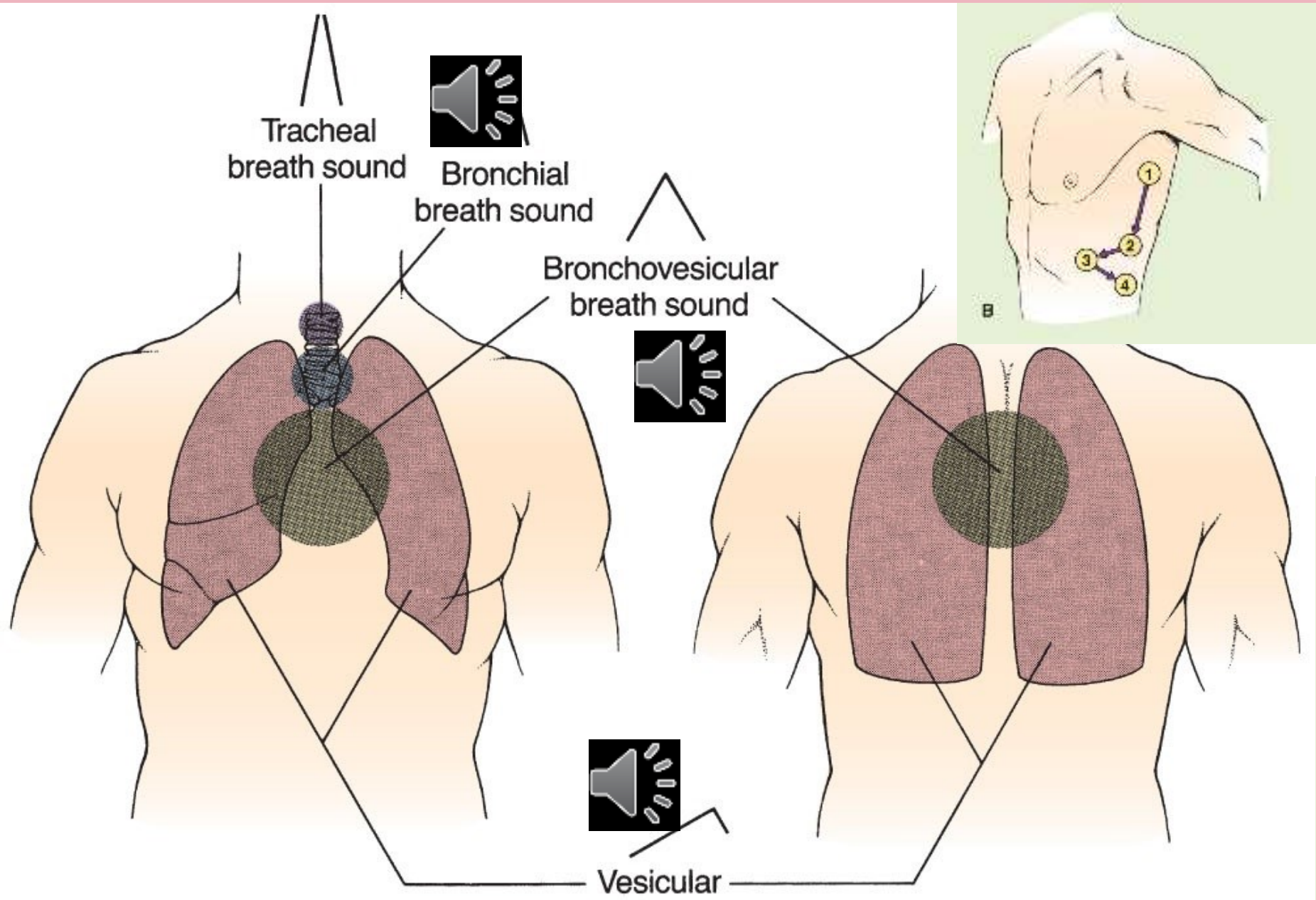
- The nurse may hear two additional sounds, S3 and S4, when auscultating the chest.
- S3, which is normal in children but abnormal in most adults, appears after S2. S3 sounds like “lub-dub-dub”.

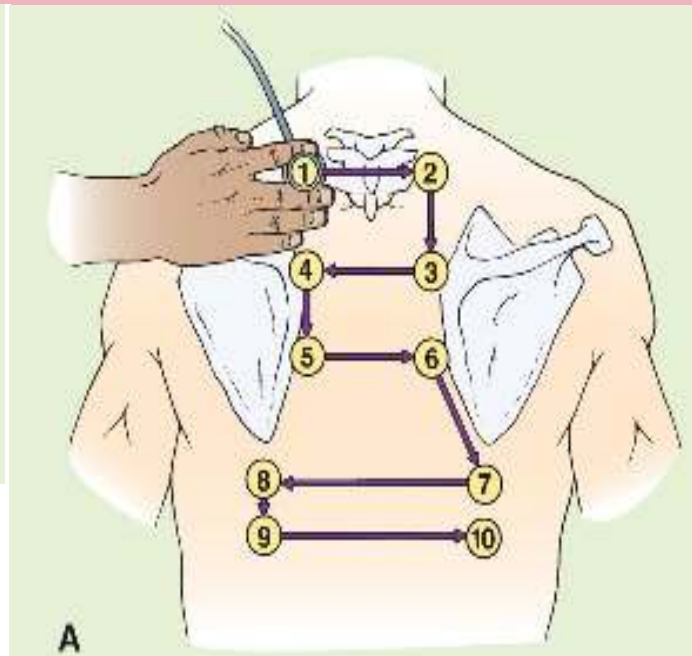
DATA COLLECTION Lung sounds

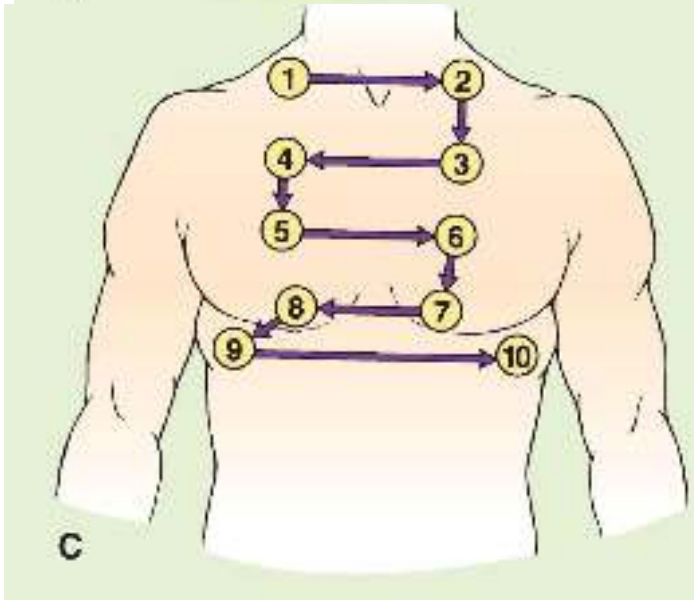
Listening to the lungs is another skill that requires frequent and repeated practice because some sounds are normal and others are abnormal.
Normal lung sounds occur when air moves in and out of passageways.
There are four normal lung sounds:

1. Tracheal sounds are loud and coarse. Are equal in length during insp. and exp.
2. Bronchial sounds, heard over the upper sternum anteriorly and between the scapulae posteriorly, are harsh and loud. They are shorter on inspiration than expiration.
3. Bronchovesicular sounds are heard on either side of the central chest or back. These medium-range sounds that are equal in length during inspiration and expiration.
4. Vesicular sounds are located in the periphery of all the lung fields. Their soft, rustling quality is longer on inspiration than expiration, with no pause

DATA COLLECTION Lung sounds sites







Stridor COVID 19

DATA COLLECTION Lung sounds sites

- *Crackles*, also called *rales*, are intermittent, high-pitched, popping, and heard in distant areas of the lungs, primarily during inspiration. They resemble the sound of crisped rice cereal when milk is added.
- *Gurgles*, also called *rhonchi*, are low-pitched, continuous, bubbling, and heard in larger airways. They are more prominent during expiration. Some describe gurgles as sounding like wet snoring. Gurgles may clear with deep breathing or coughing.
- *Wheezes* are whistling or squeaking sounds caused by air moving through a narrowed passage. Wheezes may be audible without a stethoscope.
- *Pleural friction rubs* are grating, leathery sounds caused by two

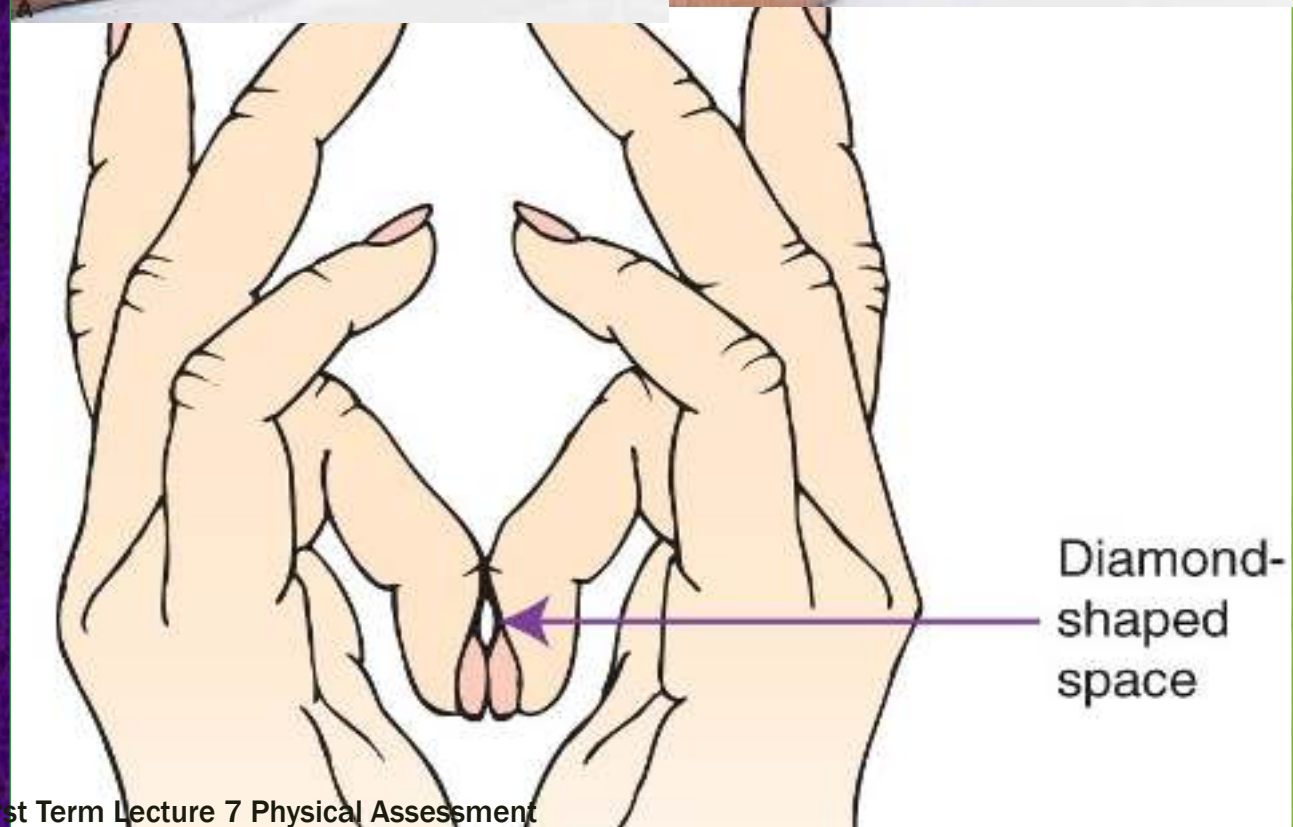
DATA COLLECTION Extremities

The nurse notes the:

- alignment, mobility, and strength of the extremities and
- compares their size.
- He or she feels the skin temperature,
- the characteristics of the nails,
- times the capillary refill, palpates local peripheral pulses,
- checks for edema,
- test the perception of skin sensations.
- assess deep tendon reflexes with a reflex hammer.







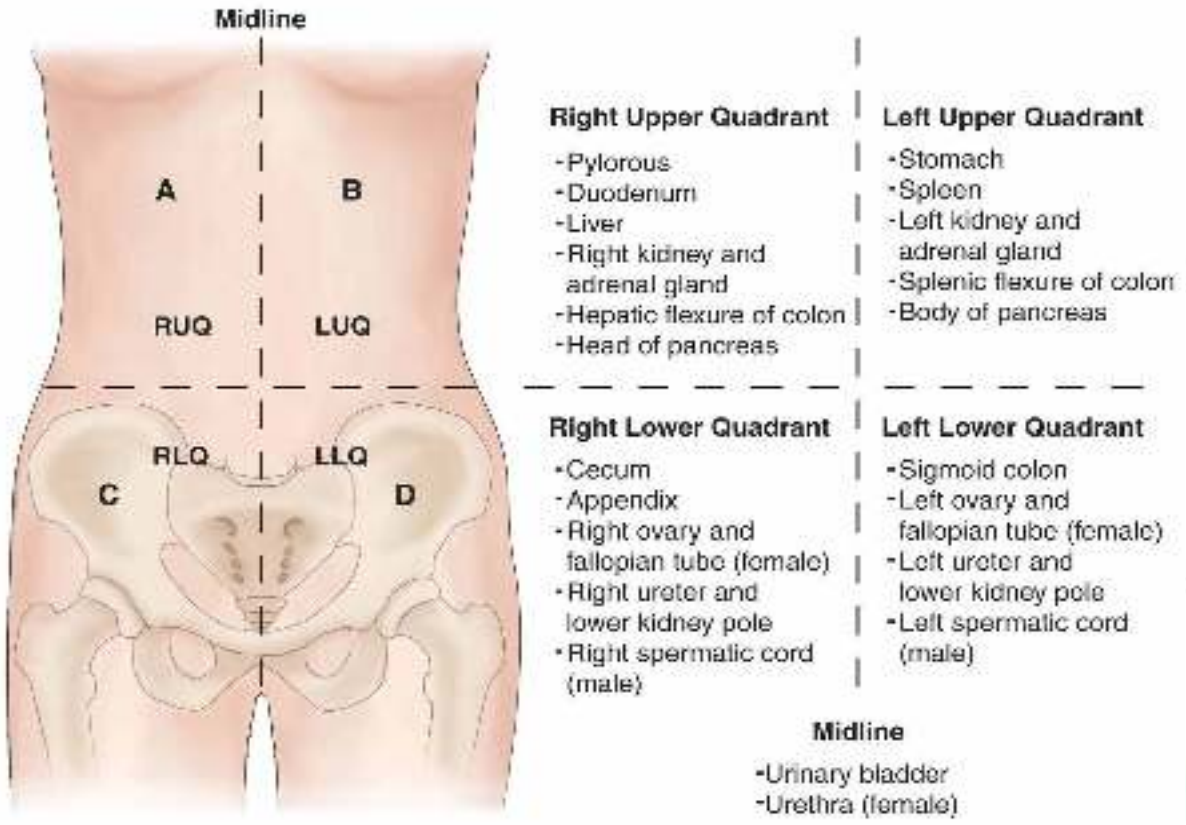
Criteria for Estimating Pitting Edema

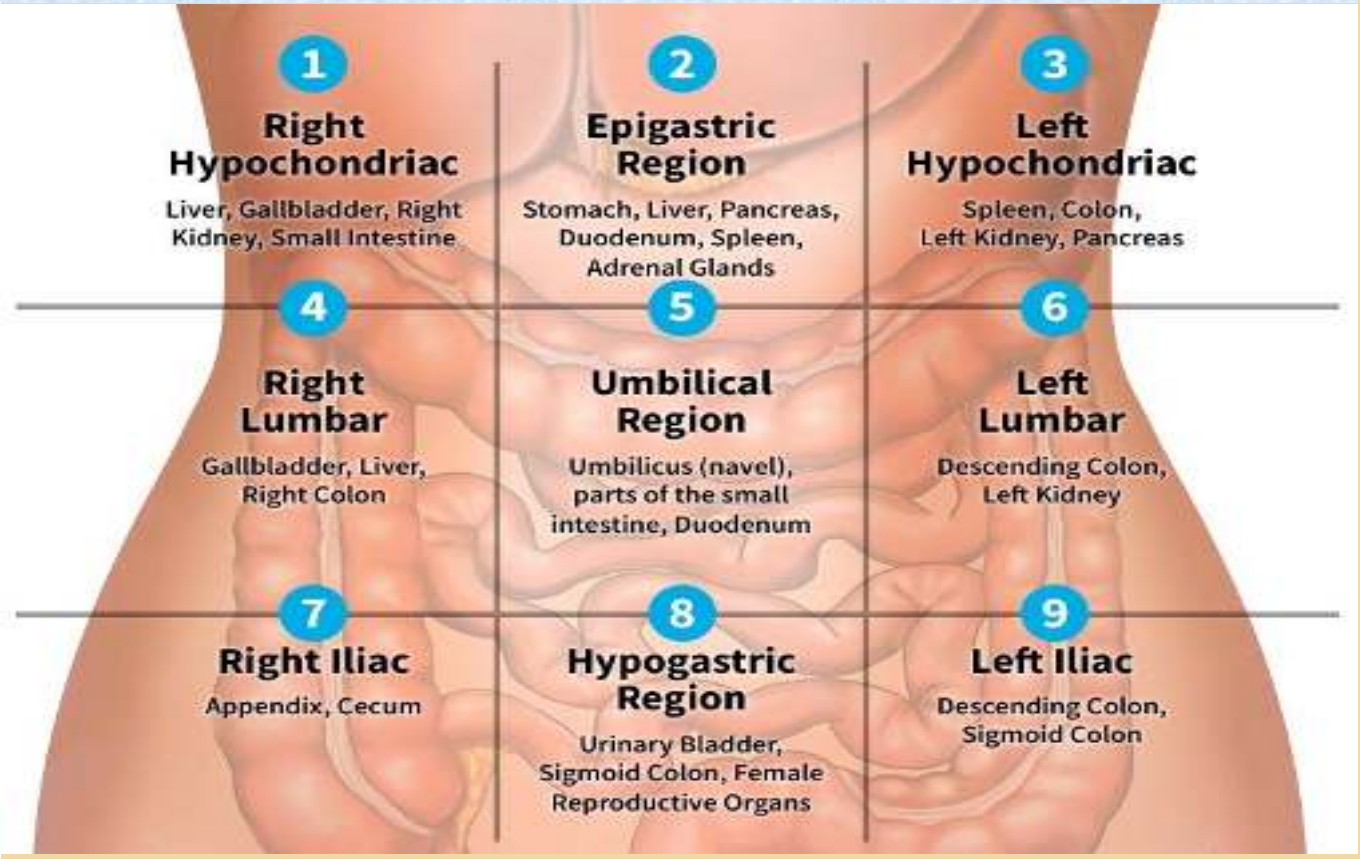
TYPE	DISCRIPTION	
Pitting Edema 1+	• Slight indentation (2 mm) • Normal contours • Associated with interstitial fluid volume 30% above normal 2+ Pitting	
Pitting Edema 2+	• Deeper pit after pressing (4 mm) • Lasts longer than 1+ • Fairly normal contour	

TYPE	DISCRIPTION	
Pitting Edema 3+	• Deep pit (6 mm) • Remains several seconds after pressing • kin swelling obvious by general inspection	
Pitting Edema 4+	• Deep pit (8 mm) • Remains for a prolonged time after pressing, possibly minutes • Frank swelling	
Brawny Edema 5+	• Fluid can no longer be displaced secondary to excessive interstitial fluid accumulation • No pitting • Tissue palpates as firm or hard • Skin surface shiny, warm, moist	

DATA COLLECTION Abdomen

- Most gastrointestinal and accessory digestive organs lie within the abdomen. The bladder, if distended, may rise into the abdomen. For assessment purposes, the abdomen is divided into four quadrants.
- *The abdomen is always inspected and then auscultated—in that sequence—before using palpation or percussion techniques*. Touching or manipulating the abdomen can alter bowel sounds, thus producing invalid findings.





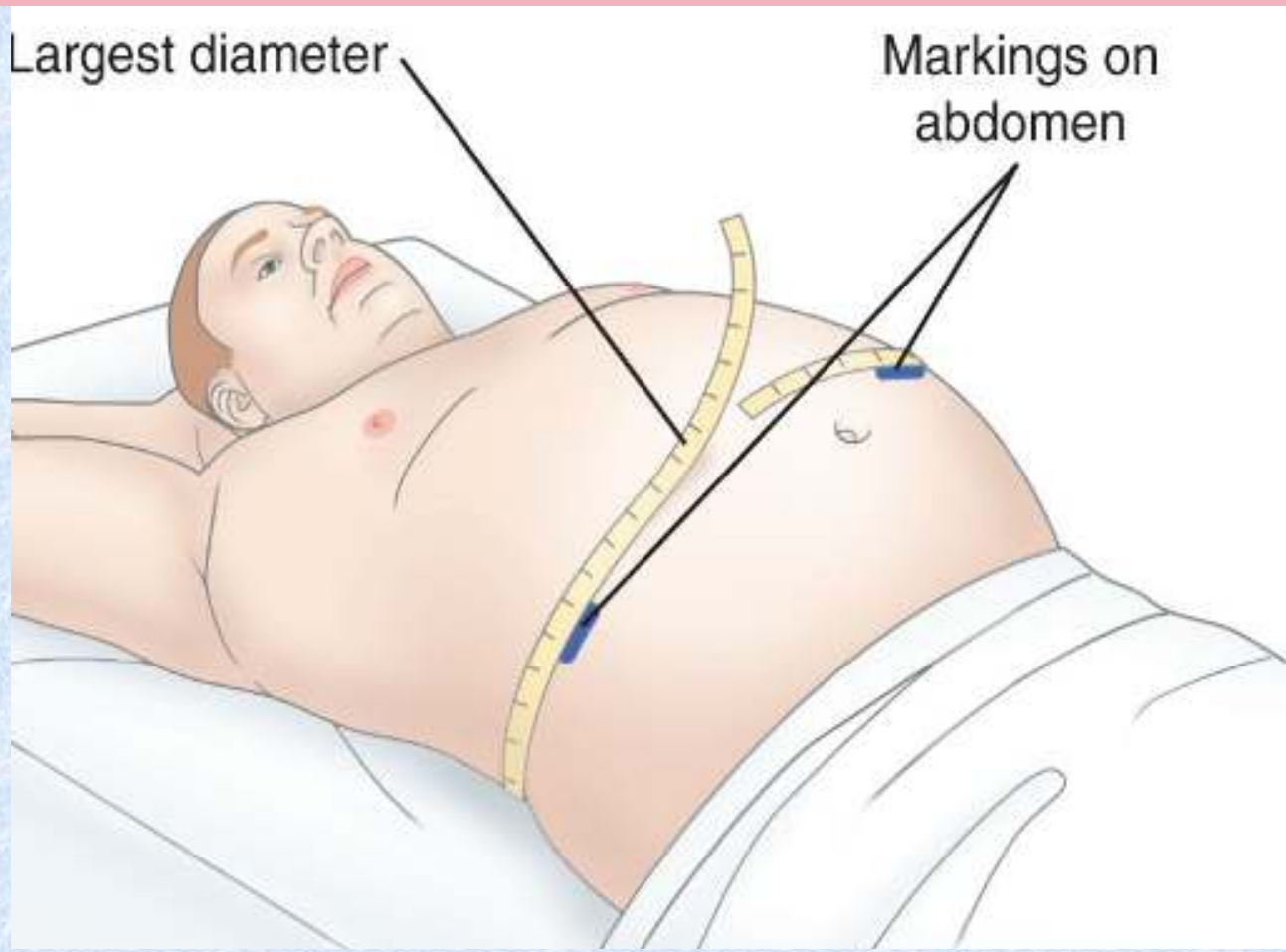
DATA COLLECTION Abdomen

Bowel Sounds

- Wave-like muscular contractions of the large and small intestines that move fluid

Abdominal Girth

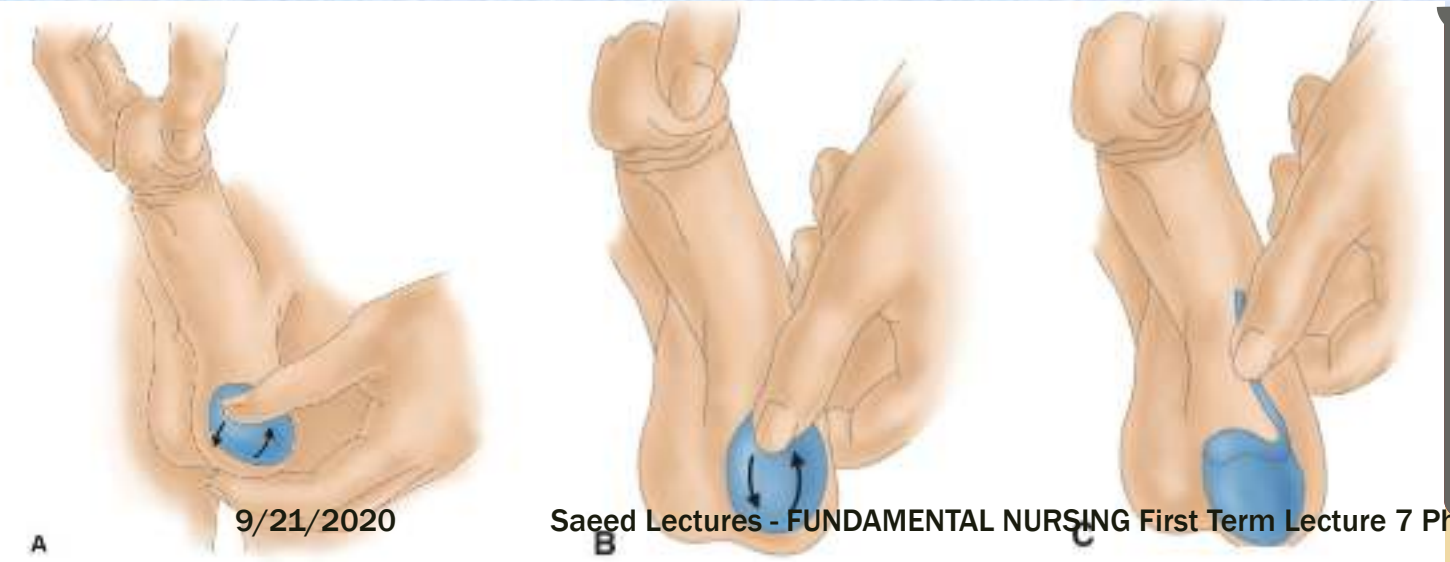
- If the abdomen appears unusually large,



- the nurse checks its girth (circumference) daily by using a tape measure around the largest diameter.

DATA COLLECTION Genitalia

- During inspection, the nurse notes the condition of the skin and the distribution and characteristics of pubic hair (lice may infest pubic hair), as well as the presence of lesions.
- The nurse teaches the client as follows:
- Examine the testes monthly when they are warm and loosely within the scrotum (eg, during bathing).
- Elevate the penis with one hand, gently roll each testicle between the thumb and index finger.
- Feel each testicle horizontally, vertically and check for any unusual lumps.
- Continue palpation following the spermatic cord from the testicle to where it ascends into the abdomen.
- Report any unusual findings at once; an early diagnosis carries a better prognosis.



Remember post bond Genitalia and anus last:

1. **To build trust with client**
2. **Assessment from clean to less clean**

DATA COLLECTION Genitalia

- Unless a client has specific symptoms, the nurse inspects only the anus. If touching is required, gloves are necessary.
- To examine the anus, the nurse positions the client on the side with the knees bent (Sims Position).
- Assess for: any unusual findings …
- External hemorrhoids (saccular protrusions filled with blood) may extend beyond the external sphincter muscle.
- Rectal fissures (cracks) if the client has a history of chronic constipation.

Remember, post bond Genitalia and anus test last:

1. **To build trust with client**
2. **Assessment course from clean to less clean**


Thank
You
Merci
Obrigado
Obrigado
Asante
Spasibo
Maake
Grazie
Kiitos
Multumesc
Vinaka
Arigato
Dank Je
Matondo
Juspaxar
Mamana
Nirringrazzjak
Ua Tsaug Rau Koj
Mochchakkeram
Kia Ora
Salamat
Welalin
Terma Kasih
Cam on ban
Danken
Chokrane
Matondo
Grazie
Asante
Kiitos
Maake
Merci
Spasibo
Obrigado
Grazie
Asante
Kiitos
Mochchakkeram
Kia Ora
Ua Tsaug Rau Koj
Salamat
Mabur Nuwun
Chokrane
Matondo
Obrigado
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Asante
Kiitos
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Multumesc
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Mamana
Grazie
Nirringrazzjak
Obrigado
Nirringrazzjak
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Terma Kasih
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Grazie
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SPECIAL EXAMINATIONS AND TESTS PART I

Objectives

- *On completion of this chapter, the audience should be able to:*
  1. *Differentiate between an examination and a test.*
  2. *List general nursing responsibilities related to assisting with special examinations and tests.*
  3. *Name five positions commonly used during tests or examinations.*
  4. *List six commonly performed categories of tests or examinations.*
  5. *Explain the following procedures: sigmoidoscopy, paracentesis, lumbar*
  6. *puncture, throat culture, and measurement of capillary blood glucose.*
  7. *Discuss at least three factors to consider when performing examinations and tests on older adults.*

EXAMINATIONS AND TESTS

- *A diagnostic examination is a procedure that involves the physical inspection of body structures and evidence of their functions.*
- *It is facilitated through the use of technical equipment and techniques, such as the following:*
  - Radiography (X-rays)
  - Endoscopy (optical scopes)
  - Radionuclide imaging (radioactive chemicals)
  - Ultrasonography (high-frequency sound waves)
  - Electrical graphic recordings

Diagnostic Terms Meaning

SUFFIX	MEANING	EXAMPLES	DESCRIPTION
–graphy	To record	Angiography	Test that records an image of blood vessels
–gram	An image	Angiogram	The actual image recorded during angiography
–scopy	To see	Sigmoidoscopy	Test in which the lower intestine is inspected
–scope	Examination instrument	Sigmoidoscope	A tube with a light and lens for looking within the lower intestine
–centesis	To puncture	Thoracentesis	Procedure in which a needle is used to puncture the thorax and withdraw fluid
–metry	To measure	Pelvimetry	Procedure in which the pelvis is measured
–meter	Instrument for obtaining measurements	Glucometer	Instrument for measuring glucose

A laboratory test is a procedure that involves the examination of body fluids or specimens. It involves comparing the components of a collected specimen with normal findings.

General Nursing Responsibilities In Diagnostic Tests

Preprocedural Care

1. Determine the client's understanding of the procedure.
2. Witness the client's signature on a consent form.
3. Teach or follow test preparation requirements.
4. Obtain equipment and supplies.
5. Arrange the examination area.

Procedural Care

6. Position and drape the client.
7. Assist the examiner.
8. Provide the client with physical and emotional support.

Postprocedural Care

9. Attending to the Client.
10. Care for specimens.
11. Record and report appropriate information.

Pre-procedural Care

1. **Clarifying Explanations.**
   In some cases, a signed consent form is required before performing examinations or tests. Legally, consent must contain three elements: capacity, comprehension, and voluntariness

2. **Preparing Client.**
   - Withholding food and fluids or modifying the diet if needed.
   - Teaching the client about procedure, benefits and risks
   - Take V.S.
   - Change into an examination gown, identification bracelet in hand
   - Report any incorrect test preparations promptly, test canceled.

Pre-procedural Care cont.

3. **Arranging the Examination Area.**
   If the procedure is performed at the bedside, the nurse removes unnecessary articles from the area and provides privacy.
   Arrange lite, warmth, enough area, receptacles …

4. **Obtaining Equipment and Supplies.**
   If an examination or test is performed at the bedside or in an examination room on the nursing unit, the nurse obtains equipment and supplies ahead of time.

Procedural Care

1. **Positioning and Draping. Seven positions are commonly used depending on the type of examination.**

2. **Assisting the Examiner.**
   The nurse must be familiar with the examination equipment and the order of its use.

3. **Providing Physical and Emotional Support..**
   - Observe the client's physical and emotional reactions.
   - Holding the client's hand
   - Offering words of encouragement help the client to endure temporary discomfort.
   - Communicate assessments of the client to the examiner, who may choose to shorten or modify the examination in some manner.

Indications for Common Examination Positions

POSITION	USES
Supine position (Fig. A).	• Examinations of the head, neck, chest, abdomen, and extremities
Prone position (Fig. B).	• Examinations of the posterior head, back, buttocks, and extremities
Dorsal recumbent position (Fig. C).	• External genitalia inspection • Vaginal examination • Rectal examination • Urinary catheter insertion



A **Supine position; client is lying flat with the face up.**



is opposite of the supine position.

B **Prone position; the client lies on his or her abdomen with the head turned to one side.**



C **Dorsal recumbent position; is a reclining position with the knees bent, hips rotated outward, and feet flat.**

Indications for Common Examination Positions

POSITION	USES
Lithotomy position (Fig. D). 	• Internal pelvic examination (female) • Obstetric delivery • Cystoscopic (bladder) examination • Rectal examination
Sims' position (Fig. E). 	• Rectal examination • Vaginal examination • Rectal temperature assessment • Suppository insertion • Enema administration

Lithotomy position; is a reclining position with the feet in metal supports called "stirrups."

D

Sims' position; the client lies on the left side, chest leaning forward, right knee bent toward the head, right arm forward, left arm extended behind the body.

E

Indications for Common Examination Positions

POSITION	USES
Knee–chest position (Fig. F).	• Rectal and lower intestinal examinations • Prostate gland examination
Modified standing position (Fig. G).	• Prostate gland examination



F **Knee–chest position; also called a "genupectoral position," the client rests on the knees and chest.**



G **Modified standing position, the client stands with the upper half of the body leaning forward.**

Post-Procedural Care

1. **Attending to the Client.**
   First, the nurse helps the client to a position of comfort. He or she rechecks vital signs to verify that the client's condition is stable.

2. **Caring for Specimens.**
   Sometimes specimens (samples of tissue or body fluids) are collected during an examination or test. To ensure their accurate analysis, the nurse does the following:
   - Collects the specimen in an appropriate container
   - Labels the specimen container with the correct information
   - Attaches the proper laboratory request form
   - Ensures that the specimen does not decompose before it can be examined
   - Delivers the specimen to the laboratory as soon as possible

Common Factors that Invalidate Examination or Test Results

- Incorrect diet preparation
- Failure to remain fasting
- Insufficient bowel cleansing
- Drug interactions
- Inadequate specimen volume
- Failure to deliver specimen in a timely manner
- Incorrect or missing test requisition

Post-Procedural Care cont.

3. **Recording and Reporting Data.**

The nurse must document certain information whenever a client undergoes a special examination or test. General information includes the following:

- Date and time
- Pertinent pre-examination assessments and preparations
- Type of test or examination
- Who performed the test or examination
- Where the test or examination was performed
- Response of client during the examination and afterward
- Type of specimen obtained, if any
- Appearance, size, or volume of specimen
- Where the specimen was transported



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God Bless You

Physical Assessment

Part II

Things to consider...

Inspection

Auscultation

Percussion

Palpation

Physical Assessment Equipment & Environment

For a basic physical assessment, the nurse needs the following:

- Gloves
- Examination gown
- Cloth or paper drapes
- Scale
- Stethoscope
- Sphygmomanometer
- Thermometer
- Pen light or flashlight
- Tongue blade
- Assessment form and pen

Nurses assess the special examination room or bedside for:

- Easy access to a restroom
- Door or curtain for privacy
- Adequate warmth
- padded, adjustable table or bed
- Sufficient area for moving to either side of the client
- Adequate lighting
- Facilities for hand hygiene
- Clean counter or surfacefor placing examination equipment

PERFORMING A PHYSICAL ASSESSMENT

- **Gathering General Data during the first contact with the client.**
- **At this time, the nurse appraises the client's overall condition.**
- **By observing and interacting with the client before the actual physical examination, the nurse notes the following:**

- *Physical appearance with regard to clothing and hygiene*
- *Level of consciousness*
- *Body size*
- *Posture*
- *Gait and coordinated movement (or lack of it)*
- *Use of ambulatory aids*

PERFORMING A PHYSICAL ASSESSMENT cont.

- Obtaining Weight and Height
- **Provide privacy**
- Draping and Positioning, preferred by two drapes (a sheet of soft cloth or paper) up and down.





Selecting an Approach for Data Collection

- **Head-to-Toe Approach**
  - A head-to-toe approach means assessing the client from the top of the body to the feet. This has three advantages:

1. It prevents overlooking some aspect of data collection.
- 2. It reduces the number of position changes.
- 3. It generally takes less time than system review appeoach.

- **Body Systems Approach**
- **A body systems approach means assessing the client according to the functional systems of the body.**
- **It involves examining the structures in each system**

DATA COLLECTION

- **Head and Neck**
  - Head

The nurse begins at the client's head by assessing his or her mental status and the symmetry and function of craniofacial structures (eyes, ears, nose, mouth).

- **Mental Status Assessment**
- **A technique for determining the level of a client's cognitive functioning) provides information about a client's attention, concentration, memory, and ability to think abstractly.**

DATA COLLECTION Mental Assessment

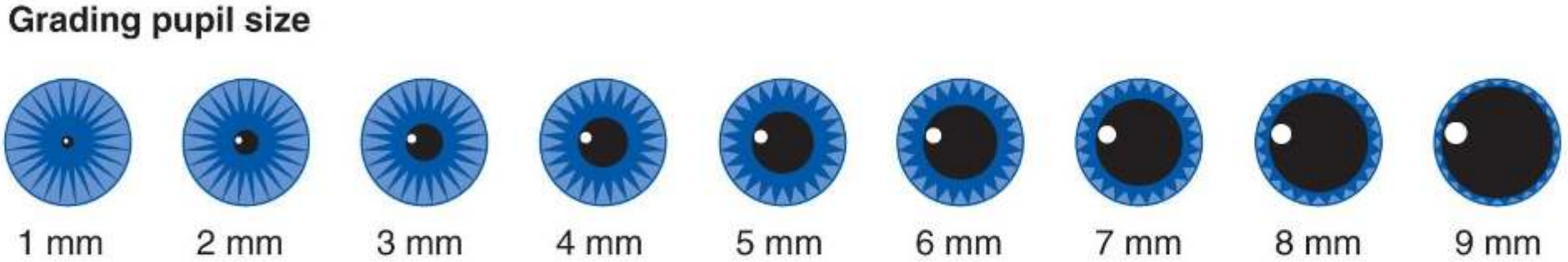
- More objective assessment data are important, however, when caring for the following clients:

- Clients who were previously unconscious
- Clients who were recently resuscitated
- Clients with periods of confusion
- Clients with a head injury
- Clients who took an overdose of drugs
- Clients with a history of chronic alcoholism
- Clients with a psychiatric diagnosis

DATA COLLECTION eyes

- Are both eyes similar
- Iris color
- Sclera color
- Use ophthalmoscope

- visual acuity,
- pupil size and response
- Use PERRLA: pupils equally round and react to light and accommodation







A gross examination of the visual field



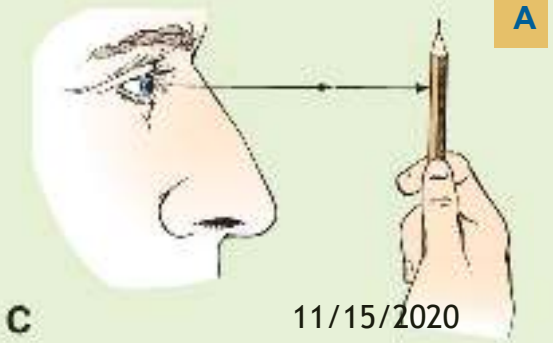
Ophthalmoscope

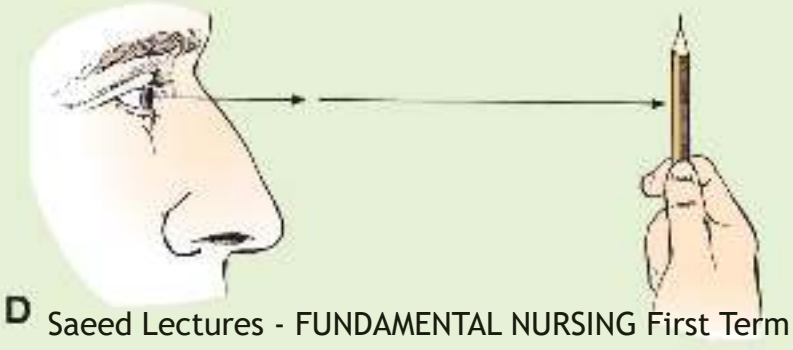


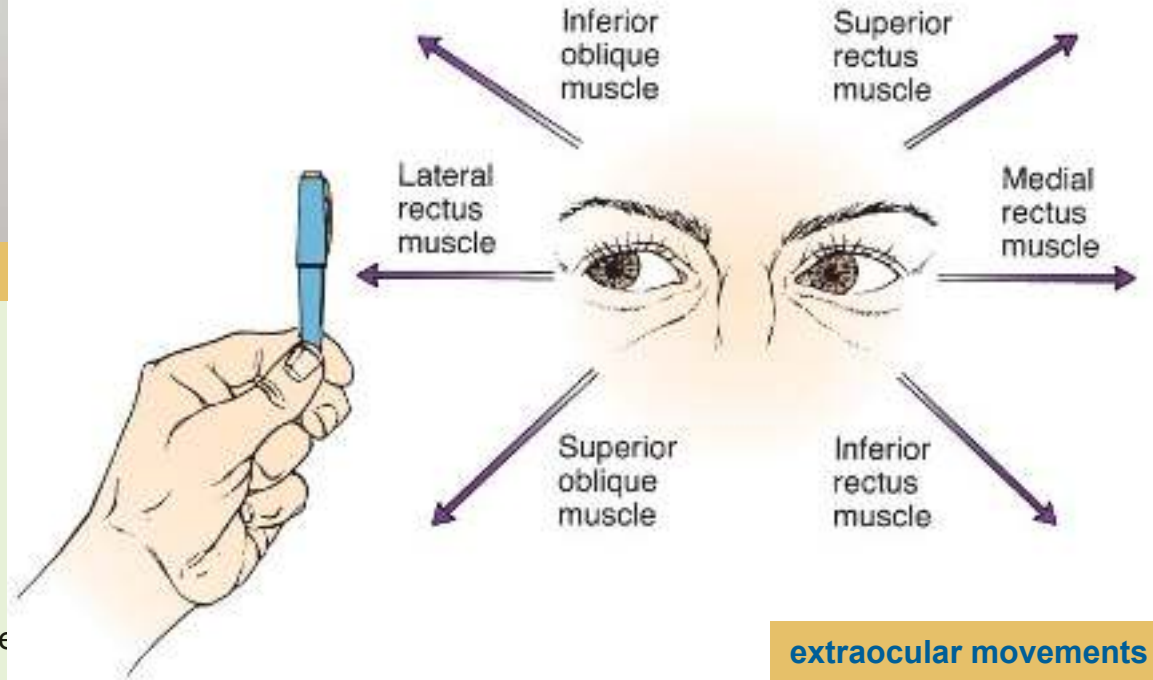
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L
A



Snellen eye chart





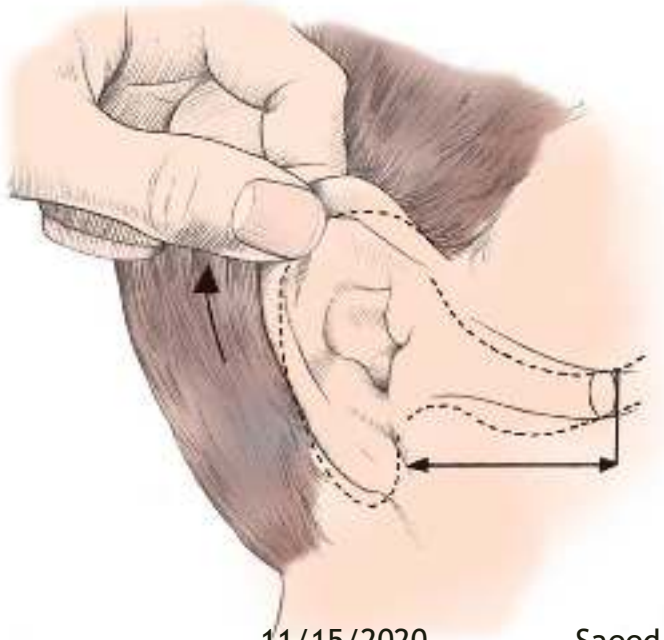


extraocular movements

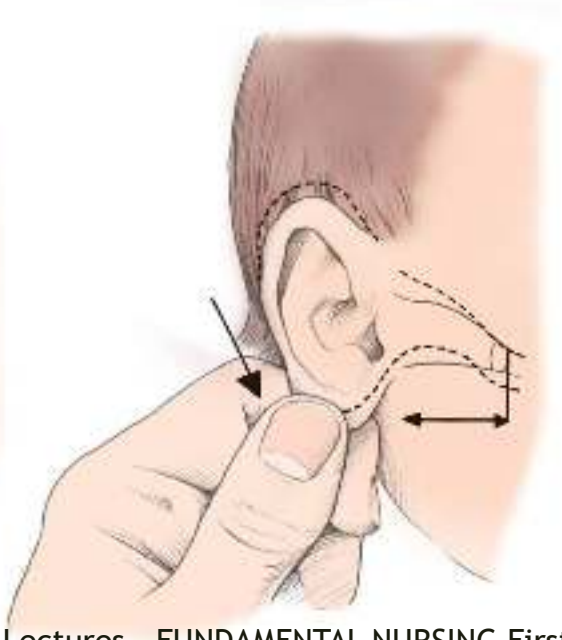
DATA COLLECTION Ears

- The nurse inspects and palpates the external ears.
- Use an otoscope, to examine the tympanic membrane, or eardrum.

- Hearing acuity
- Chick cerumen & color
- Refer client if there is any hearing problem to th audiologist

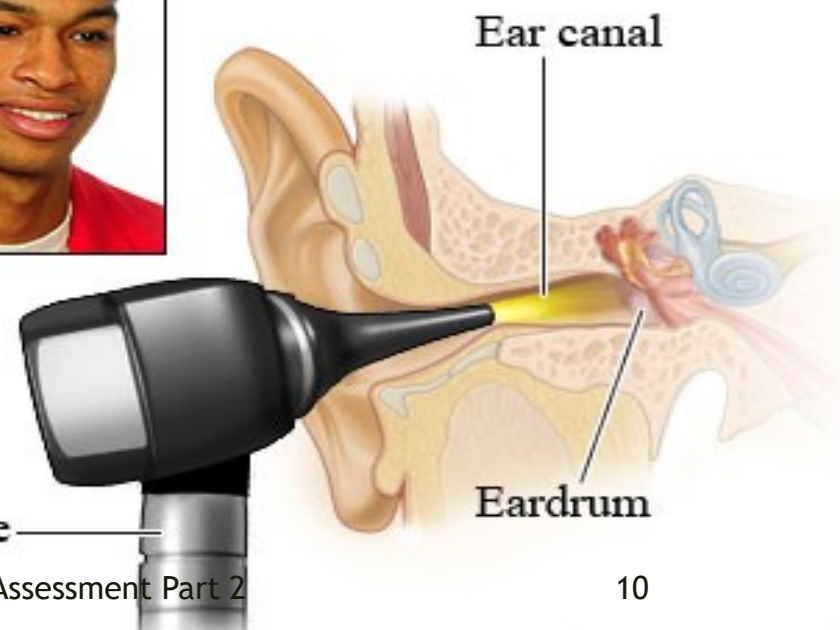


A



B

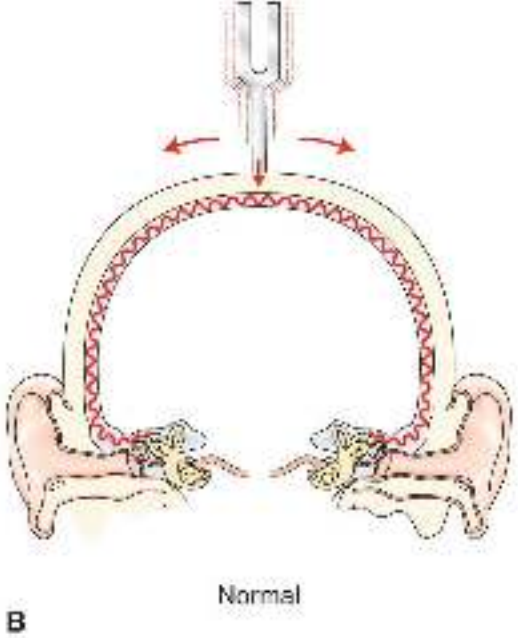






Voice/Whisper Test for Hearing Acuity



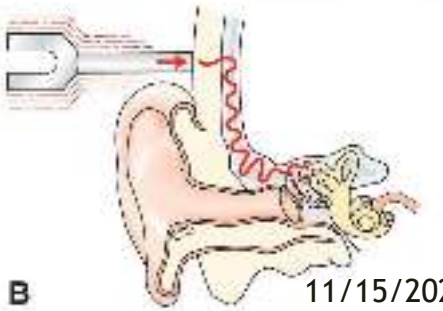


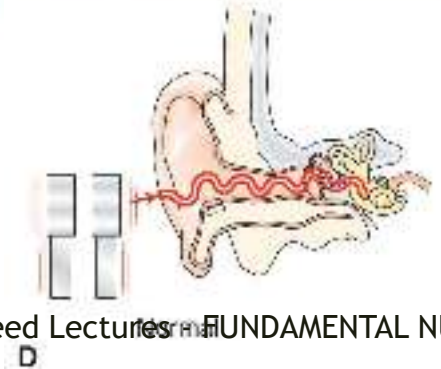
The Weber test assesses sound conducted through bone.

The Weber test assesses sound conducted through bone.









For the Rinne test, the tuning fork base is placed first on the mastoid process.

- (**A, B**) to test bone conduction of sound, after which the prongs are moved to the front of the external auditory canal.
- (**C, D**) to test air conduction of sound.

DATA COLLECTION Nose

- The nurse inspects the nose and nasal passages by having the client assume a “sniffing” position.
- The nurse documents a deviated septum, lesions, growths, flaring of the nostrils, or unusual drainage.
- Smelling acuity if impairment is suspected.
- To test smelling acuity:
- 1. Have the client occlude one nostril and close his or her eyes.
- 2. Place substances with strong odors, such as lemon, vanilla extract, coffee, peppermint, or alcohol, one at a time beneath the patient’s (open) nostril.



DATA COLLECTION Mouth and Oral Mucous Membranes

- Bu using torch and tongue depressor
- The nurse inspects, the lips, mouth, tongue and teeth.
- having the client open the mouth widely. protruding tongue is normally in midline.

- The nurse documents any dentures,
- missing or malpositioned teeth.
- Some unusual breath odors.
- Normal oral mucous membranes are pink, intact, and moist





DATA COLLECTION Facial Skin

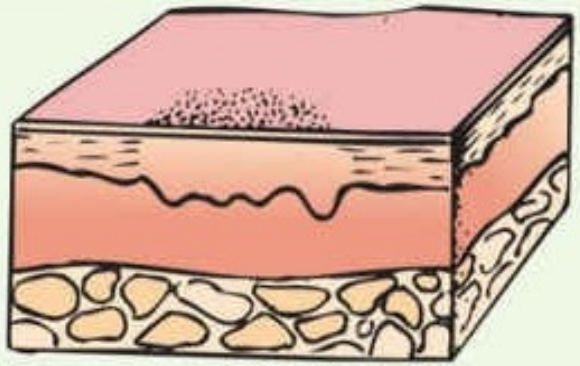
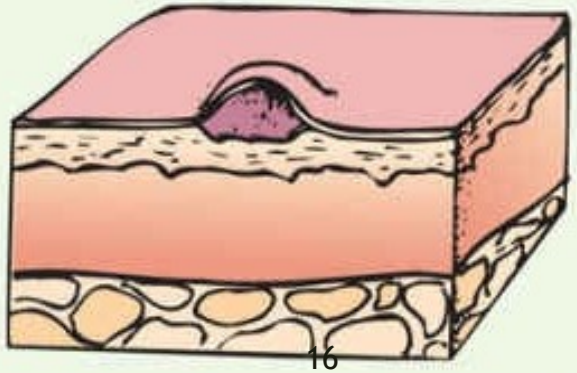
- The nurse notes characteristics of the facial skin while assessing the head.
- While examining the skin, the nurse may detect one or more alterations in its
- integrity:

- A wound
- An ulcer
- An abrasion
- A laceration
- A fissure, a crack in skin
- A scar is a mark left by the healing of a wound or lesion.

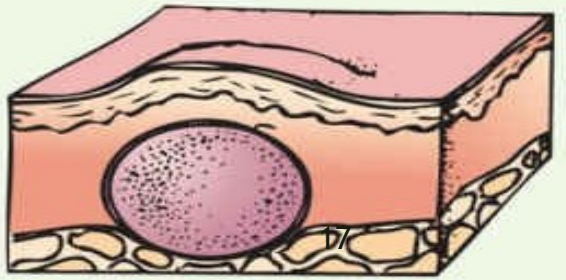
Common skin color variation

COLOR	TERM	Description
Pale	Pallor	Anemia, blood loss
Red	Erythema	Superficial burns, local inflammation
Pink	Flushed	Fever
Purple	Ecchymosis	Trauma to soft tissue
Blue	Cyanosis	Hypo oxygenation
Yellow	Jaundice	Liver disease, destruction of RBCs
Brown	Tan	Pregnancy, kidney disease

Common skin lesions

TYPE OF LESION	DESCRIPTION	EXAMPLE	ILLUSTRATION
Macule	Flat, round, colored, nonpalpable area	Freckles	
Papule	Elevated, palpable, solid	Wart	
Vesicle	Elevated, round, filled with serum	Blister	

Common skin lesions cont.

TYPE OF LESION	DESCRIPTION	EXAMPLE	ILLUSTRATION
Wheal	Elevated, irregular border, no free fluid	Hives	
Pustule	Elevated, raised border, filled with pus	Boil	
Nodule	Elevated, solid mass, deeper and firmer than papule	Enlarged lymph node	
Cyst	Encapsulated, round, fluid-filled or solid mass beneath the skin	Tissue growth	

DATA COLLECTION Hair Scalp

- Hair
- Assessment of the hair includes scalp hair, eyebrows, and eyelashes.
- The nurse notes the color, texture, and distribution (presence or absence in unusual locations for gender or age).

- Scalp
- The nurse assesses the scalp by randomly separating the hair and inspecting the skin surface. He or she looks for signs that the scalp is smooth, intact, and free of lesions.
- The nurse also palpates the skull for any unusual contour.

DATA COLLECTION Neck

- Assess the client ability to bend the head forward, backward, and to either side, as well as to rotate it 180 degrees.
- The trachea (windpipe) should be in the center of the neck.
- Pulsations in the carotid arteries are visible and easy to palpate.

- There should be no unusual bulges or fullness in the neck.
- Some nurses lightly palpate the lymph nodes in the neck area or assess anteriorly for an enlarged thyroid gland.



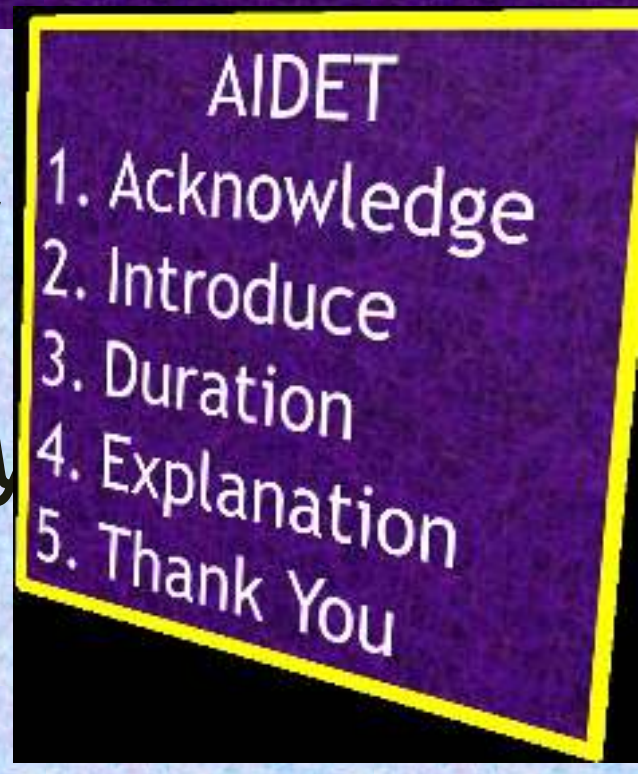

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Things to consider...

- *Always perform hand hygiene and wear gloves!*
- *Position of the patient - make sure they are comfortable!*
- *Maintain Privacy*
- *Explain the process (USE AIDET!)*
- *Be as organized and systematic as possible*



AIDET – Five Fundamentals of Communication

A	Acknowledge
I	Introduce
D	Duration
E	Explanation
T	Thank You

Decreased Anxiety

Increased Compliance

+

Improved clinical outcomes and increased patient satisfaction


Thank
You
Merci
Obrigado
Asante
Spasibo
Maake
Grazie
Kiitos
Multumesc
Matondo
Arigato
Vinaka
Dank Je
Juspaxar
Mamana
Nirringrazzjak
Terma Kasih
Welalin
Cam on ban
Kia Ora
Chokrane
Danken
Salamat
Ua Tsaug Rau Koj
Mochchakkeram
Rabh Mabh Agas
Mabur Nuwun
God Bless You
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Physical Assessment

Part III

Things to consider...

Inspection

Auscultation

Percussion

Palpation

DATA COLLECTION Chest Shape and Movement

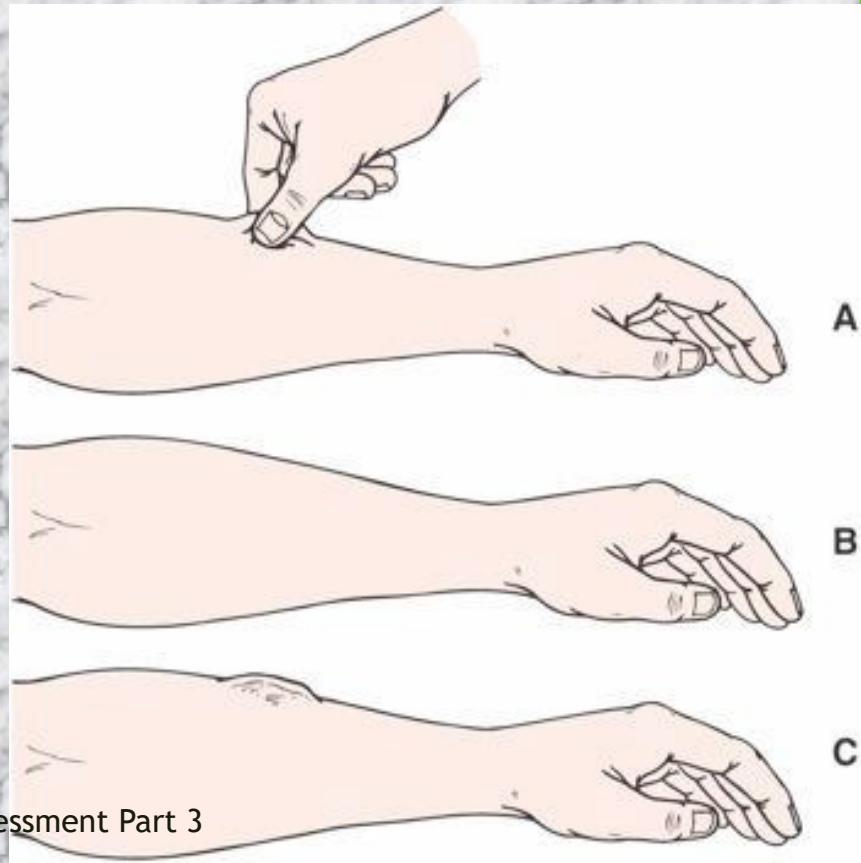
- Assess chest's shape and movement with breathing
- Notes the curved appearance of the spine
- Assesses skin turgor
- Breasts, heart sounds, and lung sounds.

- Skin turgor, the elastic quality of the skin and the pressure exerted on it by the fluid within.
- When skin is pinched will return to normal status

A pinched

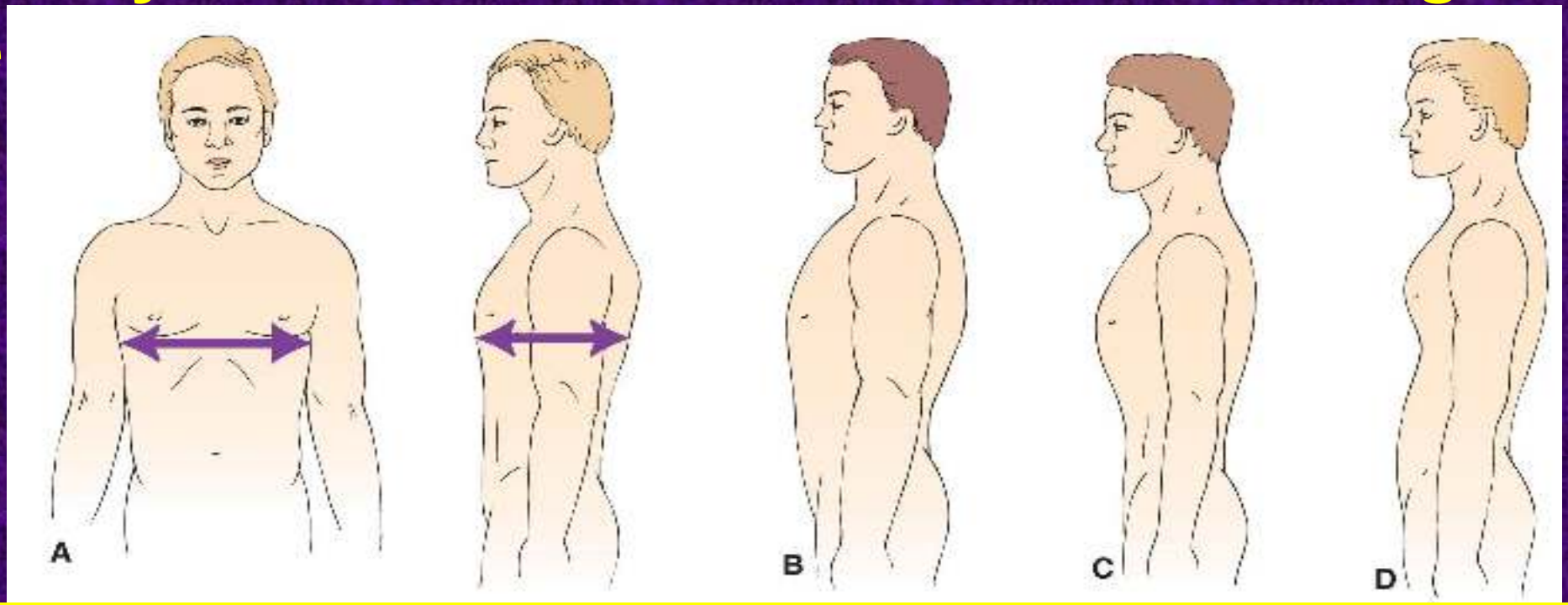
B Normal

C Pooe turgor



DATA COLLECTION Chest Shape and Movement

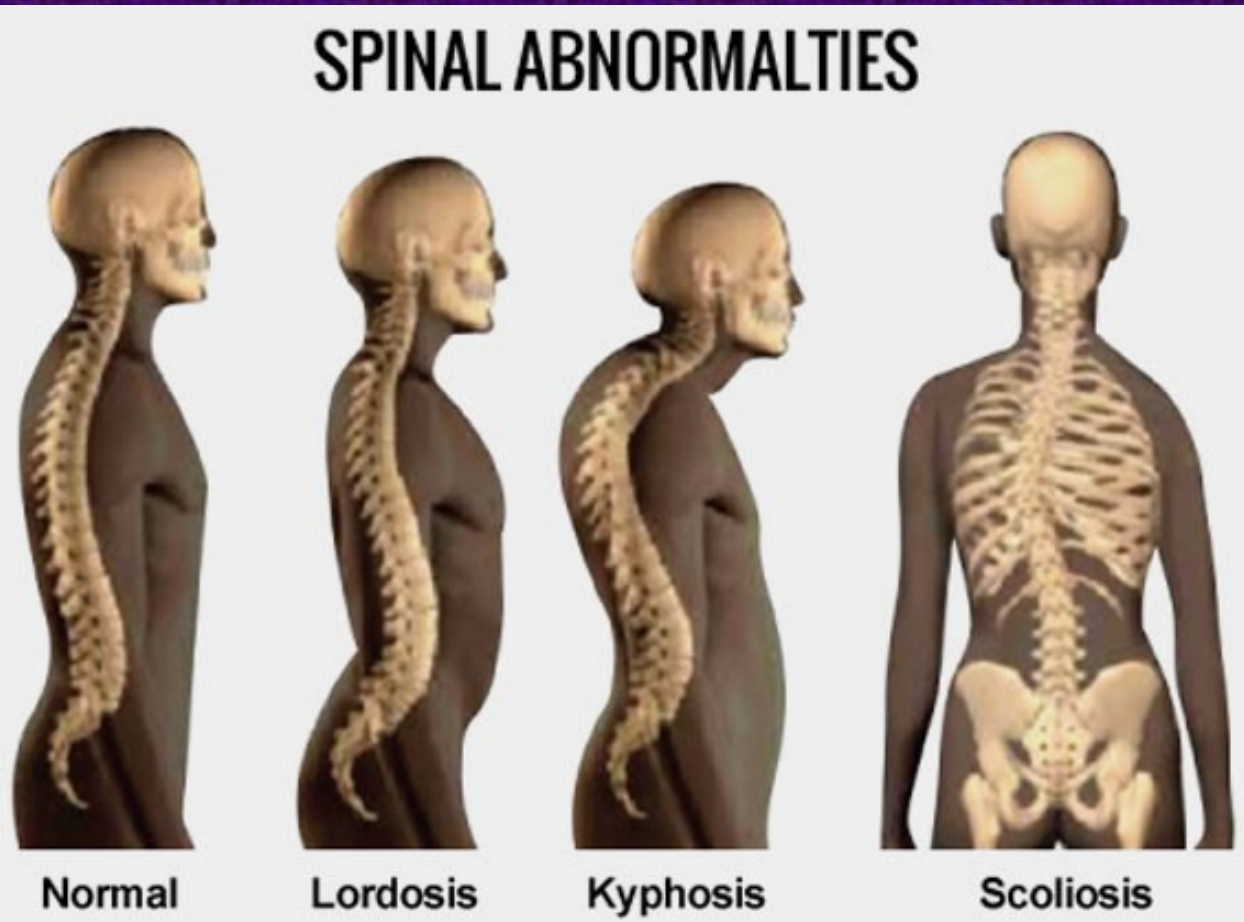
- In normal adults, the lateral dimension of the chest is approximately twice the anterior - posterior dimension.
- Various musculoskeletal abnormalities, cardiac or respiratory diseases, or trauma can cause changes in shape



A. Normal chest. **B.** Barrel chest. **C.** Pigeon chest. **D.** Funnel chest.

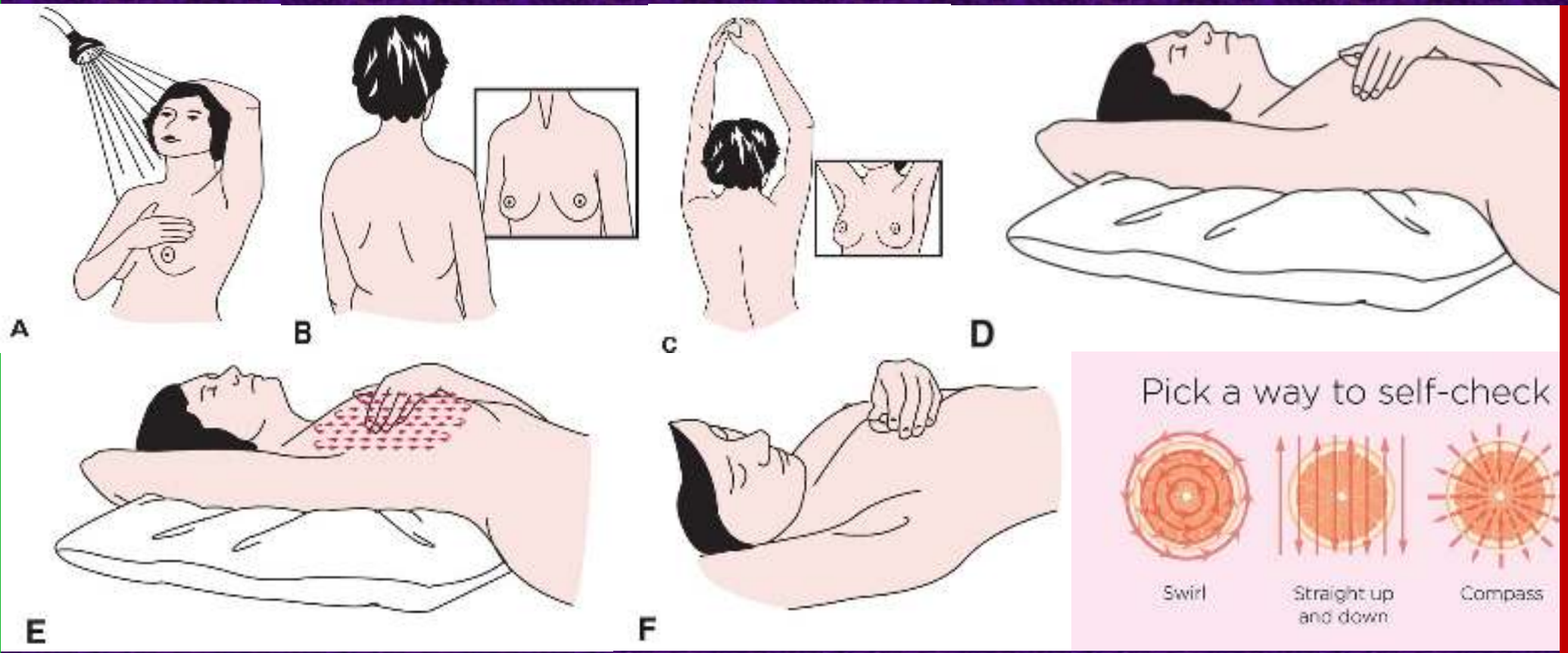
DATA COLLECTION Spine

- The spine, or vertebral column, appears in midline with gentle concave and convex curves when viewed from the side.



DATA COLLECTION Breasts

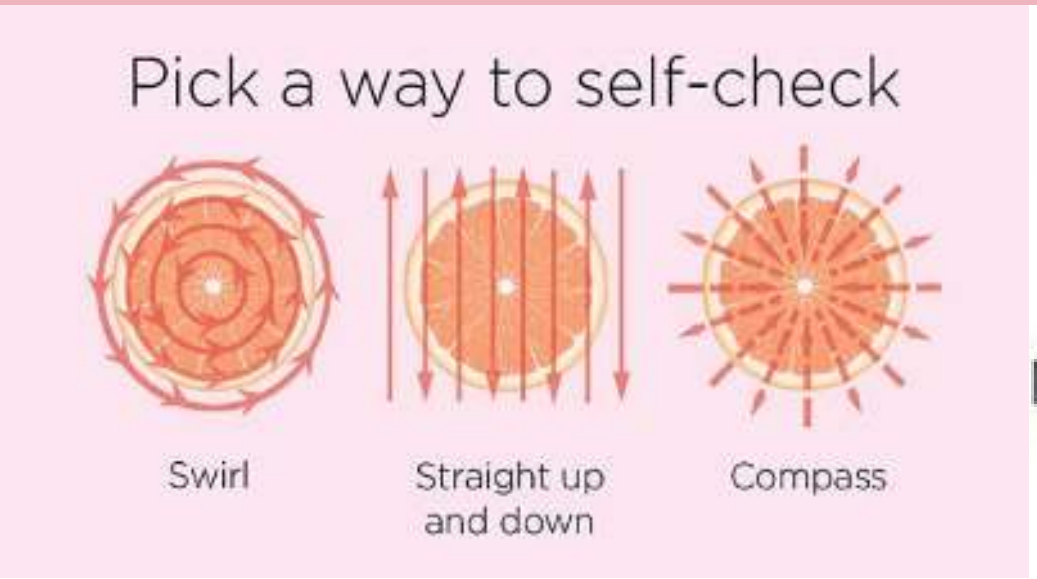
- Although breast abnormalities such as tumors can occur in men, they are more common in women.
- Teaching breast self-examination (BSE)



REPORT IF FIND

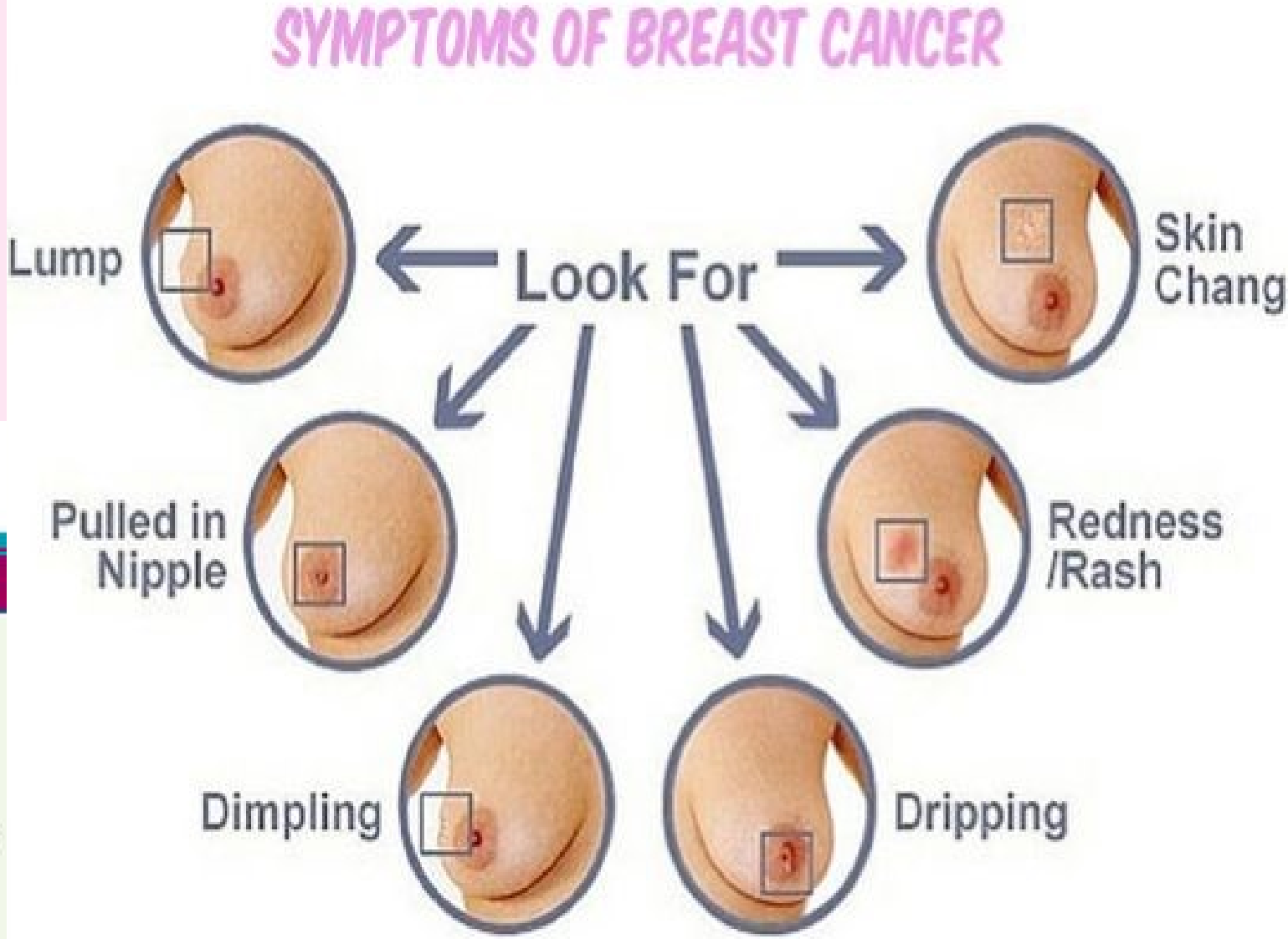
1. Lump
2. Nipple retract
3. Dripping
4. Dimpling
5. Redness Rash
6. Skin change

DATA COLLECTION Breasts cont.

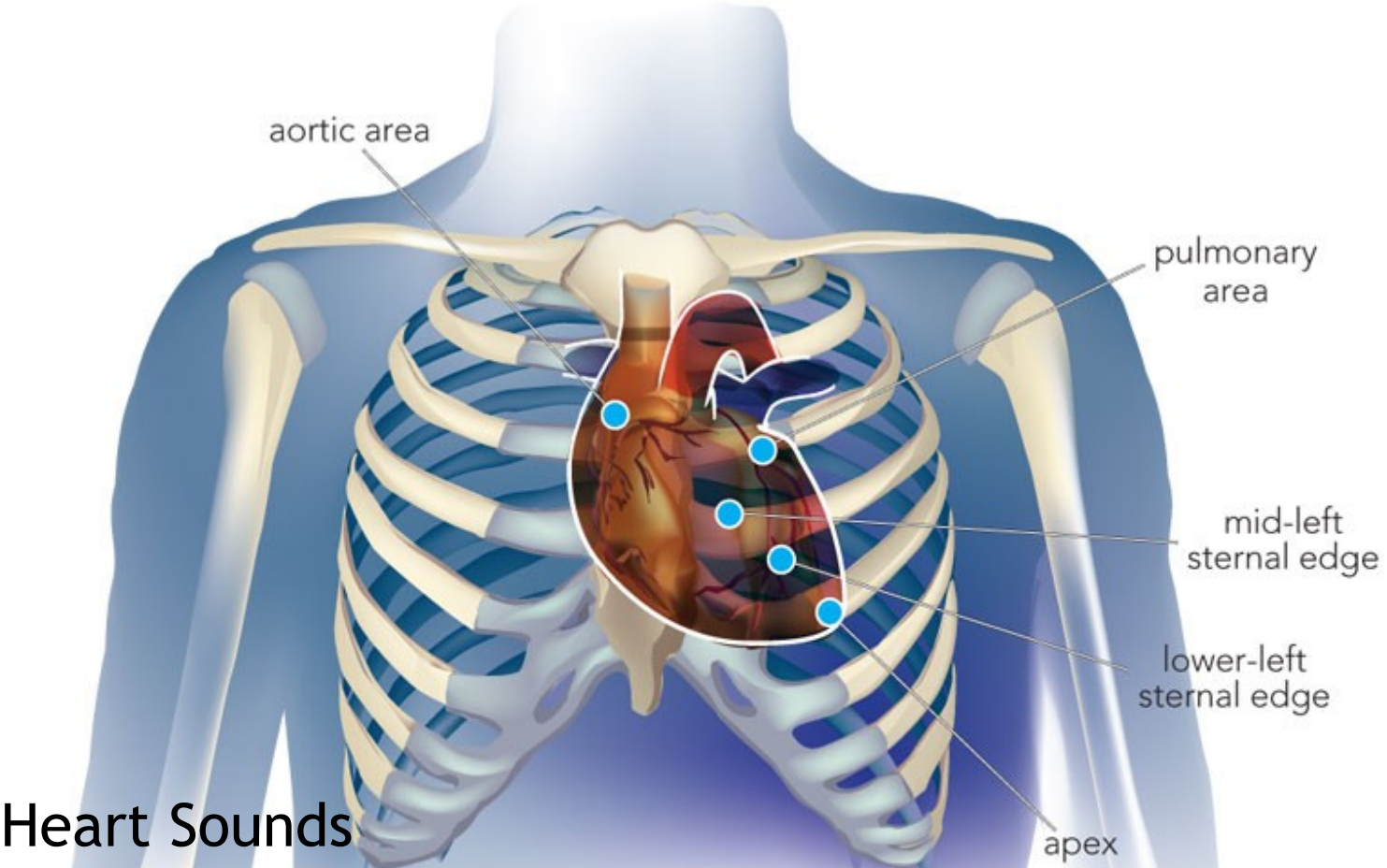


Breast Examination Guidelines

TECHNIQUE	AGE	FREQUENCY
Self-examination (optional)	≥20 years	Once per month
Clinical examination by a nurse or physician	20–39 years	Every 3 years
	≥40 years	Every year
Mammography	40 years	First examination and yearly thereafter
Clinical examination, mammography, magnetic resonance imaging (MRI)	Any age, high risk	Yearly

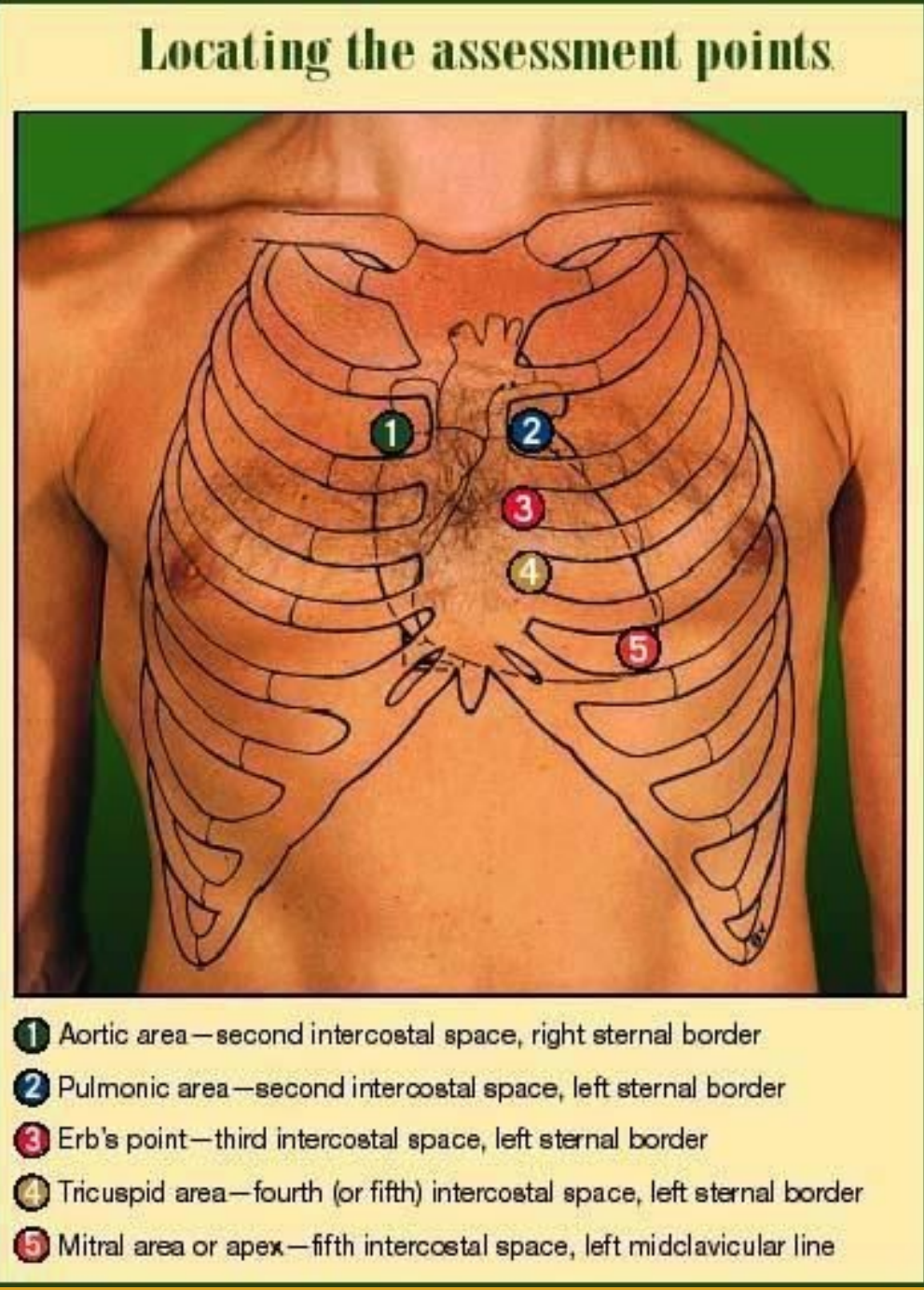


DATA COLLECTION Heart sounds



Heart Sounds

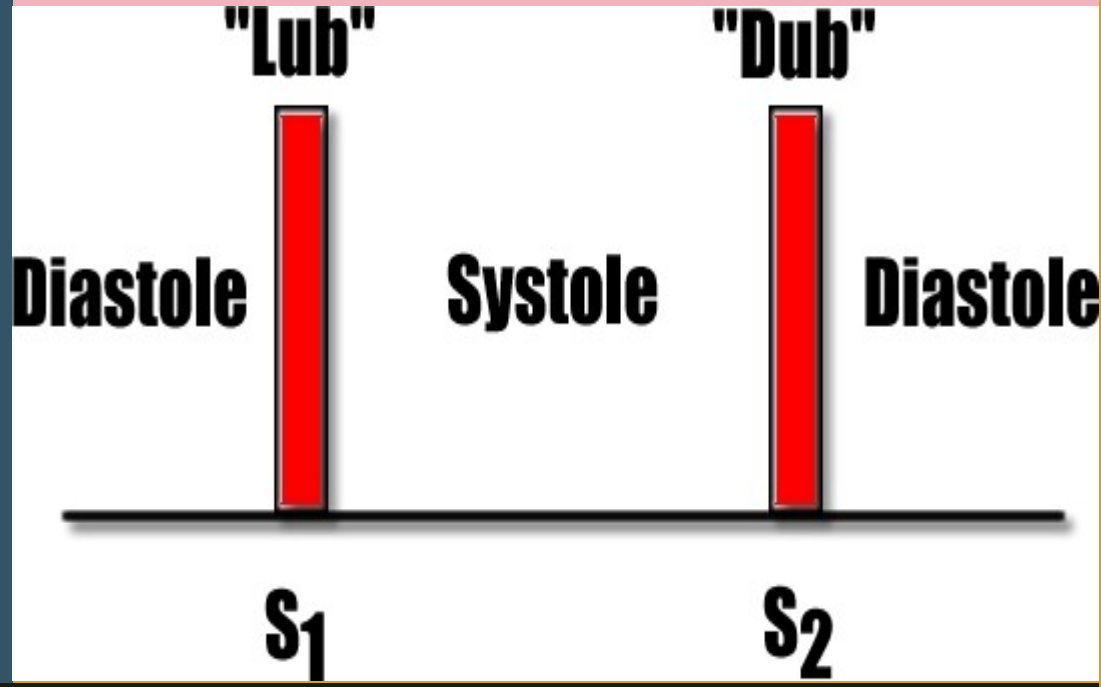
When assessing the anterior chest, the nurse listens to the heart sounds, which presumably are caused by the closing of the atrial and ventricular valves.



DATA COLLECTION Heart sounds

Normal Heart Sounds

- The two normal heart sounds are S1 and S2.
- The first heart sound, S1correlates with the “lub” sound and is louder at the apex or mitral area when using the diaphragm of a stethoscope.
- S2 or the “dub” sound, can be heard in the mitral area, it is louder over the



Abnormal Heart Sounds

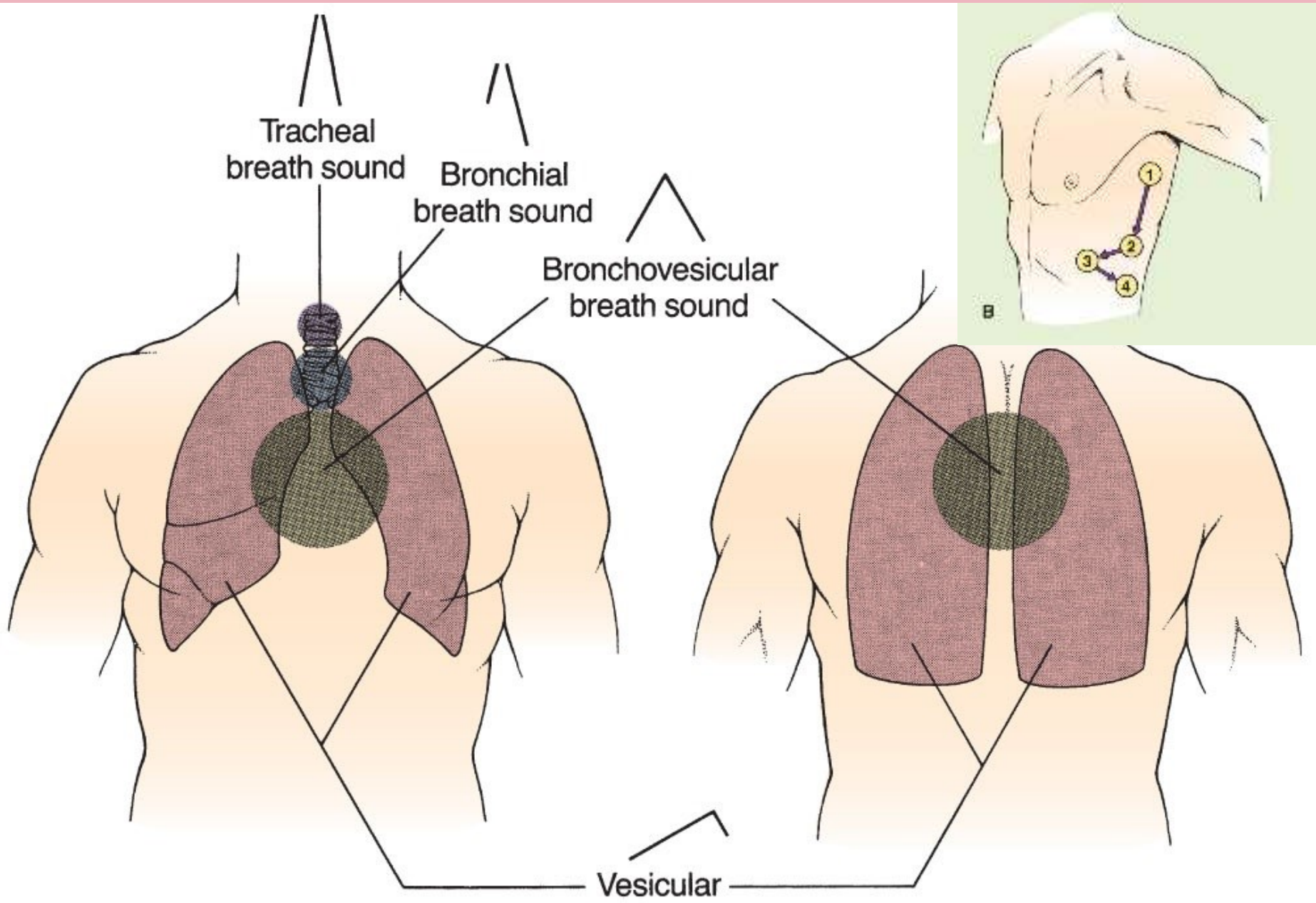
- The nurse may hear two additional sounds, S3 and S4, when auscultating the chest.
- S3, which is normal in children but abnormal in most adults, appears after S2. S3 sounds like “lub-dub-dub”.

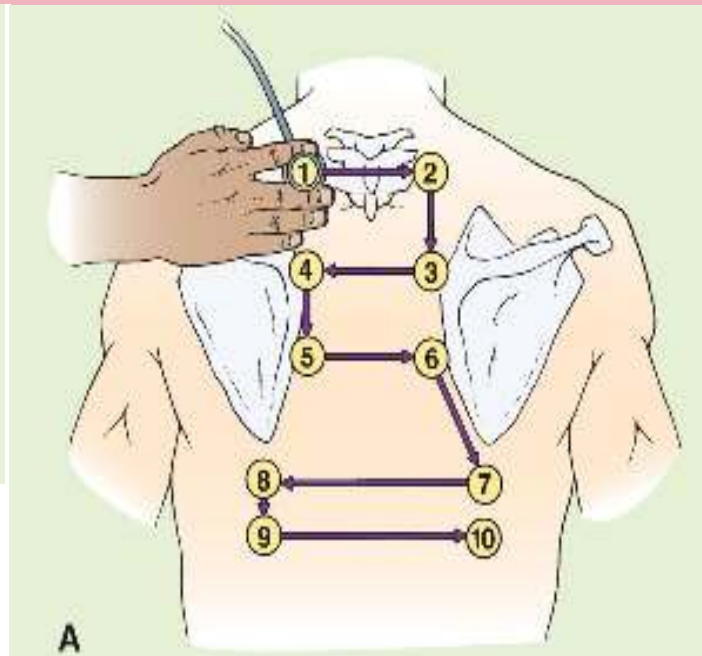
DATA COLLECTION Lung sounds

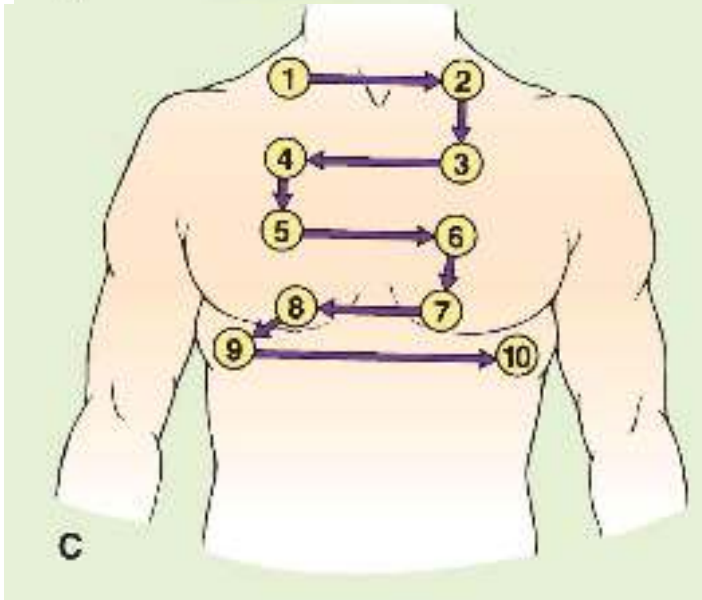
Listening to the lungs is another skill that requires frequent and repeated practice because some sounds are normal and others are abnormal.
Normal lung sounds occur when air moves in and out of passageways.
There are four normal lung sounds:

1. Tracheal sounds are loud and coarse. Are equal in length during insp. and exp.
2. Bronchial sounds, heard over the upper sternum anteriorly and between the scapulae posteriorly, are harsh and loud. They are shorter on inspiration than expiration.
3. Bronchovesicular sounds are heard on either side of the central chest or back. These medium-range sounds that are equal in length during inspiration and expiration.
4. Vesicular sounds are located in the periphery of all the lung fields. Their soft, rustling quality is longer on inspiration than expiration, with

DATA COLLECTION Lung sounds sites







DATA COLLECTION Lung sounds sites

- *Crackles*, also called *rales*, are intermittent, high-pitched, popping, and heard in distant areas of the lungs, primarily during inspiration. They resemble the sound of crisped rice cereal when milk is added.
- *Gurgles*, also called *rhonchi*, are low-pitched, continuous, bubbling, and heard in larger airways. They are more prominent during expiration. Some describe gurgles as sounding like wet snoring. Gurgles may clear with deep breathing or coughing.
- *Wheezes* are whistling or squeaking sounds caused by air moving through a narrowed passage. Wheezes may be audible without a stethoscope.
- *Pleural friction rubs* are grating, leathery sounds caused by two

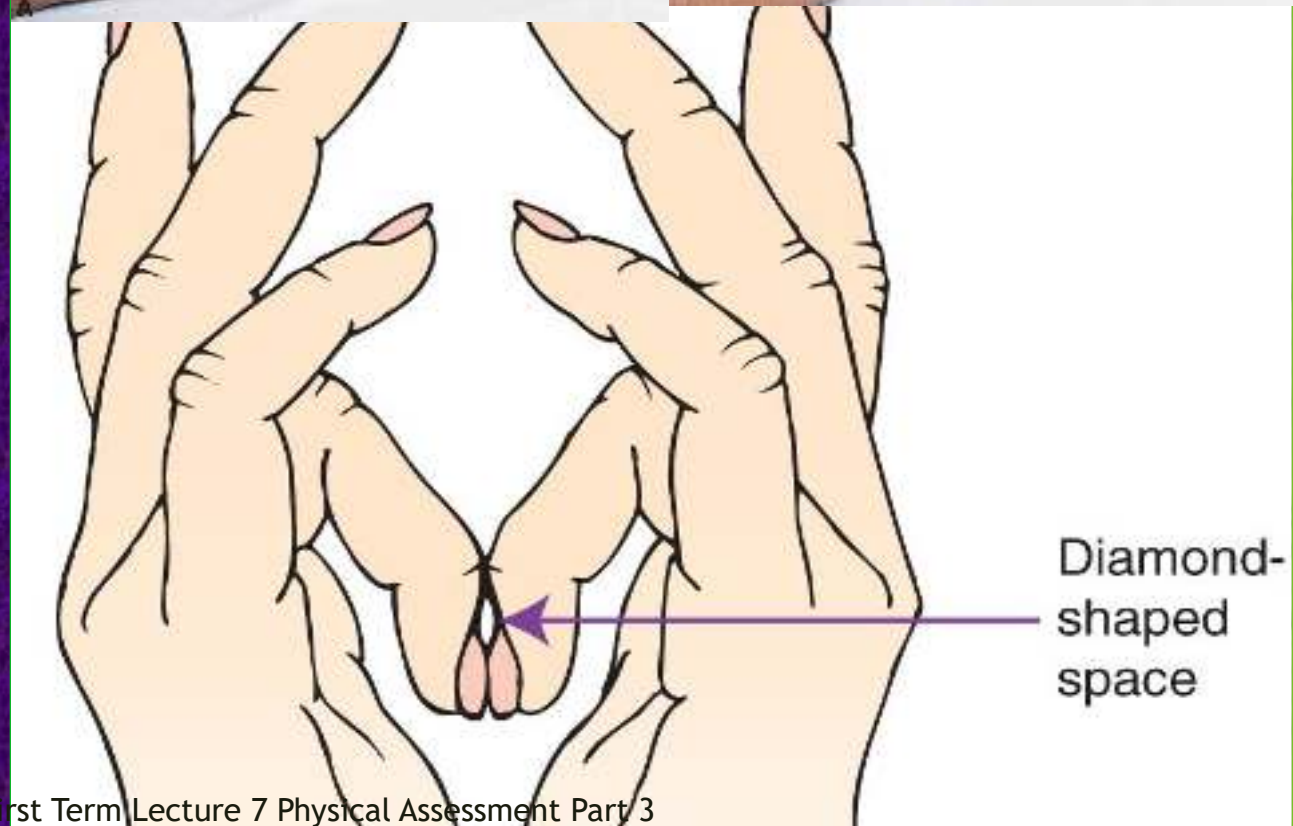
DATA COLLECTION Extremities

The nurse notes the:

- alignment, mobility, and strength of the extremities and
- compares their size.
- He or she feels the skin temperature,
- the characteristics of the nails,
- times the capillary refill, palpates local peripheral pulses,
- checks for edema,
- test the perception of skin sensations.
- assess deep tendon reflexes with a reflex hammer.







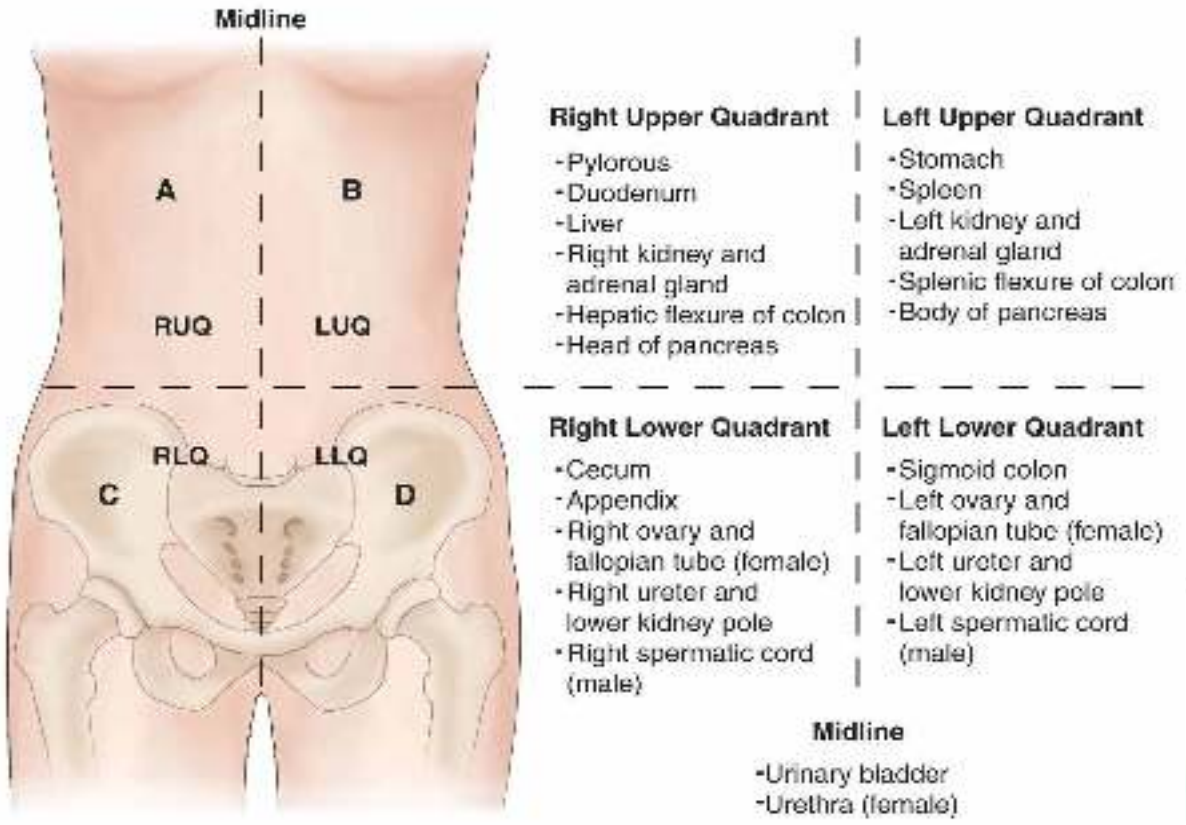
Criteria for Estimating Pitting Edema

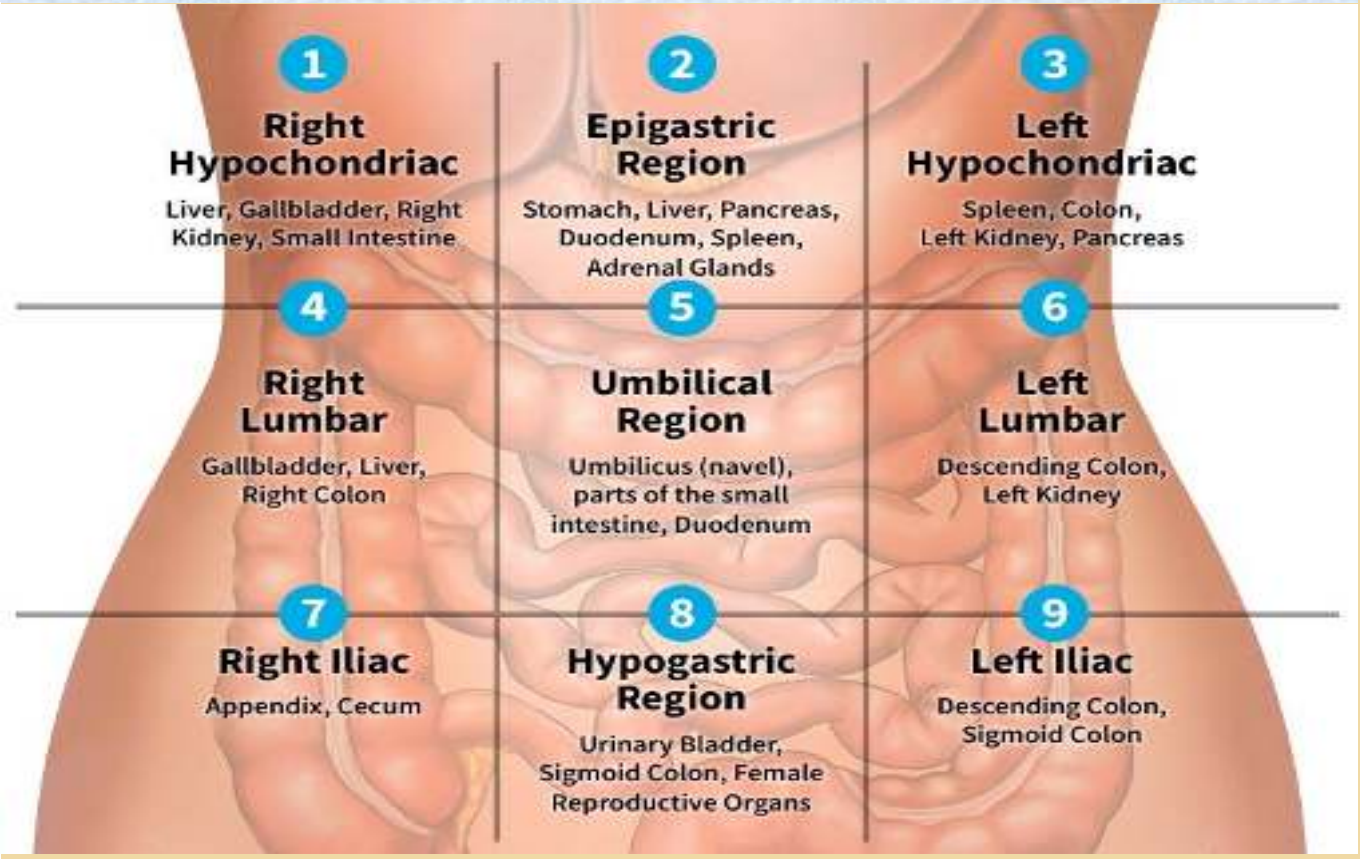
TYPE	DISCRIPTION	
Pitting Edema 1+	• Slight indentation (2 mm) • Normal contours • Associated with interstitial fluid volume 30% above normal 2+ Pitting	
Pitting Edema 2+	• Deeper pit after pressing (4 mm) • Lasts longer than 1+ • Fairly normal contour	

TYPE	DISCRIPTION	
Pitting Edema 3+	• Deep pit (6 mm) • Remains several seconds after pressing • kin swelling obvious by general inspection	
Pitting Edema 4+	• Deep pit (8 mm) • Remains for a prolonged time after pressing, possibly minutes • Frank swelling	
Brawny Edema 5+	• Fluid can no longer be displaced secondary to excessive interstitial fluid accumulation • No pitting • Tissue palpates as firm or hard • Skin surface shiny, warm, moist	

DATA COLLECTION Abdomen

- Most gastrointestinal and accessory digestive organs lie within the abdomen. The bladder, if distended, may rise into the abdomen. For assessment purposes, the abdomen is divided into four quadrants.
- *The abdomen is always inspected and then auscultated—in that sequence—before using palpation or percussion techniques*. Touching or manipulating the abdomen can alter bowel sounds, thus producing invalid findings.

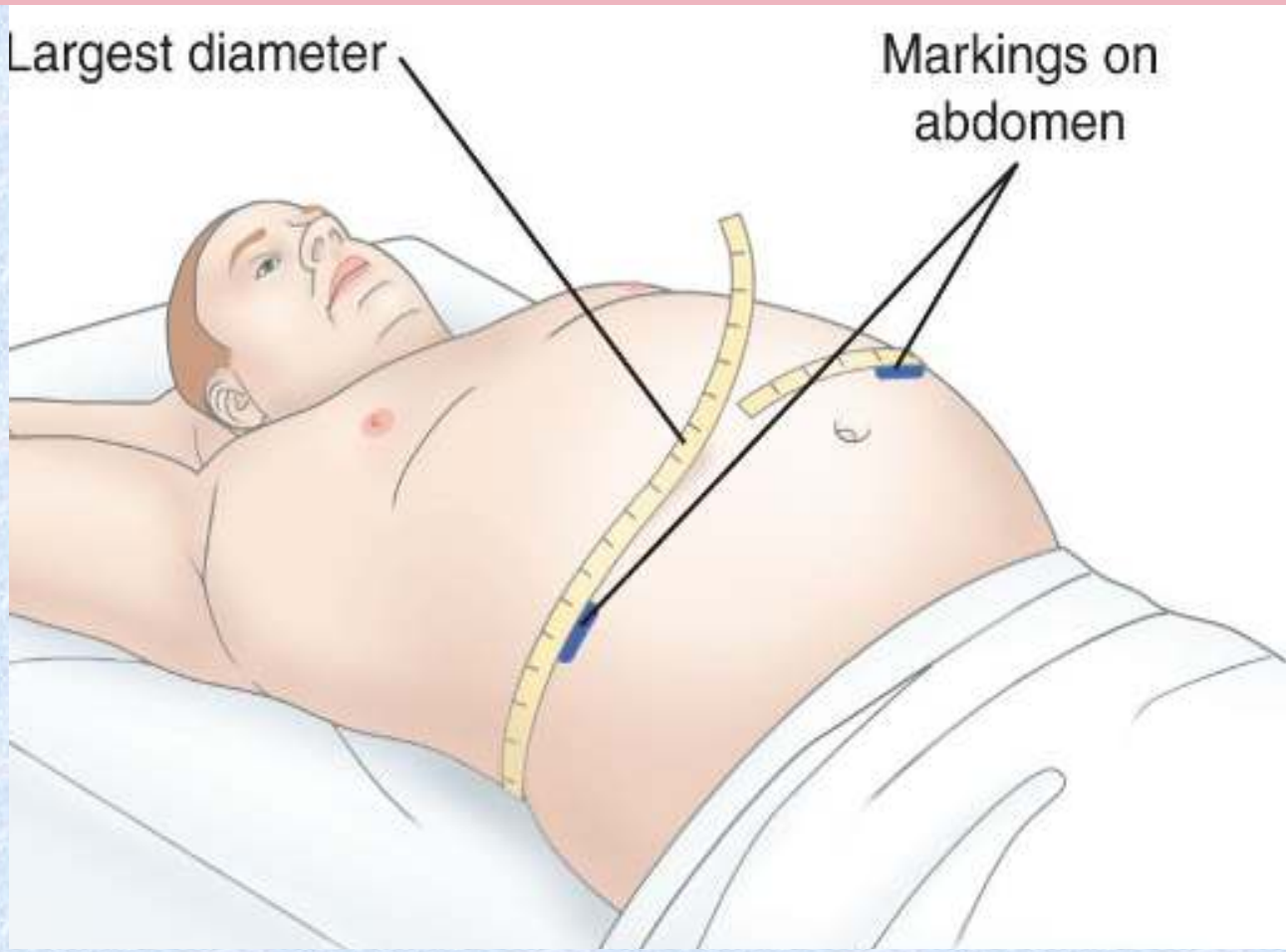




DATA COLLECTION Abdomen

Bowel Sounds

- Wave-like muscular contractions of the large and small intestines that move fluid

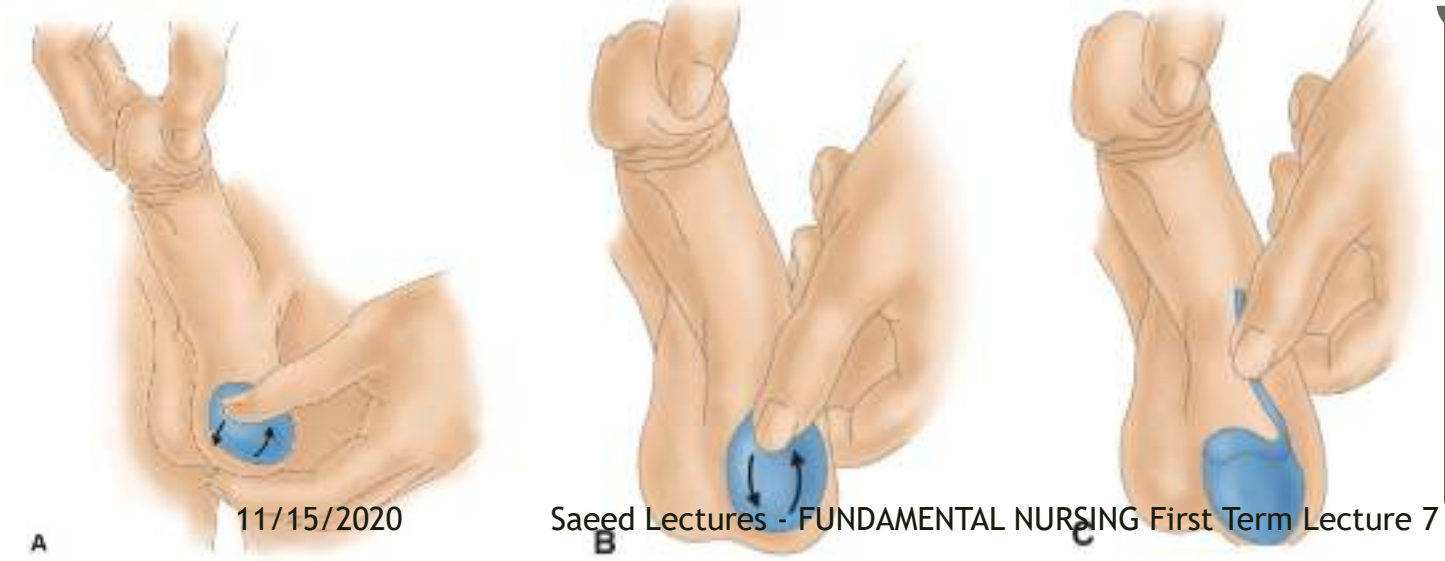


Abdominal Girth

- If the abdomen appears unusually large,
- the nurse checks its girth (circumference) daily by using a tape measure around the largest diameter.

DATA COLLECTION Genitalia

- During inspection, the nurse notes the condition of the skin and the distribution and characteristics of pubic hair (lice may infest pubic hair), as well as the presence of lesions.
- The nurse teaches the client as follows:
- Examine the testes monthly when they are warm and loosely within the scrotum (eg, during bathing).
- Elevate the penis with one hand, gently roll each testicle between the thumb and index finger.
- Feel each testicle horizontally, vertically and check for any unusual lumps.
- Continue palpation following the spermatic cord from the testicle to where it ascends into the abdomen.
- Report any unusual findings at once; an early diagnosis carries a better prognosis.



Remember post bond Genitalia and anus last:

1. To build trust with client
2. Assessment from clean to less clean

DATA COLLECTION Genitalia

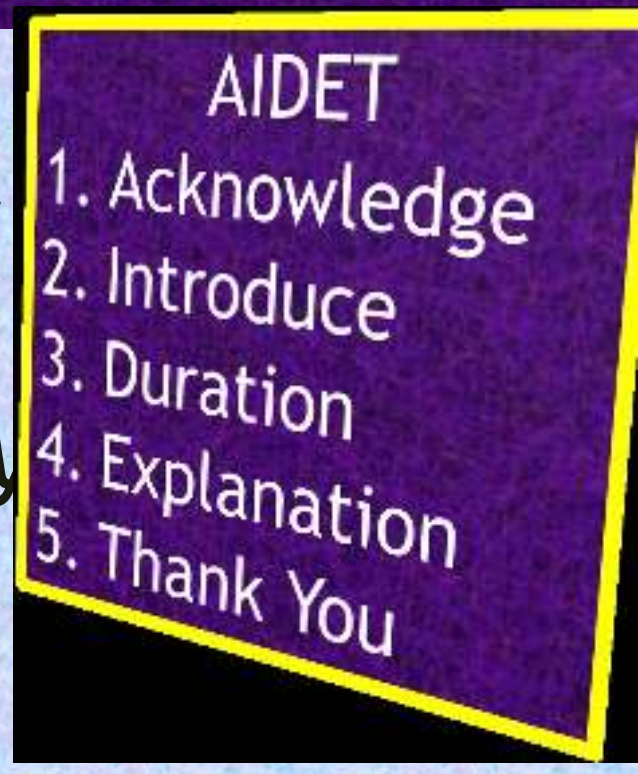
- Unless a client has specific symptoms, the nurse inspects only the anus. If touching is required, gloves are necessary.
- To examine the anus, the nurse positions the client on the side with the knees bent (Sims Position).
- Assess for: any unusual findings …
- External hemorrhoids (saccular protrusions filled with blood) may extend beyond the external sphincter muscle.
- Rectal fissures (cracks) if the client has a history of chronic constipation.

Remember, post bond Genitalia and anus test last:

1. To build trust with client
2. Assessment course from clean to less clean

Things to consider...

- *Always perform hand hygiene and wear gloves!*
- *Position of the patient - make sure they are comfortable!*
- *Maintain Privacy*
- *Explain the process (USE AIDET!)*
- *Be as organized and systematic as possible*



AIDET – Five Fundamentals of Communication

A	Acknowledge
I	Introduce
D	Duration
E	Explanation
T	Thank You

Decreased Anxiety

Increased Compliance

+

Improved clinical outcomes and increased patient satisfaction


Thank
You
Merci
Obrigado
Obrigado
Asante
Spasibo
Maake
Grazie
Kiitos
Multumesc
Vinaka
Arigato
Dank Je
Matondo
Juspaxar
Kia Ora
Mochchakkeram
Nirringrazzjak
Ua Tsaug Rau Koj
Terma Kasih
Welalin
Mamana
Danken
Chokrane
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Merci
Kia Ora
Juspaxar
Nirringrazzjak
Obrigado

الله يحفظك
God Bless You


DISEASE
SANITATION
OCTOBER 15
NAILS
PERSONAL
PREVENTION
HANDLING
CHEMICAL
WIPES
GELS
CARE
DISPENSER
HANDSOAP
METHODS
RINSE
BASIN
HEALTH
BACTERIUM
DISEASE
DANGER
IMPORTANT
WASHING
RUB
SANITIZER
ANTIBACTERIAL
SODIUM HYDROXIDE
SOAP
CONTROL
HEALTH
CLEAN
TOUCHING
FREQUENT
PUBLIC
STAY HEALTHY!
ANTIOXIDANT
FOAM
PALM
RUB
FINGERS
ALCOHOL
SOAP
CLEAN
SAVE LIVES
INFECTIONS
SANITIZERS
GELS
CLEANING
DIRTY
BACTERIA
TOWEL
FAUCET
INFECTION
HANDWASHING
CHEMICAL
HYGIENE
AIR DRY
PREVENT
PRESERVATION
FOAM
DANGER
ALCOHOL BASED
LATHER
WARNING
HANDWASHING
RUNNING WATER
CLEANER
PEOPLE
MICROBES
ANTISEPTIC






Shower
Towel
Foam
Cleanness
Bacteria
HYGIENE
Soap
SOAP
Broom
Mask
Washing









HYGIENE

Objectives

- By the end of this lecture the student will be able to:
  1. *Define hygiene.*
  2. *Name hygiene practices that most people perform regularly.*
  3. *Give reasons why a partial bath is more appropriate than a daily bath for older adults.*
  4. *Listing types of medical baths and uses.*
  5. *Name situations in which shaving with a safety razor is contraindicated.*
  6. *Name three items recommended for oral hygiene.*
  7. *Illustrate the procedure for oral hygiene to an unconscious client.*
  8. *Name two types of clients for whom nail care is provided with extreme caution.*

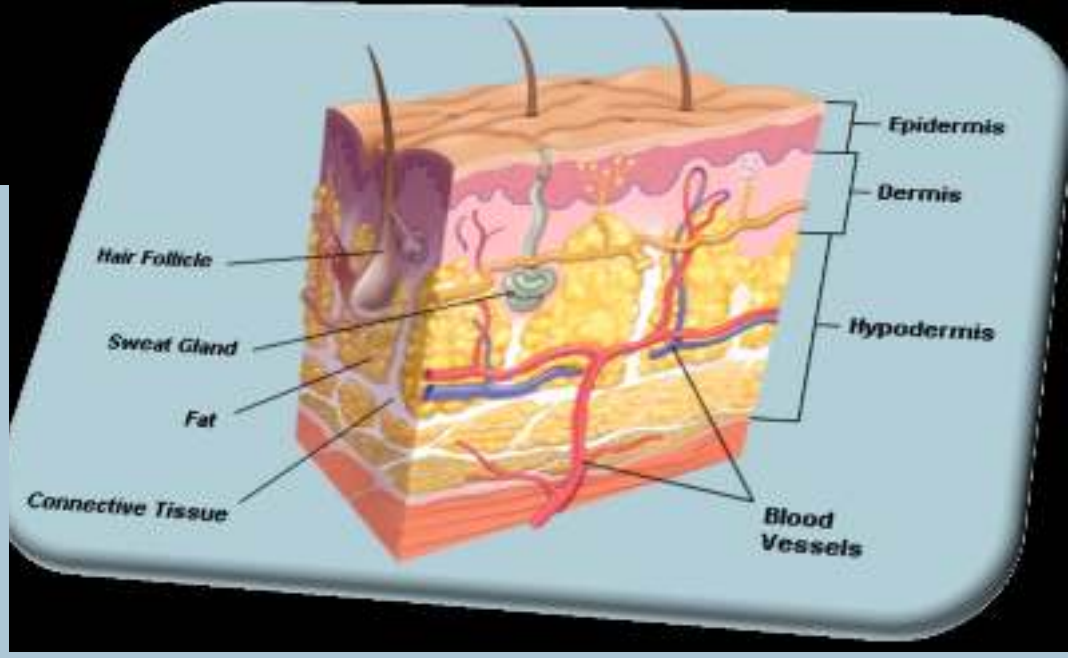
Hygiene

- Hygiene means those practices that help in promoting health through personal cleanliness.
  - *People foster hygiene through activities such as:*
    - bathing,
    - performing oral care,
    - cleaning and maintaining fingernails and toenails, and
    - shampooing and grooming hair.
  - *Hygiene also includes care and maintenance of devices such as eyeglasses and hearing aids to ensure continued and proper function.*

THE INTEGUMENTARY SYSTEM

- The word integument (covering) refers to the collective structures that cover the
- surface of the body and its openings.
- Most hygiene practices are based on maintaining or restoring a healthy integumentary system, which includes:
  1. *The skin,*
  2. *mucous membranes,*
  3. *hair, and*
  4. *nails.*

Skin



- The skin consists of the epidermis, dermis, and subcutaneous layer.
- The epidermis, or outermost layer, contains dead skin cells that form a tough protein called keratin.
- The cells in the epidermis are shed continuously

- Skin structures carry out the following functions:
  1. *Protect inner body structures from injury and infection*
  2. *Regulate body temperature*
  3. *Maintain fluid and chemical balance*
  4. *Provide sensory information such as pain, temperature, touch, and pressure*
  5. *Assist in converting precursors to vitamin D when exposed to sunlight*

Skin cont.

Types of Skin Glands and its Functions

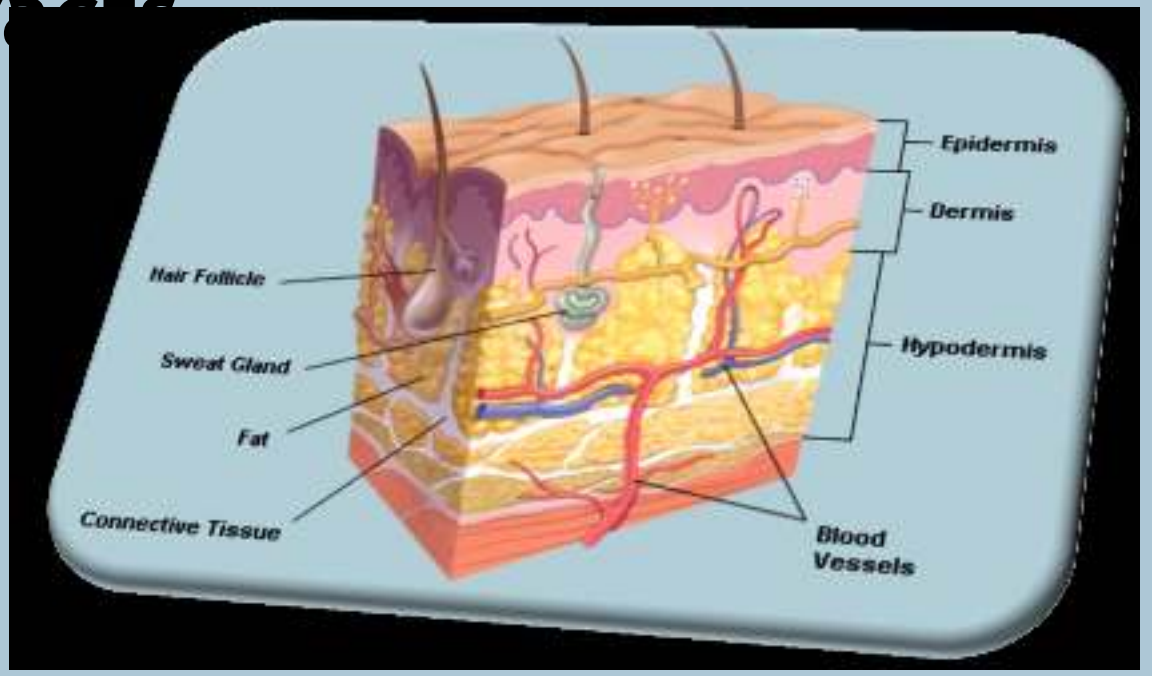
GLAND	LOCATION	SECRETION	PURPOSE
Sudoriferous	Throughout the dermis and subcutaneous layers, especially in the axilla and groin	Sweat	Regulate body temperatureExcrete body waste
Ceruminous	Ear canals	Cerumen	Perform protective functions; cerumen has antimicrobial properties
Sebaceous	Throughout the dermis	Sebum	Lubricate skin and hair
Ciliary	Eyelids	Sweat and sebum	Protect lid margin and lubricate eyelash follicles

Mucous Membranes

- The mucous membranes are continuous with the skin.
- They line body passages such as the digestive, respiratory, urinary, and reproductive systems.
- Mucous membranes also line the conjunctiva of the eye.
- Goblet cells in the mucous membranes secrete mucus, that keeps the membranes soft and moist.

Hair

- Each hair is a thread of keratin, a protein substance.
- Sebaceous glands in the hair follicles release sebum, an oily secretion that adds weight to the shafts of hair, causing them to flatten against the skull.
- Oily hair further attracts dust and debris.



Examples of Integumentary Disorders

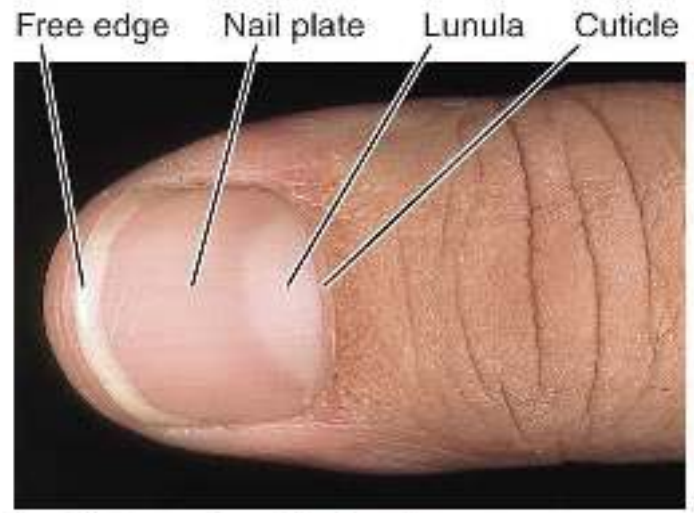
CONDITION	DESCRIPTION	CLIENT TEACHING
Acne	Inflammation of sebaceous glands and hair follicles on the face, upper chest, and back	Keep the face clean Refrain from touching or squeezing lesions Avoid the use of oily cosmetics
Contact dermatitis	Allergic sensitivity evidenced by red skin rash and itching	Avoid scratching or wearing clothing made of irritating fibers, such as wool Use tepid water and hypoallergenic or glycerin soap when bathing Pat the skin dry; do not rub
Furuncle (boil)	Raised pustule, usually in the neck, axillary, or groin area that feels hard and painful	Keep hands away from the infected lesion Use separate face cloth and towels from the rest of the family; launder personal bath items in hot water and bleach Wash hands thoroughly before and after applying medication to the skin
Psoriasis	Noninfectious chronic skin disorder that appears as elevated silvery scales that shed over elbows, knees, trunk, and scalp; acute episodes occur between periods of relief	Follow medical regimen, which may be lifelong Be wary of advertised remedies that promise a cure or quick relief, because they rarely do
Pediculosis (lice infestation)	Brown crawling insects that move over the scalp and skin and deposit yellowish-white eggs on hair shafts including pubic area; skin bite causes itching	Inspect the skin carefully as adult lice move quickly from light Look for eggs (nits) on hairs ¼ in. to ½ in. from the scalp or skin surface Do not share clothing, combs, and brushes; lice are spread by direct contact Use a pediculicide (chemical that kills lice), in addition to a lice comb and manual removal Do not use hair conditioner: it coats the hair and protects the nits

Examples of Integumentary Disorders

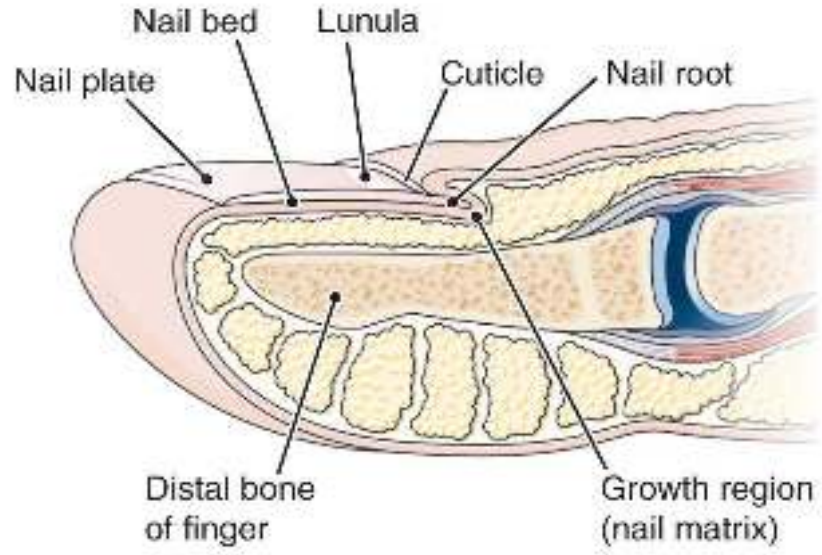
CONDITION	DESCRIPTION	CLIENT TEACHING
Scabies	Infestation with an itch mite that burrows within the webs and sides of fingers, around arms, axilla, waist, breast, lower buttocks, and genitalia	Bathe thoroughly in the morning and at night Apply prescribed medication after bathing Don clean clothes after bathing Avoid skin-to-skin contact with uninfected people
Tinea capitis, pedis, corporis, and cruris	Fungal infection in the scalp, feet, body, or groin that appears as a ring or cluster of papules or vesicles that cause itching, become scaly, cracked, and sore	Use separate bathing and grooming articles Keep body areas dry, especially in folds of skin Wear clothing that promotes evaporation of perspiration
Skin cancer	Newly pigmented growth or change in existing skin lesion, especially where skin is chronically exposed to sun	See a physician for examination and possible biopsy Avoid direct sun exposure between 10 AM and 4 PM Recommend using a sunscreen with an SPF ≥15 Wear a wide-brimmed hat on sunny days Do not use artificial tanning facilities
Fungal nail infection	Thick, yellowed, rough-appearing toenails or fingernails that can spread from one nail to others	Consult a physician about prescription drugs, which are approximately 50% effective Wear leather shoes, and alternate pairs to reduce damp shoe conditions Be aware that unsanitary utensils used in the application of artificial fingernails can spread the fungus Seek professional nail care from a podiatrist
Candidiasis	Yeast infection of the mouth or vagina; oral candidiasis appears as white patches or red spots on the tongue, gums, or throat; vaginal candidiasis appears as a thick, cottage cheese–like discharge that causes itching and burning	Follow directions for oral or topical antifungal medications Swish antifungal mouth rinses, retain the solution in the mouth as long as possible, and then swallow the rinse Avoid simple sugars and alcohol because they promote the growth of yeast Eat yogurt that contains live *Lactobacillus acidophilus* to restore a balance of helpful to harmful microbes

Nails

- *Fingernails and toenails are also made of keratin, which in concentrated amounts gives them their tough texture.*
- *Fingernails and toenails provide some protection to the digits.*
- *Normal nails are thin, pink, and smooth.*
- *Under nails considered reservoir of microbes, so must be cared well.*



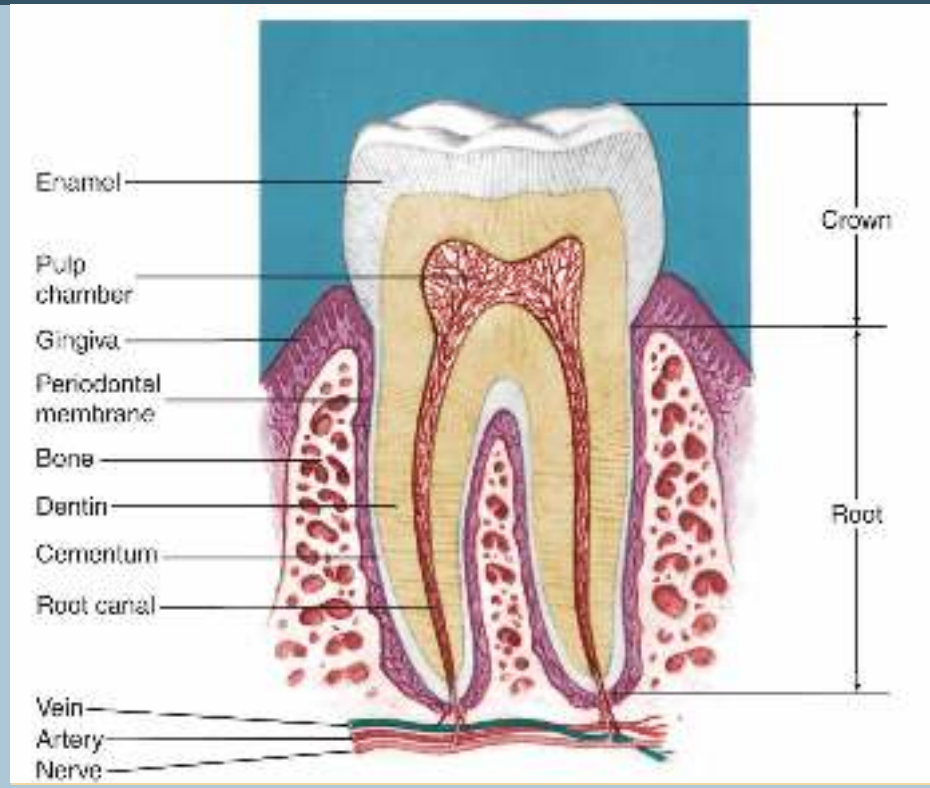
A External nail structures



B Internal and external nail structures

Teeth

- *Teeth, the enamel of which is a keratin structure, are present beneath the gums at birth.*
- *The exposed portion of each tooth is referred to as the crown; the portion within the gum is the root.*
- *Healthy teeth are firmly fixed within the gums.*



- *The combination of sugar, plaque, & bacteria may eventually erode the enamel, causing caries (cavities).*
- *Tartar (hardened plaque) is more difficult to remove and may lead to gingivitis (inflammation of the gums).*

HYGIENE PRACTICES

- *The integument contains many secretory glands that produce odors and attract debris, and the teeth are prone to decay if uncared for.*
- *Therefore, hygiene measures are beneficial for maintaining personal cleanliness and healthy integumentary structures.*
  - *Bathing*
  - Shaving
  - Oral Hygiene
  - *Hair care*
  - *Nail care*

Bathing

In addition to bathing for hygiene purposes, other types of bathing serve different functions. In general, however, most bathing is done in a tub or shower, at a sink, or at the bedside.

- ***Cleansing body***
- ***Eliminating body odor***
- ***Reducing the potential for infection***
- ***Stimulating circulation***
- ***Providing a refreshed and relaxed feeling***
- ***Improving self-image***

Bathing cont.

Bathing is a hygiene practice in which a person uses a cleansing agent such as soap and water to remove sweat, oil, dirt, and microorganisms from the skin, bathing has several benefits:

Therapeutic Baths

TYPE	DESCRIPTION	PURPOSE
Sitz bath	Immersion of the buttocks and perineum in a small basin of continuously circulating water	Removes blood, serum, stool, or urine Reduces local swelling Relieves discomfort
Sponge bath	Applications of tepid water to the skin	Reduces a fever
Medicated bath	Soaking or immersing in a mixture of water and another substance, such as baking soda (sodium bicarbonate), oatmeal, or cornstarch	Relieves itching or a rash
Whirlpool bath	Warm water that is continuously agitated within a tub or tank	Improves circulation Increases joint mobility Relieves discomfort Removes dead tissue

Bathing cont.

Baths differentiates as following:

- ***Complete baths;*** *If the safety risks are negligible and there are no contraindications, the nurse encourages clients to bathe independently in a tub or shower.*
  - *Tub bath*
  - *Shower*
  - *Complete bed bath*
- ***Partial bath;*** *A partial bath means washing only those body areas that are subject to greatest soiling or that are sources of body odor: generally, the face, hands, axillae, and perineal area. Partial bathing is done at a sink or with a basin at the bedside.*

Bathing cont.

Sometimes the perineum, the area around the genitals and rectum, requires special or frequent cleansing in addition to bathing.

It is also appropriate whenever male or female clients have bloody drainage, urine, or stool collected in this area. When providing perineal care, nurses must:

- *Prevent direct contact by*
  wearing clean gloves.
- *Cleanse so excretions are removed*
  from less soiled to more soiled areas.



These principles help prevent the transfer of infectious microorganisms to the nurse and to uncontaminated areas on or within the client

Bathing cont.

Bed Bath

Clients who cannot take a tub bath or cannot shower independently may be given any one of three types of baths:

- *a bed bath, a towel bath, or a bag bath.*
- *During a bed bath (washing with a basin of water at the bedside), the client may actively assist with some aspects of bathing.*

Bathing cont.

Bed Bath / Towel Bath

- A single large towel used to cover and wash a client.
- Towel 3 × 7.5 ft (1 × 2 m) with no basin or soap.
- Towel wetted in water heated to (40° to 43°C) and (30 mL) of no-rinse liquid cleanser.

Bed Bath / Bag bath

- A commercially packaged kit with 8 to 10 pre-moistened, disposable cloths in a plastic bag or container.
- *The nurse warms the container and its contents in a microwave.*



Shaving

- Shaving removes unwanted body hair.
- Shaving is accomplished with an electric or a safety razor.
- In some circumstances, use of a safety razor is contraindicated

Use of a safety razor is contraindicated for clients:

- *Receiving anticoagulants (drugs that interfere with clotting)*
- *Receiving thrombolytic agents (drugs that dissolve blood clots)*
- *Taking high doses of aspirin*
- *With blood disorders such as hemophilia*
- *With liver disease who have impaired clotting*
- *With rashes or elevated or inflamed skin lesions on or near the face*
- *Who are suicidal*

Shaving

- Prepare a basin of warm
- Wash skin with warm
- Lather the skin with soap or shaving cream
- Start at the upper areas of the face
- Pull the skin taut below the area to be shaved.
- Pull razor in the direction of hair growth







- Use short strokes.
- Rinse the razor after each stroke or as hair accumulates.
- Rinse remaining soap or shaving cream from the skin
- Apply direct pressure to areas that bleed
- Apply aftershave lotion

Oral Hygiene

- Oral hygiene consists of those practices used to clean the mouth, especially brushing and flossing the teeth.
- Dentures and bridges also require special cleaning and care.
- Oral Care for Unconscious Clients
- Unconscious clients are not salivating in response to seeing, smelling, and eating food.
- They need oral care even more frequently than conscious clients.
- Swords (dried crusts containing mucus, microorganisms, and epithelial cells shed from the mucous membrane) are common on the lips and teeth of unconscious clients.

Oral Hygiene / Denture Care

- Dentures (artificial teeth) substitute for a person's lower or upper set of teeth, or both.
- A bridge, a dental appliance that replaces one or several teeth is fixed permanently to other natural teeth so that it cannot be removed, or it is fastened with a clasp that allows it to be detached from the mouth.
- Dentists recommend that dentures and bridges remain in place except during cleaning.
- Keeping dentures and bridges out for long periods permits the gum lines to change, affecting the fit.

Oral Care for Conscious Clients











Dentures care





Oral Care for Unconscious Clients

- Tooth brushing is the preferred technique for providing oral hygiene to unconscious clients
- Oral Care for Unconscious Clients
- unconscious clients are not salivating in response to seeing, smelling, and eating food.
- They need oral care even more frequently than conscious clients.
- Sordes (dried crusts containing mucus, microorganisms, and epithelial cells shed from the mucous membrane) are common on the lips and teeth of unconscious clients.

Oral Care for Unconscious Clients

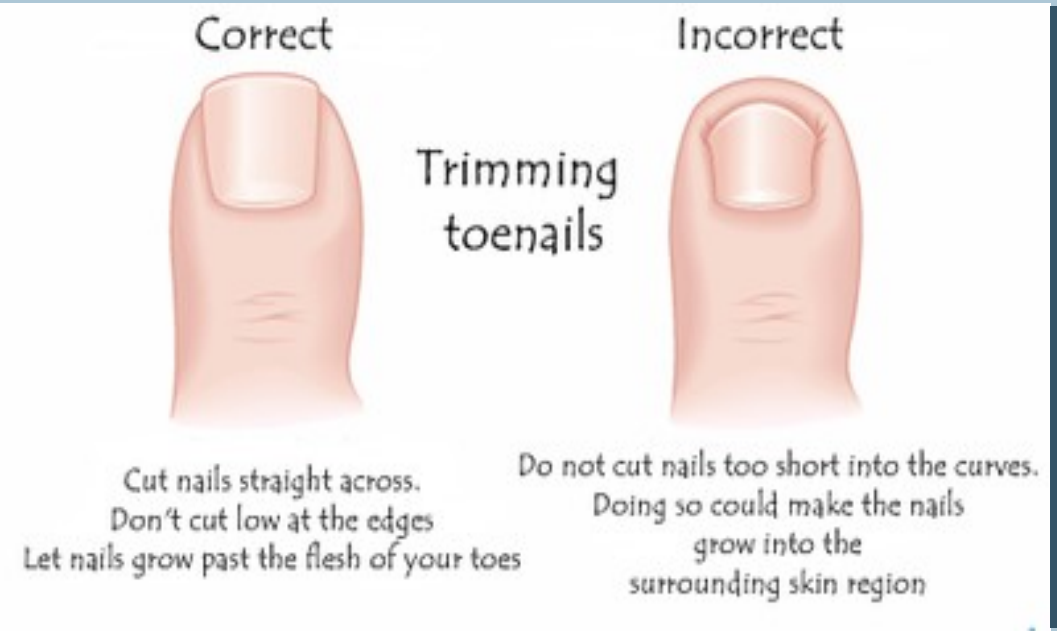
- Equipment:
  1. Tooth brush / tooth past
  2. Cup of water
  3. Sodium bicarb
  4. Tongue plaid
  5. Gauzes
  6. Lemon juice
  7. Mouth and lips lubricant (Glycerin)

Hair Care

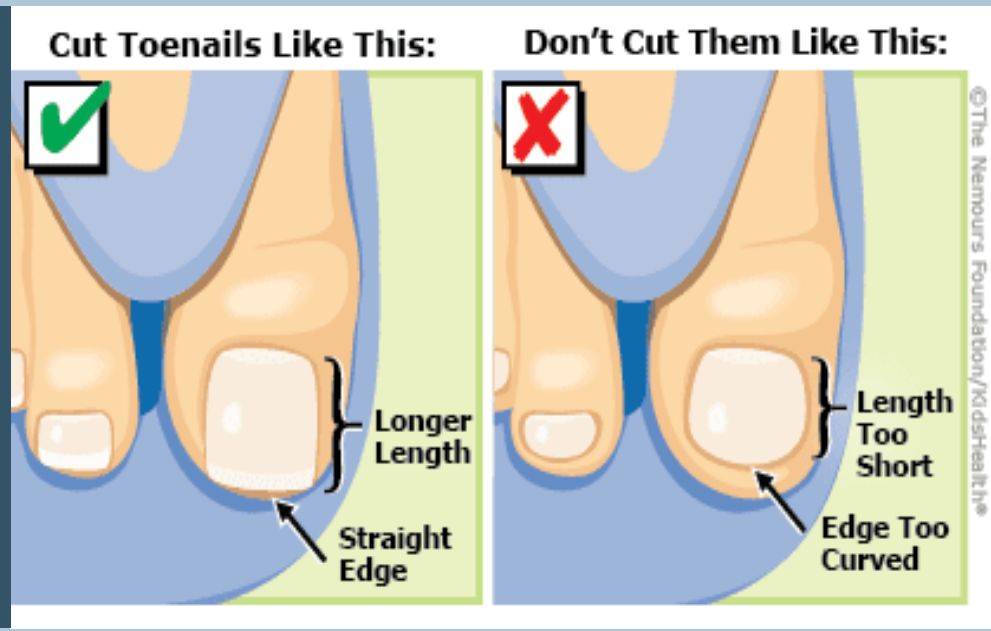
- Sometimes, clients need assistance with grooming or shampooing their hair.
  - *Try to use a hairstyle the client prefers.*
  - *Brush the hair slowly and carefully to avoid damaging the hair.*
  - *Brush the hair to increase circulation and distribution of sebum.*
  - *Use a wide-toothed comb, starting at the ends of the hair rather than from crown.*
  - *Apply a conditioner or alcohol to loosen tangles.*
  - *Use oil on the hair if it is dry. Many preparations*
- Shampooing
  - *Hair should be washed as often as necessary to keep it clean.*

Nail Care

- Nail care involves keeping the fingernails and toenails clean and trimmed.
- cares for the client's nails as follows:
- Soak the hands or feet in warm water for 15-20 ms. to soften the keratin and loosen trapped debris.
- Clean under the nails with a wooden orange.
- Push cuticles (thin edge of skin at the base of the nail) downward with a soft towel
- Cut nails in straigt line to avoid sharp or jagged points that may injure the adjacent skin.



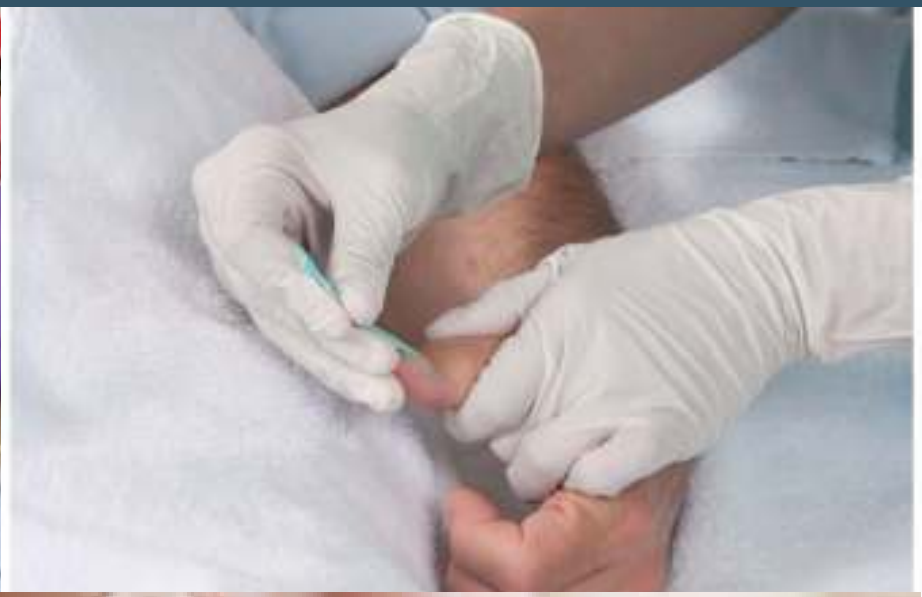
Podiatrist
(a person with special training in caring for feet) are often indicated.



Nail Care







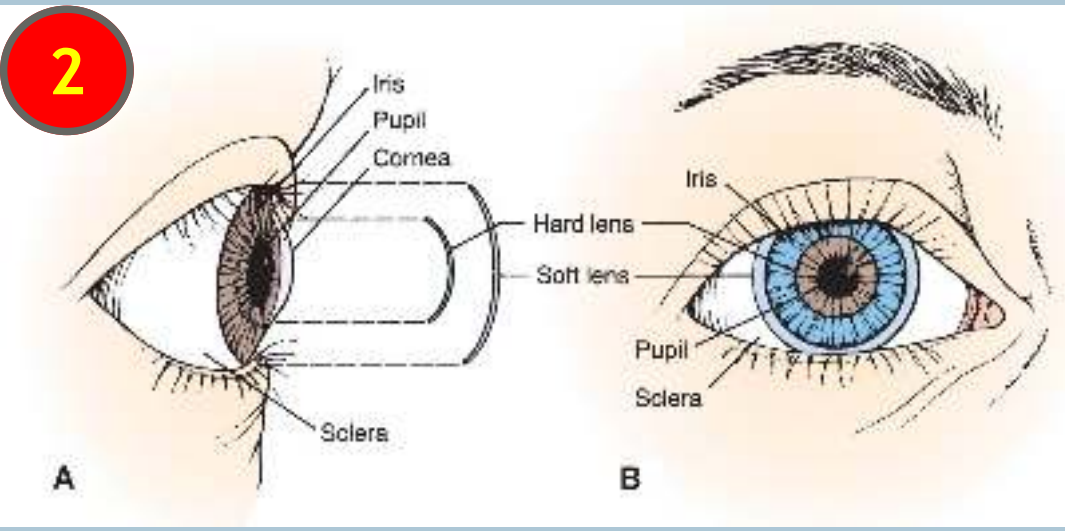
1. Soaking 15-20 ms.
2. Clean in between
3. Cut straight
4. Remove debris



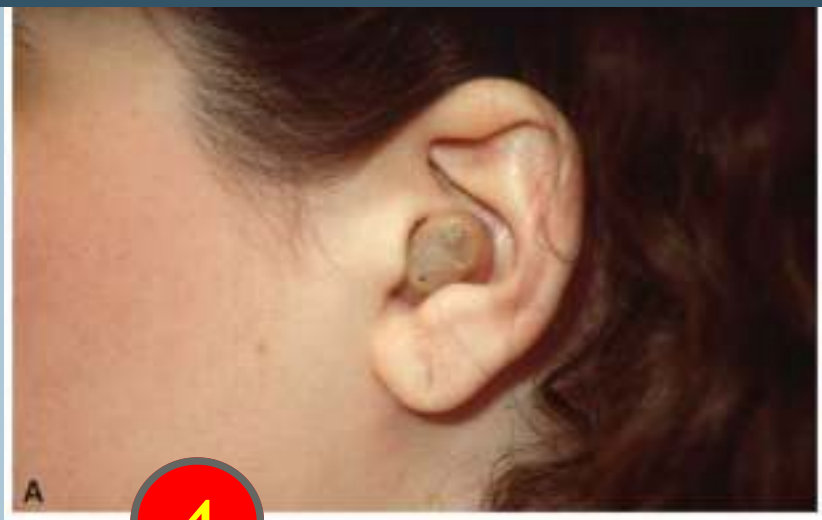


VISUAL AND HEARING DEVICES

- In Care of visual and hearing devices **follow the manufacturer's guide**:
  1. **Eyeglasses**
  2. **Contact lenses**
  3. **Prosthetic Eyes**
  4. **Hearing Aids**







4



Nursing Implications

- Clients who require assistance with personal hygiene may have a variety of health problems / nursing diagnoses:
  - *Bathing Self-Care Deficit*
  - *Dressing Self-Care Deficit*
  - *Readiness for Enhanced Self-Care*
  - *Self-neglect*
  - *Activity Intolerance*
  - *Risk for Impaired Skin Integrity*


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